IB CHEMISTRY HL

Past-Paper Review Package - Markscheme

Answers to the questions package, same labels

Multiple-choice answers are listed per lesson; structured-question answers are the official IB markscheme entries for exactly the parts included in the questions package.

	Lesson	MC	Struct.	Page
1.1	Inorganic Nomenclature	11	5	2
1.2	Stoichiometry	77	55	4
2.1	Ionization Energy	64	38	43
3.1	VSEPR Theory	49	29	71
4.1	Sigma & Pi Bonds and Delocalization	54	30	110
4.2	Hybridization	25	3	134
4.3	Covalent Network Structures	13	10	137
5.1	Molecular Polarity	9	2	147
5.2	Intermolecular Forces	39	26	149
5.3	Separation Techniques	2	20	164
6.1	Redox Reactions & Oxidation Numbers	52	18	174
6.2	Balancing Redox Reactions	5	10	191
6.3	Reactivity	14	3	196
7.1	Primary & Secondary Cells	57	36	200
7.2	Electrolytic Cells	29	29	244
7.3	Energy From Fuels	6	52	277
8.1	Thermochemical Equations	26	9	312
8.2	Calorimetry	26	20	318
9.1	Hess's Law	16	2	344
9.2	Enthalpies of Formation & Combustion	21	13	348
9.3	Bond Enthalpies	8	15	355
9.4	Born-Haber Cycles	35	4	372

LESSON 1.1

Inorganic Nomenclature

Answers for 11 multiple-choice and 5 structured questions

Part A - Multiple choice answers

1.1-M1	May 2025 TZ3 · P1A · Q6	D
1.1-M2	May 2022 TZ1 · P1 · Q9	D
1.1-M3	May 2019 TZ1 · P1 · Q10	A
1.1-M4	May 2018 TZ2 · P1 · Q9	D
1.1-M5	May 2016 · P1 · Q30	D
1.1-M6	May 2015 TZ2 · P1 · Q9	C
1.1-M7	Nov 2013 · P1 · Q9	A
1.1-M8	Nov 2013 · P1 · Q31	C
1.1-M9	May 2013 TZ1 · P1 · Q9	A
1.1-M10	Nov 2012 · P1 · Q30	B
1.1-M11	Nov 2011 · P1 · Q13	C

Part B - Structured questions markscheme

1.1-S1 May 2019 TZ1 · Paper 2 · Q4 parts (b)(v)

4.	**b**	**v**	manganese(IV) oxide ***OR*** manganese dioxide ✓	*Accept "manganese(IV) dioxide".*	**1**

1.1-S2 May 2015 TZ1 · Paper 3 · Q17 parts (b)(i)

(b) (i) $3Ca^{2+}(aq) + 2PO_4^{3-}(aq) \rightleftharpoons Ca_3(PO_4)_2(s)$; **[1]**

Ignore state symbols.
Accept single arrow sign.

1.1-S3 May 2014 TZ2 · Paper 3 · Q19 parts (a)

19. (a) $Pb^{2+}(aq) + 2Cl^{-}(aq) \rightleftharpoons PbCl_2(s)$

correct reactant ions and product;
correct state symbols; ***[2]***
Do not penalize if equilibrium sign is not given.

1.1-S4 May 2013 TZ1 · Paper 3 · QD1 parts (b)(i), (b)(iii) · also 6.1

(b) (i) $K_2Cr_2O_7$; ***[1]***

(iii) +3/III; ***[1]***
Do not allow incorrect notation such as 3+ or 3.

1.1-S5 May 2010 TZ1 · Paper 3 · QE3 parts (c)

(c) $Al^{3+} + PO_4^{3-} \rightarrow AlPO_4$; ***[1]***

LESSON 1.2

Stoichiometry

Answers for 77 multiple-choice and 55 structured questions

Part A - Multiple choice answers

1.2-M1	May 2025 TZ2 · P1A · Q23	D
1.2-M2	May 2025 TZ3 · P1A · Q1	D
1.2-M3	May 2025 TZ3 · P1A · Q30	D
1.2-M4	Nov 2024 · P1 · Q1	C
1.2-M5	Nov 2024 · P1 · Q2	D
1.2-M6	May 2024 TZ1 · P1 · Q38	A
1.2-M7	May 2024 TZ2 · P1 · Q3	A
1.2-M8	Nov 2023 TZ1 · P1 · Q1	B
1.2-M9	Nov 2023 TZ1 · P1 · Q2	D
1.2-M10	Nov 2023 TZ1 · P1 · Q3	C
1.2-M11	Nov 2023 TZ1 · P1 · Q38	A
1.2-M12	May 2023 TZ2 · P1 · Q1	D
1.2-M13	Nov 2022 · P1 · Q1	A
1.2-M14	Nov 2022 · P1 · Q3	B
1.2-M15	May 2022 TZ1 · P1 · Q1	B
1.2-M16	May 2022 TZ1 · P1 · Q2	A
1.2-M17	May 2022 TZ2 · P1 · Q1	C
1.2-M18	Nov 2021 · P1 · Q1	C
1.2-M19	Nov 2021 · P1 · Q2	B
1.2-M20	May 2021 TZ1 · P1 · Q1	D
1.2-M21	May 2021 TZ2 · P1 · Q1	C
1.2-M22	May 2021 TZ2 · P1 · Q2	B
1.2-M23	May 2021 TZ2 · P1 · Q3	D
1.2-M24	Nov 2020 · P1 · Q1	D
1.2-M25	Nov 2019 · P1 · Q1	C
1.2-M26	Nov 2019 · P1 · Q2	D
1.2-M27	Nov 2019 · P1 · Q3	A
1.2-M28	May 2019 TZ2 · P1 · Q1	C
1.2-M29	May 2019 TZ2 · P1 · Q38	B
1.2-M30	Nov 2018 · P1 · Q1	A
1.2-M31	Nov 2018 · P1 · Q3	B
1.2-M32	Nov 2018 · P1 · Q18	B
1.2-M33	May 2018 TZ1 · P1 · Q3	D
1.2-M34	May 2018 TZ2 · P1 · Q1	D
1.2-M35	May 2018 TZ2 · P1 · Q2	C
1.2-M36	May 2018 TZ2 · P1 · Q3	B
1.2-M37	Nov 2017 · P1 · Q1	D
1.2-M38	May 2017 TZ1 · P1 · Q39	D
1.2-M39	Nov 2016 · P1 · Q1	D
1.2-M40	Nov 2016 · P1 · Q3	B
1.2-M41	Nov 2015 · P1 · Q3	A
1.2-M42	May 2015 TZ1 · P1 · Q1	C
1.2-M43	May 2015 TZ1 · P1 · Q3	B
1.2-M44	Nov 2014 · P1 · Q1	D
1.2-M45	May 2014 TZ1 · P1 · Q3	A
1.2-M46	May 2014 TZ2 · P1 · Q1	D
1.2-M47	May 2014 TZ2 · P1 · Q2	C
1.2-M48	Nov 2013 · P1 · Q1	D
1.2-M49	Nov 2013 · P1 · Q2	D
1.2-M50	Nov 2013 · P1 · Q3	B
1.2-M51	Nov 2013 · P1 · Q40	C
1.2-M52	May 2013 TZ1 · P1 · Q1	C
1.2-M53	May 2013 TZ2 · P1 · Q1	C
1.2-M54	May 2013 TZ2 · P1 · Q2	B
1.2-M55	Nov 2012 · P1 · Q1	C
1.2-M56	Nov 2012 · P1 · Q2	B
1.2-M57	May 2012 TZ1 · P1 · Q1	D
1.2-M58	May 2012 TZ1 · P1 · Q2	A
1.2-M59	May 2012 TZ2 · P1 · Q1	D
1.2-M60	May 2012 TZ2 · P1 · Q2	B
1.2-M61	Nov 2011 · P1 · Q1	D
1.2-M62	Nov 2011 · P1 · Q2	D
1.2-M63	Nov 2011 · P1 · Q5	A
1.2-M64	Nov 2011 · P1 · Q40	C
1.2-M65	May 2011 TZ1 · P1 · Q1	B
1.2-M66	May 2011 TZ1 · P1 · Q2	D
1.2-M67	May 2011 TZ1 · P1 · Q4	D
1.2-M68	May 2011 TZ2 · P1 · Q1	D
1.2-M69	May 2011 TZ2 · P1 · Q2	A
1.2-M70	May 2011 TZ2 · P1 · Q3	C
1.2-M71	May 2011 TZ2 · P1 · Q40	C
1.2-M72	Nov 2010 · P1 · Q2	C
1.2-M73	May 2010 TZ1 · P1 · Q1	D
1.2-M74	May 2010 TZ1 · P1 · Q3	A
1.2-M75	May 2010 TZ2 · P1 · Q1	B
1.2-M76	May 2010 TZ2 · P1 · Q2	C
1.2-M77	May 2010 TZ2 · P1 · Q39	B

Part B - Structured questions markscheme

1.2-S1 May 2025 TZ1 · Paper 1B · Q1 parts (a)

1.	**(a)**		158.12 «g mol^{-1}» ✓ «0.5000 dm^3 * 0.1500 mol dm^{-3} * 158.12 g mol^{-1} =» 11.86 «g $Na_2S_2O_3$» ✓	*Award **[2]** for correct final answer.*	**2**

1.2-S2 May 2025 TZ2 · Paper 1B · Q1 parts (d)(iv), (d)(v)

			«% absorbed = 100 – 81.3 = » 18.7 «%» ✓		
1	**(d)**	**(iv)**	«$C_1 = \frac{C_2V_2}{V_1} = \frac{4.00\times25.00}{2.00}$ =» 50.0 «ppm» ✓		**1**
1	**(d)**	**(v)**	«$\frac{4.00\times10^{-3}}{40.08}$» 9.98×10^{-5} « mol dm^{-3}» ✓		**1**

1.2-S3 May 2025 TZ2 · Paper 1B · Q3 parts (b)

3	**(b)**		*Limiting reagent:* HCl ***AND*** *Reason:* same initial mass of solid produces different mass of CO_2 /carbon dioxide ***OR*** different concentration/initial amount in moles of HCl produced different mass of CO_2 /carbon dioxide/gas/overall mass loss ***OR*** final mass of all mixtures would be same if $NaHCO_3$ was limiting ✓	*Accept converse argument.*	**1**

1.2-S4 May 2025 TZ2 · Paper 2 · Q4 parts (b)

4	**(b)**		theoretical yield⟨⟨$= 48.73\ x\ \frac{2x334.4\ +\ 98.1}{2\ x\ 334.4}\ =\ 48.73\ x\ \frac{766.9}{668.8}$⟩⟩ = 55.87 g✓ percentage yield ⟨⟨$= 100\ x\ \frac{51.41}{55.87}$⟩⟩ «=100 x 51.41/ 55.87» = 92.00 «%» ✓	*Award **[2]** for correct final answer.*	**2**

1.2-S5 May 2025 TZ3 · Paper 1B · Q1 parts (a)(ii), (a)(iii)

1.	**(a)**	**(ii)**	«1»Cu(s) + **4**HNO_3(aq) → «1»$Cu(NO_3)_2$(aq) + **2**NO_2(g) + **2**H_2O(l)✓	*Accept any correct multiples or fractions.*	**1**
1.	**(a)**	**(iii)**	« (63.55 x 0.0100) ÷ 2» = 0.318«g»✓		**1**

1.2-S6 May 2025 TZ3 · Paper 1B · Q3 parts (b)(i), (b)(v)

3.	**(b)**	**(i)**	<table><tr><th>*Loss in mass (g)*</th><th>*Amount of H_2O /mol*</th></tr><tr><td>5.361</td><td>0.298</td></tr></table> ✓	***Both** must be to correct sig. fig.*	**1**
3.	**(b)**	**(v)**	«0.60/0.10 =» 6 ✓	*Do **not** accept non-integer values.*	**1**

1.2-S7 Nov 2024 · Paper 2 · Q7 parts (a), (b), (c) · also 6.1

7.	**(a)**		oxygen/O_2 ***AND*** gains electrons ***OR*** oxygen/ O_2 ***AND*** oxidation state decreases/changes from 0 to −2 ✓		**1**
7.	**(b)**		**«**n(O_2) = ¼ n(I^-) =**»** 1.2 x 10^{-3} **«**mol**»** ✓		**1**
7.	**(c)**		**«**$\frac{1.2\times10^{-3}\times32.00}{200\times10^{-3}}$**»** = 0.19 **«**g dm^{-3}**»** ✓	*Award **[1]** for 0.32 **«**g dm^{-3}**»** if 2.0 x 10^{-3} mol was used.*	**1**

Question			Answers	Notes	Total

1.2-S8 Nov 2024 · Paper 3 · Q1 parts (b)(i)

1.	**b**	**i**	«3.723 x 0.0052 =» 0.019 «μg» ✓	*Ignore uncertainty values.*	**1**

1.2-S9 Nov 2024 · Paper 3 · Q16 parts (a)(i), (a)(ii), (a)(iii) · also 5.3

16.	**a**	**i**	≪ $\frac{0.897\times180.17}{138.13}$ ≫= 1.17 «g» ✓		**1**
16.	**a**	**ii**	water ***AND*** incomplete drying/water retained in solid ***OR*** ethanoic acid ***AND*** not evaporated ✓	*Apply LP.* *Accept "ethanoic anhydride **AND** excess reagent".* *Accept "salicylic acid **AND** incomplete conversion/reaction".*	**1**
16.	**a**	**iii**	*Any three of:* dissolve in hot solvent ✓ filter/centrifuge «undissolved impurities» ✓ cool to reform crystals ✓ filter/centrifuge «purified product from dissolved impurities» ✓ rinse with cold solvent ✓	*Steps given must relate to the process of recrystallization.* *Do **not** award "filter" or "centrifuge" twice unless separate functions specified.* *Accept 'drying of final product'.* *Do **not** accept steps in the process of converting aspirin to sodium salt/soluble aspirin.*	**3 max**

1.2-S10 May 2024 TZ1 · Paper 2 · Q1 parts (a)(i), (a)(iii)

1.	**(a)**	**(i)**	mol (citric acid) = $\frac{0.45}{192.14}$ = 0.00234 ✓ mol ($NaHCO_3$) = $\frac{0.25}{84.01}$ = 0.00298 ✓ 0.00298÷3 = 0.000992 mol therefore $NaHCO_3$ is LR✓	*Award 1 mark for 2 correct masses of reactants or molar masses* *Accept sodium hydrogen carbonate for M3* *Only award M3 if it is correctly based on previous mole calculations*	**3**

1.	**(a)**	**(iii)**	$100 \cdot \frac{0.043}{0.0738}$ = 58.4 % ✓		**1**

1.2-S11 May 2024 TZ1 · Paper 3 · Q2 parts (a)

2.	**a**		**ALTERNATIVE 1** $n(CuSO_4) = \frac{6.4\text{ g}}{(63.55 + 32.07 + (4 \times 16.00))\text{ g mol}^{-1}} = \frac{6.4\text{ g}}{159.62\text{ g mol}^{-1}}$ $= 0.040$ «mol» ***AND*** $n(H_2O) = \frac{(10.0 - 6.4)\text{ g}}{(2 \times 1.01) + 16.00\text{ g mol}^{-1}} = \frac{3.6\text{ g}}{18.02\text{ g mol}^{-1}} = 0.20$ «mol» ✓ $n(CuSO_4) : n(H_2O) = \frac{0.20\text{ mol}}{0.04\text{ mol}} = 5.0$ ✓ **ALTERNATIVE 2** % $CuSO_4$ in $CuSO_4 \cdot 5H_2O$ $= 100 \times \frac{63.55 + 32.07 + (4 \times 16.00)\text{ g mol}^{-1}}{63.55 + 32.07 + (9 \times 16.00) + (10\text{x}1.01)\text{ g mol}^{-1}} = \frac{15962\text{ g mol}^{-1}}{249.72\text{ g mol}^{-1}}$» $= 63.92$ ✓ mass of $CuSO_4 = 10.0\text{ g} \times \frac{63.92}{100} = 6.392\text{ g}$ ✓ **ALTERNATIVE 3** mass of H_2O in $CuSO_4 \cdot 5H_2O$ $= 10\text{ g} \times \frac{(5 \times 16.00) + (10 \times 1.01)\text{ g mol}^{-1}}{(63.55 + 32.07 + (9 \times 16.00) + (10 \times 1.01))\text{ g mol}^{-1}} = \frac{900.1\text{ g}}{249.72}$» $= 3.6$ g ✓ mass of $CuSO_4$ in $CuSO_4 \cdot 5H_2O$ «$= 10.0\text{ g} - 3.6\text{ g}$» $= 6.4$ g ***OR*** comparison with graph shows mass lost is 3.6 g ✓	*Accept other valid methods.*	**2**

(continued…

1.2-S12 Nov 2023 TZ1 · Paper 2 · Q3 parts (b)

3	**b**		***ALTERNATIVE 1*** expected yield «$=2.83 \times \frac{60.06}{46.03}$» = 3.69 **«g»** ✓ percentage yield «$=100 \times \frac{1.72}{3.69}$» = 46.6 **«%»** ✓ ***ALTERNATIVE 2*** **«**amount of methanoic acid used = $\frac{2.83}{46.03}$ **=»** 0.0615 **«**mol**»** ✓ **«**expected amount of methyl methanoate = 0.0615 mol**»** **«**actual amount of methyl methanoate = $\frac{1.72}{60.06}$ = 0.0286 mol**»** percentage yield **«=**$\frac{0.0286}{0.0615} \times 100$**»** = 46.5% ✓	*Award **[2]** for correct final answer.* *Award **[0]** for 60.8% (simple ratio of starting and final masses).*	**2**

1.2-S13 May 2023 TZ1 · Paper 3 · Q14 parts (c)

14.	**(c)**		«M_r CH_4: M_r CO_2/ 16.05 ÷ 44.01 =» 0.365 ✓ «21 × 0.365 = » 7.66 «times» ✓	*Award **[2]** for correct final answer.* *Accept 7.64* «times» *(integer M_rs).* *Award **[1 max]** for 57.6* «times» if inverted ratio is used.	**2**

1.2-S14 May 2023 TZ2 · Paper 2 · Q5 parts (a)(iii)

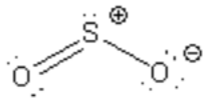
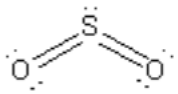
1.2-S15 May 2023 TZ2 · Paper 3 · Q2 parts (a)(ii), (c)(iii)

2.	**(a)**	**(ii)**	none / no effect ✓		**1**

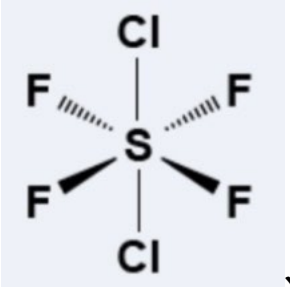
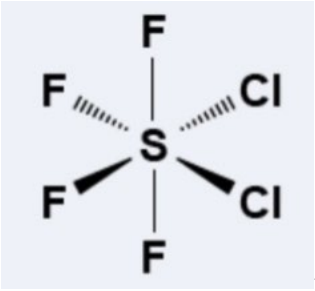
2.	**(c)**	**(iii)**	***ALTERNATIVE 1*** mass Fe in the 79.6 g kale ⟨⟨$= 8.66\text{x}10^{-4} \times \frac{500}{10.0}$⟩⟩ = 0.0433 «g» ✓ percent by mass ⟨⟨$= \frac{0.0433}{79.6} \times 100$ ⟩⟩ = 0.0544«%» ✓ ***ALTERNATIVE 2*** mass of kale in titration flask ⟨⟨$= 79.6 \times \frac{10.0}{500}$⟩⟩ = 1.592 «g» ✓ percent by mass ⟨⟨$= \frac{8.66\text{x}10^{-4}}{1.592} \times 100$ ⟩⟩ = 0.0544«%» ✓	*Award **[2]** for correct final answer.* *For **ALTERNATIVE 2:*** *Award M1 for either 1.59 «g» **OR** 1.592 «g» **and*** *Award M2 for 0.0545«%» **OR** 0.0544«%».* *M2 must be to 3 sig. fig.* *Award **[1 max]** for 0.00109«%».*	**2**

1.2-S16 Nov 2022 · Paper 2 · Q5 parts (a), (c), (f) · also 3.1, 5.1

5.	**(a)**		«$0.40\% \times 500.0g =$ » 2.0 «g» ✓ «$2.0\ g \times \frac{1\ mol\ S}{32.07\ g} = 0.062\ mol\ of\ S$» = 0.062 «mol of SO_2» ✓	*Award **[2]** for correct final answer.* *Accept 0.063 «mol».*	**2**

5.	**(c)**		 ***OR***  ✓	*Do **not** penalise missing formal charges.*	**1**

			Any two of:	*Accept same size flask*	

5.	**(f)**		*Non-polar isomer:*  ✓ *Polar isomer:*  ✓	*Accept other methods of clearly representing 3D structure.*	**2**

1.2-S17 May 2022 TZ1 · Paper 2 · Q1 parts (a)(i), (b)(i), (b)(iii), (c)(i), (c)(ii), (d)(i), (d)(iii), (e)(iv) · also 2.1, 6.1, 7.3

1.	**a**	**i**	$2\ Mg(s) + O_2(g) \rightarrow 2\ MgO(s)$ ✓	*Do not accept equilibrium arrows.* *Ignore state symbols.*	**1**

1.	**b**	**i**	$\langle\langle \frac{53.726\ g - 47.372\ g}{24.31\ g\ mol^{-1}} = \frac{6.354\ g}{24.31\ g\ mol^{-1}} \rangle\rangle$ = 0.2614 «mol» ✓		**1**

1.	**b**	**iii**	⟨⟨ $0.2614\ \text{mol x}\ (24.31\ \text{g mol}^{-1} + 16.00\ \text{g mol}^{-1}) = 0.2614\ \text{mol x}\ 40.31\ \text{g mol}^{-1}$⟩⟩ = 10.536 «g» ✓ ⟨⟨$100\ \text{x}\ \frac{\mathbf{9.569\ g}}{\mathbf{10.536g}} = 90.822$⟩⟩ = 91 «%» ✓	*Award 0.2614 mol x 40.31 g mol^{-1}.* *Accept alternative methods to arrive at the correct answer.* *Accept final answer in the range 90.5-91.5%.* ***[2]** for correct final answer.*	**2**
1.	**c**	**i**	yes ***AND*** «each Mg combines with $^2/_3$ N, so» mass increase would be 14x$^2/_3$ which is less than expected increase of 16x ***OR*** 3 mol Mg would form 101g of Mg_3N_2 but would form 3 x MgO = 121 g of MgO ***OR*** 0.2614 mol forms 10.536 g of MgO, but would form 8.796 g of Mg_3N_2 ✓	*Accept Yes **AND** "the mass of N/N_2 that combines with each g/mole of Mg is lower than that of O/O_2"* *Accept YES **AND** "molar mass of nitrogen less than of oxygen".*	**1**
1.	**c**	**ii**	incomplete reaction ***OR*** Mg was partially oxidised already ***OR*** impurity present that evaporated/did not react ✓	*Accept "crucible weighed before fully cooled".* *Accept answers relating to a higher atomic mass impurity consuming less O/O_2.* *Accept "non-stoichiometric compounds formed".* *Do **not** accept "human error", "wrongly calibrated balance" or other non-chemical reasons.* *If answer to (b)(iii) is >100%, accept appropriate reasons, such as product absorbed moisture before being weighed.*	**1**
1.	**d**	**i**	«1» $Mg_3N_2(s)$ + **6** $H_2O(l) \rightarrow$ **3** $Mg(OH)_2(s)$ + **2** $NH_3(aq)$ ✓		**1**
1.	**d**	**iii**	*Mg_3N_2:* -3 ***AND*** *NH_3:* -3 ✓	*Do not Accept 3 or 3-.*	**1**

1.	**e**	**iv**	nitrogen ***AND*** electron lost from first *«energy»* level/s sub-level/s-orbital ***AND*** magnesium from p sub-level/p-orbital/second *«energy»* level ***OR*** nitrogen ***AND*** electron lost from lower level «than magnesium» ✓	*Accept "nitrogen **AND** electron lost closer to the nucleus «than magnesium»".*	**1**

1.2-S18 May 2022 TZ2 · Paper 2 · Q1 parts (a), (b)(i), (c) · also 6.1

1.	**a**		2 Li (s) + 2 H_2O (l) → 2 LiOH (aq) + H_2 (g) ✓		**1**
1.	**b**	**i**	n_{Li}« $\frac{0.200\ g}{6.94\ g}$ = »0.0288 «mol» ✓ «n_{LiOH} = n_{Li}» [LiOH] «= $\frac{0.0288\ mol}{0.5000\ dm^3}$ = » 0.0576 «$mol\ dm^{-3}$» ✓	*Award **[2]** for correct final answer.*	**2**

1.	**c**		H_2O ***AND*** hydrogen gains electrons «to form H_2» ***OR*** H_2O ***AND*** H oxidation state changed from +1 to 0 ✓	*Accept "H_2O **AND** H/H_2O is reduced".*	**1**

1.2-S19 Nov 2021 · Paper 2 · Q6 parts (c)(i), (c)(ii), (c)(iii), (c)(iv) · also 6.2

6.	**c**	**i**	«$\frac{37.50\,\text{cm}^3}{1000} \times 5.000 \times 10^{-4}\,\text{mol dm}^{-3}$ =» 1.875×10^{-5} «mol» ✓		**1**
6.	**c**	**ii**	1:4 ✓	*Accept "4 mol $S_2O_3^{2-}$:1 mol O_2", but not just 4:1.*	**1**
6.	**c**	**iii**	«$1.875 \times 10^{-5}\,\text{mol} \times \frac{1}{4}$ =» 4.688×10^{-6} «mol» ✓ «$\frac{4.688 \times 10^{-6}\,\text{mol}}{\frac{25.00\,\text{cm}^3}{1000}}$ =» 1.875×10^{-4} «mol dm^{-3}» ✓	*Award **[2]** for correct final answer.*	**2**
6.	**c**	**iv**	$MnO_2\,(s) + 2e^- + 4H^+\,(aq) \rightarrow Mn^{2+}\,(aq) + 2H_2O\,(l)$ ✓		**1**

1.2-S20 Nov 2020 · Paper 2 · Q1 parts (a)(i), (b)(i), (b)(ii), (b)(iii), (b)(v), (b)(vi) · also 2.1, 6.1

1.	**a**	**i**	$1s^2 2s^2 2p^6 3s^2 3p^5$ ✓	*Do **not** accept condensed electron configuration.*	**1**
1.	**b**	**i**	«$\frac{2.67\,g}{86.94\,g\,mol^{-1}}$ =» 0.0307 «mol» ✓		**1**
1.	**b**	**ii**	«n_{HCl} = 2.00 mol dm^{-3} x 0.2000 dm^3» = 0.400 mol ✓ «$\frac{0.400}{4}$ =» 0.100 mol ***AND*** MnO_2 is the limiting reactant ✓	*Accept other valid methods of determining the limiting reactant in M2.*	**2**
1.	**b**	**iii**	«0.0307 mol × 4 = 0.123 mol» «0.400 mol – 0.123 mol =» 0.277 «mol» ✓		**1**
1.	**b**	**v**	MnO_2: +4 ✓ $MnCl_2$: +2 ✓		**2**
1.	**b**	**vi**	oxidizing agent ***AND*** oxidation state of Mn changes from +4 to +2/decreases ✓		**1**

1.2-S21 Nov 2019 · Paper 2 · Q2 parts (a)(i), (a)(ii), (a)(iii)

2.	**a**	**i**	4 : 1 ✓		**1**
2.	**a**	**ii**	$n_{S_2O_3^{2-}}$ = «0.0258 dm^3 × 0.010 mol dm^{-3} =» 2.58 × 10^{-4} «mol» ✓ « $\frac{2.58\times10^{-4}\text{ mol}}{4}$ =» 6.45 × 10^{-5} «mol» ✓	*Award **[2]** for correct final answer.*	**2**
2.	**a**	**iii**	«difference in moles per dm^3 = (6.45 × 10^{-5} – 5.03 × 10^{-5}) × $\frac{1000}{300.0}$ =» 4.73 × 10^{-5} «mol dm^{-3}» ✓ «convert to mg per dm^3: 4.73 × 10^{-5} mol dm^{-3} × 32.00 g mol^{-1} × 1000 mg g^{-1} = » 1.51 «ppm/mg dm^{-3}» ✓	*Award **[2]** for correct final answer.*	**2**

1.2-S22 Nov 2019 · Paper 2 · Q7 parts (a)(i), (a)(ii)

7.	**a**	**i**	$C(NH_2)_3NO_3\,(s) \rightarrow 2N_2\,(g) + 3H_2O\,(g) + C\,(s)$ ✓		**1**
7.	**a**	**ii**	moles of gas = «5 × $\frac{10.0\text{ g}}{122.11\text{ g mol}^{-1}}$ =» 0.409 «mol» ✓		**1**

1.2-S23 May 2019 TZ2 · Paper 2 · Q5 parts (d)(iii)

5.	**d**	**iii**	«molar mass of $NaHCO_3$ =» 84.01 «g mol^{-1}» ✓ «concentration = $\frac{3.0\times10^{-2}\text{ g}}{84.01\text{ g mol}^{-1}}\times\frac{1}{0.100\text{ dm}^3}$ =» 3.6 × 10^{-3} «mol dm^{-3}» ✓	*Award **[2]** for correct final answer.*	**2**

1.2-S24 May 2019 TZ2 · Paper 3 · Q1 parts (c)(ii)

1.	**c**	**ii**	«daily limit = 5.0×10^{-6} g $kg^{-1} \times 80.0$ kg =» 4.0×10^{-4} «g of lead» ✓ «volume = $\frac{4.0 \times 10^{-4}\ g}{1.5 \times 10^{-2}\ g\ dm^{-3}}$ =» 2.7×10^{-2}/0.027 «dm^3» ✓	*Award **[2]** for correct final answer*	**2**

1.2-S25 May 2019 TZ2 · Paper 3 · Q22 parts (b)(i), (b)(ii)

22.	**b**	**i**	n(aspirin) «= n(NaOH) = $\frac{16.25\ cm^3}{1000} \times 0.100\ mol\ dm^{-3}$ »= 1.625×10^{-3} «mol» ✓ m(aspirin) «= 1.625×10^{-3} mol × 180.17 g mol^{-1} »= 0.293 «g» ✓	*Award **[2]** for correct final answer.*	**2**
22.	**b**	**ii**	« $\frac{0.293\ g}{0.300\ g} \times 100\ \%$ »= 97.7 «%» ✓		**1**

1.2-S26 Nov 2018 · Paper 2 · Q1 parts (a)(i), (a)(ii), (b)(i), (b)(ii), (d) · also 7.2, 8.2

1.	**a**	**i**	n_{CuSO4} **«**$= 0.0800\ dm^3 \times 0.200\ mol\ dm^{-3}$**»** $= 0.0160$ mol ***AND*** n_{Fe} **«**$= \frac{3.26\ g}{55.85\ g\ mol^{-1}}$**»** $=$ 0.0584 mol ✓ $CuSO_4$ is the limiting reactant ✓	*Do **not** award M2 if mole calculation is not shown.*	**2**
1.	**a**	**ii**	***ALTERNATIVE 1:*** **«**$0.0160\ mol \times 63.55\ g\ mol^{-1} =$**»** 1.02 **«**g**»** ✓ **«**$\frac{0.872\ g}{1.02\ g} \times 100 =$**»** 85.5 **«**%**»** ✓ ***ALTERNATIVE 2:*** **«**$\frac{0.872\ g}{63.55\ g\ mol^{-1}} =$**»** 0.0137 **«**mol**»** ✓ **«**$\frac{0.0137\ mol}{0.0160\ mol} \times 100 =$**»** 85.6 **«**%**»** ✓	*Accept answers in the range 85–86 %.* *Award **[2]** for correct final answer.*	**2**

1.	**b**	**i**	***ALTERNATIVE 1:*** $q = $ **«**$80.0\,g \times 4.18\,J\,g^{-1}\,K^{-1} \times 7.5\,K =$**»** 2.5×10^3 **«J»**/2.5 **«kJ»** ✓ **«**per mol of $CuSO_4 = \frac{-2.5\,kJ}{0.0160\,mol} = -1.6 \times 10^2\,kJ\,mol^{-1}$**»** **«**for the reaction**»** $\Delta H = -1.6 \times 10^2$ **«**kJ**»** ✓ ***ALTERNATIVE 2:*** $q = $ **«**$80.0\,g \times 4.18\,J\,g^{-1}\,K^{-1} \times 7.5\,K =$**»** 2.5×10^3 **«J»**/2.5 **«kJ»** ✓ **«**$n_{Cu} = \frac{0.872}{63.55} = 0.0137\,mol$**»** **«**per mol of $CuSO_4 = \frac{-2.5\,kJ}{0.0137\,mol} = -1.8 \times 10^2\,kJ\,mol^{-1}$**»** **«**for the reaction**»** $\Delta H = -1.8 \times 10^2$ **«**kJ**»** ✓	*Award **[2]** for correct final answer.*	**2**
1.	**b**	**ii**	density **«**of solution**»** is $1.00\,g\,cm^{-3}$ ***OR*** specific heat capacity **«**of solution**»** is $4.18\,J\,g^{-1}\,K^{-1}$/that of **«**pure**»** water ***OR*** reaction goes to completion ***OR*** iron/$CuSO_4$ does not react with other substances ✓	*The mark for "reaction goes to completion" can only be awarded if 0.0160 mol was used in part (b)(i).* *Do **not** accept "heat loss".*	**1**

1.	**d**		*Anode (positive electrode):* $2H_2O\,(l) \rightarrow O_2\,(g) + 4H^+\,(aq) + 4e^-$ ✓ *Cathode (negative electrode):* $2H_2O\,(l) + 2e^- \rightarrow H_2\,(g) + 2OH^-\,(aq)$ ***OR*** $2H^+\,(aq) + 2e^- \rightarrow H_2\,(g)$ ✓	*Accept "$4OH^-\,(aq) \rightarrow O_2\,(g) + 2H_2O\,(l) + 4e^-$"* ***OR*** *"$Fe^{2+}\,(aq) \rightarrow Fe^{3+}\,(aq) + e^-$" for M1.* *Accept "$Fe^{2+}\,(aq) + 2e^- \rightarrow Fe\,(s)$" **OR** "$SO_4^{2-}\,(aq) + 4H^+\,(aq) + 2e^- \rightarrow 2H_2SO_3(aq) + H_2O\,(l)$" for M2.*	**2**

1.2-S27 Nov 2018 · Paper 3 · Q3 parts (b)(i)

3.	**b**	**i**	$[Pb^{2+}] = 0.50 \times 10^{-6}/5.0 \times 10^{-7}$ **«**g dm^{-3}**»** ✓ $[Pb^{2+}]$ **«**$= \frac{0.50 \times 10^{-6}\ \text{g dm}^{-3}}{207.20\ \text{g mol}^{-1}}$**»** $= 2.4 \times 10^{-9}$ **«**mol dm^{-3}**»** ✓	*Award **[2]** for correct final answer.*	**2**

1.2-S28 Nov 2018 · Paper 3 · Q18 parts (c)

18.	**c**		**«**back**»** titration ***OR*** thermal decomposition ***OR*** atomic absorption/AA ✓	*Accept "gravimetric analysis".* *Do **not** accept description of a technique without proper term given for the technique.*	**1**

1.2-S29 May 2018 TZ1 · Paper 3 · Q2 parts (a)

2.	**a**		$CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + CO_2(g) + H_2O(l)$ ✓	*Accept "$CO_2(aq)$".*	**1**

1.2-S30 May 2018 TZ2 · Paper 2 · Q1 parts (a)

1.	**a**		$n(H_2SO_4)$ **«**$= 0.0500\ dm^3 \times 0.100\ mol\ dm^{-3}$**»** $= 0.00500/5.00 \times 10^{-3}$**«**mol**»** ✓		**1**

1.2-S31 May 2018 TZ2 · Paper 3 · Q4 parts (a)(iii)

4.	**a**	**iii**	$m =$ **«**$\frac{50.94}{6.02 \times 10^{23}}$ **=»** 8.46×10^{-23} **«**g**»** ✓		**1**

1.2-S32 May 2017 TZ2 · Paper 2 · Q8 parts (b)(i), (b)(ii)

	b	**i**	**«**$0.100\ mol\ dm^{-3} \times 0.0250\ dm^3$**»** $= 0.00250$ **«**mol**»** ✓		**1**
	b	**ii**	**«**$M =$ $\frac{0.510\,g}{0.00250\,mol}$ **=»** 204 **«**g mol^{-1}**»** ✓		**1**

1.2-S33 May 2017 TZ2 · Paper 3 · Q1 parts (b)

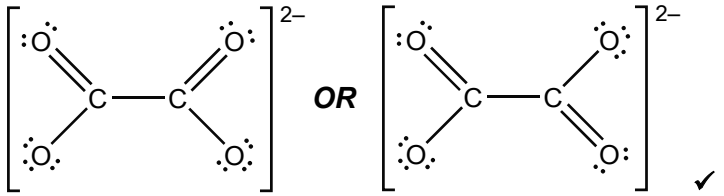
1.	**b**		**«**$18000 = 0.54x - 2000$**»** $x = 37037$ **«**million tonnes of CO_2**»** ✓ **«**$\frac{32.00}{44.01} \times 37037 = 26930$**»** 27000/2.7 x 10^4 **«**million tonnes of O_2**»** ✓	*Accept "37000* **«**million tonnes of CO_2**»***" for M1.* *Award **[2]** for correct final answer with two significant figures.* *Award **[1]** for non rounded answers in range 26903–26936* **«**million tonnes of O_2**»**.	**2**

1.2-S34 May 2017 TZ2 · Paper 3 · Q20 parts (a)(i), (a)(ii)

20.	**a**	**i**	$n(\text{salicylic acid}) = \text{«}\frac{2.65\text{ g}}{138.13\text{ g mol}^{-1}}\text{=»}\ 0.0192\ \text{«mol»}$ ***AND*** $n(\text{ethanoic anhydride}) = \text{«}\frac{2.51\text{ g}}{102.10\text{ g mol}^{-1}}\text{=»}\ 0.0246\ \text{«mol»}$ ✓	*Accept integer values for molar masses in (a) (i) and (ii).*	**1**
20.	**a**	**ii**	«mass = 0.0192 mol × 180.17 g mol^{-1} =» 3.46 «g» ✓	*Award ECF mark **only** if limiting reagent determined in (i) has been used.*	**1**

1.2-S35 Nov 2016 · Paper 2 · Q2 parts (c)(ii), (c)(iii), (c)(iv), (d), (e) · also 3.1, 4.1

2.	**c**	**ii**	«$n(NaOH) = \left(\frac{14.0}{1000}\right) dm^3 \times 0.100\,mol\,dm^{-3}$ =» 1.40×10^{-3} «mol» ✓		**1**
2.	**c**	**iii**	«$\frac{1}{2} \times 1.40 \times 10^{-3}$ =» 7.00×10^{-4} «mol» ✓		**1**
2.	**c**	**iv**	***ALTERNATIVE 1:*** «mass of pure hydrated ethanedioic acid in each titration = $7.00 \times 10^{-4}\,mol \times 126.08\,g\,mol^{-1}$ =» 0.0883 / 8.83×10^{-2} «g» ✓ mass of sample in each titration = «$\frac{25}{1000} \times 5.00\,g$ =» 0.125 «g» ✓ «% purity = $\frac{0.0883\,g}{0.125\,g} \times 100$ =» 70.6 «%» ✓ ***ALTERNATIVE 2:*** «mol of pure hydrated ethanedioic acid in 1 dm^3 solution = $7.00 \times 10^{-4} \times \frac{1000}{25}$ =» 2.80×10^{-2} «mol» ✓ «mass of pure hydrated ethanedioic acid in sample = $2.80 \times 10^{-2}\,mol \times 126.08\,g\,mol^{-1}$ =» 3.53 «g» ✓ «% purity = $\frac{3.53\,g}{5.00\,g} \times 100$ =» 70.6 «%» ✓ ***ALTERNATIVE 3:*** mol of hydrated ethanedioic acid (assuming sample to be pure) = $\frac{5.00\,g}{126.08\,g\,mol^{-1}} = 0.03966$ «mol» ✓ actual amount of hydrated ethanedioic acid = «$7.00 \times 10^{-4} \times \frac{1000}{25}$ =» 2.80×10^{-2} «mol» ✓ «% purity = $\frac{2.80 \times 10^{-2}}{0.03966} \times 100$ =» 70.6 «%» ✓	*Award suitable part marks for alternative methods.* *Award **[3]** for correct final answer.* *Award **[2 max]** for 50.4 % if anhydrous ethanedioic acid assumed.*	**3**

2.	**d**		 ***OR*** ✓	*Accept single negative charges on two O atoms singly bonded to C.* *Do not accept resonance structures.* *Allow any combination of dots/crosses or lines to represent electron pairs.*	**1**
2.	**e**		electrons delocalized **«**across the O–C–O system**»** ***OR*** resonance occurs ✓ 122 **«**pm**»** < C–O < 143 **«**pm**»** ✓	*Accept delocalized π-bond(s).* *No ECF from (d).* *Accept any answer in range 123 **«**pm**»** to 142 **«**pm**»**.* *Accept "bond intermediate between single and double bond" or "bond order 1.5".*	**2**

1.2-S36 Nov 2016 · Paper 2 · Q3 parts (a), (c)

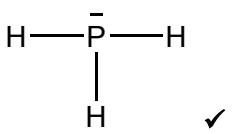
3.	**a**		H_2O ***AND*** (l) ✓	*Do **not** accept H_2O (aq).*	**1**

3.	**c**		$n(HCl) = \text{«}\frac{10.0}{1000}\,dm^3 \times 2.00\,mol\,dm^{-3} =\text{»}\ 0.0200 / 2.00 \times 10^{-2}$ **«**mol**»** ***AND*** $n(Na_2S_2O_3) = \text{«}\frac{50}{1000}\,dm^3 \times 0.150\,mol \times dm^{-3} =\text{»}\ 0.00750 / 7.50 \times 10^{-3}$ **«**mol**»** ✓ 0.0200 **«**mol**»** > 0.0150 **«**mol**»** ***OR*** 2.00×10^{-2} **«**mol**»** > $2 \times 7.50 \times 10^{-3}$ **«**mol**»** ***OR*** $\frac{1}{2} \times 2.00 \times 10^{-2}$ **«**mol**»** > 7.50×10^{-3} **«**mol**»** ✓	*Accept answers based on volume of solutions required for complete reaction.* *Award **[2]** for second marking point.* *Do **not** award M2 unless factor of 2 (or half) is used.*	**2**

1.2-S37 Nov 2016 · Paper 3 · Q25 parts (a), (b) · also 5.3

25.	**a**		***ALTERNATIVE 1:*** «theoretical yield $= \frac{1.552\,\text{g}}{138.13\,\text{g mol}^{-1}} \times 180.17\,\text{g mol}^{-1}$ =» 2.024 «g» ✓ «experimental yield $= \frac{1.124\,\text{g}}{2.024\,\text{g}} \times 100$ =» 55.53 «%» ✓ ***ALTERNATIVE 2:*** « $\frac{1.552\,\text{g}}{138.13\,\text{g mol}^{-1}}$ » = 0.01124 «mol salicylic acid/aspirin theoretical» ***AND*** « $\frac{1.124\,\text{g}}{180.17\,\text{g mol}^{-1}}$ » = 0.006239 «mol aspirin experimental» ✓ «experimental yield $= \frac{0.006239\,\text{mol}}{0.01124\,\text{mol}}$ x 100 =» 55.51 «%» ✓	*Accept answers in the range 55.4 % to 55.7 %.* *Award **[2]** for correct final answer.*	**2**
25.	**b**		low temperature gives greater difference between solubility of aspirin and impurities ***OR*** «product» crystallizes out from cold solution/«ice-cold water/lower temperature» speeds up crystallization process ***OR*** aspirin/product has low solubility «in water» at low temperatures ✓		**1**

1.2-S38 May 2016 · Paper 2 · Q1 parts (a)(i), (a)(ii), (a)(iv), (a)(v), (b)(i), (b)(iii), (b)(iv), (c)(i), (c)(ii), (c)(iii), (d)(i) · also 3.1, 4.2, 5.1, 5.2, 6.1, 8.2

1.	**a**	**i**	 ✓	*Accept structures using dots and/or crosses to indicate bonds and/or lone pair.*	**1**
1.	**a**	**ii**	sp^3 ✓	*Do not allow **ECF** from a (i).*	**1**

1.	**a**	**iv**	non-polar ***AND*** P and H have the same electronegativity ✓	*Accept "similar electronegativities".* *Accept "polar" if there is a reference to a small difference in electronegativity and apply **ECF** in 1 a (v).*	**1**
1.	**a**	**v**	PH_3 has London «dispersion» forces ✓ NH_3 forms H-bonds ✓ H-bonds are stronger ***OR*** London forces are weaker ✓	*Accept van der Waals' forces, dispersion forces and instantaneous dipole – induced dipole forces.* *Accept "dipole-dipole forces" as molecule is polar.* *H-bonds in NH_3 (only) must be mentioned to score **[2]**.* *Do **not** award M2 or M3 if:* • *implies covalent bond is the H-bond* • *implies covalent bonds break.* *Accept "dipole-dipole forces are weaker".*	**2 max**

1.	**b**	**i**	P_4 is a molecule «comprising 4P atoms» ***AND*** 4P is four/separate «P» atoms ***OR*** P_4 represents «4P» atoms bonded together ***AND*** 4P represents «4» separate/non-bonded «P» atoms ✓		**1**

1.	**b**	**iii**	P_4: 0 ✓ $H_2PO_2^-$: +1 ✓	*Do not accept 1 or 1+ for $H_2PO_2^-$.*	**2**
1.	**b**	**iv**	oxygen gained, so could be oxidation ✓ hydrogen gained, so could be reduction ***OR*** negative charge «on product/$H_2PO_2^-$»/gain of electrons, so could be reduction ✓ oxidation number increases so must be oxidation ✓	*Award **[1 max]** for M1 and M2 if candidate displays knowledge of at least two of these definitions but does not apply them to the reaction.* *Do not award M3 for "oxidation number changes".*	**3**
1.	**c**	**i**	«$\left\langle \frac{2.478}{4 \times 30.97} \right\rangle$» $= 0.02000$ «mol» ✓		**1**
1.	**c**	**ii**	n(NaOH) = «$0.1000 \times 5.00 =$» 0.500 «mol» ***AND*** P_4/phosphorus is limiting reagent ✓	*Accept $n(H_2O) = \frac{100}{18} = 5.50$ **AND** P_4 is limiting reagent.*	**1**
1.	**c**	**iii**	amount in excess «$= 0.500 - (3 \times 0.02000)$» $= 0.440$ «mol» ✓		**1**

1.	**d**	**i**	temperature rise «$= \frac{750 \times 1.00}{0.2000 \times 1.00}$» $= 3750$ «°C/K» ✓	*Do not accept –3750.*	**1**

1.2-S39 Nov 2015 · Paper 2 · Q1 parts (a)(i), (a)(ii), (b), (c)(iii), (d), (e) · also 6.1

1. (a) (i) IO_3^- *to* I_2 : V/+5 to 0;
I^- *to* I_2 : –I/–1 to 0; ***[2]***
Accept change in oxidation number –5 and +1.
Penalize incorrect notation such as 5+ or 5 once only.

(ii) *Oxidizing agent*: IO_3^-/iodate **and** *Reducing agent*: I^-/iodide; ***[1]***

(b) $0.250 \times 2.00 \times 10^{-3} / 5.00 \times 10^{-4}$ (mol of KIO_3);
$(5.00 \times 10^{-4} \times 214.00 =) 0.107\,(g)$; ***[2]***
*Award **[2]** for the correct final answer.*

(iii) $1.44 \times 10^{-5}\,(\text{mol})$; ***[1]***

(d) IO_3^- : $3C_6H_8O_6$ / 1:3 mole ratio;

$(1.44 \times 10^{-5}\,\text{mol} \times 3 =)\ 4.32 \times 10^{-5}\,(\text{mol})$; ***[2]***

*Award **[2]** for the correct final answer.*
*Award **[1 max]** for "4.80 x 10^{-6} (mol)" obtained from reversed ratio, 3:1.*

(e) $(4.32 \times 10^{-5} \times 176.14 =)\ 7.61 \times 10^{-3}\,(\text{g})$; ***[1]***

Accept $M_r = 176$ and mass $= 7.60 \times 10^{-3}$(g).

1.2-S40 Nov 2015 · Paper 2 · Q9 parts (a)(ii)

(ii) $(1.35 \times 10^{-2} - 7.36 \times 10^{-3} =)\ 6.14 \times 10^{-3}$ / $6.1 \times 10^{-3}\,(\text{mol})$; ***[1]***

1.2-S41 May 2015 TZ1 · Paper 2 · Q1 parts (a), (b)(i), (b)(ii), (b)(iii), (b)(iv), (c), (d), (e) · also 3.1, 4.1, 5.2

1. (a) $\left(\frac{(12.70+12.50)}{2}=\right)12.60\,(\text{cm}^3)$;

$(0.01260\times0.100=)1.26\times10^{-3}\,(\text{mol})$; **[2]**

*Award **[2]** for correct final answer.*

(b) (i) $\left(\frac{1.26\times10^{-3}}{2}=\right)6.30\times10^{-4}\,(\text{mol})$; **[1]**

(ii) $(6.30\times10^{-4}\times10=)6.30\times10^{-3}\,(\text{mol})$; **[1]**

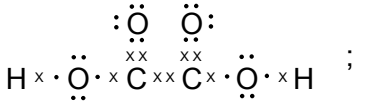
(iii) $\left(\frac{0.795}{6.30\times10^{-3}}=\right)126\,(\text{g mol}^{-1})$; **[1]**

(iv) $M_r(C_2H_2O_4)=90.04$ **and** $M_r(H_2O)=18.02$;

$x=2$; **[2]**

Accept integer values for M_r's of 90 and 18 and any reasonable calculation.
*Award **[1 max]** if no working shown.*

(c) hydrogen bonding; **[1]**

(d) 
; **[1]**

Mark cannot be scored if lone pairs are missing on oxygens.
Accept any combination of lines, dots or crosses to represent electron pairs.

(e) *Acid*:
one double and one single bond / one shorter and one longer bond;
Accept "two double and two single".

Conjugate base:
two 1.5 bonds / both bonds same length;
Accept "four / all".

electrons delocalized / resonance forms; **[3]**
Award marks for suitable diagrams.

1.2-S42 Nov 2014 · Paper 2 · Q4 parts (a), (b)

4. (a) $P_4O_{10}: \left(\frac{5.00}{283.88} =\right) 0.0176\,(\text{mol})$ **and** $H_2O: \left(\frac{1.50}{18.02} =\right) 0.0832\,(\text{mol})$;

H_2O is the limiting reactant **and** reason related to stoichiometry; ***[2]***

(b) $\frac{0.0832 \times 4}{6} / 0.0555\,(\text{mol})$;

$(0.0555 \times 98.00 =) 5.44\,\text{g}$; ***[2]***

The unit is needed for M2.
*Award **[2]** for correct final answer.*
Do not penalize slight numerical variations due to premature rounding.

1.2-S43 May 2014 TZ2 · Paper 2 · Q1 parts (a), (c)(ii), (c)(iii), (g) · also 5.2

(a) $M(CH_3COOH)\left(= (4 \times 1.01) + (2 \times 12.01) + (2 \times 16.00)\right) = 60.06\,(\text{g mol}^{-1})$;

Accept 60 (g mol^{-1}).
mass $(CH_3COOH)(= 5.00 \times 1.05) = 5.25\,(\text{g})$;

$\frac{5.25}{60.06} = 0.0874\,(\text{mol})$; ***[3]***

*Award **[3]** for correct final answer.*
Accept 0.0875 (comes from using Mr = 60 g mol^{-1}).

(ii) $26.00\,(\text{cm}^3)$; ***[1]***

(iii) $26.00 - 3.00 = 23.00\,(\text{cm}^3)$;

*If other methods used, award **M1** for calculating amount of NaOH reacting with CH_3COOH.*

$0.200 \times \frac{23.00}{5.00} = 0.920\,(\text{mol dm}^{-3})$; ***[2]***

*Award **[2]** for correct final answer.*
If (ii) given as mean titre (26.5 cm^3) then ECF answer comes to 0.94 (mol dm^{-3}).

(g) ethyl ethanoate/$CH_3COOC_2H_5$/ester;
forms only weak hydrogen bonds (to water); ***[2]***
Allow "does not hydrogen bond to water" / "hydrocarbon sections too long" / OWTTE.
M2 can only be given only if M1 correct.

1.2-S44 Nov 2013 · Paper 2 · Q5 parts (a), (c), (g), (h), (i) · also 3.1, 4.1, 5.1, 9.3

5. (a) $\underline{(l)}\,N(NO_2)_3(g) + \underline{2}\,CH_3OH(l) \rightarrow \underline{2}\,N_2(g) + \underline{2}\,CO_2(g) + \underline{4}\,H_2O(l)$; ***[1]***

(c) *bonds broken:* $(6\times305)+(3\times158)=1830+474=2304\,(kJ\,mol^{-1})$;
bonds made: $(2\times945)+(3\times498)=1890+1494=3384\,(kJ\,mol^{-1})$;
enthalpy change: $2304-3384=-1080\,(kJ\,mol^{-1})$; ***[3]***
*Award **[3]** for correct final answer.*
*Award **[2 max]** for $+1080\,(kJ\,mol^{-1})$.*

Accept –234 kJ mol^{-1} which arise from students assuming that 305 kJ mol^{-1} refers to the strength of a single N–O bond. Students may then take N=O from the data book value (587 kJ mol^{-1}).
bonds broken: $(3\times305)+(3\times587)+(3\times158)=915+1761+474=3150\,(kJ\,mol^{-1})$
bonds made: $(2\times945)+(3\times498)=1890+1494=3384\,(kJ\,mol^{-1})$
enthalpy change: $3150-3384=-234\,(kJ\,mol^{-1})$.

*Award **[2 max]** for correct calculation of the enthalpy change of reaction for the equation in part (a), which gives –2160 (kJ mol^{-1}).*
*Award **[1]** if the final answer is not –2160 but the candidate has correctly calculated the bonds broken in trinitramide as 2304 (kJ mol^{-1}).*

(g) (N–N bond in) trinitramide is longer/nitrogen (gas) is shorter / 0.145 nm in trinitramide versus 0.110 nm in nitrogen;

(h) $106^{\circ} - 108^{\circ}$;
Accept $< 109^{\circ}$.

Any two for ***[2 max]***.
4 (negative) charge centres/electron pairs/electron domains around central nitrogen;
central nitrogen has a lone/non-bonding pair;
lone/non-bonding pairs repel more than bonding pairs;
molecule will be (trigonal/triangular) pyramidal;
(negative) charge centres/electron pairs/electron domains will be tetrahedrally arranged/orientated/ have tetrahedral geometry; ***[3 max]***
Do not apply ECF.

(i) polar;
net dipole moment present in molecule / unsymmetrical distribution of charge / polar bonds do not cancel out / centre of negatively charged oxygen atoms does not coincide with positively charged nitrogen atom; ***[2]***
Marks may also be awarded for a suitably presented diagram showing net dipole moment.
Do not accept "unsymmetrical molecule".
Apply ECF from part (h).

1.2-S45 May 2013 TZ1 · Paper 2 · Q1 parts (b)

(b) *In 25.0 cm³:* $n_{HA} = 1.21 \times 10^{-3}\,(\text{mol})$;
In 100 cm³: $n_{HA} = 4.84 \times 10^{-3}\,(\text{mol})$;
$\text{M}\left(= \dfrac{0.675}{4.84 \times 10^{-3}}\right) = 139\,(\text{g mol}^{-1})$; ***[3]***
Award ***[3]*** *for correct final answer.*
Accept suitable alternative methods.

1.2-S46 May 2013 TZ2 · Paper 2 · Q1 parts (e)(i), (e)(ii), (f)(i), (f)(ii), (f)(iii), (f)(iv) · also 6.1

(e) (i) gain (of electrons); ***[1]***

(ii) VII / +7; ***[1]***
Do not accept 7 or 7+.

(f) (i) volume $= 16.80\,(\text{cm}^3) / 18.00 - 1.20\,(\text{cm}^3)$;
amount $\left(= \frac{16.80 \times 5.00 \times 10^{-3}}{1000}\right) = 8.40 \times 10^{-5}\,(\text{mol})$; ***[2]***
*Award **[2]** for correct final answer.*

(ii) $(8.40 \times 10^{-5} \times 5 \times 10) = 4.20 \times 10^{-3}\,(\text{mol per } 250\,\text{cm}^3)$; ***[1]***

(iii) $(55.85 \times 4.20 \times 10^{-3}) = 0.235\,(\text{g})$; ***[1]***
Do not penalize if 56 g mol^{-1} is used for atomic mass of iron.

(iv) $\left(\frac{0.235 \times 100}{1.65} = \right) 14.2\,\%$; ***[1]***
No ECF if answer >100 %.

1.2-S47 Nov 2012 · Paper 2 · Q1 parts (a)(i), (a)(ii), (a)(iii), (a)(iv), (b), (c)(ii), (c)(iii), (c)(iv), (c)(v), (c)(vi), (d), (e) · also 3.1, 5.2, 8.1

1. (a) (i) *$I_2(s)$*: four/4 **and** *$ICl(l)$*: three/3; ***[1]***

(ii) $n(Cl_2) = \left(\frac{2.24}{2 \times 35.45} =\right) 0.0316 / 3.16 \times 10^{-2}\,(\text{mol})$;
Allow answers such as 3.2×10^{-2}/0.032/3.15×10^{-2}/0.0315 (mol).

$n(\text{ICl}) = 2 \times 0.0316 \,/\, 0.0632 / 6.32 \times 10^{-2}\,(\text{mol})$;
Allow answers such as 6.4×10^{-2}/0.064/6.3×10^{-2}/0.063 (mol).

$m(\text{ICl}) = (0.0632 \times 162.35 =)\, 10.3\,(\text{g})$; ***[3]***
Allow answers in range 10.2 to 10.4 (g).
*Award **[3]** for correct final answer.*

(iii) $\left(\frac{8.60}{10.3} \times 100 =\right) 83.5\%$; ***[1]***
Allow answers in the range of 82.5 to 84.5%.

(iv) negative/–/minus/ < 0; ***[1]***

(b) Br_2 has London/dispersion/van der Waals' forces/vdW **and** ICl has (London/dispersion/van der Waals' forces/vdW and) dipole–dipole forces;
dipole–dipole forces are stronger than London/dispersion/van der Waals'/vdW forces ; ***[2]***
Allow induced dipole-induced dipole forces for London forces.
Allow interactions instead of forces.
Do not allow ICl polar and Br_2 non-polar for M1.
Name of IMF in both molecules is required for M1 and idea of dipole-dipole stronger than vdW is required for M2.

(ii) $(25.25 - 1.05) = 24.20\,(\text{cm}^3)$; ***[1]***
Accept 24.2 (cm^3) but not 24 (cm^3).

(iii) $\left(\frac{24.20\times5.00\times10^{-2}}{1000}\right)=1.21\times10^{-3}/0.00121(\text{mol});$ ***[1]***

(iv) $(0.5\times1.21\times10^{-3})=6.05\times10^{-4}/0.000605(\text{mol});$ ***[1]***

Accept alternate method e.g. $(0.384/126.9\times0.2015)=6.10\times10^{-4}/0.000610$ (mol).

(v) $(126.90\times6.05\times10^{-4})=7.68\times10^{-2}/0.0768(\text{g});$ ***[1]***

Accept alternate method e.g. $(6.10\times10^{-4}\times126.9)$ or $(0.2015\times0.384)=7.74\times10^{-2}/0.00774$ (g).

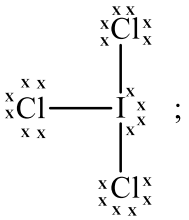
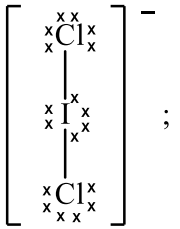
(vi) $\left(\frac{7.68\times10^{-2}}{0.2015}\times100\right)=38.1\%$; ***[1]***

Answer must be given to <u>three</u> significant figures.

(d) ICl / iodine monochloride; ***[1]***

Do not accept iodine or bromine.

(e)

	ICl_3	ICl_2^-
Lewis (electron dot) structure	 ;	 ;
	Do not penalize for an incorrect shape for Lewis structures. *Accept lines, dots or crosses for electron pairs for both Lewis structures.* *Penalize missing lone pairs <u>on Cl</u> once only.* *Square brackets and negative charge must be shown for Lewis structure of $[ICl_2]^-$.*	
Name of shape	T-shaped;	linear;

[4]

No ECF for shape if Lewis structure is incorrect.

1.2-S48 May 2012 TZ1 · Paper 2 · Q7 parts (a)(i), (a)(ii), (a)(iii)

7. (a) (i) molar mass = 102.20 ($g\ mol^{-1}$);

amount $\left(= \frac{5.00}{102.20}\right)$ = 0.0489 (mol); ***[2]***

(ii) theoretical yield = $(84.18 \times 0.0489 =)$ 4.12 (g);

percentage yield $= \left(\frac{2.62}{4.12} \times 100 =\right) 63.6\ \%$; ***[2]***

Accept alternative calculation method.

(iii) yield above 100% not possible / experimental yield > theoretical yield / *OWTTE*;
Must have reference to a final yield.

sample contaminated with hexan-1-ol/water / inadequate drying / *OWTTE*; ***[2]***
Do not accept error in reading balance/weighing scale.

1.2-S49 May 2012 TZ2 · Paper 2 · Q2 parts (a), (b), (c)(i) · also 6.3

2. (a) $Mg(s) + 2HCl(aq) \rightarrow H_2(g) + MgCl_2(aq)$; ***[1]***

(b) $n(Mg) = \left(\frac{0.0740}{24.31}\right) = 3.04 \times 10^{-3}\ (mol)$;

Accept range 3.04×10^{-3} *to* 3.08×10^{-3}.

$n(HCl) = (2.00 \times 15.0 \times 10^{-3}) = 3.00 \times 10^{-2}\ (mol)$;
Mg; ***[3]***

(c) (i) $n(H_2) = n(Mg) = 3.04 \times 10^{-3}\ (mol)$; ***[1]***
Accept same value as in 2(b).
Answer must be in range 3.04×10^{-3} *to* 3.08×10^{-3} *and must have 2, 3 or 4 significant figures.*

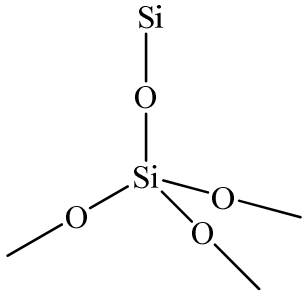
1.2-S50 Nov 2011 · Paper 2 · Q1 parts (b)(i), (b)(ii), (c)(ii) · also 4.3

(b) (i) *T*: 4 **and** *m*: 3 **and** *p*: 3; ***[1]***

(ii) $n = (65.0 / 65.02) = 1.00\ (\text{mol})$; ***[1]***

No penalty for using whole number atomic masses.

(ii) *Structure:*
drawing of giant structure showing tetrahedrally arranged silicon;
Minimum information required for mark is Si and 4 O atoms, in a tetrahedral arrangement (not 90° bond angles) but with each of the 4 O atoms showing an extension bond.



Bonding:
(giant/network/3D) covalent; ***[2]***

1.2-S51 May 2011 TZ1 · Paper 3 · QE2 parts (b)

(b) 162 g of $C_6H_{10}O_5$ requires 192 g of O_2;
0.010 g requires 0.012 g of O_2;

OR

$\text{n(organic matter)} = \frac{0.010}{162}$ **and** $\text{n}(O_2) = 6 \times \text{n(organic matter)}$;

$\text{m}(O_2) = 6\text{n(organic matter)} \times 32.0 = 0.012\text{g}$; ***[2]***

*Award **[2]** for correct final answer.*

1.2-S52 Nov 2010 · Paper 2 · Q6 parts (d)(i)

(d) (i) (nitric acid) $7.5\ \text{cm}^3$; ***[1]***

1.2-S53 May 2010 TZ1 · Paper 2 · Q1 parts (a)(i), (a)(ii), (b)(i), (b)(ii), (b)(iii), (b)(iv), (c)(i) · also 2.1, 6.1

1. (a) (i) *Copper:*
0 to +2 / increases by 2 / +2 / 2+;
Allow zero/nought for 0.

Nitrogen:
+5 to +4 / decreases by 1 / –1 / 1–; ***[2]***
Penalize missing + sign or incorrect notation such as 2+, 2^+ or II, once only.

(ii) nitric acid/HNO_3 / NO_3^-/nitrate; ***[1]***
Allow nitrogen from nitric acid/nitrate but not just nitrogen.

(b) (i) 0.100×0.0285 ;
2.85×10^{-3} (mol); ***[2]***
*Award **[2]** for correct final answer.*

(ii) 2.85×10^{-3} (mol); ***[1]***

(iii) $(63.55 \times 2.85 \times 10^{-3}) = 0.181$ g ; ***[1]***
Allow 63.5.

(iv) $\left(\frac{0.181}{0.456} \times 100 =\right)$ 39.7 % ; ***[1]***

(c) (i) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^9$; ***[1]***
Do not allow [Ar]$3d^9$.

1.2-S54 May 2010 TZ2 · Paper 2 · Q1 parts (a), (b), (c), (g)

1. (a) $n(\text{HCl})\,(= 0.200\ \text{mol dm}^{-3} \times 0.02720\ \text{dm}^3) = 0.00544 / 5.44 \times 10^{-3}\,(\text{mol})$; ***[1]***

(b) $n(\text{HCl})$ excess $(= 0.100\ \text{mol dm}^{-3} \times 0.02380\ \text{dm}^3) = 0.00238 / 2.38 \times 10^{-3}\,(\text{mol})$; ***[1]***

Penalize not dividing by 1000 once only in (a) and (b).

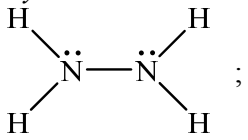
(c) $n(\text{HCl})$ reacted $(= 0.00544 - 0.00238) = 0.00306 / 3.06 \times 10^{-3}\,(\text{mol})$; ***[1]***

(g) only $CaCO_3$ reacts with acid / impurities are inert/non-basic / impurities do not react with the acid / nothing else in the eggshell reacts with acid / no other carbonates; ***[1]***
Do not accept "all calcium carbonate reacts with acid".

1.2-S55 May 2010 TZ2 · Paper 2 · Q4 parts (a), (b), (c) · also 3.1, 9.3

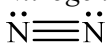
4. (a) $N_2H_4(g) + 2F_2(g) \rightarrow N_2(g) + 4HF(g)$
*Award **[1]** for reactants and products.*
*Award **[1]** if this equation is correctly balanced.* ***[2]***
Ignore state symbols.

(b) *Hydrazine:*



;

Nitrogen:



; ***[2]***
Accept lines, dots and crosses to show electron pairs.
Penalize missing lone pairs once only.

(c) $\sum \text{BE (bonds broken)} = (4 \times 391) + 158 + 2(158) / 2038\,(\text{kJ})$;
$\sum \text{BE (bonds formed)} = (945) + 4(568) / 3217\,(\text{kJ})$;
$\Delta H^{\ominus} = 2038 - 3217 = -1179\,(\text{kJ})$; ***[3]***
*Award **[3]** for correct final answer.*
*Award **[2]** for (+)1179(kJ).*

LESSON 2.1

Ionization Energy

Answers for 64 multiple-choice and 38 structured questions

Part A - Multiple choice answers

2.1-M1	May 2025 TZ1 · P1A · Q4	A
2.1-M2	May 2025 TZ1 · P1A · Q7	D
2.1-M3	May 2025 TZ1 · P1A · Q15	A
2.1-M4	May 2025 TZ2 · P1A · Q3	C
2.1-M5	May 2025 TZ2 · P1A · Q4	A
2.1-M6	May 2025 TZ2 · P1A · Q6	A
2.1-M7	May 2025 TZ2 · P1A · Q12	D
2.1-M8	May 2025 TZ2 · P1A · Q16	C
2.1-M9	May 2025 TZ3 · P1A · Q13	D
2.1-M10	Nov 2024 · P1 · Q7	D
2.1-M11	Nov 2024 · P1 · Q10	D
2.1-M12	May 2024 TZ1 · P1 · Q5	A
2.1-M13	May 2024 TZ2 · P1 · Q6	A
2.1-M14	May 2024 TZ2 · P1 · Q9	B
2.1-M15	Nov 2023 TZ1 · P1 · Q5	C
2.1-M16	May 2023 TZ1 · P1 · Q5	B
2.1-M17	May 2023 TZ2 · P1 · Q4	A
2.1-M18	May 2023 TZ2 · P1 · Q5	C
2.1-M19	Nov 2022 · P1 · Q4	C
2.1-M20	May 2022 TZ1 · P1 · Q5	A
2.1-M21	May 2022 TZ2 · P1 · Q5	B
2.1-M22	Nov 2021 · P1 · Q5	D
2.1-M23	May 2021 TZ1 · P1 · Q6	A
2.1-M24	May 2021 TZ1 · P1 · Q7	B
2.1-M25	May 2021 TZ2 · P1 · Q5	A
2.1-M26	Nov 2020 · P1 · Q5	B
2.1-M27	Nov 2019 · P1 · Q5	C
2.1-M28	May 2019 TZ1 · P1 · Q4	C
2.1-M29	May 2019 TZ1 · P1 · Q5	B
2.1-M30	May 2019 TZ1 · P1 · Q8	B
2.1-M31	Nov 2018 · P1 · Q5	B
2.1-M32	Nov 2018 · P1 · Q7	B
2.1-M33	May 2018 TZ1 · P1 · Q5	A
2.1-M34	May 2018 TZ1 · P1 · Q7	B
2.1-M35	May 2018 TZ2 · P1 · Q5	D
2.1-M36	Nov 2017 · P1 · Q6	B
2.1-M37	Nov 2017 · P1 · Q7	D
2.1-M38	May 2017 TZ1 · P1 · Q5	B
2.1-M39	May 2017 TZ2 · P1 · Q5	A
2.1-M40	Nov 2016 · P1 · Q6	A
2.1-M41	May 2016 · P1 · Q5	D
2.1-M42	May 2016 · P1 · Q6	B
2.1-M43	Nov 2015 · P1 · Q5	D
2.1-M44	Nov 2015 · P1 · Q6	B
2.1-M45	May 2015 TZ1 · P1 · Q4	A
2.1-M46	May 2015 TZ1 · P1 · Q5	C
2.1-M47	May 2015 TZ1 · P1 · Q7	B
2.1-M48	May 2015 TZ2 · P1 · Q5	D
2.1-M49	Nov 2014 · P1 · Q5	B
2.1-M50	Nov 2014 · P1 · Q7	D
2.1-M51	May 2014 TZ1 · P1 · Q5	C
2.1-M52	May 2014 TZ2 · P1 · Q4	A
2.1-M53	May 2014 TZ2 · P1 · Q6	B
2.1-M54	May 2013 TZ1 · P1 · Q5	C
2.1-M55	May 2013 TZ2 · P1 · Q5	B
2.1-M56	Nov 2012 · P1 · Q5	A
2.1-M57	May 2012 TZ1 · P1 · Q8	D
2.1-M58	Nov 2011 · P1 · Q6	D
2.1-M59	May 2011 TZ1 · P1 · Q5	B
2.1-M60	May 2011 TZ2 · P1 · Q6	A
2.1-M61	Nov 2010 · P1 · Q5	B
2.1-M62	May 2010 TZ1 · P1 · Q5	B
2.1-M63	May 2010 TZ1 · P1 · Q6	B
2.1-M64	May 2010 TZ2 · P1 · Q5	B

Part B - Structured questions markscheme

2.1-S1 May 2025 TZ1 · Paper 2 · Q1 parts (d)

1.	**(d)**		«*IE*(Fe) = 762 kJ mol^{-1} » *E* for 1 atom «= $\frac{762 \times 10^3\ J\ mol^{-1}}{6.02 \times 10^{23}\ mol^{-1}}$ » = 1.266 x 10^{-18} «J» ✓ f «= $\frac{E}{h} = \frac{1.266 \times 10^{-18}\ J}{6.63 \times 10^{-34}\ J\ s}$ » = 1.91 x 10^{15} «s^{-1}» ✓ λ «= $\frac{c}{f} = \frac{3.0 \times 10^8\ m\ s^{-1}}{1.91 \times 10^{15}\ s^{-1}}$ »= 1.57 x 10^{-7} «m» ✓	*Award [3] for correct final answer.*	**3**

2.1-S2 May 2025 TZ1 · Paper 2 · Q2 parts (a)

2.	**(a)**		1s^2 2s^{2}2p^6 3s^{2}3p^{6}3d^7 / [Ar] 3d^7 ✓		**1**

2.1-S3 May 2025 TZ2 · Paper 2 · Q2 parts (b)(iii), (b)(vii), (b)(viii), (c)(i), (c)(ii), (c)(iii) · also 6.3, 7.2, 9.4

2	**(b)**	**(iii)**	lattice enthalpy ✓		**1**

2	**(b)**	**(vii)**	***ALTERNATIVE 1:*** place thallium in a solution of copper sulfate ✓ if reaction occurs then thallium is more reactive ***OR*** if no reaction occurs then copper is more reactive ✓ ***ALTERNATIVE 2:*** place copper in a solution of thallium sulfate ✓ if reaction occurs then copper is more reactive ***OR*** if no reaction occurs then thallium is more reactive ✓		**2**

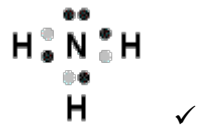
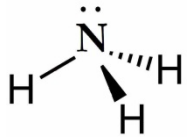
2	**(b)**	**(viii)**	*Anode:* oxygen ***AND*** water is more easily oxidized than sulfate ion ✓ *Cathode:* thallium ***AND*** thallium is more easily reduced than water ✓		**2**
2	**(c)**	**(i)**	limit of convergence «of continuum» ✓	*Accept high energy limit instead of limit of convergence.*	**1**

2	**(c)**	**(ii)**	IE ⟨⟨$= \frac{589\ x\ 10^3\ Jmol^{-1}}{6.02\ x\ 10^{23}\ mol^{-1}}$⟩⟩ = 9.78 x 10^{-19}«J $atom^{-1}$» ✓ frequency ⟨⟨$= \frac{E}{h} = \frac{9.78\ x\ 10^{-19}\ J}{6.63\ x\ 10^{-34}\ Js}$⟩⟩ = 1.48 x 10^{15} «s^{-1}» ✓ wavelength ⟨⟨$= \frac{c}{f} = \frac{3.00x10^8 m\ s^{-1}}{1.48\ x\ 10^{15}\ s^{-1}}$⟩⟩ = 2.03 x 10^{-7} «m» / 203 «nm» ✓	*Award **[3]** for correct final answer.*	**3**
2	**(c)**	**(iii)**	nuclear charge of Tl less than that of Pb / increasing «effective» nuclear charge from Tl to Pb ✓ shielding effect/occupied energy levels stay the same ✓	*Do **not** accept the atomic radius of lead is less than that of thallium for M1.* *Do **not** accept that the loss of an electron from thallium leaves a complete subshell.*	**2**

2.1-S4 May 2024 TZ1 · Paper 2 · Q5 parts (a), (b)(i) · also 3.1

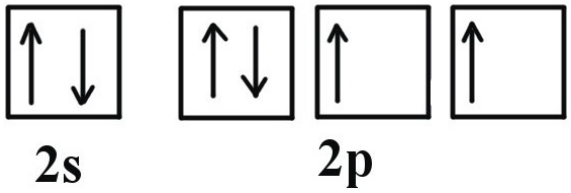
5.	**(a)**		16 ***AND*** 7th IE much higher ✓	*Accept VI/6*	**1**
5.	**(b)**	**(i)**	Structure 1: E «6-0-4 = » +2 2x O «6-6-1 = » –1 O «6-4-2 = » 0 ✓ Structure 2: E «6-0-6 = » 0 3x O «6-4-2 = » 0 ✓ structure 2 most likely ✓	*M1 requires two FCs for different oxygens* *accept notations on diagram for M1, M2* *Do not award M3 without explanation based on formal charge.*	**3**

2.1-S5 May 2023 TZ1 · Paper 2 · Q2 parts (a)(i), (b), (c)(i), (c)(ii), (c)(iii) · also 3.1, 5.2

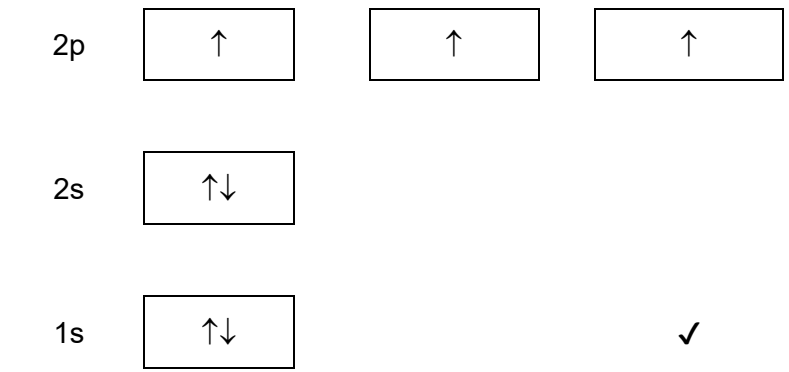
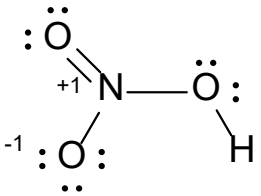
2.	**(a)**	**(i)**	 arrows ***AND*** *identifies 2s* ***AND*** *2p sub orbitals* ✓	*Accept "hooks" to represent the electrons.*	**1**
2.	**(b)**		valence electron further from nucleus/«atomic» radius larger «down the group» ✓ «electron» more shielded/ less attractive force/easier to remove ✓		**2**
2.	**(c)**	**(i)**	tetrahedral ✓		**1**
2.	**(c)**	**(ii)**	 ✓  ✓	*Accept a combination of dots /crosses /lines in the Lewis structure* *Lone pair not required for shape*	**2**

2.	**(c)**	**(iii)**	ammonia has intermolecular/IMF hydrogen bonds «phosphine does not» ✓ phosphine «and ammonia» dipole-dipole/London dispersion forces/instantaneous dipole attractions/Van der Waals forces ✓ hydrogen bonds stronger ✓	*Accept converse argument.* *Award 1 for* stating that NH_3 is more polar than phosphine so the dipole-dipole forces are stronger	**3**

2.1-S6 May 2023 TZ1 · Paper 3 · Q1 parts (a), (b), (c) · also 9.4

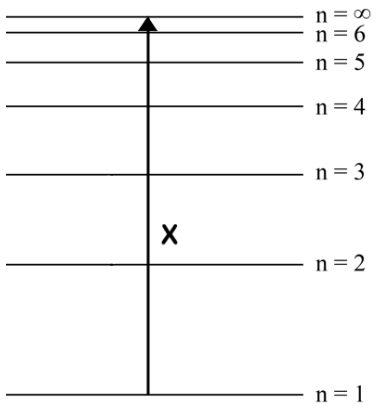
1.	**(a)**		ionization energy breaks bond/attractive force between nucleus and electron ***AND*** electron affinity forms bond/attractive force between nucleus and electron ✓	*Accept for ionization energy "energy needed/endothermic to remove an electron".* ***AND*** *for electron affinity "energy released/exothermic adding an electron".*	**1**
1.	**(b)**		electron «removed» from 2p in B ***AND*** 2s in Be ✓ shielding effect of 2s «reduces energy needed to remove 2p» ***OR*** 2p at higher energy level/further from nucleus ***OR*** full 2s more stable «than single electron in p» ✓	*Accept electron configurations of **both** B and Be for M1.*	**2**
1.	**(c)**		 ✓	*Accept second arrow, representing opposite spin, in any 2p box.* *Accept half-arrows.*	**1**

2.1-S7 May 2022 TZ1 · Paper 2 · Q6 parts (a)(i), (a)(ii), (a)(iii) · also 3.1, 4.1

6.	**a**	**i**	 ✓	*Accept **all** 2p electrons pointing downwards.* *Accept half arrows instead of full arrows.*	**1**
6.	**a**	**ii**	 bonds and non-bonding pairs correct ✓ formal charges correct ✓	*Accept dots, crosses or lines to represent electron pairs.* *Do **not** accept resonance structures with delocalised bonds/electrons.* *Accept + and – sign respectively.* *Do not accept a bond between nitrogen and hydrogen.* *For an incorrect Lewis structure, allow ECF for non-zero formal charges.*	**2**
6.	**a**	**iii**	*Any three of:* two N-O same length/order ✓ delocalization/resonance ✓ N-OH longer «than N-O» ***OR*** N-OH bond order 1 ***AND*** N-O bond order 1½ ✓	*Award **[2 max]** if bond strength, rather than bond length discussed.* *Accept N-O between single and double bond **AND** N-OH single bond.*	**3**

2.1-S8 May 2022 TZ2 · Paper 2 · Q2 parts (a), (b)(ii)

2.	**a**		increasing number of protons/nuclear charge/Z_{eff} ✓ **«**atomic**»** radius/size decreases ***OR*** same number of energy levels ***OR*** similar shielding **«**by inner electrons**»** ✓		**2**

2.	**b**	**ii**	 upward arrow X ***AND*** starting at n = 1 extending to n = ∞ ✓		**1**

2.1-S9 May 2022 TZ2 · Paper 2 · Q5 parts (a)(i), (a)(ii), (b) · also 6.1

5.	**a**	**i**	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^6$ ✓		**1**
5.	**a**	**ii**	*Any two of:* IE_9: electron in lower energy level ***OR*** IE_9: more stable/full electron level ✓ IE_9: electron closer to nucleus ***OR*** IE_9: electron more tightly held by nucleus ✓ IE_9: less shielding by **«**complete**»** inner levels ✓		**2 max**
5.	**b**		–1 ✓	*Accept "– *l*".*	**1**

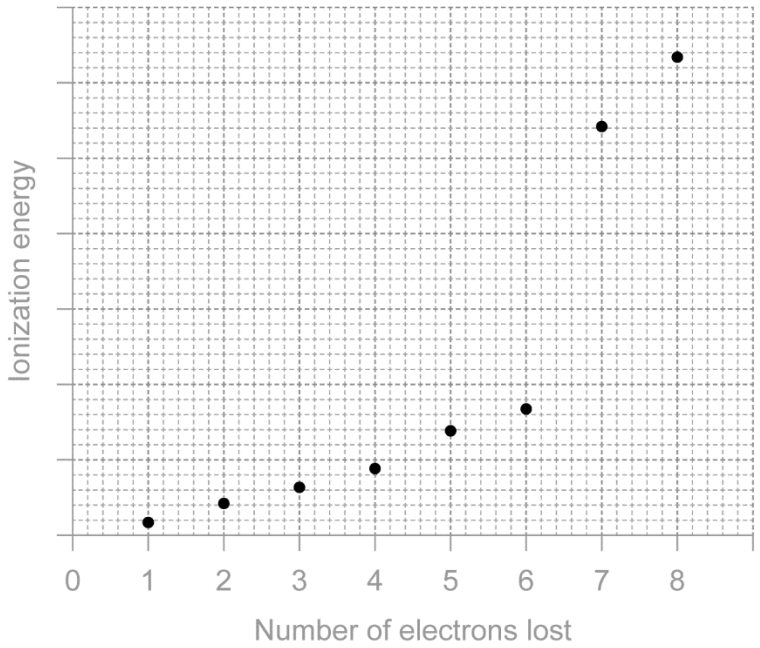
2.1-S10 Nov 2021 · Paper 2 · Q2 parts (a)

2.	**a**		increasing number of protons ***OR*** increasing nuclear charge ✓ **«**atomic**»** radius/size decreases ***OR*** same number of shells/electrons occupy same shell ***OR*** similar shielding **«**by inner electrons**»** ✓		**2**

2.1-S11 Nov 2021 · Paper 2 · Q9 parts (b)(ii)

9.	**b**	**ii**	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^{10}$ ***OR*** [Ar] $4s^1 3d^{10}$ ✓	*Accept configuration with 3d before 4s.*	**1**

2.1-S12 May 2021 TZ1 · Paper 2 · Q1 parts (c), (d)(iii), (e)(i), (e)(ii) · also 1.2, 6.1

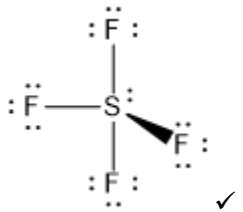
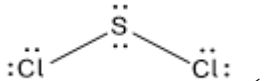
1.	**c**		 two regions of small increases ***AND*** a large increase between them✓ large increase from 6th to 7th ✓	*Accept line/curve showing these trends.*	**2**

1.	**d**	**iii**	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$ ✓	*Do **not** accept "[Ne] $3s^2\ 3p^6$".*	**1**

1.	**e**	**i**	$4FeS(s) + 7O_2(g) \rightarrow 2Fe_2O_3(s) + 4SO_2(g)$ ✓	*Accept any correct ratio.*	**1**
1.	**e**	**ii**	+6 ***OR*** –2 to +4 ✓	*Accept "6/VI".* *Accept "–II, 4/+4/IV".* Do **not** accept 2- to 4+.	**1**

2.1-S13 May 2021 TZ2 · Paper 2 · Q2 parts (a)(ii), (b)(i), (c), (d) · also 3.1, 5.2

2.	**a**	**ii**	P has «three» unpaired electrons in 3p sub-level ***AND*** S has one full 3p orbital «and two 3p orbitals with unpaired electrons» ***OR*** P: $[Ne]3s^2 3p_x^1 3p_y^1 3p_z^1$ ***AND*** S: $[Ne]3s^2 3p_x^2 3p_y^1 3p_z^1$ ✓ repulsion between paired electrons in sulfur **«**and therefore easier to remove**»** ✓	*Accept orbital diagrams for 3p sub-level for M1. Ignore other orbitals or sub-levels.* *Accept "removing electron from S gives more stable half-filled sub-level" for M2.*	**2**
2.	**b**	**i**	*Cr:* [Ar] $4s^1 3d^5$ ✓ *Cr^{3+}:* [Ar] $3d^3$ ✓	*Accept "[Ar] $3d^5 4s^1$".* *Accept "[Ar] $3d^3 4s^0$".* *Award **[1 max]** for two correct full electron configurations "$1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^5$ **AND** $1s^2 2s^2 2p^6 3s^2 3p^6 3d^3$".* *Award **[1 max]** for $4s^1 3d^5$ **AND** $3d^3$.*	**2**

<table>
<tr>
<td>2.</td>
<td>c</td>
<td></td>
<td>
<table>
<tr><th>Species</th><th>SF_4</th><th>SCl_2</th></tr>
<tr><td>Lewis structure</td><td>  ✓</td><td>  ✓</td></tr>
<tr><td>Molecular geometry</td><td>saw horse/
see saw ✓</td><td>bent/V
shaped/angular ✓</td></tr>
</table>
</td>
<td>SF_4/SCl_2 structure does not have to be 3-D for mark.

Penalize missing lone pairs of electrons on halogens once only.

Accept any combination of dots, lines or crosses for bonds/lone pairs.

Accept “non-linear” for SCl_2 molecular geometry.

Award [1] for two correct electron domain geometries, e.g. trigonal bipyramidal for SF_4 and tetrahedral for SCl_2.</td>
<td>4</td>
</tr>
<tr>
<td>2.</td>
<td>d</td>
<td></td>
<td>H_2O forms hydrogen bonding «while SCl_2 does not» ✓

SCl_2 «much» stronger London/dispersion/«instantaneous» induced dipole-induced dipole forces ✓

Alternative 1:
H_2O less volatile AND hydrogen bonding stronger «than dipole–dipole and dispersion forces» ✓

Alternative 2:
SCl_2 less volatile AND effect of dispersion forces «could be» greater than hydrogen bonding ✓</td>
<td>Ignore reference to Van der Waals.

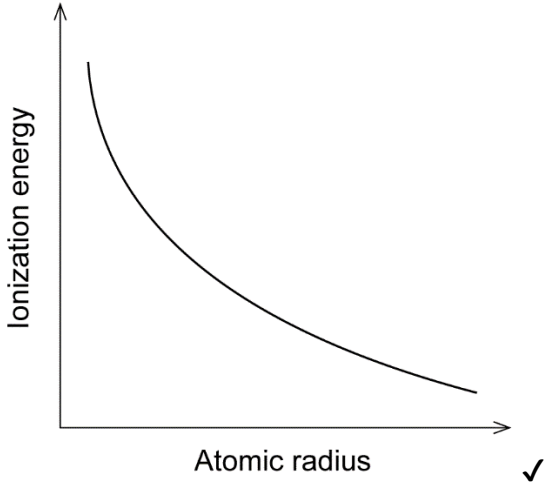
Accept “SCl_2 has «much» larger molar mass/electron density” for M2.</td>
<td>3</td>
</tr>
</table>

2.1-S14 Nov 2020 · Paper 2 · Q6 parts (a), (c) · also 7.2

6.	**a**		«$E = \frac{745\,000\ \text{J mol}^{-1}}{6.02 \times 10^{23}\ \text{mol}^{-1}}$ =» 1.24×10^{-18} J ✓ «$E = hv$» «$1.24 \times 10^{-18}\ \text{J} = 6.63 \times 10^{-34}\ \text{J s} \times \text{v}$» $\text{v} = 1.87 \times 10^{15}$ «s^{-1}/Hz» ✓	*Award **[2]** for correct final answer.* *Award **[1]** for 1.12x10^{39} «Hz».*	**2**

6.	**c**		cathode/negative «electrode» ***AND*** Cu^{2+} reduced «at that electrode» ✓	*Accept cathode/negative «electrode» **AND** copper forms «at that electrode».*	**1**

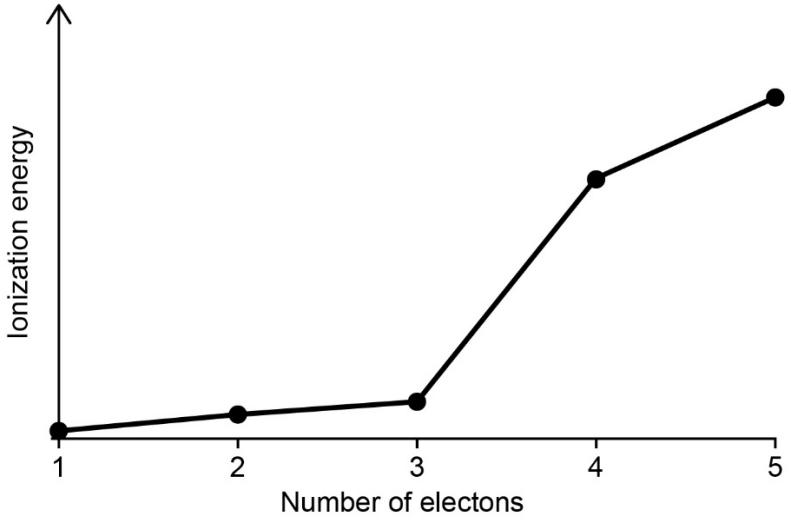
2.1-S15 May 2019 TZ1 · Paper 3 · Q1 parts (b)(iv)

1.	**b**	**iv**	 ✓	*Accept any negative sloping line.* *Do **not** award mark if line touches either axis.*	**1**

2.1-S16 May 2019 TZ2 · Paper 2 · Q3 parts (c), (d)(i), (d)(ii), (d)(iii) · also 3.1, 4.1, 4.2

3.	**c**		*Nitrogen and carbon:* N has greater nuclear charge/**«**one**»** more proton **«**and electrons both lost from singly filled p-orbitals**»** ✓ *Nitrogen and oxygen:* O has a doubly filled **«**p-**»**orbital ***OR*** N has only singly occupied **«**p-**»**orbitals ✓	*Accept "greater e– ⁻ e⁻ repulsion in O" or "lower e– ⁻ e⁻ repulsion in N".* *Accept box annotation of electrons for M2.*	**2**
3.	**d**	**i**	delocalization ***OR*** delocalized π-electrons ✓	*Accept "resonance".*	**1**
3.	**d**	**ii**	linear ***AND*** 2 electron domains ***OR*** linear ***AND*** 2 regions of electron density ✓	*Accept "two bonds **AND** no lone pairs" for reason.*	**1**
3.	**d**	**iii**	sp ✓		**1**

2.1-S17 Nov 2018 · Paper 2 · Q4 parts (c)

4.	**c**		 *Sketch showing:* largest increase between third and fourth ionization energies ✓ $IE_1 < IE_2 < IE_3 < IE_4 < IE_5$ ✓		**2**

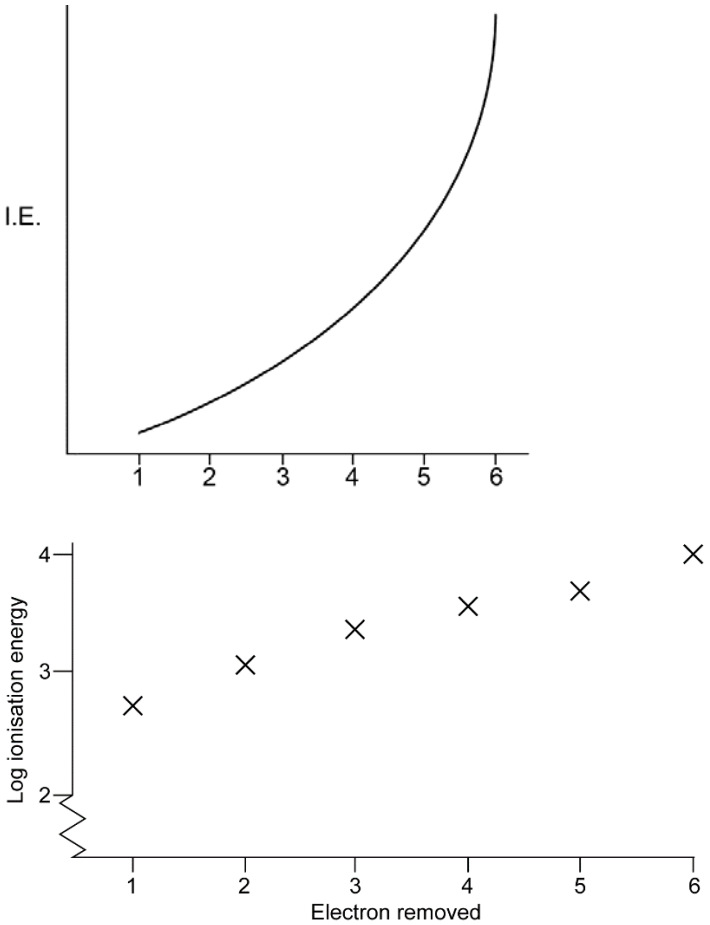
2.1-S18 Nov 2017 · Paper 2 · Q3 parts (a), (d)(ii) · also 6.1

3.	**a**		increasing number of protons ***OR*** increasing nuclear charge ✓ «atomic» radius/size decreases ***OR*** same number of shells ***OR*** similar shielding «by inner electrons» ✓ **«**greater energy needed to overcome increased attraction between nucleus and electrons**»**		**2**

3.	**d**	**ii**	*Charge on complex ion:* 1+/+ ✓ *Oxidation state of cobalt:* +2 ✓		**2**

2.1-S19 May 2017 TZ1 · Paper 2 · Q2 parts (d)(i), (d)(iii)

2.	**d**	**i**	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^2$ ✓		**1**

2.	**d**	**iii**	 regular increase for first five ***AND*** sharp increase to the 6th ✓	*A log graph is acceptable.* *Accept log plot on given axes (without amendment of y-axis).* *Award mark if gradient of 5 to 6 is greater than "best fit line" of 1 to 5.*	**1**

2.1-S20 May 2016 · Paper 2 · Q4 parts (a)(i), (b)(ii), (e)(i), (e)(ii), (e)(iii) · also 7.1, 9.4

4.	**a**	**i**	same charge ***AND*** same/similar ionic radius ✓		**1**

4.	**b**	**ii**	yes ***AND*** $\Delta G^{\ominus}$ is negative ***OR*** yes ***AND*** $E^{\ominus}$ for the cell is positive ***OR*** yes ***AND*** $Sn^{2+}(aq)$ is a stronger reducing agent than $V^{3+}(aq)$ ***OR*** yes ***AND*** $Sn^{2+}(aq)$ is more negative than $E^{\ominus}$ for $VO^{2+}(aq)$ ***OR*** yes ***AND*** $VO^{2+}(aq)$ is a stronger oxidizing agent than $Sn^{4+}(aq)$ ***OR*** yes ***AND*** $E^{\ominus}$ for $VO^{2+}(aq)$ is more positive than $E^{\ominus}$ for $Sn^{4+}(aq)$ ✓	*Do **not** accept reference to anti-clockwise rule.*	**1**

4.	**e**	**i**	*First*: 4s ***AND*** *Second*: 4s ***AND*** *Third*: 3d ***AND*** *Fourth*: 3d ✓	*Do **not** apply **ECF** from (c).*	**1**
4.	**e**	**ii**	**«**in the same sub-shell and a**»** decrease in electron-electron repulsion ***OR*** **«**in the same sub-shell and**»** as more electrons removed, the pull of of the nucleus/positive ions holds the remaining electrons more tightly ✓	*Do **not** accept "greater nuclear charge/ effective nuclear charge".*	**1**
4.	**e**	**iii**	electron 5 is lost from the 3d orbital ***OR*** electron 5 is lost from the valence shell ✓ electron 6 is lost from a 3p orbital ***OR*** electron 6 is lost from a **«**complete**»** inner shell ✓ 3p orbital/complete inner shell experiences a much larger effective nuclear charge ***OR*** 3p orbital/complete inner shell is less well shielded ***OR*** 3p orbital/complete inner shell is nearer the nucleus ✓	*Award **[1 max]** (for M1/M2) (**ECF**) if candidate recognises electrons 5 and 6 are from different levels.*	**3**

2.1-S21 Nov 2015 · Paper 2 · Q2 parts (a)

2. (a) (Cu) $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^{10}$ / $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^1$;
Do not accept [Ar] $4s^1 3d^{10}$.

(Cu^{2+}) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^9$; ***[2]***
Do not accept [Ar] $3d^9$.

2.1-S22 May 2015 TZ1 · Paper 2 · Q4 parts (a)

4. (a) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^1$ / $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^{10}$; **[1]**

2.1-S23 May 2015 TZ1 · Paper 2 · Q8 parts (e), (f) · also 3.1

(e) Cl has 7 valence electrons **and** is in group 7;
Accept "group 17" as suggested by IUPAC.

Cl has 3 occupied (electron) shells/energy levels **and** so is in period 3; **[2]**

(f)

Molecule	Shape	Bond angle	Polarity
SCl_2	bent/angular/ v-shaped	< 109.5° *Accept 100° − 108°. Literature value is 103°.*	polar
$SClF_5$	Octahedral *Accept square bipyramidal.*	90° (180°)	polar

;;; **[3]**

Do not accept ECF for bond angles and polarities from incorrect shapes.
*Award **[3]** for all six correct.*
*Award **[2]** for four or five correct.*
*Award **[1]** for two or three correct.*

2.1-S24 May 2015 TZ1 · Paper 3 · Q1 parts (a)

1. (a) $E = \frac{hc}{\lambda}$ / $E = \alpha\frac{1}{\lambda}$ / energy inversely proportional to wavelength / the higher the energy the shorter the wavelength / *OWTTE*; **[1]**

Do not accept $E = h\upsilon/hf$.

2.1-S25 May 2015 TZ1 · Paper 3 · Q18 parts (a)(i)

18. (a) (i) $\left(\lambda = \frac{3.0\times10^{8}}{3.5\times10^{13}} =\right) 8.6\times10^{-6}\,(\text{m})$; **[1]**

2.1-S26 Nov 2014 · Paper 2 · Q8 parts (b)(i), (b)(ii), (c)(ii), (d)(i), (d)(ii), (d)(iii) · also 3.1, 4.2, 5.1

(b) (i) decreasing repulsion between electrons / radius decreases as electrons are removed; ***[1]***

Accept increasing positive charge on ion attracts electrons more strongly.

(ii) 10^{th} electron is in second energy level/shell while 11^{th} electron is in first energy level/shell / 10^{th} is removing electron from electronic arrangement 2,1 while 11^{th} ionization energy is removing electron from electronic arrangement 2;
11^{th} electron removed is much closer to the nucleus / 11^{th} electron removed from a (much) lower energy level/shell; ***[2]***

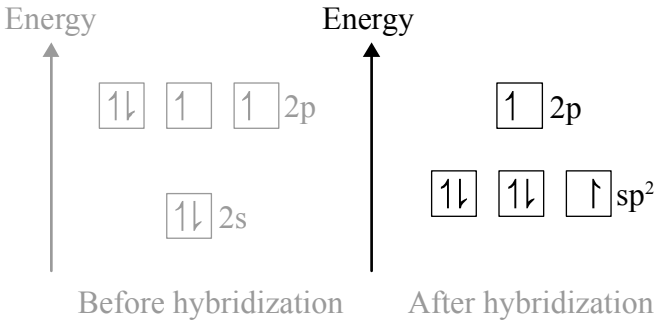
Accept opposite statement for 10^{th} electron.

(ii) electrostatic attraction between a pair of electrons and (positively charged) nuclei; ***[1]***

Accept a/two pairs of shared electrons.

(d) (i) C: sp hybridization;
O: sp^2 hybridization; ***[2]***
*Award **[1]** if the answer is sp without specifying C or O atoms.*

(ii)



three sp^2 orbitals **and** one p-orbital at higher energy;
sp^2 orbitals contain: two, two and one electron **and** p-orbital contains one electron; ***[2]***
Do not allow ECF from (d)(i).

(iii) ability of atom/nucleus to attract bonding/shared pair of electrons / attraction of nucleus for bonding/shared pair of electrons / *OWTTE*; ***[1]***

2.1-S27 May 2014 TZ1 · Paper 2 · Q5 parts (b)(ii), (c)(i), (c)(ii), (d)(i), (e) · also 5.2, 6.3

(ii) $2Br^-(aq) \rightarrow Br_2(aq) + 2e^-$;
$Cl_2(g) + 2e^- \rightarrow 2Cl^-(aq)$; ***[2]***
Ignore state symbols.
Accept e instead of e^-.

(c) (i) HF has hydrogen bonds (between molecules); ***[1]***

(ii) strength of van der Waals'/London/dispersion forces increases;
as mass/size/number of electrons of halogen atom/molecule increases; ***[2]***

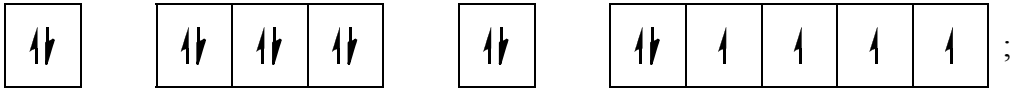
(d) (i) *Cr:* $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^5$ / $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^1$;
Cr^{3+}: $1s^2 2s^2 2p^6 3s^2 3p^6 3d^3$; ***[2]***

(e) successive ionization energy values increase with removal of each electron;
large increase in ionization energy when sixth electron is removed;
as electron is one energy level/shell closer to the nucleus; ***[2 max]***
Accept a suitably annotated diagram.

2.1-S28 Nov 2013 · Paper 2 · Q4 parts (a)

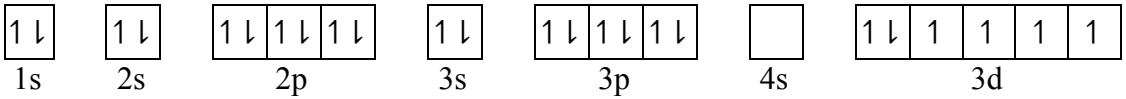
4. (a) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^4$ / [Ar]$3d^4$; ***[1]***

2.1-S29 May 2013 TZ2 · Paper 2 · Q3 parts (e)

(e) 

; ***[1]***
Allow full arrows instead of half-arrows in orbital diagram.
Sub-levels must be labelled for mark.

2.1-S30 Nov 2012 · Paper 2 · Q2 parts (b)(ii)

(ii) 

[1]

Allow full arrows instead of half-arrows for example ↑↓.
Do not allow arrows with the same spin for example ↑↑ or ↓↓ in the same orbital.
Do not allow an orbital diagram with a $4s^1 3d^5$ configuration.

2.1-S31 May 2012 TZ1 · Paper 2 · Q8 parts (b)(i), (b)(ii)

(b) (i) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^2$;
4s; ***[2]***

(ii) 4th electron removed from 3d **and** 5th electron removed from 3p;
10th electron removed from 3p **and** 11th electron removed from 3s;

Accept either of the following answers for the third mark:
electrons removed from lower energy level / energy level closer to nucleus are attracted more strongly;
greater effective nuclear charge / s electrons more penetrating; ***[3 max]***

2.1-S32 May 2012 TZ2 · Paper 2 · Q9 parts (a)(i), (a)(ii)

9. (a) (i) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^6 4s^2$ / $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^6$; ***[1]***

(ii) $[Ar]\,3d^5$; ***[1]***

2.1-S33 Nov 2011 · Paper 2 · Q2 parts (c)

(c) $1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,3d^6$; ***[1]***
Do not award mark for $[Ar]3d^6$.
Do not allow $1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,4s^1\,3d^5$.

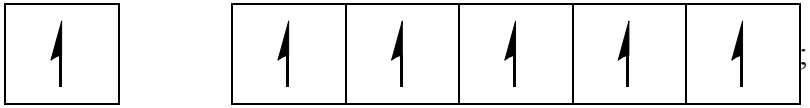
2.1-S34 May 2011 TZ1 · Paper 2 · Q3 parts (c)

(c) $1s^2 2s^2 2p^6 3s^2 3p^6 3d^7$ / $[Ar]\ 3d^7$; ***[1]***

2.1-S35 May 2011 TZ2 · Paper 2 · Q3 parts (a)(i), (a)(ii), (a)(iii)

(a) (i) the electron configuration (of argon) / $1s^2 2s^2 2p^6 3s^2 3p^6$; ***[1]***

(ii) $x = 1$ **and** $y = 5$; ***[1]***

(iii) 

[1]

Accept all six arrows pointing down rather than up.

2.1-S36 Nov 2010 · Paper 2 · Q3 parts (d)

(d) *Cu*:
$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^1$;

Cu^+:
$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10}$; ***[2]***

Ignore relative order of 3d and 4s.
Penalize only once if noble gas core is given.

2.1-S37 Nov 2010 · Paper 2 · Q4 parts (a)(i), (a)(ii), (a)(iii), (b)(i), (b)(ii), (b)(iii), (c)(i), (c)(ii), (c)(iv) · also 7.2, 9.4

4. (a) (i) energy (per mole) needed to remove one/first/most loosely bound electron from a (neutral) atom;
in the gaseous state;
$Mg(g) \rightarrow Mg^{+}(g) + e^{-}$; ***[3]***
Gaseous state symbols needed.
Accept e instead of e^{-}.
Only penalize omission of gas phase once in either the second marking point or the third marking point.

(ii) successive electrons (are more difficult to remove because each is) taken from more positively charged ion/ *OWTTE*;
increased electrostatic attraction; ***[2]***

(iii) 10th electron comes from 2nd energy level/$n = 2$ **and** 11th electron comes from 1st first energy level/$n = 1$ / *OWTTE*;
electron in 1st energy level closer to nucleus;
electron in 1st energy level not shielded by inner electrons / exposed to greater effective nuclear charge; ***[3]***

(b) (i) contains ions which are free to move (only) in molten state;
Mg^{2+} move to cathode/negative electrode and Cl^{-} move to anode/positive electrode / *OWTTE*; ***[2]***

(ii) anode/positive electrode;
$2Cl^{-} \rightarrow Cl_2 + 2e^{-}$ / $Cl^{-} \rightarrow \frac{1}{2}Cl_2 + e^{-}$; ***[2]***
Accept e instead of e^{-}.
Do not accept $Cl^{-} \rightarrow Cl + e^{-}$.
Ignore state symbols.

(iii) magnesium has large negative electrode potential / $E^{\ominus}$;
reduction of H_2O/H^{+} to H_2 has less negative electrode potential;
Mg^{2+} not readily reduced (in comparison to H_2O);
if formed, magnesium would (immediately) react with water to form Mg^{2+};
magnesium more reactive than hydrogen; ***[1 max]***
Do not accept Mg too reactive.

(c) (i) *I*:
atomization/sublimation (of Mg) / $\Delta H^{\ominus}_{\text{atomization}}(\text{Mg})$ / $\Delta H^{\ominus}_{\text{sublimation}}(\text{Mg})$;

V:
enthalpy change of formation of $(MgCl_2)$ / $\Delta H^{\ominus}_{\text{formation}}(MgCl_2)$; ***[2]***

(ii) *Energy value for II*:
$+243$;

Energy value for III:
$738 + 1451 = 2189$;

Energy value for IV:
$2(-349)$;
$\Delta H^{\ominus}_{\text{lat}}(MgCl_2) = 642 + 148 + 243 + 2189 = (+)2252(KJ)$; ***[4]***

(iv) oxide greater charge;
oxide smaller radius; ***[2]***
Accept opposite arguments.

2.1-S38 May 2010 TZ2 · Paper 2 · Q6 parts (b)(i), (d), (e)(i), (e)(ii) · also 5.1

(b) (i) power/strength/ability of an atom to attract electrons/shared electron pair / *OWTTE*;
in a (covalent) bond; ***[2]***
Accept the word "element" in place of "atom".
Do not accept electron (singular).

(d) $1s^2 2s^2 2p^6 3s^1$;
Do not accept [Ne] $3s^1$.

first electron easy/easiest to remove / 1 electron in outermost/n = 3 energy level / furthest from nucleus;
large increase between 1st and 2nd IE as electron now removed from n = 2;
next 8 electrons more difficult to remove / show (relatively) small increase as these electrons are in the same energy level/second energy level/n = 2;
large increase between 9th and 10th IE as electron now removed from n = 1 / 2 electrons very hard/most difficult to remove / innermost/lowest/closest to the nucleus/energy level/n = 1 / *OWTTE*;
electron 11 also comes from 1s, so shows a small increase; ***[4 max]***

(e) (i) outer electron in Al is in 3p/p orbital/sub-shell/sub-level;
higher orbital/sub-shell / e^- further from nucleus / shielded by 3s electrons; ***[2]***

(ii) in S, electron paired in 3p/p orbital/sub-shell/sub-level;

repulsion between paired electrons (and therefore easier to remove); ***[2]***

LESSON 3.1

VSEPR Theory

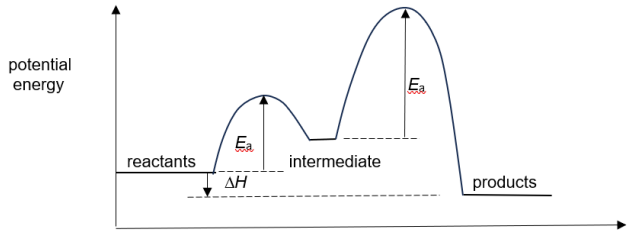
Answers for 49 multiple-choice and 29 structured questions

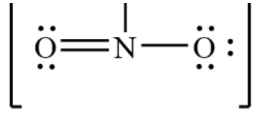
Part A - Multiple choice answers

3.1-M1	May 2025 TZ1 · P1A · Q9	B
3.1-M2	May 2025 TZ1 · P1A · Q10	A
3.1-M3	May 2025 TZ1 · P1A · Q11	C
3.1-M4	May 2025 TZ2 · P1A · Q9	B
3.1-M5	May 2025 TZ3 · P1A · Q9	C
3.1-M6	May 2024 TZ1 · P1 · Q11	C
3.1-M7	May 2024 TZ1 · P1 · Q12	C
3.1-M8	May 2024 TZ2 · P1 · Q12	C
3.1-M9	Nov 2023 TZ1 · P1 · Q12	D
3.1-M10	May 2023 TZ1 · P1 · Q11	D
3.1-M11	May 2023 TZ1 · P1 · Q12	B
3.1-M12	May 2023 TZ2 · P1 · Q9	A
3.1-M13	May 2023 TZ2 · P1 · Q11	A
3.1-M14	May 2023 TZ2 · P1 · Q12	C
3.1-M15	Nov 2022 · P1 · Q10	C
3.1-M16	Nov 2022 · P1 · Q11	D
3.1-M17	May 2022 TZ1 · P1 · Q13	C
3.1-M18	May 2022 TZ2 · P1 · Q11	D
3.1-M19	May 2022 TZ2 · P1 · Q12	C
3.1-M20	May 2021 TZ2 · P1 · Q12	B
3.1-M21	Nov 2020 · P1 · Q11	C
3.1-M22	Nov 2019 · P1 · Q13	D
3.1-M23	May 2019 TZ2 · P1 · Q11	B
3.1-M24	Nov 2018 · P1 · Q9	C
3.1-M25	Nov 2018 · P1 · Q10	D
3.1-M26	May 2018 TZ1 · P1 · Q10	B
3.1-M27	May 2018 TZ2 · P1 · Q11	B
3.1-M28	May 2018 TZ2 · P1 · Q12	D
3.1-M29	May 2017 TZ2 · P1 · Q9	B
3.1-M30	Nov 2016 · P1 · Q11	B
3.1-M31	Nov 2016 · P1 · Q14	B
3.1-M32	May 2016 · P1 · Q11	C
3.1-M33	May 2015 TZ1 · P1 · Q6	D
3.1-M34	May 2015 TZ1 · P1 · Q12	D
3.1-M35	May 2015 TZ2 · P1 · Q10	C
3.1-M36	May 2015 TZ2 · P1 · Q12	B
3.1-M37	Nov 2014 · P1 · Q9	D
3.1-M38	May 2014 TZ2 · P1 · Q12	C
3.1-M39	May 2013 TZ1 · P1 · Q12	D
3.1-M40	Nov 2012 · P1 · Q12	B
3.1-M41	May 2012 TZ1 · P1 · Q9	B
3.1-M42	May 2012 TZ2 · P1 · Q9	B
3.1-M43	May 2012 TZ2 · P1 · Q11	A
3.1-M44	May 2011 TZ1 · P1 · Q14	A
3.1-M45	May 2011 TZ2 · P1 · Q11	B
3.1-M46	Nov 2010 · P1 · Q10	C
3.1-M47	Nov 2010 · P1 · Q12	A
3.1-M48	May 2010 TZ2 · P1 · Q9	B
3.1-M49	May 2010 TZ2 · P1 · Q12	D

Part B - Structured questions markscheme

3.1-S1 May 2025 TZ1 · Paper 2 · Q6 parts (a)(i), (a)(iv), (b)(ii), (b)(iii), (b)(iv) · also 4.1, 6.1, 6.2

6.	**(a)**	**(i)**	+2/II ✓	*Do not accept 2, 2+,.*	**1**
			Any three of: partially filled d-orbitals ✓ «ligand causes» d-orbitals «to» split ✓ light is absorbed as electrons transit to a higher energy level «in d-d transition»	*Accept "energy absorbed" for M3 only.* *Accept a stated wavelength in the range of 575nm to 585nm for M4.*	
6.	**(a)**	**(iv)**	*Overall redox equation:* $S_2O_3^{2-} + 4Fe^{3+} + H_2O \rightarrow 2SO_2 + 4Fe^{2+} + 2H^+$ ✓	*Allow ECF for M2*	**2**
				*Accept any combination of peak heights.* *Accept double-headed arrows or lines with no heads.*	

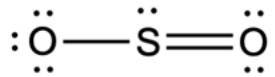
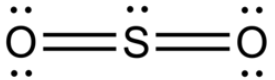
6.	**(b)**	**(ii)**	 ✓	*Accept single negative charges on two O atoms **AND** a single positive charge on the N atom.* Accept any combination of lines, dots or crosses	**1**
6.	**(b)**	**(iii)**	trigonal planar ✓	*Accept triangular planar.* *Apply ECF for any N containing species in 6(b)(ii).*	**1**
6.	**(b)**	**(iv)**	124 «x 10^{-12} m» ✓ delocalized electrons / resonance structure ✓	*Accept any length between 115 «x 10^{-12} m» and 135 «x 10^{-12} m» for M1.* *Accept 'intermediate between single and double', or 'bond order greater than 1 but less than 2' for M2.*	**2**
			n(C) «$= n(CO_2) = \frac{9.49\ g}{44.01\ g\ mol^{-1}}$» = 0.216 «mol» ***AND*** *n*(H) «$2n(H_2O) = 2 \times \frac{5.18\ g}{18.02\ g\ mol^{-1}}$» = 0.575 «mol» ✓ ***OR*** *H = 0.581* «g» ***AND***	*M2 for finding mass of oxygen*	

3.1-S2 May 2025 TZ2 · Paper 2 · Q1 parts (a)(i), (a)(ii), (a)(iii), (a)(iv), (a)(v), (b)(vi), (c)(ii), (c)(v) · also 4.2, 5.1, 5.2, 6.1, 7.1, 8.2

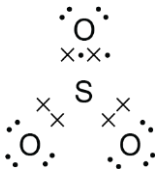
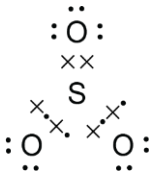
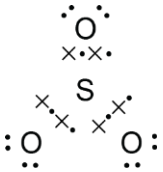
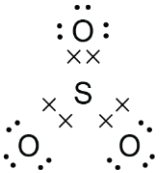
1	**(a)**	**(i)**	✓	*Accept any combination of dots or crosses to represent electrons, or lines to represent electron pairs.*	**1**
1	**(a)**	**(ii)**	*Hybridization:* sp ✓ *Sigma bonds:* 2 ***AND*** *Pi bonds:* 2 ✓		**2**
1	**(a)**	**(iv)**	*Partial positive charge:* hydrogen/H ***AND*** *Partial negative charge:* nitrogen/N ✓	*Accept carbon/C for "Partial positive charge".*	**1**
1	**(a)**	**(v)**	N_2 «only» has London/dispersion forces ***AND*** HCN has dipole-dipole «and London(dispersion) »forces ***OR*** HCN has dipole-dipole force ***AND*** N_2 does not ✓ intermolecular forces are weaker in N_2 ***OR*** dipole-dipole forces stronger «than London(dispersion) forces»✓	*Accept instantaneous / transient / induced dipole attractions for London forces* *Do **not** accept van der Waals' forces for M1.*	**2**

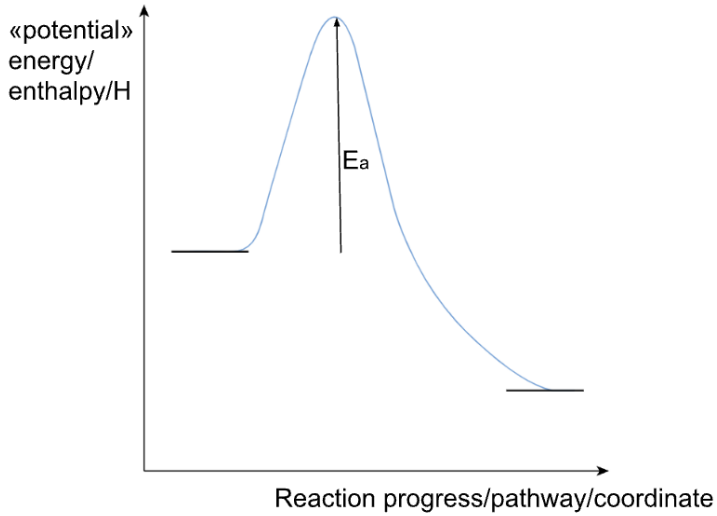
1	**(b)**	**(vi)**	amount neutralised «= c x V = 0.100 x 0.0200» = 0.00200 «mol» ✓ ΔT ⟨⟨= $\frac{0.00200\ x\ 13300}{50.0\ x\ 4.18}$⟩⟩ = 0.127 «K / °C» ✓	*Award **[2]** for correct final answer.*	2
1	**(c)**	**(ii)**	+2 ✓	*Accept II for +2 but do **not** accept 2+ or 2.*	**1**
1	**(c)**	**(v)**	$E^{Ö}$ less positive/lower than «+0.77 V for» Fe^{3+}/Fe^{2+} ✓ the less positive $E^{Ö}$ the weaker the oxidizing agent / stronger the reducing agent ***OR*** thus, a weaker oxidizing agent✓		**2**

3.1-S3 May 2025 TZ3 · Paper 2 · Q10 parts (a)(i), (a)(ii), (a)(iii), (a)(iv), (b) · also 1.2, 2.1, 5.1

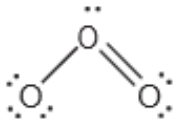
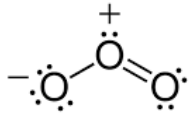
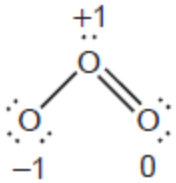
10.	**(a)**	**(i)**	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^4$ ***OR*** $[Ne]\ 3s^2\ 3p^4$ ✓		**1**
10.	**(a)**	**(ii)**	 ***OR***  ✓	*Accept the Lewis formulas with the formal charges.* *Accept any combination of dots or crosses to represent electrons, or lines to represent electron pairs.*	**1**
10.	**(a)**	**(iii)**	*Electron domain geometry:* trigonal planar ✓ *Molecular domain geometry:* bent / V-shaped / angular ✓	*Apply ECF from Lewis formula.*	**2**
10.	**(a)**	**(iv)**	«both» bonds are polar / electronegativity difference «between O and S» ✓ «bond» dipoles do not cancel each other / there is a net dipole «because bonds are at an angle less than 180° » ✓	*Accept unsymmetrical distribution of charge **OR** dipoles add to give a «partial» positive «charge» on S and a «partial» negative «charge» on the O atoms for M2.* *Apply ECF from molecular geometry.*	**2**
10.	**(b)**		$2SO_2(g) + 2H_2O(l) + O_2(g) \rightarrow 2H_2SO_4(aq)$ correct product ***AND*** state symbols for reactants and product ✓ correct balancing ✓		**2**

3.1-S4 Nov 2024 · Paper 2 · Q4 parts (a)(i), (a)(ii), (a)(iii), (d)(i), (d)(iv) · also 4.1, 6.1, 8.1

4.	**(a)**	**(i)**	*Obeys octet rule:*  ***OR*** other resonance structures ✓ *Does not obey octet rule:*  ***OR***  ***OR*** other resonance structures ✓	*Accept any combination of dots or crosses to represent electrons, or lines to represent electron pairs.* *Do **not** accept delocalized structures.* *For M2 accept:* 	**2**

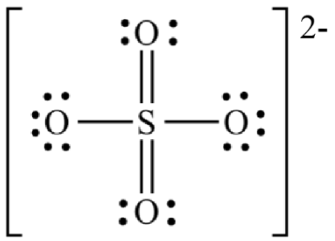
4.	**(a)**	**(ii)**	**«**lowest**»** formal charge ✓		**1**
4.	**(a)**	**(iii)**	**«**all bonds**»** less than 161 **«**pm**»** ✓	*Accept any value less than 161 pm and above 121 pm (actual value is 142 pm).* *Do **not** award the mark if two bond length values are given.* *Do **not** apply ECF from the Lewis structures in (a)(i).*	**1**
4.	**(d)**	**(i)**	 x-axis labelled reaction progress/pathway/coordinate ***AND*** y-axis labelled **«**potential**»** energy/enthalpy/H ✓ curve showing exothermic reaction ✓ labelled arrow/line showing E_a ✓	*Do **not** accept just "reaction" or "time" for x-axis label.* *Accept double-headed arrow for E_a but do **not** accept a downward arrow.* *Accept more complex diagrams showing an intermediate.*	**3**
4.	**(d)**	**(iv)**	V_2O_5 : +5 ***AND*** V_2O_4 : +4 ✓	*Do **not** accept 5+ or 4+.*	**1**

3.1-S5 May 2024 TZ2 · Paper 2 · Q2 parts (a), (b)(i), (b)(ii), (b)(iii), (b)(iv), (c), (d) · also 2.1, 4.1, 4.2

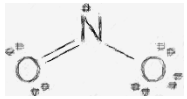
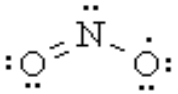
2.	**(a)**		*Electron configuration:* $1s^2 2s^2 2p^4$ ✓ *Unpaired electrons:* 2 ✓		**2**
2.	**(b)**	**(i)**	 ✓	*Accept any combination of dots, crosses and lines for electron pairs, including an arrow for the coordinate bond.* *Do **not** accept delocalized structures.* *Accept*  .	**1**
2.	**(b)**	**(ii)**	 formal charge of central O = +1 ✓ formal charge of single bonded O = −1 ***AND*** double bonded O = 0 ✓		**2**

2.	**(b)**	**(iii)**	bent/angular/V-shaped ✓ any estimated value in the range *110* ° *– 119* ° ✓	*Accept a bent diagram for M1.*	**2**
2.	**(b)**	**(iv)**	sp^2 ✓		**1**
2.	**(c)**		**«**both equal and**»** any estimated value in the range 122–147 **«**pm**»** ✓ resonance ***OR*** delocalization ***OR*** bond order is 1.5 ✓		**2**
2.	**(d)**		shorter wavelength radiation/UV needed to break the bond in O_2 ✓ O_2 has stronger bond ✓		**2**

3.1-S6 May 2023 TZ1 · Paper 2 · Q5 parts (a)(i), (b)(i) · also 1.2

5.	**(a)**	**(i)**	 ✓	*Accept any combination of dots, crosses and lines.* *Double bonds do not have to be opposite each other.* *Do **not** penalise missing square brackets.*	**1**
5.	**(b)**	**(i)**	$Ba^{2+}(aq) + SO_4^{2-}(aq) \rightleftharpoons BaSO_4(s)$ ✓	*Accept single arrow in place of equilibrium sign.*	**1**

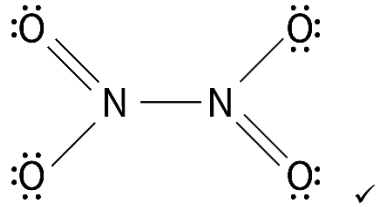
3.1-S7 May 2023 TZ1 · Paper 2 · Q9 parts (b)(iii), (b)(iv)

9.	**(b)**	**(iii)**	 1 2  1 2 ✓		**1**
9.	**(b)**	**(iv)**	No ***AND*** Structure B has all atoms of formal charge 0✓		**1**

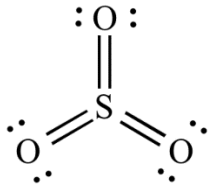
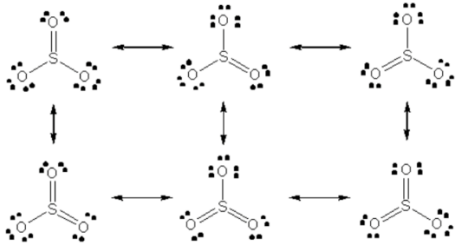
	Structure A	Structure B
Oxygen 1	0	0
Nitrogen	+1	0
Oxygen 2	-1	0

3.1-S8 May 2023 TZ2 · Paper 2 · Q2 parts (a)(iv), (b), (c)(i), (c)(ii) · also 4.1, 9.2

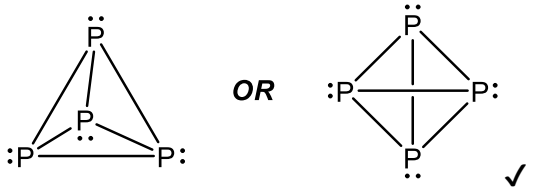
2.	**(a)**	**(iv)**	« $\Delta H^{\ominus}$ = 2(33.18) – 9.16 =» «+» 57.20 «kJ mol^{-1}» ✓		**1**

2.	**(b)**		 ✓	*Accept any combination of dots, crosses or lines to represent electron pairs.*	**1**
2.	**(c)**	**(i)**	it has resonance structures ✓	*Accept bond order = 1.5.* *Accept delocalized electrons «in NO bonds».*	**1**
2.	**(c)**	**(ii)**	110-120° ✓	*Accept any value in the range given.* *(Bond angle is actually 112.3°.)*	**1**

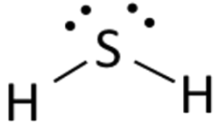
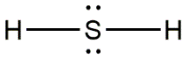
3.1-S9 May 2022 TZ2 · Paper 2 · Q6 parts (c)(i), (c)(ii)

6.	**c**	**i**	 ✓	 *Accept any of the above structures as formal charge is not being assessed.*	**1**
6.	**c**	**ii**	three electron domains **«**attached to the central atom**»** ✓ repel/as far away as possible /120° «apart» ✓		**2**

3.1-S10 Nov 2021 · Paper 2 · Q3 parts (a)(i), (a)(ii), (b)(i), (b)(ii), (c)(i) · also 1.2, 5.1, 9.2

3.	**a**	**i**	 ***OR*** ✓	*Accept any diagram with each P joined to the other three.* *Accept any combination of dots, crosses and lines.*	**1**
3.	**a**	**ii**	$P_4\,(s) + 6Cl_2\,(g) \rightarrow 4PCl_3\,(l)$ ✓		**1**
3.	**b**	**i**	*Electron domain geometry:* tetrahedral ✓ *Molecular geometry:* trigonal pyramidal ✓ *Bond angle:* 100**«°»** ✓	*Accept any value or range within the range 91–108**«°»** for **M3**.*	**3**
3.	**b**	**ii**	*PCl_5 is non-polar:* symmetrical ***OR*** dipoles cancel ✓ *PCl_4F is polar:* P–Cl has a different bond polarity than P–F ✓ non-symmetrical **«**dipoles**»** ***OR*** dipoles do not cancel ✓	*Accept F more electronegative than /different electronegativity to Cl for **M2.***	**3**
3.	**c**	**i**	**«**$-398.9\ kJ\ mol^{-1} - (-306.4\ kJ\ mol^{-1}) =$**»** -92.5 **«**$kJ\ mol^{-1}$**»** ✓		**1**

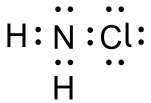
3.1-S11 May 2021 TZ1 · Paper 2 · Q2 parts (a)(i), (a)(ii)

2.	**a**	**i**	 ***OR***  ✓	*Accept any combination of lines, dots or crosses to represent electrons.*	**1**
2.	**a**	**ii**	bent/non-linear/angular/v-shaped✓		**1**

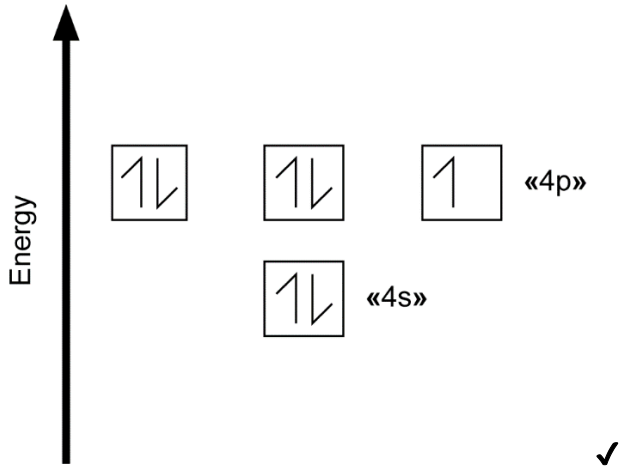
3.1-S12 Nov 2020 · Paper 2 · Q2 parts (a), (b), (c) · also 4.1, 4.2

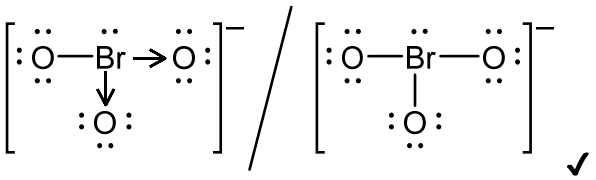
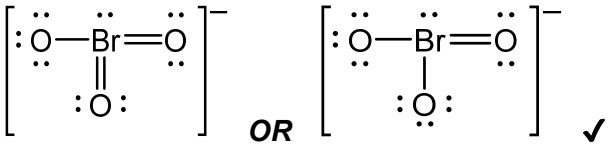
2.	**a**		*Electron domain geometry:* tetrahedral ✓ *Molecular geometry:* bent/V-shaped ✓		**2**
2.	**b**		sp^2 ✓		**1**
2.	**c**		σ-bonds: 3 ***AND*** π-bonds: 1 ✓		**1**

3.1-S13 May 2019 TZ1 · Paper 2 · Q5 parts (c)(i), (c)(ii), (c)(iii), (c)(iv) · also 4.2

5.	**c**	**i**	 ✓	*Accept any combination of dots/crosses or lines to represent electron pairs.*	**1**
5.	**c**	**ii**	sp^3 ✓		**1**
5.	**c**	**iii**	*Molecular geometry:* **«**trigonal**»** pyramidal ✓ *H–N–H bond angle:* 107° ✓	*Accept angles in the range of 100–109.*	**2**
5.	**c**	**iv**	covalent/dative/coordinate ✓		**1**

3.1-S14 Nov 2018 · Paper 2 · Q3 parts (a)(i), (a)(ii), (b)(i), (b)(ii), (c), (d)(i), (d)(ii) · also 2.1, 6.2

3.	**a**	**i**	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^2\ 3d^{10}\ 4p^5$ ***OR*** $[Ar]\ 4s^2\ 3d^{10}\ 4p^5$ ✓	*Accept 3d before 4s.*	**1**
3.	**a**	**ii**	 ✓	*Accept double-headed arrows.*	**1**

3.	**b**	**i**	*Structure I – follows octet rule:*  ✓ *Structure II – does not follow octet rule:*  ***OR*** ✓	*Accept dots, crosses or lines to represent electron pairs.*	**2**
3.	**b**	**ii**	**«**structure I**»** formal charge on Br = +2 ***OR*** **«**structure II**»** formal charge on Br = 0/+1 ✓ structure II is preferred ***AND*** it produces formal charge closer to 0 ✓	*Ignore any reference to formal charge on oxygen.*	**2**

3.	**c**		*Geometry:* trigonal/pyramidal ✓ *Reason:* three bonds ***AND*** one lone pair ***OR*** four electron domains ✓ *O–Br–O angle:* 107° ✓	*Accept "charge centres" for "electron domains".* *Accept answers in the range 104–109°.*	**3**
3.	**d**	**i**	$BrO_3^-(aq) + 6e^- + 6H^+(aq) \rightarrow Br^-(aq) + 3H_2O(l)$ correct reactants and products ✓ balanced equation ✓	*Accept reversible arrows.*	**2**
3.	**d**	**ii**	$BrO_3^-(aq) + 6Fe^{2+}(aq) + 6H^+(aq) \rightarrow Br^-(aq) + 3H_2O(l) + 6Fe^{3+}(aq)$ ✓		**1**

3.1-S15 May 2018 TZ2 · Paper 2 · Q7 parts (a)(i), (a)(ii), (b), (c)(i), (c)(ii), (d), (e) · also 4.1, 4.3, 5.1, 5.2

7.	**a**	**i**	polar bonds **«**between H and group 16 element**»** ***OR*** difference in electronegativities «between H and group 16 element» ✓ uneven distribution of charge/electron cloud ***OR*** non-linear/bent/V-shaped/angular shape «due to lone pairs» ***OR*** polar bonds/dipoles do not cancel out ✓	*M2:* *Do **not** accept "net/overall dipole moment" without further explanation.* *Accept "non-symmetrical **«**shape/distribution of charge**»**".*	**2**
7.	**a**	**ii**	number of electrons increases ✓ London/dispersion/instantaneous induced dipole-induced dipole forces increase ✓	*M1: Accept "M_r/A_r increases" or "molecules become larger in size/mass/surface area".*	**2**
7.	**b**		*Electron domain geometry:* tetrahedral ✓ *Molecular geometry:* bent/V-shaped/angular ✓	*Both marks can be awarded for clear diagrams. Electron domain geometry requires a 3-D diagram showing the tetrahedral arrangement.*	**2**

<table>
<tr>
<td>7.</td>
<td>c</td>
<td>i</td>
<td>
<table>
<tr><th>Structure:</th><th>I</th><th>II</th></tr>
<tr><td>O atom labelled (1)</td><td>0</td><td>«+»1</td></tr>
<tr><td>O atom labelled (2)</td><td>0</td><td>−1</td></tr>
</table>
✓
✓
</td>
<td>Award [1] for any two correctly filled cells.</td>
<td>2</td>
</tr>
<tr>
<td>7.</td>
<td>c</td>
<td>ii</td>
<td>structure I AND no formal charges
OR
structure I AND no charge transfer «between atoms» ✓</td>
<td></td>
<td>1</td>
</tr>
<tr>
<td>7.</td>
<td>d</td>
<td></td>
<td>O_3 has bond between single and double bond AND O_2 has double bond
OR
O_3 has bond order of 1.5 AND O_2 has bond order of 2
OR
bond in O_3 is weaker/longer than in O_2 ✓

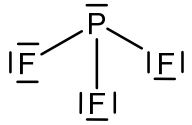
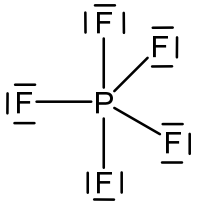
O_3 requires longer wavelength ✓</td>
<td>M1: Do not accept “ozone has one single and one double bond”.</td>
<td>2</td>
</tr>
<tr>
<td>7.</td>
<td>e</td>
<td></td>
<td>CO_2 «non-polar» «weak» London/dispersion forces/instantaneous induced dipole-induced dipole forces between molecules ✓

SiO_2 network/lattice/3D/giant «covalent» structure ✓</td>
<td>M1: The concept of “between” is essential.</td>
<td>2</td>
</tr>
</table>

3.1-S16 May 2018 TZ2 · Paper 3 · Q2 parts (a)

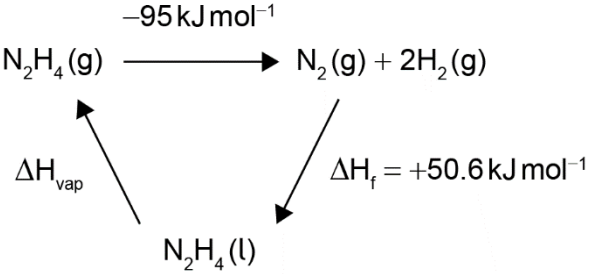
2.	**a**		*Any two of:* *Ethene:* **«**carbon–carbon**»** double bond ***AND*** *Ethane:* **«**carbon–carbon**»** single bond ✓ ethene has a shorter carbon–carbon bond **«**than ethane**»** ✓ *Ethene*: planar/two-dimensional/2-D ***AND*** *Ethane:* tetrahedral **«**carbons**»**/ three-dimensional/3-D ***OR*** *Ethene:* each carbon surrounded by three electron domains ***AND*** *Ethane:* each carbon surrounded by four electron domains ***OR*** different molecular geometries/shapes ✓ rotation about carbon–carbon inhibited/blocked in ethene ***AND*** not in ethane ✓ **«**H–C–C/H–C–H**»** bond angles different ***OR*** *Ethene:* **«**bond angles approximately**»** 120° ***AND*** *Ethane:* 109.5/109°✓	*Do **not** accept "different number of atoms/hydrogens/bonds" etc.* *Accept "Ethene: unsaturated **AND** Ethane: saturated" **OR** "Ethene: has a double bond **AND** Ethane: does not" **OR** "Ethene: two flexible bonds between carbon atoms **AND** Ethane: one".* *Accept any reasonable physical description of the two different molecular models based on a variety of kits for M1.* *For ethene, accept any bond angle in the range 117–122°.* *Award **[2]** if any two of the concepts listed are shown in a correctly labelled or annotated diagram.* *Award **[1 max]** for two correct statements for either molecule but with no comparison given to the other.* *Award **[1 max]** for suitable unlabeled diagrams of both compounds.*	**2 max**

3.1-S17 Nov 2017 · Paper 2 · Q4 parts (a), (b), (c) · also 4.2, 5.1

<table>
<tr>
<td>4.</td>
<td>a</td>
<td></td>
<td>
<table>
<tr><th></th><th>PF_3</th><th>PF_5</th></tr>
<tr><td>Lewis structure</td><td>  ✓</td><td>  ✓</td></tr>
<tr><td>Molecular geometry</td><td>trigonal pyramidal ✓</td><td>trigonal bipyramidal ✓</td></tr>
<tr><td>Bond angles</td><td>≤109«°» ✓</td><td>90«°» AND 120«°», «180«°» » ✓</td></tr>
</table>
</td>
<td>Accept any combination of dots, crosses and lines.
Penalize missing lone pairs once only.
Do not apply ECF for molecular geometry.
Accept values in the range 95–109 for PF_3.</td>
<td>6</td>
</tr>
<tr>
<td>4.</td>
<td>b</td>
<td></td>
<td>PF_3 polar AND PF_5 non-polar ✓</td>
<td>Apply ECF from part (a) molecular geometry.</td>
<td>1</td>
</tr>
<tr>
<td>4.</td>
<td>c</td>
<td></td>
<td>sp^3 ✓</td>
<td></td>
<td>1</td>
</tr>
</table>

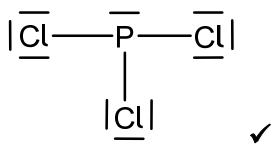
3.1-S18 May 2017 TZ1 · Paper 2 · Q5 parts (a), (b), (f), (g), (h)(i) · also 1.2, 4.2, 9.1, 9.3

5.	**a**		107° ✓	*Accept 100° to <109.5°.* *Literature value = 105.8°*	**1**
5.	**b**		tetrahedral ✓ sp^3 ✓	*No ECF allowed.*	**2**

5.	**f**		*bonds broken*: E(N–N) + 4E(N–H) ***OR*** 158 **«**kJ mol^{-1}**»** + 4 × 391 **«**kJ mol^{-1}**»** / 1722 **«**kJ**»** ✓ *bonds formed*: E(N≡N) + 2E(H–H) ***OR*** 945 **«**kJ mol^{-1}**»** + 2 × 436 **«**kJ mol^{-1}**»** / 1817 **«**kJ**»** ✓ **«**ΔH = bonds broken – bonds formed = 1722 – 1817 =**»** –95 **«**kJ**»** ✓	*Award [3] for correct final answer.* *Award [2 max] for +95 «kJ».*	**3**
5.	**g**		 ***OR*** ΔH_{vap} = –50.6 kJ mol^{-1} – (–95 kJ mol^{-1}) ✓ **«**ΔH_{vap} =**»** +44 **«**kJ mol^{-1}**»** ✓	*Award [2] for correct final answer. Award [1 max] for –44 «kJ mol^{-1}».* *Award [2] for:* ΔH_{vap} = – 50.6 kJ mol^{-1} – (–85 J mol^{-1}) = +34 «kJ mol^{-1}». *Award [1 max] for –34 «kJ mol^{-1}».*	**2**
5.	**h**	**i**	total mass of oxygen **«**= 8.0 × 10^{-3} g dm^{-3} × 1000 dm^3**»** = 8.0 **«**g**»** n(O_2) **«**= $\frac{8.0\,g}{32.00\,g\,mol^{-1}}$ =**»** 0.25 **«**mol**»** ***OR*** n(N_2H_4) = n(O_2) ✓ **«**mass of hydrazine = 0.25 mol × 32.06 g mol^{-1} =**»** 8.0 **«**g**»** ✓	*Award [3] for correct final answer.*	**3**

continued…

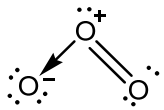
3.1-S19 May 2017 TZ2 · Paper 2 · Q3 parts (b)

3	**b**		*Lewis structure:*  ✓ *Molecular geometry:* trigonal/triangular pyramidal ✓ *Bond angles:* < 109.5° ✓	*Penalize missing lone pairs once only between this question and 4(b)(ii).* *Accept any combination of lines, dots or crosses to represent electrons.* *Do not apply ECF.* *Do not accept answer equal to or less than 90°.* *Literature value is 100.1°.*	**3**

3.1-S20 Nov 2016 · Paper 3 · Q22 parts (c)(i)

22.	**c**	**i**	octahedral ✓	*Accept "square bipyramidal".*	**1**

3.1-S21 May 2014 TZ2 · Paper 2 · Q4 parts (a), (b), (c), (d), (e) · also 4.1, 4.2

(a) 

[1]

Accept lines, dots or crosses to represent electron pairs.
*Formal charges and arrow for dative bond **not** required.*
*If Lewis structure incorrect remember to take into account possible ECF on parts (b)–(d) based on the number of electron domains and bond types in the Lewis diagram in part (a) and do **not** award marks for these if they are inconsistent with the structure given in (a).*

(b) *Shape:* non-linear / bent / v-shaped / angular;

Bond angle: 117° ; ***[2]***
Accept values from 115° to 119° / <u>just/slightly</u> less than 120°.

(c) sp^2; ***[1]***

(d) one is just one σ **and** one has one σ and one π; ***[1]***
Accept "both bonds comprise one σ and a shared π "/ OWTTE.

(e) delocalization occurs / delocalized π-bond / (has two) resonance structures / it is a resonance hybrid;
length intermediate between H_2O_2 and O_2 / *OWTTE*; ***[2]***

SECTION B

3.1-S22 Nov 2013 · Paper 2 · Q3 parts (a)(ii), (b) · also 4.1

(ii) nitrogen/N **and** oxygen/O; ***[1]***
Do not accept nitrate/NO_3^-.
Accept atoms in nitrate/NO_3^-.

(b) pi/π-electrons shared by more than two atoms/nuclei / a pi/π-bond/overlapping p-orbitals that extends over more than two atoms/nuclei;
all (N–O) bonds equal length/strength/bond-order / charge on all oxygen/O atoms equal / increases stability/lowers PE (of the ion); ***[2]***
Accept a diagram that clearly shows one or both points.

3.1-S23 May 2013 TZ1 · Paper 2 · Q6 parts (c)(i), (c)(ii), (c)(iii), (c)(iv), (c)(v), (e) · also 2.1, 4.1, 4.2, 5.2

(c) (i) $1s^2 2s^2 2p^6 3s^2 3p^3$; ***[1]***

(ii) *PCl_3* ; *PCl_5* ; ***[2]***

Penalize missing lone pairs on chlorine only once.
Accept any combination of lines, dots or crosses to represent electron pairs.

(iii)

	PCl_3	PCl_5
Shape	trigonal/triangular pyramidal;	trigonal/triangular bipyramidal;
Bond angles	any angle between 99° and 108°;	90° **and** 120°; *ignore* 180°

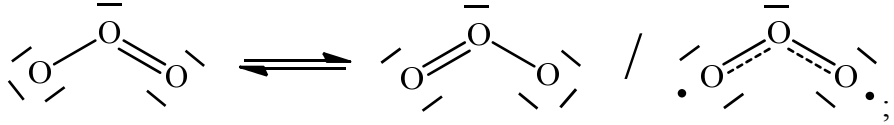
[4]

Shape and bond angle must be consistent with the number of electron domains given in the diagram in (ii).

(iv) sp^3 (hybridization); ***[1]***

(v) PCl_5 has higher melting point than PCl_3;
PCl_5 has stronger intermolecular/London/dispersion/van der Waals' forces;
(because of) more electrons/greater mass; ***[3]***
Accept the opposite argument for PCl_3.
*Award **[1 max]** for answers suggesting PCl_3 has higher melting point because it is polar and PCl_5 is not.*

(e) overlap of *p* orbitals / *p* electrons of double/π bond and non-bonding/lone pair on oxygen interact / *OWTTE*;
π electrons not localized / different resonance structures possible /



both bonds/O–O and O=O have equal length / *OWTTE*;
both bonds/O–O and O=O have equal bond energy / *OWTTE*; ***[4]***

3.1-S24 May 2013 TZ2 · Paper 2 · Q5 parts (a)(iii), (a)(iv), (b)(i), (b)(ii), (c)(i), (c)(ii), (c)(iii), (c)(iv) · also 5.1, 9.2

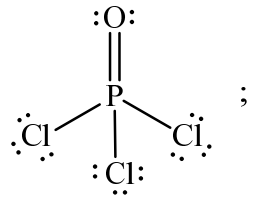
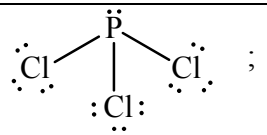
(iii) heat/enthalpy change/required/absorbed when 1 mol of a compound is formed from its elements in their standard states/at 100 kPa/10^5 Pa/1 bar; ***[1]***

Allow $1.01\times10^5\ Pa$ / $101\,kPa$ / $1\,atm$.

Allow under standard conditions or standard temperature and pressure. Temperatures not required in definition, allow if quoted (for example, 298 K/25 °C – most common) but pressure value must be correct if stated.

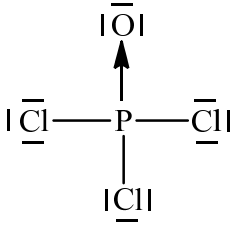
(iv) $\left(\Delta H^{\ominus} = [(2)(-288.1)] - [(2)(-542.2)]\right) = (+)508.2\ (\text{kJ mol}^{-1})$; ***[1]***

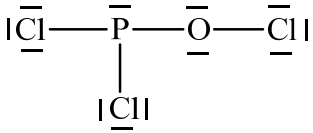
(b) (i)

	$POCl_3$	PCl_3
Lewis (electron dot) Structure	 ; *Accept legitimate alternatives for $POCl_3$ as shown below*.*	 ;
	Allow any combination of dots/crosses or lines to represent electron pairs. *Penalise missing lone pair on P in PCl_3 but penalize missing lone pairs once only on terminal O or Cl atoms.*	
Shape	tetrahedral;	trigonal/triangular pyramidal; *Do not allow tetrahedral.* *Do not allow just pyramidal.*
	Do not allow ECF from Lewis structures with incorrect number of negative charge centres. *Only allow shapes based on legitimate structures below.*	

[4]

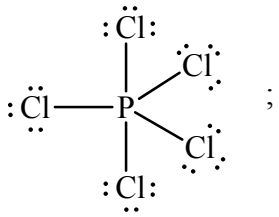
**Legitimate alternatives for $POCl_3$:*





(ii) allow any bond angle in the range 100° to less than 109° (experimental value is 100°);
due to four negative charge centres/four electron pairs/four electron domains (one of which is a lone pair)/tetrahedral arrangement of electron pairs/domains;
extra repulsion due to lone pair electrons / lone pairs occupy more space (than bonding pairs) so Cl–P–Cl bond angle decreases from 109.5° / *OWTTE*;

[3]

(c) (i)  ;

[1]

Allow any combination of dots/crosses or lines to represent electron pairs.
Do not penalise missing lone pairs on Cl if already penalised in (b)(i).

(ii) trigonal/triangular bipyramidal; ***[1]***

Do not allow ECF from Lewis structures with incorrect number of negative charge centres.

(iii) 120° **and** 90°/180°; ***[1]***

Ignore other bond angles such as 240° and 360°.
Apply list principle if some correct and incorrect angles given.

(iv)

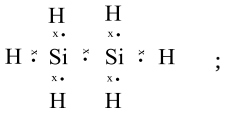
	Isomer 1	Isomer 2	Isomer 3
Structure	both Br's in axial positions, all Cl's in equatorial positions	both Br's in equatorial positions, two Cl's in axial positions, one Cl in equatorial position	one Br in equatorial position, one Br in axial position, one Cl in axial position and two Cl's in equatorial positions
Molecular polarity	non-polar	polar	polar

[3]

*Award **[1]** for correct structure **and** molecular polarity.*
*Award **[1 max]** for correct representations of all three isomers.*
Lone pairs not required.

3.1-S25 May 2012 TZ1 · Paper 2 · Q5 parts (a)(i), (a)(ii), (a)(iii), (a)(iv), (a)(v), (a)(vi), (b)(i), (b)(ii), (b)(iii), (c) · also 4.2, 4.3, 5.1, 5.2, 9.2, 9.3

5. (a) (i)



;

[1]

Accept any combination of lines, dots or crosses to represent electron pairs.

(ii) 109° / 109.5° / 109° 28';
four/tetrahedrally arranged negative charge centres/electron domains/electron pairs (around central/silicon atom) / equal repulsion between bonding pairs (around central/silicon atom) / *OWTTE*; ***[2]***
M2 is an independent marking point.
Reference must be made to negative or electron.
Do not accept tetrahedral molecule.

(iii) sp^3; ***[1]***

(iv) C–H;
larger difference in electronegativity (for C–H bond) / smaller difference in electronegativity (for Si–H bond) / ΔEN (CH) = 0.4 **and** ΔEN (SiH) = 0.3; ***[2]***

(v) both (molecules) non-polar;
both (molecules) symmetrical / polar bond effects cancel out / *OWTTE*; ***[2]***

(vi) stronger/larger/greater van der Waals'/London/dispersion forces;
Do not accept stronger/larger/greater intermolecular forces.

more electrons / stronger instantaneous dipole; ***[2]***
Do not accept larger mass.

(b) (i) ($\sum \Delta H_f^\circ$ (products) =) –5360 (kJ);
($\sum \Delta H_f^\circ$ (reactants) =) +160 (kJ);
= –5520 (kJ); ***[3]***

(ii) (–1560 × 2 =) –3120 (kJ); ***[1]***

(iii) *Structure:*
CO_2 molecular **and** SiO_2 three-dimensional/network/giant lattice/giant covalent/macromolecular/repeating tetrahedral units;
CO_2 linear **and** SiO_2 tetrahedral;

Intramolecular Bonding:
covalent bonds in CO_2 **and** SiO_2;
double bonds in CO_2 **and** single bonds in SiO_2; ***[3 max]***
Accept diagrams showing bonding types (double and single) within the structures.

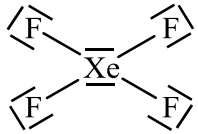
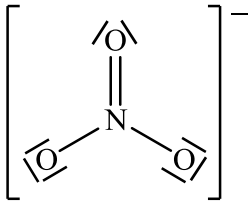
(c) *Bonds broken:*
6Si–H, Si–Si, H–H / (+)2570 (kJ);
Bonds formed:
8Si–H / (–)2544 (kJ);
+26 (kJ);

OR

Bonds broken:
Si–Si, H–H / (+)662 (kJ);
Bonds formed:
2Si–H / (–)636 (kJ);
+26 (kJ); ***[3]***

3.1-S26 May 2012 TZ2 · Paper 2 · Q4

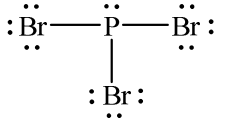
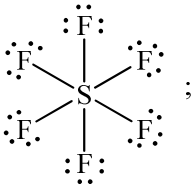
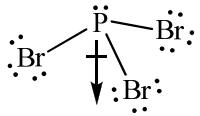
4.

Species	Lewis structure	Shape	Bond angle
XeF_4	 ;	square planar;	90° ;
NO_3^-	 ;	trigonal/triangular planar;	120° ;

[6]

Accept any combination of lines, dots or crosses to represent electron pairs.

3.1-S27 Nov 2011 · Paper 2 · Q6 parts (c)(i), (c)(ii), (c)(iii), (d)(i) · also 4.1, 5.1

(i) *Lewis structure:*  ; *Allow* x*'s, dots or lines to represent electrons.* *Penalize missing lone pairs on terminal atoms once only for the two Lewis structures.*	(i) *Lewis structure:*  ; *Allow* x*'s, dots or lines to represent electrons.* *Penalize missing lone pairs on terminal atoms once only for the two Lewis structures.*
(ii) *Shape:* trigonal/triangular pyramidal; *Bond angle:* less than 109.5°; *Allow any angle less than 109.5° but greater than or equal to 100° (experimental value is 101°).*	(ii) *Shape:* octahedral; *Bond angle:* 90°; *Ignore extra correct bond angles (e.g. 90° and 180° scores but not 90° and 120°).*
(iii) *Polarity:* polar **and** *Explanation:* net dipole (moment) / polar PBr bonds **and** molecule not-symmetrical / bond dipoles do not cancel / asymmetric distribution of electron cloud /  / *OWTTE*;	(iii) *Polarity:* non-polar **and** *Explanation:* no net dipole (moment) / polar SF bonds but molecule symmetrical / bond dipoles cancel / symmetric distribution of electron cloud / *OWTTE*;

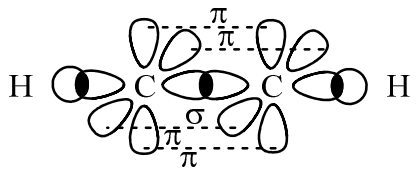
[8]

Do not allow ECF in this question from incorrect Lewis structure.
*Allow **[1 max]** for stating that PBr_3 is polar and SF_6 is non-polar without giving a reason or if explanations are incorrect.*
Allow polar bonds do not cancel for PBr_3 and polar bonds cancel for SF_6.
Do not allow asymmetric molecule as reason for PBr_3 or symmetric molecule for SF_6 as reason alone.

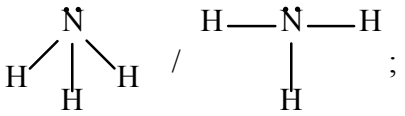
(d) (i) σ *bond:*
end-on/axial overlap with electron density between the two carbon atoms/nuclei / end-on/axial overlap of orbitals so shared electrons are between atoms / *OWTTE*;

π *bond:*
sideways/parallel overlap of p orbitals with electron density above **and** below internuclear axis/σ bond / sideways/parallel overlap of p orbitals so shared electrons are above **and** below internuclear axis/σ bond / *OWTTE*; ***[2]***

Marks can be scored from a suitable diagram.
*Award **[1 max]** for stating end-on/axial overlap for σ and sideways/parallel overlap for π only i.e. without mentioning electron density **OR** stating electron density between the two atoms/nuclei for σ and above and below internuclear axis for π.*



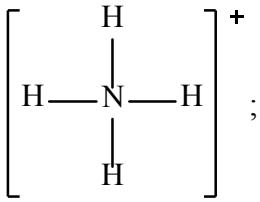
3.1-S28 May 2011 TZ2 · Paper 2 · Q5 parts (a)(i), (a)(ii)

(a) (i)  / ;

Accept any combination of dots/crosses and lines to represent electron pairs.

trigonal/triangular pyramid;
Allow 3D representation using wedges and dotted bonds of trigonal pyramidal molecule.
107°; ***[3]***

Accept any angle between 105° and 108.5°.
No ECF for shape based on incorrect Lewis structure.

(ii)  ;

Charge needed for mark.

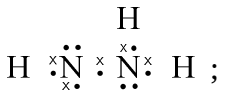
tetrahedral;
Allow a 3D representation using wedges and dotted bonds of tetrahedral molecule.

109.5° / 109° / 109° 28'; ***[3]***

No ECF for shape based on incorrect Lewis structure.

3.1-S29 Nov 2010 · Paper 2 · Q7 parts (a)(i), (a)(ii), (b), (c)(i), (c)(ii), (c)(iii), (d)(ii), (d)(iii) · also 4.2, 5.2, 9.2, 9.3

7. (a) (i)



[1]

Accept x's, dots or lines for electron pairs

(ii) H–N–H < 109 / any angle between 104° and 109°;
due to four centres of electron/negative charge / four electron pairs (one of which is a lone e- pair) / four electron domains;
extra repulsion due to lone electron pairs; ***[3]***
Do not allow ECF for wrong Lewis structures.

(b) weaker van der Waals'/London/dispersion/intermolecular forces in ethene;
stronger (intermolecular) hydrogen bonding in hydrazine; ***[2]***
*If no comparison between strengths then **[1 max]**.*

(c) (i) $\Delta H_r^{\ominus} = \Sigma \Delta H_f^{\ominus}$ products – $\Sigma \Delta H_f^{\ominus}$ reactants;
Can be implied by working.

$\Delta H_f^{\ominus}(H_2O(l)) = -286\,(kJ)$;

$\Delta H_r^{\ominus} = 2(-286) - 50.6 = -622.6\,(kJ)$;

*Award **[3]** for correct final answer.*

[3]

(ii) bonds broken: 4 N–H, N–N, O=O / $+2220\,(kJ\,mol^{-1})$;
bonds formed: N≡N, 4 O–H / $-2801\,(kJ\,mol^{-1})$;
$-581\,(kJ\,mol^{-1})$; ***[3]***
*Award **[3]** for correct final answer.*

(iii) value based on ΔH_f more accurate;
ΔH_f accurate for compounds in reaction;
bond energy calculation assumes average bond energies;
(bond energy calculation) only applies to gaseous states / ignores intermolecular bonds; ***[3 max]***

(ii) $109^{\circ}/109.5^{\circ}$; ***[1]***

(iii) sp^3; ***[1]***

No ECF if bond angle incorrect in (ii).

LESSON 4.1

Sigma & Pi Bonds and Delocalization

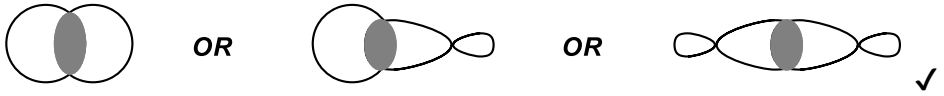
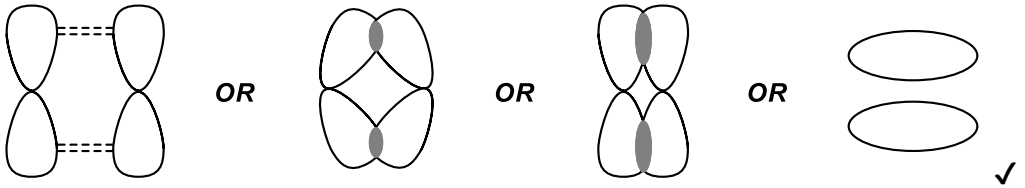
Answers for 54 multiple-choice and 30 structured questions

Part A - Multiple choice answers

4.1-M1	May 2025 TZ1 · P1A · Q13	C
4.1-M2	Nov 2024 · P1 · Q13	C
4.1-M3	May 2024 TZ1 · P1 · Q10	C
4.1-M4	May 2024 TZ2 · P1 · Q14	C
4.1-M5	Nov 2023 TZ1 · P1 · Q9	D
4.1-M6	May 2023 TZ1 · P1 · Q15	C
4.1-M7	May 2023 TZ1 · P1 · Q33	C
4.1-M8	May 2023 TZ2 · P1 · Q13	C
4.1-M9	Nov 2022 · P1 · Q9	B
4.1-M10	Nov 2022 · P1 · Q13	C
4.1-M11	May 2022 TZ1 · P1 · Q11	B
4.1-M12	Nov 2021 · P1 · Q12	B
4.1-M13	May 2021 TZ1 · P1 · Q12	D
4.1-M14	May 2021 TZ2 · P1 · Q10	A
4.1-M15	Nov 2020 · P1 · Q9	D
4.1-M16	Nov 2020 · P1 · Q13	A
4.1-M17	Nov 2019 · P1 · Q9	B
4.1-M18	Nov 2019 · P1 · Q10	A
4.1-M19	May 2019 TZ1 · P1 · Q12	D
4.1-M20	May 2019 TZ1 · P1 · Q36	C
4.1-M21	May 2019 TZ2 · P1 · Q12	B
4.1-M22	Nov 2018 · P1 · Q12	A
4.1-M23	May 2018 TZ1 · P1 · Q13	A
4.1-M24	May 2018 TZ2 · P1 · Q10	B
4.1-M25	May 2018 TZ2 · P1 · Q13	B
4.1-M26	Nov 2017 · P1 · Q14	C
4.1-M27	May 2017 TZ1 · P1 · Q12	B
4.1-M28	May 2017 TZ1 · P1 · Q13	B
4.1-M29	May 2017 TZ2 · P1 · Q10	C
4.1-M30	May 2017 TZ2 · P1 · Q13	C
4.1-M31	May 2017 TZ2 · P1 · Q15	D
4.1-M32	May 2016 · P1 · Q14	C
4.1-M33	Nov 2015 · P1 · Q13	C
4.1-M34	Nov 2014 · P1 · Q10	B
4.1-M35	Nov 2014 · P1 · Q12	C
4.1-M36	Nov 2014 · P1 · Q13	C
4.1-M37	May 2014 TZ1 · P1 · Q10	D
4.1-M38	May 2014 TZ1 · P1 · Q12	C
4.1-M39	May 2014 TZ1 · P1 · Q13	B
4.1-M40	Nov 2013 · P1 · Q13	C
4.1-M41	May 2013 TZ2 · P1 · Q12	C
4.1-M42	May 2013 TZ2 · P1 · Q13	C
4.1-M43	Nov 2012 · P1 · Q13	C
4.1-M44	May 2012 TZ1 · P1 · Q12	C
4.1-M45	May 2012 TZ1 · P1 · Q13	B
4.1-M46	May 2011 TZ1 · P1 · Q10	B
4.1-M47	May 2011 TZ1 · P1 · Q11	A
4.1-M48	May 2011 TZ2 · P1 · Q10	C
4.1-M49	May 2011 TZ2 · P1 · Q13	C
4.1-M50	May 2011 TZ2 · P1 · Q14	D
4.1-M51	Nov 2010 · P1 · Q13	A
4.1-M52	May 2010 TZ1 · P1 · Q10	D
4.1-M53	May 2010 TZ1 · P1 · Q13	B
4.1-M54	May 2010 TZ2 · P1 · Q13	D

Part B - Structured questions markscheme

4.1-S1 May 2022 TZ2 · Paper 2 · Q7 parts (b)(i), (b)(ii), (b)(iii), (c) · also 4.2, 5.2

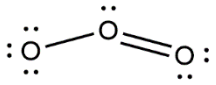
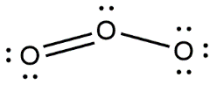
7.	**b**	**i**	*Sigma (σ):*  ***OR*** ***OR*** ✓ *Pi (π):*  ***OR*** ***OR*** ***OR*** ✓	*Accept overlapping p-orbital(s) with both lobes of equal size/shape.* *Shaded areas are not required in either diagram.*	**2**
7.	**b**	**ii**	Sigma (σ): 2 ***AND*** Pi (π): 2 ✓		**1**
7.	**b**	**iii**	sp ✓		**1**
7.	**c**		HCN has stronger dipole–dipole attraction ✓	*Do **not** accept reference to H-bonds.*	**1**

4.1-S2 Nov 2021 · Paper 2 · Q10 parts (a)(i), (a)(ii) · also 4.2

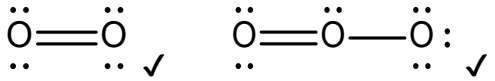
<table>
<tr>
<td>10.</td>
<td>a</td>
<td>i</td>
<td>Sigma (σ) bond:
overlap «of atomic orbitals» along the axial / intermolecular axis / electron density is between nuclei
OR
head-on/end-to-end overlap «of atomic orbitals» ✓

Pi (π) bond:
overlap «of p-orbitals» above and below the internuclear axis/electron density above and below internuclear axis
OR
sideways overlap «of p-orbitals» ✓</td>
<td>Accept a suitable diagram.</td>
<td>2</td>
</tr>
<tr>
<td>10.</td>
<td>a</td>
<td>ii</td>
<td>
<table>
<tr><td></td><td>Ethane</td><td>Ethene</td><td>Ethyne</td></tr>
<tr><td>Hybridization of carbon</td><td>sp^3</td><td>sp^2</td><td>sp</td></tr>
</table>
✓
</td>
<td>All 3 required for mark.</td>
<td>1</td>
</tr>
</table>

4.1-S3 May 2021 TZ1 · Paper 2 · Q7 parts (a)(i), (a)(ii), (b) · also 3.1

7.	**a**	**i**	 ***OR***  ✓	*Accept any combination of lines, dots or crosses to represent electrons.* *Do **not** accept structures that represent 1.5 bonds.*	**1**
7.	**a**	**ii**	both equal ✓ delocalization/resonance ✓	*Accept bond length between 121 and 148 pm/ that of single O-O bond and double O=O bond for M1.*	**2**
7.	**b**		bond in O_3 is weaker ***OR*** O_3 bond order 1.5/< 2 ✓ lower frequency/longer wavelength «UV light» has enough energy to break the O–O bond in O_3 «but not that in O_2» ✓	*Do **not** accept bond in O_3 is longer for M1.* *Accept "lower frequency/longer wavelength «UV light» has lower energy".*	**2**

4.1-S4 Nov 2019 · Paper 2 · Q1 parts (a), (b), (c), (d), (e)(i), (e)(ii) · also 3.1

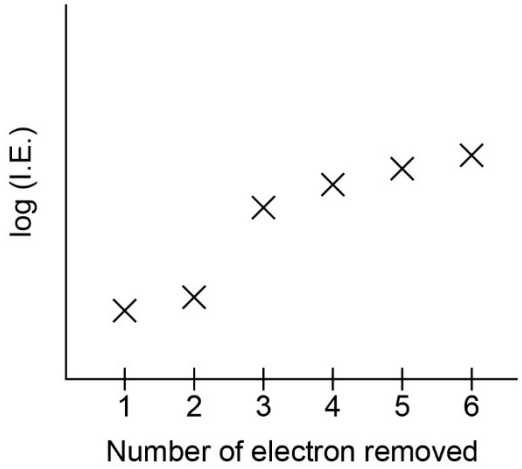
1.	**a**		 ✓ ✓	*Coordinate bond may be represented by an arrow.* *Do **not** accept delocalized structure for ozone.*	**2**
1.	**b**		resonance **«**structures**»** ***OR*** delocalization of **«**the double/pi bond**»** electrons ✓ 121 **«**pm**»** < length < 148 **«**pm**»** ✓	*Accept any length between these two values.*	**2**
1.	**c**		any value from 110°–119° ✓		**1**
1.	**d**		**«**bond**»** in O_2 stronger than in O_3 ✓ ozone absorbs lower frequency/energy **«**radiation than oxygen**»** ***OR*** ozone absorbs longer wavelength **«**radiation than oxygen**»** ✓	*Accept ozone **«**layer**»** absorbs a range of frequencies.*	**2**
1.	**e**	**i**	steps 1 ***AND*** 3 ✓		**1**

1.	**e**	**ii**	***ALTERNATIVE 1:*** for oxygen: $E = \text{«}\frac{498\,000\ \text{J mol}^{-1}}{6.02\times10^{23}\ \text{mol}^{-1}}\ \text{»}\ 8.27\times10^{-19}\ \text{«J»}$ ✓ $\lambda = \text{«}\frac{6.63\times10^{-34}\ \text{J s}\times3.00\times10^{8}\ \text{m s}^{-1}}{8.27\times10^{-19}\ \text{J}}\ \text{»}\ 2.40\times10^{-7}\ \text{«m»}$ ✓ ***ALTERNATIVE 2:*** for ozone: similar calculation using 200 < bond enthalpy < 400 for ozone, such as $E = \text{«}\frac{300\,000\ \text{J mol}^{-1}}{6.02\times10^{23}\ \text{mol}^{-1}}\text{»}\ 4.98\times10^{-19}\ \text{«J»}$ ✓ $\lambda = \text{«}\frac{6.63\times10^{-34}\ \text{J s}\times3.00\times10^{8}\ \text{m s}^{-1}}{4.98\times10^{-19}\ \text{J}}\ \text{»}\ 3.99\times10^{-7}\ \text{«m»}$ ✓	*Award **[2]** for correct final answer.*	**2**

4.1-S5 May 2019 TZ1 · Paper 2 · Q2 parts (c), (f)(ii), (g), (h) · also 5.2, 6.1, 7.3

2.	**c**		*Any one of:* «regular» hexagon ***OR*** all «H–C–C/C-C-C» angles equal/120º ✓ all C–C bond lengths equal/intermediate between double and single ***OR*** bond order 1.5 ✓		**1**
2.	**f**	**ii**	$2C_6H_5COOH\,(s) + 15O_2\,(g) \rightarrow 14CO_2\,(g) + 6H_2O\,(l)$ correct products ✓ correct balancing ✓		**2**
2.	**g**		*Oxidized:* C/carbon **«**in C_6H_5COOH**»** ***AND*** *Reduced:* O/oxygen **«**in O_2**»** ✓		**1**
2.	**h**		«intermolecular» hydrogen bonding ✓	*Accept diagram showing hydrogen bonding.*	**1**

4.1-S6 May 2018 TZ1 · Paper 2 · Q2 parts (e), (g)(i), (g)(ii) · also 2.1

2.	**e**		 general increase ✓ only one discontinuity between “IE2” and “IE3” ✓		**2**

2.	**g**	**i**	*sigma (σ):* overlap **«**of atomic orbitals**»** along the axial/internuclear axis ***OR*** head-on/end-to-end overlap **«**of atomic orbitals**»** ✓ *pi (π):* overlap **«**of p-orbitals**»** above and below the internuclear axis ***OR*** sideways overlap **«**of p-orbitals**»** ✓	*Award marks for suitable diagrams.*	**2**
2.	**g**	**ii**	*sigma (σ):* 3 ***AND*** *pi (π):* 2 ✓		**1**

4.1-S7 May 2018 TZ2 · Paper 2 · Q8 parts (a)

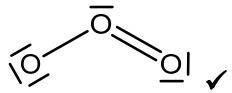
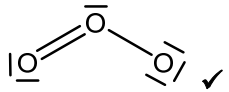
8.	**a**		*Physical evidence:* equal C–C bond «lengths/strengths» ***OR*** regular hexagon ***OR*** **«**all**»** C–C have bond order of 1.5 ***OR*** «all» C–C intermediate between single and double bonds ✓ *Chemical evidence:* undergoes substitution reaction **«**more readily than addition**»** ***OR*** does not discolour/react with bromine water ***OR*** substitution forms only one isomer for 1,2-disubstitution **«**presence of alternate double bonds would form two isomers**»** ***OR*** more stable than expected **«**compared to hypothetical molecule cyclohexa-1,3,5-triene**»** ***OR*** enthalpy change of hydrogenation/combustion is less exothermic than predicted **«**for cyclohexa-1,3,5-triene**»** ✓	*M1:* *Accept "all C–C–C bond angles are equal".*	**2**

4.1-S8 May 2017 TZ1 · Paper 2 · Q7 parts (a), (b)

7.	**a**		*Any two of:* planar **«**X-ray**»** ✓ C to C bond lengths all equal ***OR*** C to C bonds intermediate in length between C–C and C=C ✓ all C–C–C bond angles equal ✓		**2 max**
7.	**b**		*benzene:* **«**electrophilic**»** substitution/S_E ***AND*** *cyclohexene:* **«**electrophilic**»** addition/A_E ✓	*Accept correct equations.*	**1**

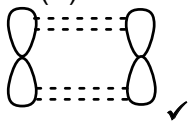
4.1-S9 May 2017 TZ2 · Paper 2 · Q4 parts (a)(i), (a)(ii), (a)(iii), (a)(iv), (b)(i), (b)(ii), (c) · also 2.1, 3.1, 5.2, 6.1

4.	**a**	**i**	triple bond in nitrogen **«**molecule**»** ***AND*** single bond in hydrazine ✓ triple bond is stronger than single bond ***OR*** more shared **«**pairs of**»** electrons make bond stronger/attract nuclei more ✓	*Accept bond enthalpy values from data booklet (158 and 945 kJ mol^{-1}).*	**2**
4	**a**	**ii**	hydrogen bonding **«**between molecules, dinitrogen tetraoxide does not**»** ✓		**1**
4	**a**	**iii**	N_2H_4: –2 ***AND*** N_2O_4: +4 ✓		**1**
4	**a**	**iv**	N_2H_4 ***AND*** oxidized/oxidation state increases ***OR*** N_2H_4 ***AND*** loses hydrogen ***OR*** N_2H_4 ***AND*** reduces/removes oxygen from N_2O_4 ✓	*Accept "N_2H_4* ***AND*** *gives electrons* ***«****to N_2O_4****»"*.	**1**

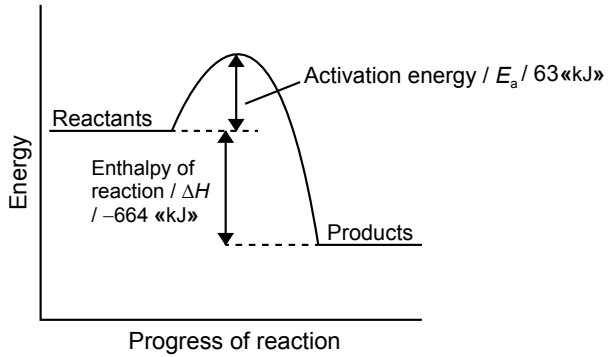
4	**b**	**i**	lone pair on p orbital **«**of O atom**»** overlaps/delocalizes with pi electrons **«**from double bond**»** ✓ both O–O bonds have equal bond length ***OR*** both O–O bonds have same/1.5 bond order ***OR*** both O–O are intermediate between O–O ***AND*** O=O ✓ both O–O bonds have equal bond energy ✓	*Accept "p/pi/π electrons are delocalized/not localized".*	**3**
4	**b**	**ii**	***ALTERNATIVE 1:***  ✓ FC: −1 ***AND*** +1 ***AND*** 0 ✓ ***ALTERNATIVE 2:***  ✓ FC: 0 ***AND*** +1 ***AND*** −1 ✓	*Accept any combination of lines, dots or crosses to represent electrons.* *Do not accept structure that represents 1.5 bonds.* *Do not penalize missing lone pairs if already penalized in 3(b).* *If resonance structure is incorrect, no ECF.* *Any one of the structures with correct formal charges for **[2 max]**.*	**2**

4	**c**		*Any two of:* IE_4: electron in lower/inner shell/energy level ***OR*** IE_4: more stable/full electron shell ✓ IE_4: electron closer to nucleus ***OR*** IE_4: electron more tightly held by nucleus ✓ IE_4: less shielding by complete inner shells ✓	*Accept "increase in effective nuclear charge" for M2.*	**2 max**

4.1-S10 Nov 2016 · Paper 2 · Q5 parts (b)(i)

5.	**b**	**i**	*Sigma* (σ): ***OR*** ✓ *Pi* (π):  ✓		**2**

4.1-S11 May 2016 · Paper 2 · Q3 parts (b)(i), (c)(i), (c)(ii) · also 8.1

3.	**b**	**i**	 products lower than reactants ***AND*** enthalpy of reaction correctly marked and labelled with name or value ✓ activation energy correctly marked and labelled with name or value ✓	*Accept other clear ways of indicating energy/ enthalpy changes.*	**2**
3.	**c**	**i**	**ALTERNATIVE 1:** σ-bond from N to N ***AND*** from N to O ✓ π-bond from N to N ✓ <u>delocalized</u> π-bond/π-electrons «extending over the oxygen and both nitrogens» ✓ **ALTERNATIVE 2:** both have 2 σ-bonds «from N to N and from N to O» ***AND*** π-bond from N to N ✓ one structure has second π-bond from N to N and the other has π-bond from N to O ✓ delocalized π-bond/π-electrons ✓	*Award **[1 max]** if candidate has identified both/either structure having 2 σ-bonds and 2 π-bonds.*	**3**
3.	**c**	**ii**	more than one possible position for a multiple/π-/pi- bond ✓	*Accept "more than one possible Lewis structure".* *Accept reference to delocalisation if M3 not awarded in c (i).* *Accept reference to fractional bond orders.*	**1**

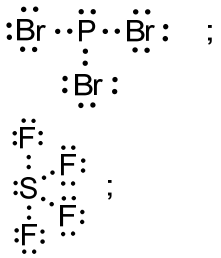
4.1-S12 May 2015 TZ1 · Paper 3 · Q20 parts (a)

20. (a) O_3 needs less energy/has weaker bonding/has lower bond enthalpy;
Accept converse argument for O_2.

O_3 has bond order 1.5/intermediate between double and single **and** O_2 has bond order 2/double bond; **[2]**

4.1-S13 May 2015 TZ2 · Paper 2 · Q9 parts (c)(i), (c)(ii), (c)(iii), (d)(i), (d)(ii), (d)(iii), (d)(iv) · also 3.1, 4.2, 5.1

(c) (i)



; ; **[2]**

Penalize lone pairs missing on Br and F once only.
Accept any combination of lines, dots or crosses to represent electron pairs.

(ii)

PBr_3	SF_4
Shape: trigonal pyramidal; *Accept triangular pyramidal.*	*Shape:* see-saw/K-shaped; *Accept distorted tetrahedral.*
No ECF for shape if Lewis structure is incorrect.	
Bond angle: *Any value* 99°<109°; *Literature value = 101°.*	*Bond angle:* *Any two for **[1]** of:* (F_{eq}–S–F_{eq}): *Any value* 100°<120° *Literature value = 103°* (F_{ax}–S–F_{ax}): *Any value* 175°<185° *Literature value = 179°/180°* (F_{ax}–S–F_{eq}): *Any value* 85°< 95° *Literature value = 89°*
No ECF for angle if shape is incorrect. *Do not award mark for correct angles if shape is incorrect.*	

[4]

(iii) P–Br and S–F bonds are polar / bonds in both molecules are polar;
non-symmetrical distribution of electron cloud / polar bonds/dipoles do not cancel because of non-symmetrical shape; **[2]**
M2 may also be scored with a suitable diagram showing the vectorial addition of the individual S–F dipole moments to show a net dipole moment centred along the axis between the F_{eq}–S–F_{eq} bond.

(d) (i) *EITHER*
(electrostatic) attraction between (positively charged) nuclei and a pair of electrons;
formed as a result of electron sharing (between the carbon and hydrogen nuclei);

OR

sigma bond formed by overlap of atomic orbitals;
s orbital from H and p/sp^2 from carbon; **[2]**

(ii) *α:* sp^3 **and** *β:* sp^2; **[1]**
Accept if numbers are given as subscripts.

(iii) *σ bond:*
end-on / axial overlap of two orbitals;

π bond:
sideways overlap of two (parallel) p orbitals; **[2]**

Accept suitable diagrams for both marks.

(iv) 11 σ **and** 3 π; **[1]**

alize missing hydrogens only once in Question 10.

4.1-S14 May 2015 TZ2 · Paper 3 · Q23 parts (a)

23. (a) O_2 has double bond/bond order of 2 **and** O_3 has bond length between O–O and O=O/bond order of 1.5;
Accept "bond in O_2 stronger than in O_3" / OWTTE.

higher frequency (UV radiation) absorbed by O_2 / low frequency (UV radiation) absorbed by O_3; **[2]**
Accept correct answers phrased in terms of energy/wavelength rather than frequency.

4.1-S15 May 2015 TZ2 · Paper 3 · Q30 parts (a), (b)

30. (a) *Structure:*
hexagon with all sides equal / <u>regular</u> hexagon;
*Accept "**two** of angles equal" OR "planar/flat" OR "hexagonal". (True for Kekulé structure!)*

Bonding:
all (C–C) bonds equal strength / all (C–C) bonds have a strength between single and double bond / (C–C) bond orders all 1.5;
Accept "length" instead of "strength".
delocalized π-bond/electrons / resonance hybrid between two equivalent structures; **[3]**
Accept all C-atoms have one π-bond and three σ-bonds / sp^2 hybridization.

Give credit for correct responses given in the wrong section.

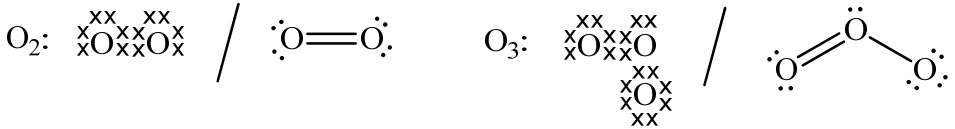
(b) (enthalpy of) hydrogenation much less (negative/exothermic)/more endothermic than predicted (for alternate single and double C–C bonds);
enthalpy of formation much less (positive/endothermic)/more exothermic than predicted (for alternate single and double C–C bonds);
(enthalpy of) combustion much less (negative/exothermic)/more endothermic than predicted (for alternate single and double C–C bonds); **[1 max]**
Accept opposite statements for alternate single and double C–C bonds.

4.1-S16 Nov 2014 · Paper 3 · Q22 parts (a)

22. (a) O_2 double bond **and** O_3 bond order 1.5/between single and double bond;
O_2 stronger/shorter / O_3 weaker/longer bond;
O_2 absorbs shorter λ (242 nm) / O_3 absorbs longer λ (330 nm); ***[3]***

4.1-S17 May 2014 TZ1 · Paper 3 · Q16 parts (d)

(d) *Oxygen:* correctly drawn Lewis structure **and** *Ozone:* correctly drawn Lewis structure;



Allow any combination of dots, x's or lines to represent electron pairs.
Allow representation of two resonance structures for ozone.

oxygen has a higher bond order than ozone **and** oxygen needs more energy to dissociate / *OWTTE*; ***[2]***
*Exact bond orders of 2 for oxygen and 1.5/1 **and** 2 for ozone may be given for first statement in M2.*
Do not award M2 if incorrect bond orders are stated for either species.

4.1-S18 Nov 2013 · Paper 3 · Q27 parts (a)(i), (a)(ii)

27. (a) (i) (circle represents six) delocalized electrons (over 6 C–C bonds); ***[1]***
Allow "delocalized pi-bonds".
Accept "resonance".

(ii) C–C/bond length/order is intermediate between double and single/1.5; ***[1]***

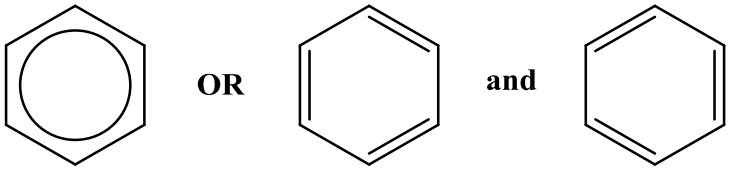
4.1-S19 May 2013 TZ1 · Paper 3 · QE1 parts (c)

(c) ozone depletion requires higher wavelength / oxygen dissociation requires lower wavelength;
M1 must refer to wavelength.
wavelength inversely proportional to energy / wavelength increases as energy decreases;
oxygen has double bond/bond order of 2 **and** ozone has bond order of 1.5/intermediate bond between single and double;
Stating that ozone has resonance or delocalization alone is not sufficient.
ozone weaker bond / O–O bond energy in ozone lower; ***[3 max]***
Accept opposite argument of oxygen for M4.

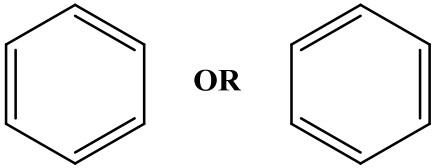
4.1-S20 May 2013 TZ1 · Paper 3 · QG1 parts (a)

G1. (a) six-membered ring (of carbon atoms) / hexagonal;
planar;
Allow flat.

all carbon to carbon bond lengths equal / all bonds have bond order of 1.5 / all carbon to carbon bonds intermediate between single and double;
resonance/delocalization;
all bond angles 120°; ***[3 max]***
Accept all carbons sp^2.
*Award **[2 max]** for representation of full delocalized benzene structure or two Kekulé structures without any explanations.*
For example:



*Award **[1 max]** for one Kekulé structure or for the name cyclohexatriene without any explanations.*
For example:



4.1-S21 May 2013 TZ2 · Paper 3 · QG2 parts (a)

G2. (a) all C–C bond lengths/strengths are the same/between the different lengths/strengths for C–C single bond and a C=C double bond / *OWTTE*;
Do not accept "bonds are the same" without any qualification.

only one 1,2-disubstituted isomer exists whereas there would be two if there were alternate double and single bonds / only three disubstituted benzene compounds rather than four / *OWTTE*;
benzene (mainly) undergoes (electrophilic) substitution reactions rather than addition reactions (which would be expected if it contains C=C double bonds) / *OWTTE*;
Accept does not undergo addition reactions / decolourize bromine water.

the enthalpy change of hydrogenation/combustion is less exothermic than would be expected if benzene contained three double bonds/not 3 times that of cyclohexene / the enthalpy change of formation is less endothermic / *OWTTE*; ***[3 max]***

4.1-S22 May 2012 TZ1 · Paper 3 · QE1 parts (a)

E1. (a) oxygen has double bond/bond order 2;
ozone has delocalized bond/bond order 1.5;
oxygen bond/bond in oxygen (molecule) is stronger/shorter / ozone bond is weaker/longer;
ozone absorbs light of lower energy/lower frequency/longer wavelength; ***[3 max]***
Accept suitable diagram.

4.1-S23 May 2012 TZ2 · Paper 2 · Q7 parts (a)(vi), (a)(vii), (b)(i), (b)(ii), (b)(iii) · also 4.2, 4.3

(vi) *Stronger bond:*
C=O /double bond;

Longer bond:
C–O /single bond; ***[2]***

(vii) bond length and bond strength identical for both carbon to oxygen bonds;
intermediate between single and double bond length and strength;
due to delocalization of the electrons (in the p orbitals); ***[3]***
Accept use of Data Booklet values of bond lengths and bond enthalpies.
Accept diagram of delocalization or the two resonance structures for M3.

(b) (i) mixing/joining together/combining/merging of (atomic) orbitals to form molecular/new orbitals (of equal energy); ***[1]***

(ii) sp^3;
isolated C atom electron configuration $1s^2 2s^2 2p^2$ / excited state C electron configuration is $1s^2\ 2s^1 2p^3$;
$2s^1 2p^3$ electrons blend to form four identical hybrid orbitals;
hybrid orbitals lower in total energy than atomic orbitals;
repulsion of (identical hybrid) orbitals creates a tetrahedral shape; ***[3 max]***
Accept suitably annotated diagram for M2, M3 and M4.

(iii) *diamond*:
sp^3;

graphite:
sp^2;
(p) electrons delocalized (around layer); ***[3]***

4.1-S24 May 2012 TZ2 · Paper 3 · QE3 parts (a)

E3. (a) bonds in oxygen are double **and** bonds in ozone are (intermediate) between single and double;
bonds in oxygen are stronger / bonds in ozone are weaker;
oxygen absorbs higher energy UV light / ozone absorbs lower energy UV light; ***[3]***

4.1-S25 May 2011 TZ1 · Paper 3 · QE4 parts (a)

E4. (a) reaction II (requires a shorter wavelength);
O_2 has double bond/bond order 2 **and** O_3 intermediate between double and single bonds/bond order of $1\frac{1}{2}$; ***[2]***
Do not accept stronger/weaker bonding without justification for the second marking point.

4.1-S26 May 2011 TZ2 · Paper 3 · QG1 parts (b)

(b) addition reactions not favoured energetically since this would involve disruption of cloud of delocalized electrons / stabilization energy would need to be supplied and product would lack delocalized ring of electrons making it less stable / *OWTTE*; ***[1]***

4.1-S27 Nov 2010 · Paper 2 · Q2 parts (a)(i), (a)(ii), (d)(iv) · also 4.2, 5.2

2. (a) (i) (bond formed by) sideways overlap;
(of) p orbitals; ***[2]***
Marks awarded either from sketch or from explanation.

(ii) C(l) is sp^3 **and** C(2) is sp^2; ***[1]***

Markscheme for 2(d)(iv) not located automatically - see the official Nov 2010 · Paper 2 markscheme.

4.1-S28 Nov 2010 · Paper 3 · QE4 parts (a)(i)

E4. (a) (i) shorter wavelength / higher energy radiation / UV is needed to break the bond in O_2;
because O_2 has double/stronger bond; ***[2]***
Accept converse argument for O_3.

4.1-S29 Nov 2010 · Paper 3 · QG1 parts (a), (b)

G1. (a) hexagonal / ring of six carbon atoms (each with one hydrogen);
planar;
all carbon-carbon bond lengths equivalent / all carbon-carbon bond lengths intermediate between single and double bonds/carbon-carbon bond order of 1.5;
all C–C–C bond angles 120°;
Allow sp^2 (hybridization for C's).

delocalization / resonance; ***[3 max]***

(b) enthalpy change of hydrogenation not equal to three times enthalpy change of hydrogenation of cyclohexene;
electron density map (of benzene) showing equal electron density / all carbon-carbon bond lengths equivalent/*OWTTE*;
Allow diffraction pattern or contour map for electron density map.

only one isomer exists for 1,2-disubstituted benzene compounds / only three disubstituted benzene compounds (rather than four);
undergoes (electrophilic) substitution reactions/does not undergo addition reactions/ does not decolorize bromine water; ***[2 max]***

4.1-S30 May 2010 TZ2 · Paper 3 · QE3 parts (d)

(d) bond in O_2 is <u>stronger</u> than the bonds in O_3 / bond in O_3 is <u>weaker</u> than the bond in O_2;
in O_2 bond is double/bond order is 2 **and** bonds in O_3 are intermediate between single and double/have an order of 1.5;
energy of light/photon depends on wavelength / *OWTTE*;
energy of light/photon with $\lambda = 300–330$ nm is insufficient to break bonds in O_2 but enough to break bonds in O_3 / *OWTTE*; ***[3 max]***

LESSON 4.2

Hybridization

Answers for 25 multiple-choice and 3 structured questions

Part A - Multiple choice answers

4.2-M1	May 2025 TZ2 · P1A · Q10	B
4.2-M2	May 2025 TZ3 · P1A · Q10	B
4.2-M3	Nov 2024 · P1 · Q12	A
4.2-M4	Nov 2023 TZ1 · P1 · Q13	C
4.2-M5	May 2023 TZ1 · P1 · Q13	A
4.2-M6	May 2022 TZ1 · P1 · Q12	D
4.2-M7	Nov 2021 · P1 · Q13	D
4.2-M8	May 2021 TZ1 · P1 · Q13	B
4.2-M9	Nov 2019 · P1 · Q12	A
4.2-M10	May 2019 TZ1 · P1 · Q13	A
4.2-M11	May 2019 TZ2 · P1 · Q13	A
4.2-M12	Nov 2018 · P1 · Q13	D
4.2-M13	May 2018 TZ1 · P1 · Q12	D
4.2-M14	Nov 2017 · P1 · Q13	A
4.2-M15	May 2017 TZ1 · P1 · Q11	A
4.2-M16	Nov 2016 · P1 · Q15	D
4.2-M17	May 2016 · P1 · Q15	A
4.2-M18	May 2015 TZ1 · P1 · Q13	B
4.2-M19	May 2015 TZ2 · P1 · Q13	A
4.2-M20	May 2014 TZ2 · P1 · Q11	C
4.2-M21	Nov 2013 · P1 · Q14	A
4.2-M22	May 2013 TZ1 · P1 · Q13	B
4.2-M23	Nov 2011 · P1 · Q14	A
4.2-M24	May 2011 TZ2 · P1 · Q15	A
4.2-M25	May 2010 TZ1 · P1 · Q14	C

Part B - Structured questions markscheme

4.2-S1 Nov 2024 · Paper 2 · Q2 parts (a)(i), (a)(ii) · also 4.1

2.	**(a)**	**(i)**	***A:*** sp^2 ***AND*** ***B:*** sp^3 ✓		**1**
2.	**(a)**	**(ii)**	σ *bonds*: orbital overlap/high electron density along internuclear axis/between nuclei ***OR*** head-on/end-to-end overlap **«**of atomic/hybridized orbitals**»** ✓ π *bonds*: **«**orbital**»** overlap/high electron density above and below internuclear/bond axis ***OR*** sideways overlap **«**of parallel p orbitals**»** ✓	*Award **[2 max]** for appropriate diagrams.*	**2**

4.2-S2 Nov 2022 · Paper 2 · Q2 parts (b), (d), (e)(ii) · also 2.1, 4.1

2.	**(b)**		nine/9 ✓		**1**
2.	**(d)**		**«**bond in ring is**»** shorter ***AND*** more electrons are shared ***OR*** **«**bond in ring is**»** shorter ***AND*** partial double/multiple bonding/bond order 1.5 ✓		**1**
2.	**(e)**	**(ii)**	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^{10}$ / $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^1$ ***OR*** $[Ar] 4s^1 3d^{10}$ / $[Ar] 3d^{10} 4s^1$ ✓		**1**

4.2-S3 May 2013 TZ2 · Paper 2 · Q8 parts (b)

(b) **1**: sp^2 **and 2**: sp^3 ; ***[1]***

LESSON 4.3

Covalent Network Structures

Answers for 13 multiple-choice and 10 structured questions

Part A - Multiple choice answers

4.3-M1	May 2025 TZ1 · P1A · Q8	**A**
4.3-M2	May 2024 TZ1 · P1 · Q9	**D**
4.3-M3	May 2024 TZ2 · P1 · Q11	**D**
4.3-M4	May 2021 TZ2 · P1 · Q13	**D**
4.3-M5	Nov 2019 · P1 · Q11	**A**
4.3-M6	May 2018 TZ1 · P1 · Q9	**C**
4.3-M7	May 2017 TZ1 · P1 · Q9	**A**
4.3-M8	Nov 2016 · P1 · Q13	**D**
4.3-M9	May 2014 TZ1 · P1 · Q11	**C**
4.3-M10	May 2014 TZ2 · P1 · Q13	**A**
4.3-M11	May 2013 TZ1 · P1 · Q11	**A**
4.3-M12	May 2013 TZ2 · P1 · Q10	**A**
4.3-M13	Nov 2011 · P1 · Q11	**A**

Part B - Structured questions markscheme

4.3-S1 Nov 2023 TZ1 · Paper 3 · Q3 parts (b)

3.	**b**		*Good conductors along the tube:* pi/delocalized electrons move freely **«**within electron cloud along tube**»** ***OR*** atoms bonded/in contact ✓ *Poor conductors across the width of the tube:* pi/delocalized electrons/electron cloud does not extend across walls/tubes ***OR*** atoms not bonded ✓		**2**

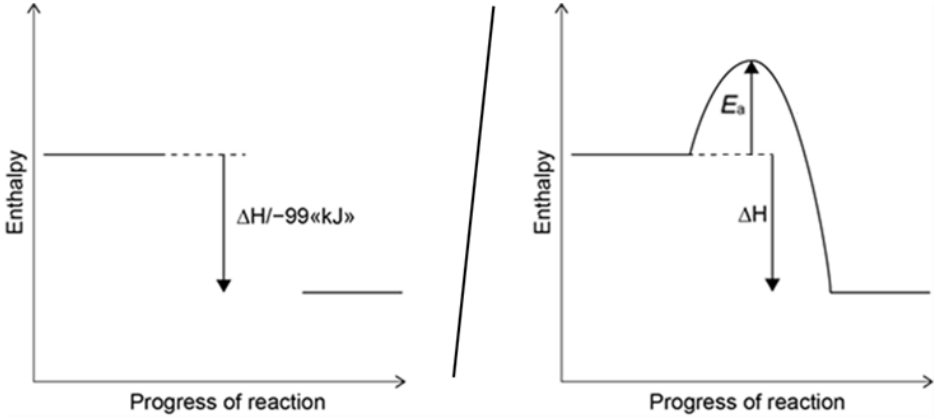
4.3-S2 May 2023 TZ2 · Paper 3 · Q1 parts (c)

1.	**(c)**		«different» allotropes ✓	*Accept "different structural forms"* ***OR*** *"oxygen forms different molecules".* *Accept correct formulas or names of allotropes.* *Accept monatomic oxygen/O only if mentioned with respect to other allotropic form or explanation provided.* *Do* ***not*** *accept "different isotopes" alone.*	**1**

4.3-S3 May 2022 TZ2 · Paper 2 · Q8 parts (a)(i), (a)(ii), (f)(i), (f)(ii) · also 8.1, 9.3

8.	**a**	**i**	Any **two** of: C_{60} fullerene: bonded to 3 C ***AND*** diamond: bonded to 4 C ✓ C_{60} fullerene: delocalized/resonance ***AND*** diamond: not delocalized / no resonance ✓ C60 fullerene: sp^2 ***AND*** diamond: sp^3✓ C_{60} fullerene: bond angles between 109–120° ***AND*** diamond: 109° ✓	*Accept "bonds in fullerene are shorter/stronger/have higher bond order **OR** bonds in diamond longer/weaker/have lower bond order".*	**2 max**
8.	**a**	**ii**	diamond giant/network covalent ***AND*** sublimes at higher temperature ✓ C_{60} molecular/London/dispersion/intermolecular **«**forces**»** ✓	. *Accept "diamond has strong covalent bonds **AND** require more energy to break **«**than intermolecular forces**»" for M1.*	**2**

8.	**f**	**i**	bond breaking: C–H + Cl–Cl / 414 **«**kJ mol^{-1}**»** + 242 **«**kJ mol^{-1}**»**/656 **«**kJ**»** ***OR*** bond breaking: 4C–H + Cl–Cl / 4 × 414 **«**kJ mol^{-1}**»** + 242 **«**kJ mol^{-1}**»** / 1898 **«**kJ**»** ✓ bond forming: **«**C–Cl + H–Cl / 324 kJ mol^{-1} + 431 kJ mol^{-1}**»** / 755 **«**kJ**»** ***OR*** bond forming: **«**3C–H + C–Cl + H–Cl / 3 × 414 **«**kJ mol^{-1}**»** + 324 **«**kJ mol^{-1}**»** + 431 kJ mol^{-1}**»** / 1997 **«**kJ**»** ✓ **«**ΔH = bond breaking – bond forming = 656 kJ – 755 kJ**»** = –99 **«**kJ**»** ✓	*Award **[3]** for correct final answer.* *Award **[2 max]** for 99 **«**kJ**».***	**3**

8.	**f**	**ii**	 reactants at higher enthalpy than products ✓ ΔH/-99 **«**kJ**»** labelled on arrow from reactants to products ***OR*** activation energy/E_a labelled on arrow from reactant to top of energy profile ✓	*Accept a double headed arrow between reactants and products labelled as ΔH for M2.*	**2**

4.3-S4 Nov 2020 · Paper 3 · Q4 parts (a)

4.	**a**		*Excellent strength:* defect-free ***AND*** rigid/regular 2D/3D ✓ *Excellent conductivity:* delocalized electrons ✓	*Accept "carbons/atoms are all covalently bonded to each other" for M1.*	**2**

4.3-S5 May 2019 TZ1 · Paper 3 · Q6 parts (a), (b)

6.	**a**		*Structure:* giant covalent/network covalent ✓ *Bonding:* each carbon covalently bonded to 3 other carbons ***OR*** each bond has order of 1.5 ✓	*Accept "cylindrical/tube shaped".* *Accept "has delocalized electrons" **OR** "has sp^2 hybridization".*	**2**
6.	**b**		*Any one of:* 3D electrodes ✓ catalysts ✓ biosensors ✓ molecular stents ✓ body armour ✓ synthetic muscles ✓ micro transistors/circuitry/capacitors/electrodes ✓ reinforcing phase in a matrix/composite material «such as concrete» ✓ micro antenna ✓ stealth technology ✓ water/air filtration ✓ solar cells ✓ tennis racquets ✓ microelectronic circuits ✓	*Do **not** accept just general answers such as "medicine" or "defence".*	**1 max**

4.3-S6 May 2018 TZ2 · Paper 3 · Q1 parts (a)(i), (a)(ii), (b), (c)

1.	**a**	**i**	consists of single/one sheet/layer **«**of carbon atoms**»** ✓ graphene has no density measurement ***OR*** graphene has no distance between layers data ***OR*** graphene has large specific surface area **«**compared to graphite**»** ✓	*Do **not** accept "sp^2" alone without reference to single/one sheet/layer.* *Accept "thickness of one atom" **OR** "consists of a plane" for M1.*	**2**
1.	**a**	**ii**	*Any one of these alternatives:* ***ALTERNATIVE 1*** **«** $\frac{1.3\times10^{11}}{76\times10^{6}}$ **»** 1.7×10^3/1711 ✓ ***ALTERNATIVE 2*** $1600 \times 76 \times 10^6 = 1.2 \times 10^{11}$ **«**is less than tensile strength of graphene**»** ✓ ***ALTERNATIVE 3*** $\frac{1.3\times10^{11}}{1600} = 8.1 \times 10^7$ **«**is greater than upper end of tensile strength for graphite**»** ✓	*Accept any value in the range 1700–27 083. Answer may be expressed in scientific notation or otherwise.* *Accept any value calculated which is less than the graphene tensile strength based on a value chosen from within the 4.8–76 $\times$ 10^6 range.*	**1**

1.	**b**		smaller/zero ✓ no delocalized electrons/electrons are bound/electrons not free to move/electrons not free to roam ***OR*** localized electrons **«**in sigma bonds**»** ***OR*** large band gap ✓	*Accept "diamond is a dielectric"* ***OR*** *"diamond does* ***not*** *conduct electricity" for M2.* *Award* ***[1 max]*** *for just "immobile/less mobile".* *Award* ***[2]*** *for "electrons immobile* ***«****in diamond****»*** *due to the large band gap"* ***OR*** *"electrons* ***«****in diamond****»*** *immobile since electrons are localized* ***«****in the sigma bonds****»****".*	**2**
1.	**c**		shorter bonds in graphene ***OR*** bonds in graphene intermediate between single and double ***OR*** bond order in graphene is 1.33 ***OR*** delocalization creates stronger bonds ***OR*** shorter bonds are stronger ✓ stronger/shorter bonds require higher temperature/faster thermal motion to be altered ***OR*** stronger/shorter bonds require greater energy to be broken ✓		**2**

4.3-S7 Nov 2015 · Paper 3 · Q13 parts (a)

13. (a) in a nanotube all atoms are held together by (strong) covalent bonds;
in graphite there are (weak) intermolecular/London/dispersion/instantaneous induced dipole-induced dipole forces between layers; ***[2]***
Accept "vdW/van der Waals' forces" for "London forces".

4.3-S8 May 2015 TZ2 · Paper 3 · Q16 parts (c)

(c) strong covalent/C–C bonding (in the walls of the nanotubes) / *OWTTE*; **[1]**

4.3-S9 Nov 2014 · Paper 2 · Q5 parts (a), (b)

5. (a) van der Waals'/vdW/London/dispersion (forces)/LDF / temporary/instantaneous/ induced dipoles; ***[1]***

(b) *Two of the following pairs:*

used as pencil (lead);
layers can flake off/slide off/break off/stick to paper / *OWTTE*;
M2 must contain concept of separation of layers, so do not award mark for phrases like "layers can slide over each other" on their own.

OR

used as carbon fibre / *OWTTE*;
bonding within layer is strong / layers are extensive / layers are strong;

OR

used as electrodes/conductor/in batteries;
has mobile/free/delocalized electrons (between layers) / electricity flows parallel to layers;

OR

used for thermal insulation;
vibrations are not easily passed between layers; ***[4 max]***

Accept other valid uses of graphite along with a suitable explanation.

4.3-S10 May 2011 TZ1 · Paper 2 · Q6 parts (a), (b)(i), (b)(ii), (c), (d), (e) · also 4.1, 4.2

6. (a) *Award **[2 max]** for three of the following features:*

Bonding

*Graphite **and** C_{60} fullerene:* covalent bonds **and** van der Waals'/London/dispersion forces;
Diamond: covalent bonds (and van der Waals'/London/dispersion forces);

Delocalized electrons

*Graphite **and** C_{60} fullerene:* delocalized electrons;
Diamond: no delocalized electrons;

Structure

Diamond: network/giant structure / macromolecular / three-dimensional structure **and** *Graphite:* layered structure / two-dimensional structure / planar;
C_{60} fullerene: consists of molecules / spheres made of atoms arranged in hexagons/pentagons;

Bond angles

Graphite: 120^{o} **and** *Diamond:* 109^{o};
C_{60} fullerene: bond angles between $109–120^{o}$;
*Allow Graphite: sp^2 **and** Diamond: sp^3.*
*Allow C_{60} fullerene: sp^2 **and** sp^3.*

Number of atoms each carbon is bonded to

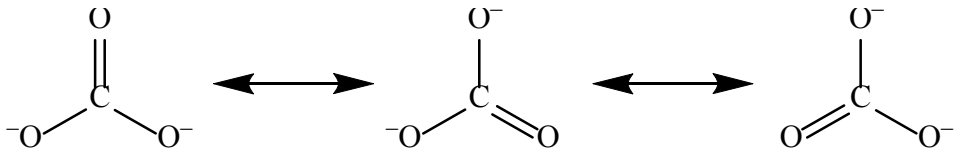
*Graphite **and** C_{60} fullerene:* each C atom attached to 3 others;
Diamond: each C atom attached to 4 atoms / tetrahedral arrangement of C (atoms); ***[6 max]***

(b) (i) network/giant structure / macromolecular;
each Si bonded covalently to 4 oxygen atoms **and** each O atom bonded covalently to 2 Si atoms / single covalent bonds; ***[2]***
*Award **[1 max]** for answers such as network-covalent, giant-covalent or macromolecular-covalent.*
Both M1 and M2 can be scored by a suitable diagram.

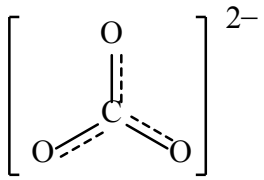
(ii) *Silicon dioxide:* strong/covalent bonds in network/giant structure/macromolecule;
Carbon dioxide: weak/van der Waals'/dispersion/London forces between molecules; ***[2]***

(c) triple (covalent) bond;
one electron pair donated by oxygen to carbon atom / dative (covalent)/coordinate (covalent) bond; ***[2]***
*Award **[1 max]** for representation of* C≡O.
Award [2] if CO shown with dative covalent bond

(d) delocalization/spread of pi/π electrons over more than two nuclei;
equal bond order/strength/length / spreading charge (equally) over all three oxygens;
gives carbonate ion a greater stability/lower potential energy; ***[3]***
M3 can be scored independently.
Accept suitable labelled diagrams for M1 and M2 e.g.



OR



Do not penalize missing brackets on resonance structure but 2– charge must be shown.
Allow –2 for charge on resonance structure.

(e) mixing/combining/merging of (atomic) orbitals to form new orbitals (for bonding);
Allow molecular or hybrid instead of new.
Do not allow answers such as changing shape/symmetries of atomic orbitals.

Carbon dioxide: sp;
Diamond: sp^3;
Graphite: sp^2;
Carbonate ion: sp^2; ***[5]***

LESSON 5.1

Molecular Polarity

Answers for 9 multiple-choice and 2 structured questions

Part A - Multiple choice answers

5.1-M1	May 2025 TZ2 · P1A · Q7	C
5.1-M2	May 2024 TZ2 · P1 · Q15	A
5.1-M3	Nov 2020 · P1 · Q10	A
5.1-M4	Nov 2017 · P1 · Q12	C
5.1-M5	Nov 2015 · P1 · Q11	B
5.1-M6	May 2015 TZ2 · P1 · Q7	A
5.1-M7	Nov 2012 · P1 · Q10	D
5.1-M8	May 2012 TZ2 · P1 · Q10	D
5.1-M9	May 2010 TZ2 · P1 · Q11	A

Part B - Structured questions markscheme

5.1-S1 Nov 2015 · Paper 2 · Q5 parts (a)(i)

5. (a) (i) ability of an atom to attract (a pair of) electrons in a covalent bond/molecule / ability of an atom to attract a shared pair of electrons; ***[1]***
Do not accept element/nucleus instead of atom.

5.1-S2 May 2015 TZ1 · Paper 3 · Q10 parts (b)(i)

(b) (i) polar/dipole moment due to the presence of C≡N (bond) / difference in electronegativity between C and N; **[1]**

LESSON 5.2

Intermolecular Forces

Answers for 39 multiple-choice and 26 structured questions

Part A - Multiple choice answers

5.2-M1	May 2025 TZ3 · P1A · Q8	D
5.2-M2	Nov 2024 · P1 · Q11	A
5.2-M3	Nov 2023 TZ1 · P1 · Q11	A
5.2-M4	May 2023 TZ2 · P1 · Q10	B
5.2-M5	May 2022 TZ2 · P1 · Q10	A
5.2-M6	Nov 2021 · P1 · Q11	B
5.2-M7	May 2021 TZ1 · P1 · Q32	B
5.2-M8	May 2021 TZ2 · P1 · Q9	A
5.2-M9	May 2021 TZ2 · P1 · Q11	C
5.2-M10	May 2019 TZ1 · P1 · Q9	B
5.2-M11	May 2019 TZ2 · P1 · Q32	D
5.2-M12	Nov 2018 · P1 · Q11	D
5.2-M13	Nov 2017 · P1 · Q11	A
5.2-M14	Nov 2016 · P1 · Q7	B
5.2-M15	Nov 2016 · P1 · Q12	D
5.2-M16	May 2016 · P1 · Q13	B
5.2-M17	Nov 2015 · P1 · Q35	C
5.2-M18	May 2015 TZ1 · P1 · Q10	D
5.2-M19	May 2015 TZ1 · P1 · Q11	A
5.2-M20	May 2015 TZ2 · P1 · Q11	A
5.2-M21	Nov 2014 · P1 · Q11	D
5.2-M22	May 2014 TZ2 · P1 · Q24	A
5.2-M23	Nov 2013 · P1 · Q11	A
5.2-M24	Nov 2013 · P1 · Q24	D
5.2-M25	May 2013 TZ2 · P1 · Q11	D
5.2-M26	May 2013 TZ2 · P1 · Q25	D
5.2-M27	Nov 2012 · P1 · Q11	C
5.2-M28	Nov 2012 · P1 · Q34	C
5.2-M29	May 2012 TZ1 · P1 · Q10	D
5.2-M30	May 2012 TZ1 · P1 · Q11	A
5.2-M31	May 2012 TZ2 · P1 · Q12	A
5.2-M32	May 2012 TZ2 · P1 · Q34	D
5.2-M33	Nov 2011 · P1 · Q12	A
5.2-M34	May 2011 TZ1 · P1 · Q12	A
5.2-M35	May 2011 TZ1 · P1 · Q25	D
5.2-M36	May 2011 TZ2 · P1 · Q24	D
5.2-M37	May 2010 TZ1 · P1 · Q11	A
5.2-M38	May 2010 TZ1 · P1 · Q12	B
5.2-M39	May 2010 TZ2 · P1 · Q10	B

Part B - Structured questions markscheme

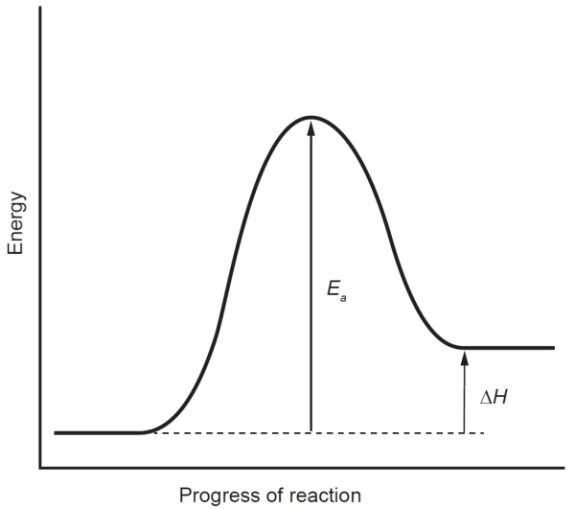
5.2-S1 May 2025 TZ3 · Paper 2 · Q7 parts (b)(i), (b)(ii), (c) · also 4.3

7.	**(b)**	**(i)**	O of C=O with H of H-N on opposite chain ✓	*bonds.*	**1**
7.	**(b)**	**(ii)**	H that is «covalently» bonded to a «highly» electronegative atom ✓ is attracted to the electronegative atom of neighbouring molecule ✓	*Accept a labelled diagram for both marks.* *Accept "F, O or N" for "electronegative atom".*	**2**
7.	**(c)**		covalent bonds between silicon atoms ✓ London/dispersion forces between poly(ethene) «chains/molecules» ✓ covalent bonds «much» stronger than London/dispersion/intermolecular forces «hence Si has higher melting point» ✓	*M3: Do **not** accept "bonding in silicon is stronger than poly(ethene)" without named bonding/forces.*	**3**

5.2-S2 Nov 2024 · Paper 2 · Q1 parts (a)(i), (a)(ii) · also 8.1

1.	**(a)**	**(i)**	London/dispersion forces **«**only**»** ✓ strength **«**of intermolecular forces**»** increases as size of electron cloud/number of electrons increases ✓	*Accept strength of intermolecular forces increases as mass/size of molecule increases for M2.*	**2**
1.	**(a)**	**(ii)**	boils/vaporizes ***OR*** molecules have sufficient energy to overcome intermolecular forces ✓ gaseous molecules are further apart **«**than in liquid**»** ✓	*Accept gases have a **«**much**»** lower density than liquids for M2.* *Do **not** award M2 unless a change of state is indicated.*	**2**

5.2-S3 May 2024 TZ2 · Paper 2 · Q5 parts (b), (c)(i), (c)(ii) · also 8.1, 9.2

5.	**(b)**		both have **«**similar**»** London/dispersion forces **«**due to having same number of electrons/similar M_r**»** ✓ ethenone has dipole-dipole forces **«**and carbon dioxide does not**»** ✓		**2**
5.	**(c)**	**(i)**	**«**$\Delta H° = \sum \Delta H°_{f\ products} - \sum \Delta H°_{f\ reactants}$ **»** = -74.0 - 87.2 + 248 ✓ = **«**+**»**86.8 **«**kJ**»** ✓	*Award **[2]** for correct final answer.* *Award **[1]** for –86.8 **«**kJ**»**.*	**2**
5.	**(c)**	**(ii)**	 endothermic graph ***AND*** labelled arrow/line showing ΔH ✓ labelled arrow/line showing E_a ✓		**2**

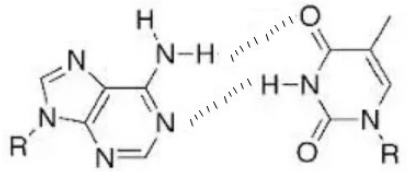
5.2-S4 Nov 2023 TZ1 · Paper 2 · Q1 parts (c)(i), (c)(ii), (d)(i), (d)(iv) · also 1.2, 4.1

1	**c**	**i**	«strongest intermolecular forces in» methanoic acid are hydrogen/H-bonds ***AND*** ethanal dipole-dipole forces ✓ hydrogen/H-bonds stronger «than dipole-dipole forces so methanoic acid has higher boiling point» ✓	*Do **not** award marks for answers based on difference in polarity or molar mass.* *Do **not** accept van der Waals' forces for dipole-dipole forces.*	**2**
1	**c**	**ii**	«both can» form hydrogen bonds with water «molecules» ✓		**1**

1	**d**	**i**	$[\mathrm{HCOOH}]$ «$=0.100\ mol\,dm^{-3} \times \frac{20.7\ cm^3}{25.0\ cm^3}$» = 0.0828 «mol dm^{-3}» ✓ mass HCOOH «=2.00 dm^3 x 0.0828 mol dm^{-3} x 46.03 g mol^{-1}» = 7.62 «g» ✓	*Accept answers in the range 7.60-7.65.* *Award **[0]** for 0.096 g - mass of acid in 25 cm^3.*	**2**

1	**d**	**iv**	the double/pi/π bond «in the methanoate ion» is delocalized ***OR*** resonance occurs ✓ shorter than the single bond ***AND*** longer than the double bond «in methanoic acid» / intermediate between single and double bond ✓	*Accept drawing showing delocalization of the bond or resonance structures for M1.*	**2**

5.2-S5 May 2023 TZ1 · Paper 3 · Q9 parts (b)

9.	**(b)**		 Adenine Thymine ✓	*Only 2 H bonds must be shown.* *Accept H bond from either H in NH_2 of Adenine, but not from both.* *Accept solid lines instead of dashes.*	**1**

5.2-S6 May 2022 TZ1 · Paper 2 · Q5 parts (a)(ii), (a)(iii), (a)(iv) · also 4.1, 4.2

5.	**a**	**ii**	dipole-dipole ✓	*Do not accept van der Waals' forces.*	**1**
5.	**a**	**iii**	*σ:* 9 ***AND*** *π:* 1 ✓		**1**
5.	**a**	**iv**	sp^2 ✓		**1**

5.2-S7 Nov 2019 · Paper 3 · Q15 parts (c)

15.	**c**		«solubility depends on forming many» H-bonds with water ✓ maltose has many hydroxyl/OH/oxygen atoms/O «and forms many H-bonds» ✓	*Reference to "with water" required.* *Accept "hydroxy" for "hydroxyl" but **not** "hydroxide".* *Reference to many/several OH groups/O atoms required for M2.*	**2**

5.2-S8 May 2019 TZ1 · Paper 3 · Q10 parts (a)

10.	**a**		coconut oil has higher content of lauric/short-chain «saturated» fatty acids ***OR*** cocoa butter has higher content of stearic/palmitic/longer chain «saturated» fatty acids ✓ longer chain fatty acids have greater surface area/larger electron cloud ✓ stronger London/dispersion/instantaneous dipole-induced dipole forces «between triglycerides of longer chain saturated fatty acids» ✓	*Do **not** accept arguments that relate to melting points of saturated and unsaturated fats.*	**3**

5.2-S9 May 2019 TZ2 · Paper 2 · Q1 parts (a), (b)(i), (b)(ii), (b)(iii), (c)(ii), (c)(iii), (d)(ii), (d)(iii) · also 3.1, 4.1, 6.1, 7.3, 8.1, 9.3

1.	**a**		$C_2H_2\,(g) + 2.5O_2\,(g) \rightarrow 2CO_2\,(g) + H_2O\,(l)$ ***OR*** $2C_2H_2\,(g) + 5O_2\,(g) \rightarrow 4CO_2\,(g) + 2H_2O\,(l)$ ✓		**1**
1.	**b**	**i**	H:C⋮⋮C:H / H−C≡C−H ✓	*Accept any valid combination of lines, dots and crosses.*	**1**
1.	**b**	**ii**	**«**ethyne**»** shorter ***AND*** a greater number of shared/bonding electrons ***OR*** **«**ethyne**»** shorter ***AND*** stronger bond ✓		**1**
1.	**b**	**iii**	London/dispersion/instantaneous dipole-induced dipole forces ✓	*Do **not** accept just "intermolecular forces" or "van der Waals' forces".*	**1**

1.	**c**	**ii**	**«**sum of bond enthalpies of reactants =**»** 2(C–H) + C≡C + 2(O–H) ***OR*** 2 × 414 **«**kJ mol^{-1}**»** + 839 **«**kJ mol^{-1}**»** + 2 × 463 **«**kJ mol^{-1}**»** ***OR*** 2593 **«**kJ**»** ✓ **«**sum of bond enthalpies of A =**»** 3(C–H) + C=C + C–O + O–H ***OR*** 3 × 414 **«**kJ mol^{-1}**»** + 614 **«**kJ mol^{-1}**»** + 358 **«**kJ mol^{-1}**»** + 463 **«**kJ mol^{-1}**»** ***OR*** 2677 **«**kJ**»** ✓ **«**enthalpy of reaction = 2593 kJ − 2677 kJ**»** = –84 **«**kJ**»** ✓	*Award **[3]** for correct final answer.*	**3**

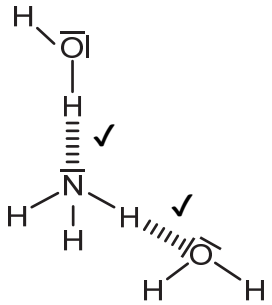
1.	**c**	**iii**	B ***AND*** it has a more negative/lower enthalpy/**«**potential**»** energy ***OR*** B ***AND*** more exothermic **«**enthalpy of reaction from same starting point**»** ✓		**1**

1.	**d**	**ii**	–1 ✓		**1**
1.	**d**	**iii**	*Any three of:* has an oxygen/O atom with a lone pair ✓ that can form hydrogen bonds/H-bonds **«**with water molecules**»** ✓ hydrocarbon chain is short **«**so does not disrupt many H-bonds with water molecules**»** ✓ **«**large permanent**»** dipole-dipole interactions with water ✓		**3 max**

5.2-S10 May 2019 TZ2 · Paper 3 · Q11 parts (c)

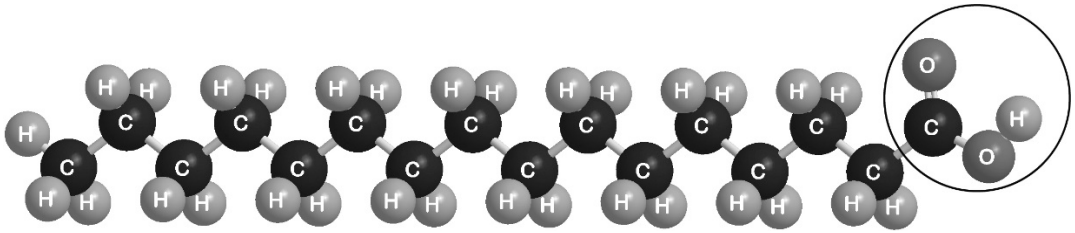
11.	**c**		long non-polar/hydrocarbon chain **«**and only one hydroxyl group**»** ***OR*** forms London/dispersion/van der Waals/vdW interactions with fat ✓	*Accept "alcohol/hydroxy/OH" for "hydroxyl" but **not** "hydroxide".*	**1**

5.2-S11 May 2018 TZ1 · Paper 2 · Q1 parts (b), (c), (e)(i), (e)(ii), (f), (i) · also 1.2, 3.1, 4.1, 7.3

1.	**b**		<table><tr><th></th><th>Electron geometry</th><th>Molecular geometry</th></tr><tr><td>Nitrogen</td><td>tetrahedral ✓</td><td>trigonal pyramidal ✓</td></tr><tr><td>Carbon</td><td>trigonal planar ✓</td><td>trigonal planar</td></tr></table>	*Note: Urea's structure is more complex than that predicted from VSEPR theory.*	**3**
1.	**c**		$n(\text{KNCO})$ «$= 0.0500\ \text{dm}^3 \times 0.100\ \text{mol dm}^{-3}$» $= 5.00 \times 10^{-3}$ «mol» ✓ «mass of urea $= 5.00 \times 10^{-3}\ \text{mol} \times 60.07\ \text{g mol}^{-1}$» $= 0.300$ «g» ✓	*Award **[2]** for correct final answer.*	**2**
1.	**e**	**i**	*Any one of:* urea has greater molar mass ✓ urea has greater electron density/greater London/dispersion ✓ urea has more hydrogen bonding ✓ urea is more polar/has greater dipole moment ✓	*Accept "urea has larger size/greater van der Waals forces".* *Do **not** accept "urea has greater intermolecular forces/IMF".*	**1**
1.	**e**	**ii**		*Award **[1]** for each correct interaction.* *If lone pairs are shown on N or O, then the lone pair on N or one of the lone pairs on O **MUST** be involved in the H-bond.* *Penalize solid line to represent H-bonding only once.*	**2**
1.	**f**		$2(H_2N)_2CO\,(s) + 3O_2\,(g) \rightarrow 4H_2O\,(l) + 2CO_2\,(g) + 2N_2\,(g)$ correct coefficients on LHS ✓ correct coefficients on RHS ✓	*Accept* $(H_2N)_2CO\,(s) + \frac{3}{2}O_2\,(g) \rightarrow 2H_2O\,(l) + CO_2\,(g) + N_2\,(g)$. *Accept any correct ratio.*	**2**

1.	**i**		lone pairs on nitrogen atoms can be donated to/shared with C–N bond ***OR*** C–N bond partial double bond character ***OR*** delocalization «of electrons occurs across molecule» ***OR*** slight positive charge on C due to C=O polarity reduces C–N bond length ✓		**1**

5.2-S12 May 2018 TZ1 · Paper 3 · Q1 parts (a)(i), (a)(ii), (b)(ii) · also 1.2

1.	**a**	**i**		*Must cut CH_2–CO bond **AND** enclose all of the –COOH group.*	**1**
1.	**a**	**ii**	*Any two of:* –COOH/CO/OH/carboxylate/carboxyl/hydroxyl/hydroxy group forms hydrogen bonds/H-bonds to water ✓ London/dispersion/instantaneous induced dipole-induced dipole forces occur between hydrocarbon chains ✓ hydrocarbon chain cannot form hydrogen bonds/H-bonds to water ✓ strong hydrogen bonds/H-bonds between water molecules exclude hydrocarbon chains «from the body of the water» ✓	*Accept "hydrophilic part/group forms hydrogen bonds/H-bonds to water".* *Accept "hydrophobic section" instead of "hydrocarbon chain".* *Award **[1 max]** for answers based on "the –COOH group being polar **AND** the hydrocarbon chain being non-polar".*	**2 max**

1.	**b**	**ii**	amount of acid = «$5.0 \times 10^{-5}\,dm^3 \times 0.0034\,mol\,dm^{-3}$» = 1.7×10^{-7} «mol» ✓ number of molecules = «$1.7 \times 10^{-7}\,mol \times 6.02 \times 10^{23}\,mol^{-1}$ =» 1.0×10^{17} ✓	*Award **[2]** for correct final answer.* *Award **[1]** for "1.0×10^{20}".*	**2**

5.2-S13 May 2017 TZ1 · Paper 3 · Q12 parts (a)(i)

12.	**a**	**i**	«water/aqueous solubility depends on forming many» H-bonds with water ✓ raffinose has many hydroxyl/O–H/oxygen atoms/O «and forms many H-bonds» ***AND*** linoleic acid has few/one hydroxyl/O–H/oxygen atom/O/carboxyl group/ COOH/is largely non-polar «and cannot form many H-bonds» ✓	*Accept statement which implies comparison.*	**2**

5.2-S14 Nov 2016 · Paper 3 · Q16 parts (b)

16.	**b**		in aqueous solution ***AND*** hydroxyl/OH/ionic/oxonium/O^+ «groups» ✓	*Accept "polar/hydroxy" for "hydroxyl".* *Do **not** accept "OH^-/hydroxide/ oxygen".*	**1**

5.2-S15 Nov 2015 · Paper 2 · Q6 parts (c)(i), (c)(ii), (c)(iii), (d)(i), (d)(ii) · also 3.1, 4.1, 4.2

(c) (i)

	NF_4^+	N_2H_2	N_2H_4
Hybridization	sp^3;	sp^2;	sp^3;

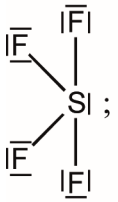
[3]

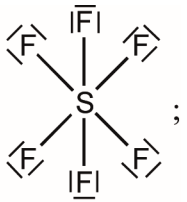
Do not penalize if it is not superscript.

(ii) *sigma bond:* overlapping of orbitals end to end/head on / overlapping of orbitals along internuclear axis;
Accept "axial overlapping of orbitals".

pi bond: side-ways/parallel overlapping of <u>p orbitals</u> (above and below internuclear axis);
Accept "overlapping of <u>p orbitals</u> above and below the internuclear axis".
Accept suitable labelled diagrams.

[2]

(iii)  ;

 ;

See saw; Octahedral; ***[4]***

Accept any combination of lines, dots or crosses to represent electron pairs.
Penalize missing lone pairs on fluorine atoms once only.
For SF_4, lone pair on S required for the mark.
Do not allow ECF for a wrong Lewis structure.
Accept saw horse/distorted tetrahedron instead of see-saw.

(d) (i) $CH_3CH_2CH_3 < CH_3CHO < CH_3CH_2OH < CH_3COOH$;; ***[2]***
*Award **[2]** for correct order.*
*Award **[1]** for $CH_3COOH > CH_3CH_2OH > CH_3CHO > CH_3CH_2CH_3$ as compounds are not listed in order of increasing boiling point.*
*Award **[1]** if one error in the order.*

(ii) $CH_3CH_2CH_3$ /London/dispersion/instantaneous induced dipole-induced dipole forces
CH_3CHO dipole-dipole forces (and London/dispersion forces)
CH_3CH_2OH H-bonding (and dipole-dipole and London/dispersion forces)
CH_3COOH H-bonding (and dipole-dipole and London/dispersion forces);;
*Award **[2]** for all four correct.*
*Award **[1]** for two or three correct.*

H-bonding strongest / London/dispersion forces weakest / dipole-dipole stronger than London/dispersion / dipole-dipole weaker than H-bonding;

Accept van der Waals' forces for London/dispersion forces.

CH_3COOH forms more/stronger H-bonds than CH_3CH_2OH / CH_3COOH is more polar than CH_3CH_2OH; ***[4]***
Accept CH_3COOH has more electrons/higher molar mass than CH_3CH_2OH.

5.2-S16 May 2015 TZ1 · Paper 3 · Q19 parts (b)(i)

(b) (i) has many (polar) hydroxyl/OH/carboxyl/COOH (groups);
Accept "alcohol/hydroxy" for hydroxyl and "carboxylic acid" for carboxyl but not hydroxide for hydroxyl.

can form (many) hydrogen bonds (with water molecules in the soil); **[2]**

5.2-S17 May 2015 TZ2 · Paper 2 · Q10 parts (f), (g) · also 7.3

(f) hydrogen bonding in compound **C**;
dipole-dipole forces in **C** / **C** is more polar;
C has greater molar mass/more dispersion/London/instantaneous induced dipole-induced dipole forces/van der Waal forces; **[2 max]**
Accept converse argument.
*Award **[1 max]** for stronger intermolecular forces.*

(g) $C_4H_9OH(l) + 6O_2(g) \rightarrow 4CO_2(g) + 5H_2O(l)$; **[1]**
Ignore state symbols.

5.2-S18 May 2014 TZ1 · Paper 3 · Q6 parts (a)

6. (a) molecule contains polar/OH/hydroxyl/NH/amino/COOH/carboxyl groups / molecule forms hydrogen bonds with water / *OWTTE*; ***[1]***
Accept polar molecule so dissolves in water.
Allow class (eg, carboxylic acid, amine etc.) if stated instead of functional groups.

5.2-S19 Nov 2013 · Paper 2 · Q8 parts (d)

(d) 2-methylbutan-2-ol has hydroxyl/OH group;
Do not accept "hydroxide group".
Allow 2-methylbutan-2-ol is an alcohol.

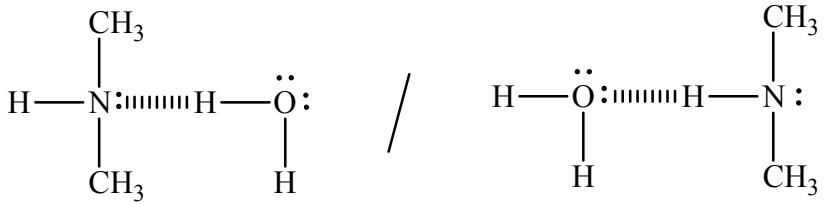
2-methylbutan-2-ol can form <u>H-bonds</u> (to water) / 2-methylbut-2-ene cannot form <u>H-bonds</u> (to water); ***[2]***

5.2-S20 May 2013 TZ2 · Paper 2 · Q4 parts (b), (c)(i), (c)(ii)

(b) van der Waals'/London/dispersion **and** dipole-dipole; ***[1]***
Allow abbreviations for van der Waals' as vdW or for London/dispersion as FDL.

(c) (i) $(CH_3)_2NH$;
(intermolecular) attraction between hydrogen (atom) in O–H/N–H (polar) bond and (lone pair on) electronegative N/O / hydrogen between two very electronegative elements (nitrogen and oxygen) / *OWTTE*; ***[2]***
Accept hydrogen bonded to nitrogen which is electronegative/has lone pair.
Do not allow ECF if M1 is incorrect.

(ii) representative drawing showing hydrogen bond between $(CH_3)_2NH$ and H_2O; ***[1]***



Do not penalize if lone pair as part of hydrogen bond is not shown.
Allow any representation of hydrogen bond (for example, dashed lines, dots etc.) which differs from full stick representation of the other covalent bonds in amine and water molecules.
Allow full line if labelled as hydrogen bond.
Lone pairs on oxygen not necessary.
Award mark if two hydrogen bonds drawn between the molecules from the lone pair and the H on the N.

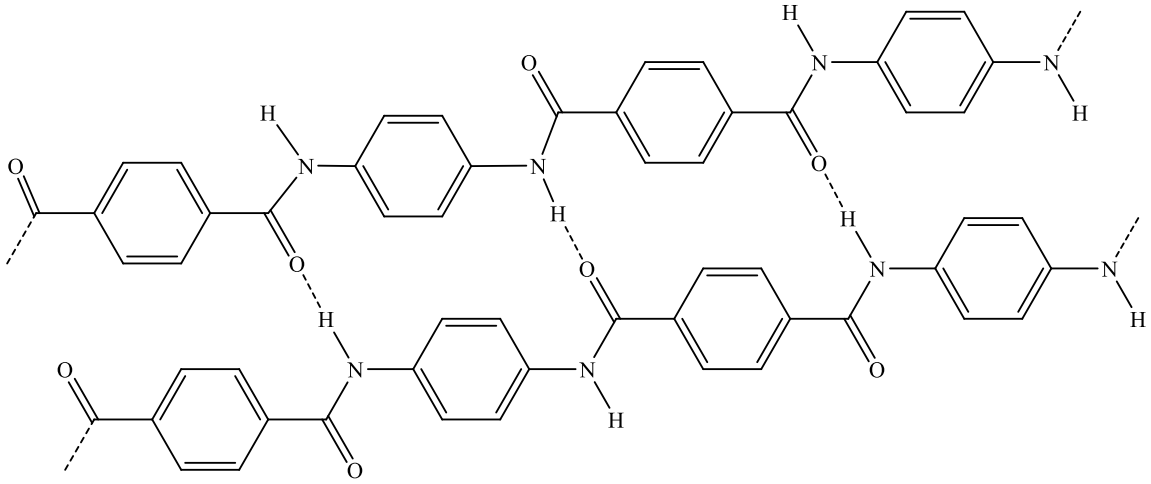
5.2-S21 Nov 2012 · Paper 3 · QC3 parts (a)(i)

C3. (a) (i) **A** has greater van der Waals/London/dispersion/intermolecular forces than **B**; ***[1]***
Penalize incorrect forces.

5.2-S22 Nov 2012 · Paper 3 · QC4 parts (b)(i), (b)(ii)

(b) (i) hydrogen bonding; ***[1]***

(ii)



dotted lines showing intermolecular forces; ***[1]***

5.2-S23 Nov 2012 · Paper 3 · QF3 parts (a)

F3. (a) many polar/–OH/hydroxyl groups;
which can hydrogen bond to water; ***[2]***

5.2-S24 Nov 2011 · Paper 2 · Q9 parts (b)(iv)

(iv) *Volatility*:
E more volatile than **F**;
hydrogen bonding in carboxylic acid/**F**; ***[2]***
Accept converse argument.

5.2-S25 May 2011 TZ2 · Paper 2 · Q4

methoxymethane is very weakly polar / weak van der Waals'/dipole-dipole forces exist between methoxymethane molecules;
Accept alternatives to van der Waals' such as London and dispersion forces.

ethanol contains a hydrogen atom bonded directly to an electronegative oxygen atom / hydrogen bonding can occur between two ethanol molecules / intermolecular hydrogen bonding in ethanol;
the forces of attraction between molecules are stronger in ethanol than in methoxymethane / hydrogen bonding stronger than van der Waals'/dipole-dipole attractions; ***[3 max]***
*Award **[2 max]** if covalent bonds breaking during boiling is mentioned in the answer.*
Penalize only once if no reference given to intermolecular nature of hydrogen bonding or van der Waal is missing.

SECTION B

5.2-S26 May 2010 TZ2 · Paper 2 · Q3 parts (b), (c) · also 4.3

(b) larger molecule / higher M_r/M / greater number of electrons;
Do not accept "larger/higher/greater mass".

greater van der Waals'/dispersion/London forces; ***[2]***

(c) *Si:* giant/network/macromolecular/3-D covalent bonding;
No mark for strong bonding without reference to covalent and network.
No mark for molecular.

Ar: (simple) atomic / (only weak) van der Waals'/dispersion/London forces; ***[2]***
No mark for (simple) molecular.

LESSON 5.3

Separation Techniques

Answers for 2 multiple-choice and 20 structured questions

Part A - Multiple choice answers

5.3-M1	May 2025 TZ1 · P1A · Q12	**C**
5.3-M2	May 2025 TZ2 · P1A · Q8	**B**

Part B - Structured questions markscheme

5.3-S1 May 2025 TZ1 · Paper 1B · Q2 parts (a), (e)

2.	**(a)**		*Any four of:* use magnet to remove iron/Fe ✓ add «excess» water to dissolve salt/NaCl ✓ filter/wash sand «into saltwater filtrate»✓ allow/heat sand to dry ✓ boil off/evaporate water from salt ***OR*** allow salt to dry overnight ✓	*Accept other reasonable orders to separations and names or formulas.* *Accept references to NaCl(aq) for M2.* *Do **not** accept responses that include sifting, separating with a sieve, **OR** by particle size.* *Accept "distill off water from salt" for M5.*	**4 max**
2.	**(e)**		*Separation method:* «fractional» distillation ✓ *Physical property:* boiling point ✓		**2**

5.3-S2 May 2024 TZ1 · Paper 3 · Q9 parts (a)

9.	**a**		yes ***AND*** third spot «from bottom» $R_f = 0.36$ ✓	*Accept values in the range of 0.35–0.37 for R_f of third spot.* *Accept answers with YES implied, as in the third spot from the bottom has an R_f of 0.36.* *Accept alternative working such as determination of d using Rf and D, where D = 64 mm, to identify the third spot as lutein.*	**1**

5.3-S3 May 2023 TZ1 · Paper 3 · Q8 parts (a)(i)

8.	**(a)**	**(i)**	*Identity of spot C:* leucine ✓		**1**

5.3-S4 May 2023 TZ2 · Paper 3 · Q7 parts (b)

7.	**(b)**		*Any two of:* sample spotted on paper/stationary phase ***AND*** solvent moves up the paper ***OR*** continuous cycles of adsorption and desorption/dissolution ***OR*** analyte moves when in solvent ***AND*** does not move when on paper ✓ different/depends on attractions to mobile phase ***AND*** stationary phase/paper ***OR*** «amino acids» separated based on solubilities in/affinity to the two phases ***OR*** separated based on polarities/polar attractions/molar masses of «amino acids» ✓ developed with ninhydrin/reagent/locating agent ***OR*** become identified with UV «light» ✓ «amino acids» identified by R_f/retention factor «values» ***OR*** R_f/retention factors «values» compared with known samples ✓		**2 max**

5.3-S5 Nov 2020 · Paper 3 · Q1 parts (a), (b), (c)

1.	**a**		oil is non-polar **«**and dissolves best in non-polar solvents**»** ***OR*** oil does not dissolve in polar solvents ✓	*Do **not** accept "like dissolves like" only.*	**1**
1.	**b**		solvent/oil is flammable ***OR*** solvent/oil must be kept below its flash point ***OR*** oxidation/decomposition of oil ***OR*** mixture has a low boiling point ✓	*Accept "to prevent evaporation of oil".*	**1**
1.	**c**		distillation **«**instead of evaporation**»** ✓	*Accept "pass vapour through a condenser and collect liquid".* *Do **not** accept "condensation" without experimental details.*	**1**

5.3-S6 Nov 2020 · Paper 3 · Q6 parts (a)(i)

6.	**a**	**i**	0.70 ✓	*Accept any value within the range "0.67–0.73".*	**1**

5.3-S7 Nov 2020 · Paper 3 · Q15 parts (b)

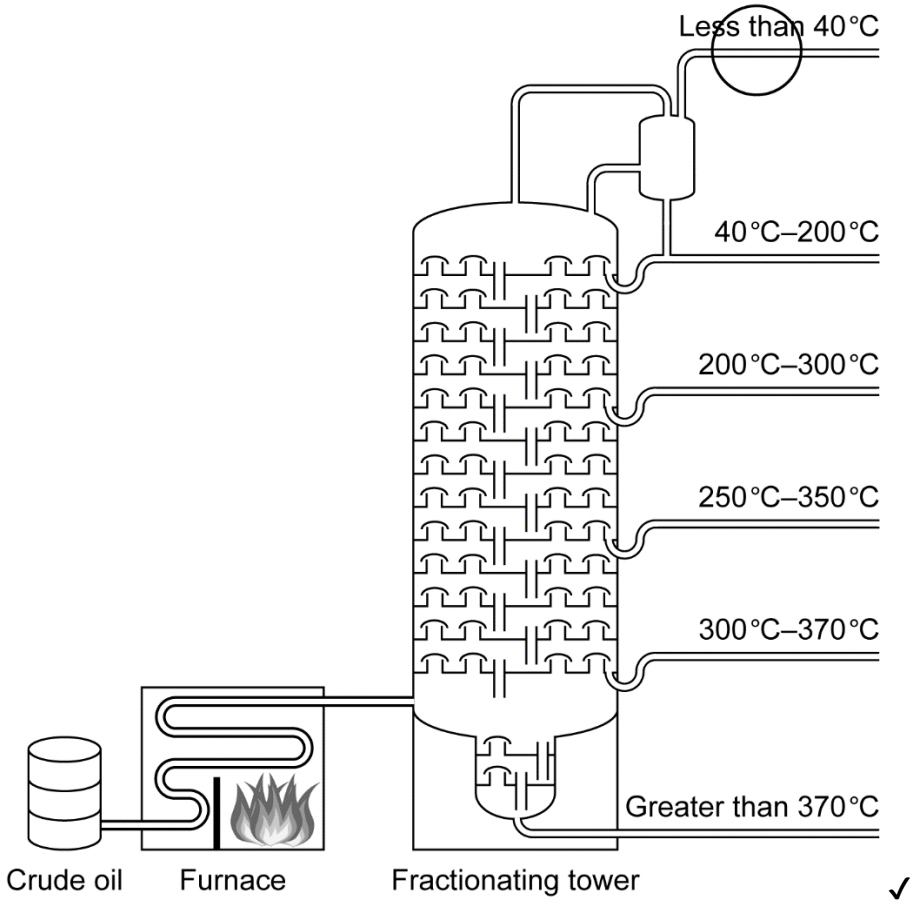
15.	**b**		to avoid dissolving the crystals/aspirin ✓	*Accept "to avoid loss of product" **OR** "aspirin is less soluble in cold water".*	**1**

5.3-S8 Nov 2020 · Paper 3 · Q18 parts (b)

18.	**b**		*Any two of:* continuous evaporation and condensation ***OR*** increased surface area in column helps condensation ✓ temperature decreases up the fractionating column ✓ liquids condense at different heights ***OR*** liquid of lowest boiling point collected first ***OR*** liquid with weakest intermolecular forces collected first ***OR*** most volatile component collected first ***OR*** fractions/liquids collected in order of boiling point/volatility ✓	*Accept "glass **«**beads**»** aid condensation **«**in fractionating column**»".*** *Accept "liquids collected in order of molar mass".*	**2 max**

5.3-S9 May 2019 TZ1 · Paper 3 · Q15 parts (a), (c)(i), (c)(ii) · also 7.3

15.	**a**		$\text{«}\frac{891\,\text{kJ mol}^{-1}}{16.05\,\text{g mol}^{-1}} = 55.5\,\text{kJ g}^{-1}\text{ =» }55.5\text{ «MJ kg}^{-1}\text{»}$ ✓		**1**

15.	**c**	**i**	 ✓ [Source: Image used with kind permission of science-resources.co.uk]		**1**
15.	**c**	**ii**	gasoline > diesel > lubricating motor oil > asphalt ✓	*Accept products written in this order whether separated by >, comma, or nothing.*	**1**

5.3-S10 May 2018 TZ1 · Paper 3 · Q6 parts (c)(ii)

6.	**c**	**ii**	compare R_f with known amino acids ***OR*** compare distance moved with known amino acids ✓	*Accept "from R_f".*	**1**

5.3-S11 May 2018 TZ1 · Paper 3 · Q20 parts (b)

20.	**b**		*Any three of:* different molar masses ***OR*** different strength of intermolecular forces ✓ different boiling points ✓ temperature in «fractionating» column decreases upwards ✓ «components» condense at different temperatures/heights *OR* «component with» lower boiling point leaves column first ✓		**3 max**

5.3-S12 May 2016 · Paper 3 · Q13 parts (b)

13.	**b**		0.47 ***AND*** chlorophyll b ✓	*Accept any R_f value in the range 0.44–0.50.*	**1**

5.3-S13 May 2015 TZ2 · Paper 3 · Q6 parts (a)(ii)

(ii) 0.50; **[1]**
Accept answers in the range 0.49–0.51.
Accept d/t.

5.3-S14 Nov 2014 · Paper 3 · Q3 parts (a), (b)

3. (a) different components have different attractions/affinities/bond strengths/solubilities for two phases / *OWTTE*;

components strongly absorbed/adsorbed by stationary phase move less / components weakly absorbed/adsorbed by stationary phase move more / components not very soluble in mobile phase move less / components very soluble in mobile phase move more / *OWTTE*; ***[2]***

(b) (nature of) solvent (used) / (type of) paper / temperature / pH; ***[1]***

Option B — Human biochemistry

Penalize missing hydrogens or incorrect bond linkages (eg, CO–HN) once only in this option.

5.3-S15 May 2014 TZ1 · Paper 3 · Q4 parts (a), (b), (c)

4. (a) *Spot 1:* $R_f = 0.75$ **and** *Spot 2:* $R_f = 0.25$; ***[1]***

(b) two amino acids have the same R_f value; ***[1]***

(c) change the polarity/make-up of the solvent / use a different solvent; ***[1]***
Accept two way (paper) chromatography.

Option B — Human biochemistry

Penalise missing hydrogens or incorrect bond linkages (eg, NH–C) once only in this option.

5.3-S16 May 2014 TZ2 · Paper 3 · Q2 parts (a)(i), (a)(ii)

2. (a) (i)

Compound	***R_f value***
A	0.28
B	0.81

; ***[1]***

*Award **[1]** for both correct.*

(ii) B is more soluble in solvent/mobile phase / B is less polar than A / B is less strongly adsorbed onto stationary phase; ***[1]***

Accept B is non-polar.

Do not allow "greater attraction/affinity to solvent" without reference to solubility.

5.3-S17 May 2013 TZ2 · Paper 3 · QA1 parts (a), (b)

A1. (a) The ratio between the distance moved by the spot and the distance moved by the solvent front / *OWTTE*; ***[1]***

Accept this expressed as a correct equation.

(b) R_f value depends on the intermolecular forces that the component has with the mobile phase compared to the stationary phase / relative attraction of the component to mobile phase compared to the stationary phase / partition of the component between the mobile phase and the stationary phase / *OWTTE*;
if polarity of the solvents is different the intermolecular forces/attraction to mobile phase/partition will be different / *OWTTE*; ***[2]***

Accept "Components have different solubilities in different solvents" / OWTTE.

5.3-S18 May 2012 TZ1 · Paper 3 · QA3 parts (a), (b)

A3. (a) place a spot of the liquid on a thin-layer chromatography/TLC plate;
place plate in a suitable solvent/eluent;
wait until solvent/eluent has almost reached the top of the plate;
spray/immerse plate in concentrated sulfuric acid/H_2SO_4/reagent to show spots / *OWTTE*;
*Do **not** accept ninhydrin.*

observe the plate to see how many spots are there on it; ***[3 max]***

(b) put spots of the pure sugars on the same plate as the mixture / produce chromatograms of the pure sugars under the same conditions as the mixture;
see if the spots in the mixture have moved the same distance/have same R_f as the pure sugars; ***[2]***

5.3-S19 May 2012 TZ2 · Paper 3 · QA2 parts (a)

A2. (a) components are <u>adsorbed</u> onto (solid) stationary phase/alumina/Al_2O_3;
components <u>dissolve</u> in mobile phase/propanone / components partition between stationary phase and mobile phase/propanone;
separation depends on different abilities of the components to adsorb and dissolve/partition / *OWTTE*; ***[3]***

5.3-S20 May 2011 TZ1 · Paper 3 · QA3 parts (a), (b)

A3. (a) silica/silicon dioxide/SiO_2;
alumina/aluminium oxide/Al_2O_3; ***[1 max]***
Accept either of the above.

(b) a spot drawn with its centre 1.4 cm from the start line; ***[1]***
Accept in the range 1.3 to 1.5 cm from the start line.

LESSON 6.1

Redox Reactions & Oxidation Numbers

Answers for 52 multiple-choice and 18 structured questions

Part A - Multiple choice answers

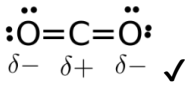
6.1-M1	May 2025 TZ1 · P1A · Q34	D
6.1-M2	May 2025 TZ2 · P1A · Q14	D
6.1-M3	May 2025 TZ2 · P1A · Q36	C
6.1-M4	May 2025 TZ2 · P1A · Q38	B
6.1-M5	May 2025 TZ3 · P1A · Q15	D
6.1-M6	Nov 2024 · P1 · Q28	B
6.1-M7	May 2024 TZ1 · P1 · Q8	B
6.1-M8	May 2024 TZ1 · P1 · Q28	D
6.1-M9	May 2024 TZ2 · P1 · Q10	D
6.1-M10	May 2024 TZ2 · P1 · Q32	A
6.1-M11	May 2023 TZ1 · P1 · Q28	A
6.1-M12	May 2023 TZ2 · P1 · Q28	C
6.1-M13	May 2022 TZ1 · P1 · Q28	D
6.1-M14	Nov 2021 · P1 · Q8	D
6.1-M15	Nov 2021 · P1 · Q28	C
6.1-M16	May 2021 TZ1 · P1 · Q9	B
6.1-M17	May 2021 TZ1 · P1 · Q29	A
6.1-M18	May 2021 TZ2 · P1 · Q28	B
6.1-M19	Nov 2020 · P1 · Q28	C
6.1-M20	Nov 2019 · P1 · Q29	D
6.1-M21	May 2019 TZ2 · P1 · Q8	C
6.1-M22	May 2019 TZ2 · P1 · Q28	D
6.1-M23	Nov 2018 · P1 · Q8	D
6.1-M24	Nov 2018 · P1 · Q28	D
6.1-M25	May 2018 TZ1 · P1 · Q28	C
6.1-M26	May 2018 TZ2 · P1 · Q28	A
6.1-M27	Nov 2017 · P1 · Q29	A
6.1-M28	Nov 2017 · P1 · Q30	B
6.1-M29	Nov 2017 · P1 · Q32	B
6.1-M30	May 2017 TZ1 · P1 · Q8	B
6.1-M31	May 2017 TZ1 · P1 · Q28	C
6.1-M32	May 2017 TZ1 · P1 · Q29	C
6.1-M33	May 2017 TZ2 · P1 · Q28	C
6.1-M34	May 2017 TZ2 · P1 · Q29	D
6.1-M35	Nov 2016 · P1 · Q30	B
6.1-M36	Nov 2015 · P1 · Q30	A
6.1-M37	Nov 2015 · P1 · Q31	C
6.1-M38	May 2015 TZ1 · P1 · Q30	A
6.1-M39	May 2015 TZ2 · P1 · Q30	C
6.1-M40	Nov 2014 · P1 · Q30	A
6.1-M41	May 2014 TZ1 · P1 · Q31	D
6.1-M42	May 2014 TZ2 · P1 · Q31	D
6.1-M43	Nov 2013 · P1 · Q32	C
6.1-M44	May 2013 TZ1 · P1 · Q30	C
6.1-M45	May 2013 TZ2 · P1 · Q30	B
6.1-M46	May 2012 TZ2 · P1 · Q28	B
6.1-M47	May 2011 TZ1 · P1 · Q31	B
6.1-M48	May 2011 TZ2 · P1 · Q30	C
6.1-M49	Nov 2010 · P1 · Q8	D
6.1-M50	Nov 2010 · P1 · Q30	C
6.1-M51	May 2010 TZ1 · P1 · Q30	B
6.1-M52	May 2010 TZ2 · P1 · Q30	A

Part B - Structured questions markscheme

6.1-S1 May 2024 TZ1 · Paper 2 · Q4 parts (a)(i), (a)(ii), (b)(i), (b)(ii), (b)(iii), (b)(iv), (b)(v), (b)(vi) · also 3.1, 4.1, 4.2, 5.1, 9.3

<table>
<tr><td>4.</td><td>(a)</td><td>(i)</td><td>+2 AND +4 ✓</td><td>Accept II/2 AND IV/4

Do not accept 2+ and 4+</td><td>1</td></tr>
<tr><td>4.</td><td>(a)</td><td>(ii)</td><td><table><tr><td></td><td>CO(g)</td><td>$H_2O(g)$</td></tr><tr><td>Oxidising or reducing agent?</td><td>reducing</td><td>oxidising</td></tr><tr><td>Species oxidised or reduced?</td><td>oxidised</td><td>reduced</td></tr></table>
✓✓</td><td>Award [1] for every two correct.</td><td>2</td></tr>
<tr><td>4.</td><td>(b)</td><td>(i)</td><td> ✓</td><td>Accept any combination of dots and crosses or lines.

Ignore non linear drawing of correct Lewis structure.</td><td>1</td></tr>
<tr><td>4.</td><td>(b)</td><td>(ii)</td><td>  ✓</td><td>Accept only one correct δ+ and δ-</td><td>1</td></tr>
<tr><td>4.</td><td>(b)</td><td>(iii)</td><td>linear AND two domains «repel» ✓

non-polar AND polar bonds cancel / symmetrical distribution of charge ✓</td><td>Accept OWTTE two domains

Accept non polar and no net dipole

Award [1] mark if correctly states “linear and non-polar" without correct explanations.</td><td>2</td></tr>
<tr><td>4.</td><td>(b)</td><td>(iv)</td><td>carbon monoxide: sp
AND
carbon dioxide: sp ✓</td><td></td><td>1</td></tr>
</table>

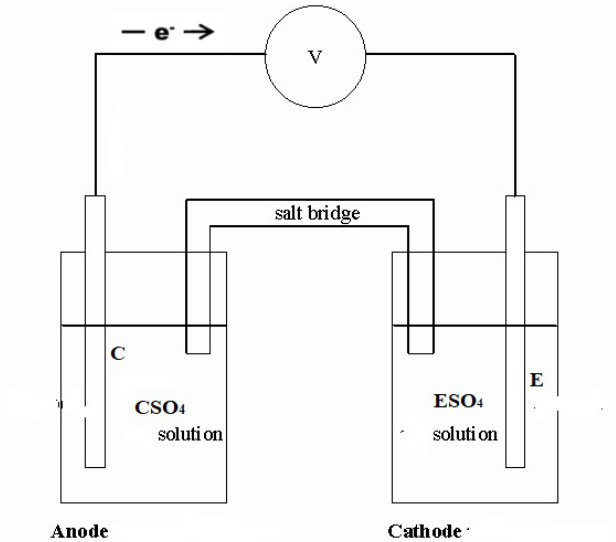
4.	**(b)**	**(v)**	1σ ***AND*** 2π ✓		**1**
4.	**(b)**	**(vi)**	CO + 2(O-H) – 2(C=O) – H-H = –41 ***OR*** *x* + 2(463) – 2(804) – 436 = –41 ✓ « *x* = –41 – 926 + 2044 = » 1077 « kJ mol^{-1} » ✓	*Award **[2]** for correct final answer.* *Award **[1]** for –1077*	**2**

Question	Answers	Notes	Total

6.1-S2 Nov 2023 TZ1 · Paper 3 · Q1 parts (d)(i)

1.	**d**	**i**	iodate/IO_3^-«aq» ✓		**1**

6.1-S3 May 2023 TZ1 · Paper 3 · Q2 parts (a)(i), (a)(ii), (b), (c) · also 6.3, 7.1, 7.2

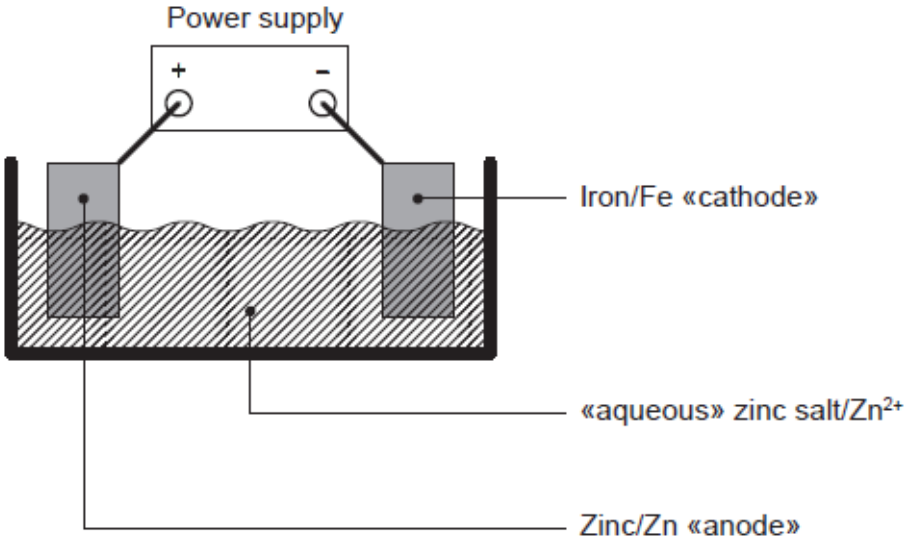
2	**(a)**	**(i)**	+2/II ✓	*Do **not** accept A^{2+}, A^{+2}, 2 **OR** 2+.*	**1**
2.	**(a)**	**(ii)**	C A B D E ✓	*Do **not** accept reverse order unless the answer makes it clear which is the most reactive.*	**1**
2.	**(b)**		 two half cells with electrodes, solutions, salt bridge and connecting wire ✓ C as anode ***AND*** E as cathode ✓ CSO_4 ***AND*** ESO_4 «solutions» labelled ✓ electron flow shown «on wire» from more reactive to less reactive metal ✓	*Voltmeter not needed.* *Accept salt bridge even if not labelled.* *Do **not** award M1 if a battery is included.* *Do **not** award M1 if liquid level not shown in container, or if electrodes and ends of salt bridge are not in solutions.* *Accept salt of C/C^{2+}«(aq)» **AND** salt of E/E^{2+}«(aq)» for M3.*	**4**
2.	**(c)**		copper/Cu ***OR*** silver/Ag ✓	*Accept lead/Pb, tin/Sn or nickel/Ni.*	**1**

6.1-S4 May 2023 TZ1 · Paper 3 · Q3 parts (a)(i)

3.	**(a)**	**(i)**	$2CO(g) \rightarrow C(s) + CO_2(g)$ ✓	*Accept reversible arrows.*	**1**

6.1-S5 May 2021 TZ1 · Paper 2 · Q3 parts (a), (c), (d), (e) · also 7.1, 7.2, 8.2

3.	**a**		1:2 ✓	*Accept 2 Fe3+: 1 Fe2+* *Do **not** accept 2:1 only*	**1**
3.	**c**		specific heat capacity « $= \frac{q}{m \times \Delta T} / \frac{1000\ \text{J}}{50\ \text{g} \times 44\ \text{K}}$ » $= 0.45$ «J g^{-1} K^{-1}» ✓		**1**
3.	**d**		*Equation:* $2Fe^{3+}(aq) + Fe(s) \rightarrow 3Fe^{2+}(aq)$ ✓ *Cell potential:* «+0.77 V − (−0.45 V) = +»1.22 «V» ✓	*Do **not** accept reverse reaction or equilibrium arrow.* *Do **not** accept negative value for M2.*	**2**

3.	**e**		 left electrode/anode labelled zinc/Zn ***AND*** right electrode/cathode labelled iron/Fe ✓ electrolyte labelled as «aqueous» zinc salt/Zn^{2+} ✓	*Accept an inert conductor for the anode.* *Accept specific zinc salts such as $ZnSO_4$.*	**2**

6.1-S6 Nov 2019 · Paper 3 · Q2 parts (a)(ii), (a)(iii), (b)

2.	**a**	**ii**	*From ethanol to ethanal:* –2 to –1 ***OR*** +1/increases by 1 ✓ *From ethanol to carbon dioxide:* –2 to +4 ***OR*** +6/increases by 6 ✓	*Do **not** accept "2– to 1–".* *Do **not** accept "2– to 4+".* *Do **not** penalize incorrect notation twice.* *Penalize incorrect oxidation state value of carbon in ethanol once only.*	**2**
2.	**a**	**iii**	ethanal < ethanoic acid < carbon dioxide ✓	*Accept formulas.* *No ECF from 2aii calculations.*	**1**
2.	**b**		Pt/platinum/PtC ***AND*** highest yield of CO_2 «at all voltages» ✓	*ECF from 2aiii.*	**1**

6.1-S7 Nov 2018 · Paper 2 · Q6 parts (a)(iii), (d) · also 5.2

6.	**a**	**iii**	–1 ✓		**1**

6.	**d**		*Any two of:* butanoic acid forms more/stronger hydrogen bonds ✓ butanoic acid forms stronger London/dispersion forces ✓ butanoic acid forms stronger dipole–dipole interaction/force ✓	*Accept "butanoic acid forms dimers".* *Accept "butanoic acid has larger M_r/hydrocarbon chain/number of electrons" for M2.* *Accept "butanoic acid has larger **«permanent»** dipole/more polar" for M3.*	**2 max**

6.1-S8 Nov 2016 · Paper 3 · Q1 parts (a)(i)

1.	**a**	**i**	*HOCl:* +1 ***AND*** *ClO_2:* +4 ✓	*Accept "I" and "IV" but **not** "1+/1" and "4+/4" notations.*	**1**

6.1-S9 Nov 2015 · Paper 3 · Q9 parts (a)(ii)

Markscheme for 9(a)(ii) not located automatically - see the official Nov 2015 · Paper 3 markscheme.

6.1-S10 May 2015 TZ1 · Paper 3 · Q9 parts (a)(i), (a)(ii) · also 6.2

9. (a) (i) $+\frac{8}{3}$ / $+2\frac{2}{3}$;

Accept +2.7 but not +3.

+2 **and** +3 / contains two (or more) iron ions with different oxidation states / contains Fe^{2+} **and** Fe^{3+}; **[2]**

Accept II and III oxidation number notation for oxidation states but not 2+ and 3+ unless ions are referred to explicitly.
*Accept "contains different iron compounds/FeO **and** Fe_2O_3".*

(ii) $Fe_3O_4(s) + 4CO(g) \rightarrow 3Fe(l) + 4CO_2(g)$; **[1]**

*Accept "$Fe_3O_4(s) + CO(g) \rightarrow 3FeO(s) + CO_2(g)$ **and** $FeO(s) + CO(g) \rightarrow Fe(l) + CO_2(g)$".*
Ignore state symbols.

6.1-S11 Nov 2014 · Paper 2 · Q10 parts (a)(i), (a)(ii), (a)(iii), (a)(iv), (b), (c), (f)(i), (f)(ii), (g) · also 2.1, 5.2, 6.2, 7.2

10. (a) (i) oxidation **and** (iron/Fe) loses electrons/increases in oxidation number/state; ***[1]***

(ii) $O_2(aq) + 4e^- + 2H_2O(l) \rightarrow 4OH^-(aq)$

$\boxed{0}$ $\boxed{I}$ $\boxed{-II}$ $\boxed{-II}$ $\boxed{I}$;;

[2]

*Award **[2]** for five correct.*
*Award **[1]** for four correct.*
Accept use of oxidation states (0, +1, –2, –2, +1) for oxidation numbers.
Penalize once for incorrect notation (eg, 2, 2–).

(iii) $O_2(aq) + 2H_2O(l) + 2Fe(s) \rightarrow 2Fe^{2+}(aq) + 4OH^-(aq)$; ***[1]***

Ignore state symbols.

(iv) Fe/iron; ***[1]***

(b) oxygen is non-polar;
needs to break strong <u>hydrogen bonds/H–bonds</u> between water molecules (to dissolve) / oxygen cannot form <u>hydrogen bonds/H–bonds</u> with water;
oxygen can only form (weak) van der Waals'/vdW/LDF/London/dispersion forces with water; ***[2 max]***

(c) groups indicate the number of electrons in the highest energy level/outer/valence shell;
periods indicate the number of (occupied) energy levels/shells (in the atom); ***[2]***

(f) (i) chlorine/Cl_2 (is produced at the positive electrode/anode);
according to electrochemical series/ E° values/ease of oxidation OH^-/H_2O reacts/oxygen is released / *OWTTE* / at low chloride concentration OH^-/H_2O reacts/oxygen is released;
high concentration makes Cl^- oxidize/react in preference to OH^-/H_2O / *OWTTE*; ***[3]***

(ii) *Negative electrode (cathode):*
$2H^+(aq) + 2e^- \rightarrow H_2(g)$ / $H^+(aq) + e^- \rightarrow \frac{1}{2}H_2(g)$ / $2H_2O(l) + 2e^- \rightarrow H_2(g) + 2OH^-(aq)$;

Positive electrode (anode):
$2Cl^-(aq) \rightarrow Cl_2(g) + 2e^-$ / $Cl^-(aq) \rightarrow \frac{1}{2}Cl_2(g) + e^-$ / $2Cl^-(aq) - 2e^- \rightarrow Cl_2(g)$ /
$Cl^-(aq) - e^- \rightarrow \frac{1}{2}Cl_2(g)$; ***[2]***
Ignore state symbols.

Accept e instead of e^-.
*Award **[1]** if half-equations are correct but placed at the wrong electrodes.*

(g) bracelet/object to be electroplated is the cathode/negative electrode;
silver anode/positive electrode;
Accept Pt anode.
Electrolyte: liquid $Na[Ag(CN_2)]$/sodium dicyanoargentate/$[Ag(CN)_2]^-$/ solution of an appropriate silver salt; ***[3]***
Accept $AgNO_3$/silver nitrate.
All marks can be scored with a labelled diagram.

6.1-S12 Nov 2013 · Paper 3 · Q9 parts (b)(iv)
Markscheme for 9(b)(iv) not located automatically - see the official Nov 2013 · Paper 3 markscheme.

6.1-S13 May 2013 TZ1 · Paper 2 · Q4 parts (a), (b), (c), (d), (e)(i), (e)(ii) · also 6.2, 7.1

4. (a) loss of electrons; ***[1]***

(b) *Carbon:*
III to IV / +3 to +4 / (+)1;

Manganese:
VII to II / +7 to +2 / –5; ***[2]***
Penalize incorrect notation such as 3+ once only in all the paper.

(c) *Oxidizing agent:* MnO_4^- **and** *Reducing agent:* $(COOH)_2$; ***[1]***
Accept correct names instead of formulas.
Do not accept Mn and C.

(d) $(COOH)_2 \rightleftharpoons 2CO_2 + 2H^+ + 2e^-$; ***[1]***
Accept either $\rightarrow$ *or* $\rightleftharpoons$.
Allow equation times 5.
Allow e instead of e^-.

(e) (i) $MnO_4^-(aq) + 8H^+(aq) + 5e^- \rightleftharpoons Mn^{2+}(aq) + 4H_2O(l) \quad E^\ominus = 1.51\,V$;
$\Delta E^\ominus (= 1.51 + 0.49) = 2.00\,V$; ***[2]***
First mark may be implied in the calculation.
Allow e instead of e^-.
Accept either $\rightarrow$ *or* $\rightleftharpoons$.

(ii) positive sign, spontaneous reaction; ***[1]***
Allow ECF from (e) (i).

6.1-S14 Nov 2012 · Paper 2 · Q4 parts (a)(i), (a)(ii), (a)(iii), (a)(iv), (b)(i), (b)(ii), (b)(iii) · also 6.2, 7.2

4. (a) (i) *Oxidation:* loss of electrons **and** *Reduction:* gain of electrons; ***[1]***

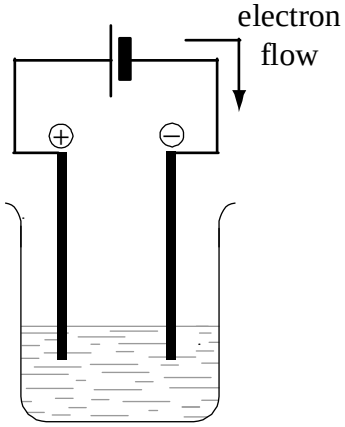
(ii) As_2O_3: +3;
NO_3^-: +5;
H_3AsO_3: +3;
N_2O_3: +3; ***[4]***
Penalize incorrect notation e.g. III, V, 3+, 5+, 3, 5 once only.

(iii) *Oxidizing agent:* substance reduced / removes electrons from another substance / causes some other substance to be oxidized / *OWTTE* **and** *Reducing agent*: substance oxidized / gives electrons to another substance / causes some other substance to be reduced / *OWTTE*; ***[1]***
*Accept Oxidizing agent: electron/e/e^- acceptor / causes oxidation / oxidation number/state decreases **and** Reducing agent: electron/e/e^- donor / causes reduction / oxidation number/state increases.*

(iv) $As_2O_3(s) + 2NO_3^-(aq) + 2H^+(aq) + 2H_2O(l) \rightarrow 2H_3AsO_4(aq) + N_2O_3(aq)$
correct coefficients for As_2O_3, H_3AsO_4 **and** NO_3^-, N_2O_3;
correct balanced equation;
Ignore state symbols.
M1 must be correct to award M2.

Oxidizing agent: $NO_3^-(aq)$/nitrate **and** *Reducing agent:* $As_2O_3(s)$/ arsenic(III) oxide; ***[3]***
Accept HNO_3(aq)/nitric acid.
Accept arsenic oxide.
Species must be fully correct to score M3.
Ignore state symbols.

(b) (i) *Diagram to show*:



Labels are not required.
one container, two electrodes, battery (and electrolyte);

Allow + and – for representation of battery (could be long and short lines for example) but M1 is not scored if a voltmeter/V if shown or labelled.
Ignore designation of electrodes (e.g. do not penalize Cu electrodes).

correct direction of electron flow (from negative pole to positive pole); ***[2]***
Allow arrow without stating e^- explicitly.
To score M2, the polarity of the battery or the cathode and anode must be shown.
If a voltmeter/V is shown, M1 is not awarded but M2 may be scored if the cathode and anode are identified with the correct direction of electron flow.

(ii) *Half-equation at positive electrode (anode)*:
$4OH^-(aq) \rightarrow 2H_2O(l) + O_2(g) + 4e^-$ / $2H_2O(l) \rightarrow O_2(g) + 4H^+(aq) + 4e^-$ /
$2OH^-(aq) \rightarrow \frac{1}{2}O_2(g) + H_2O(l) + 2e^-$;
Allow e instead of e^-.

Half-equation at negative electrode (cathode):
$Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$;
Allow e instead of e^-.
*Award **[1 max]** for M1 and M2 if correct equations are given but at wrong electrodes.*
Penalize $\rightleftharpoons$ once only in (b)(ii) and (b)(iii).

correct state symbols in **all** equations;
M3 can only be scored if the correct species are given in M1 and M2 (i.e. do not award ECF from M1 and M2 for incorrect species).

Observation at positive electrode (anode):
bubbles / gas;
Award mark for observation even if type of gas is incorrect (e.g. hydrogen).

Observation at negative electrode (cathode):
red/brown/copper/metal (deposit);
Allow mass increases / gets thicker/larger / OWTTE.

Change in colour (in any) of the solution:
solution loses blue colour/becomes paler;

Change in acidity (if any) of the solution:
becomes (more) acidic / pH decreases; ***[7]***

(iii) *Half-equation at positive electrode (anode)*:
$Cu(s) \rightarrow Cu^{2+}(aq) + 2e^-$;
Allow e instead of e^-.

Half-equation at negative electrode (cathode):
$Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$;
Allow e instead of e^-.
*Award **[1 max]** for M1 and M2 if correct equations are given but at wrong electrodes.*
Penalize $\rightleftharpoons$ once only in (b)(ii) and (b)(iii).

correct state symbols in **all** equations;
M3 can only be scored if the correct species are given in M1 and M2 (i.e. do not award ECF from M1 and M2 for incorrect species).

Observation at positive electrode (anode):
(slowly) dissolves / *OWTTE*;
Allow mass decreases / gets smaller/thinner / OWTTE;

Observation at negative electrode (cathode):
red/brown/copper/metal deposit;
Allow mass increases / gets thicker/larger / OWTTE.

Change in colour (if any) of the solution:
blue colour remains;
Allow no change or solution remains same colour.

Change in acidity (if any) of the solution:
solution does not become (more) acidic / no change / *OWTTE*; ***[7]***

6.1-S15 Nov 2012 · Paper 3 · QD2 parts (b)(i)

(b) (i) *Dichromate(VI) ions:*
accept electrons **and** (thus) reduced;

Ethanol:
loses electrons **and** (thus) oxidized; ***[2]***
*Award **[1 max]** for stating that dichromate is reduced **and** ethanol is oxidized without reference to electrons.*
No ECF from dichromate to ethanol.

6.1-S16 May 2012 TZ1 · Paper 2 · Q6 parts (e)(i), (e)(ii), (f)(i)

(e) (i) *Br_2:* 0
HBr: –1
HOBr: +1 ***[2]***
*Award **[2]** for three correct.*
*Award **[1]** for any two correct.*

(ii) bromine is oxidized **and** reduced / disproportionation; ***[1]***

(f) (i) F_2/fluorine; ***[1]***
Do not allow F.

6.1-S17 May 2012 TZ2 · Paper 2 · Q1 parts (c)(i), (c)(ii) · also 6.2

(c) (i)

Species	Oxidation number of oxygen
H_2O_2	–1
H_2O	–2
O_2	0

[2]

*Award **[2]** for three correct.*
*Award **[1]** for two correct.*

(ii) *Oxidation*:

$H_2O_2 \rightarrow O_2 + 2H^+ + 2e^-$;

Reduction:

$H_2O_2 + 2H^+ + 2e^- \rightarrow 2H_2O$; ***[2]***

6.1-S18 May 2011 TZ2 · Paper 2 · Q2 parts (a)(ii), (b)(i) · also 1.2

(ii) +3; ***[1]***

Do not accept 3, 3+ or the use of Roman numerals.

(b) (i) $Sb_2S_3 + 4\frac{1}{2}O_2 \rightarrow Sb_2O_3 + 3SO_2$ / $2Sb_2S_3 + 9O_2 \rightarrow 2Sb_2O_3 + 6SO_2$;

$2Sb_2O_3 + 3C \rightarrow 4Sb + 3CO_2$ / $Sb_2O_3 + 3C \rightarrow 2Sb + 3CO$; ***[2]***

Ignore state symbols.

LESSON 6.2

Balancing Redox Reactions

Answers for 5 multiple-choice and 10 structured questions

Part A - Multiple choice answers

6.2-M1	Nov 2022 · P1 · Q29	D
6.2-M2	May 2022 TZ1 · P1 · Q29	C
6.2-M3	May 2015 TZ2 · P1 · Q31	D
6.2-M4	May 2014 TZ1 · P1 · Q30	A
6.2-M5	May 2012 TZ1 · P1 · Q30	C

Part B - Structured questions markscheme

6.2-S1 May 2018 TZ2 · Paper 2 · Q4 parts (a)

4.	**a**		$BrO_3^-(aq) + 6H^+(aq) + 6I^-(aq) \rightleftharpoons Br^-(aq) + 3I_2(s) + 3H_2O(l)$ ✓	*Accept* $\rightarrow$ *for* $\rightleftharpoons$.	**1**

6.2-S2 Nov 2017 · Paper 3 · Q14 parts (a), (b) · also 6.1

14.	**a**		$O_2 + 4H^+(aq) + 4e^- \rightarrow 2H_2O(l)$ ✓	*Accept any balanced equation with any integer or fractional coefficients.*	**1**
14.	**b**		+2 to +3 ***OR*** +1 ***OR*** increases «by 1» ✓		**1**

6.2-S3 May 2017 TZ2 · Paper 2 · Q2 parts (a)(i), (a)(ii), (a)(iii), (a)(iv), (b)(i) · also 1.2, 7.1

2.	**a**	**i**	$Sn^{2+}(aq) \rightarrow Sn^{4+}(aq) + 2e^-$ ✓	*Accept equilibrium sign.* *Accept* $Sn^{2+}(aq) - 2e \rightarrow Sn^{4+}(aq)$	**1**
2	**a**	**ii**	$Cr_2O_7^{2-}(aq) + 14H^+(aq) + 3Sn^{2+}(aq) \rightarrow 2Cr^{3+}(aq) + 7H_2O(l) + 3Sn^{4+}(aq)$ ✓	*Accept equilibrium sign.*	**1**
2	**a**	**iii**	« $[K_2Cr_2O_7] = \frac{13.239\ g}{294.20\ g\ mol^{-1} \times 0.100\ dm^3}$ =» $0.450\ «mol\ dm^{-3}»$ ✓		**1**
2	**a**	**iv**	$n(Sn^{2+}) = «0.450\ mol\ dm^{-3} \times 0.01324\ dm^3 \times \frac{3\ mol}{1\ mol} =» 0.0179\ «mol»$ ✓ $«[Sn^{2+}] = \frac{0.0179\ mol}{0.0100\ dm^3} =» 1.79\ «mol\ dm^{-3}»$ ✓	*Award* ***[2]*** *for correct final answer.*	**2**
2	**b**	**i**	$«E^{\ominus} =» +0.85\ «V»$ ✓	*Accept 0.85 V.*	**1**

6.2-S4 May 2016 · Paper 3 · Q3 parts (a)

3.	**a**		$Fe_2O_3(s) + 3CO(g) \rightarrow 2Fe(l) + 3CO_2(g)$ ✓		**1**

6.2-S5 May 2015 TZ1 · Paper 2 · Q7 parts (a)(ii), (a)(iii), (a)(iv), (f) · also 4.1, 6.1

(ii) *Ethanol:* –2/–II;
Ethanal: –1/–I; **[2]**
Do not accept 2–, 1– but penalize once only.

(iii) $CH_3CH_2OH \rightarrow CH_3CHO + 2H^+ + 2e^-$; **[1]**

Half-equation required. Do not accept $C_2H_5OH + 2[O] \rightarrow CH_3CHO + H_2O$.

Accept e for e^-.

(iv) $3CH_3CH_2OH(aq) + Cr_2O_7^{2-}(aq) + 8H^+(aq) \rightarrow 2Cr^{3+}(aq) + 3CH_3CHO(l) + 7H_2O(l)$

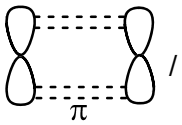
correct reactants and products;
correct balancing; **[2]**
M2 can only be scored if M1 correct.
Ignore state symbols.

(f) *Sigma bonds*:

/ /

result from head-on/end-on overlap of orbitals / *OWTTE*;
Accept axial overlap of orbitals.
Accept "symmetric orbital" with respect to same plane / OWTTE.

Pi bonds:



/

result from sideways overlap of orbitals / *OWTTE*; **[2]**
Accept "antisymmetric orbitals" with respect to (defining) plane (containing at least one atom) / OWTTE.

6.2-S6 May 2015 TZ2 · Paper 3 · Q19 parts (a)(i), (a)(ii) · also 6.1

19. (a) (i) $CH_3CH_2OH(aq) \rightarrow CH_3CHO(aq) + 2H^+(aq) + 2e^-$; **[1]**

Accept equilibrium sign, e for e^- and different representations of organic compounds (such as C_2H_6O and C_2H_4O).
Ignore state symbols.
*Do **not** accept $CH_3CH_2OH + [O] \rightarrow CH_3CHO + H_2O$ (since half-equation requested).*

(ii) (orange) dichromate(VI) ion/$Cr_2O_7^{2-}$/Cr(VI)/Cr^{6+} is reduced/converted to (green) chromium(III) ion/Cr(III)/Cr^{3+}; **[1]**

6.2-S7 Nov 2014 · Paper 3 · Q24 parts (a)

24. (a) $NO_3^- + 10H^+ + 8e^- \rightarrow NH_4^+ + 3H_2O$

correct reactants;
correct products;
correct balancing; ***[3]***
M3 can only be scored if M1 and M2 correct.

6.2-S8 May 2012 TZ2 · Paper 3 · QF3 parts (a)

F3. (a) $C_6H_8O_6 \rightarrow C_6H_6O_6 + 2H^+ + 2e^-$ ***[2]***

6.2-S9 May 2011 TZ1 · Paper 3 · QB5 parts (a), (b)

B5. (a) glucose/$C_6H_{12}O_6$;
$C_6H_{12}O_6 + 6H_2O \rightarrow 6CO_2 + 24H^+ + 24e^-$; ***[2]***

(b) oxygen/O_2;
$6O_2 + 24H^+ + 24e^- \rightarrow 12H_2O$; ***[2]***

Accept equation divided by 6.
*Award **[2 max]** if (a) and (b) are reversed and all 4 marking points are correct.*

6.2-S10 May 2010 TZ1 · Paper 3 · QC1 parts (b)(i), (b)(ii)

(b) (i) $Fe_3O_4 + 4CO \rightarrow 3Fe + 4CO_2$; ***[1]***

(ii) $Fe_3O_4 + 4H_2 \rightarrow 3Fe + 4H_2O$; ***[1]***

LESSON 6.3

Reactivity

Answers for 14 multiple-choice and 3 structured questions

Part A - Multiple choice answers

6.3-M1	May 2025 TZ1 · P1A · Q35	D
6.3-M2	May 2025 TZ3 · P1A · Q36	C
6.3-M3	Nov 2023 TZ1 · P1 · Q29	A
6.3-M4	May 2021 TZ1 · P1 · Q28	D
6.3-M5	Nov 2019 · P1 · Q30	B
6.3-M6	May 2017 TZ2 · P1 · Q24	B
6.3-M7	May 2013 TZ1 · P1 · Q7	B
6.3-M8	Nov 2012 · P1 · Q7	D
6.3-M9	May 2012 TZ1 · P1 · Q25	B
6.3-M10	Nov 2011 · P1 · Q8	A
6.3-M11	Nov 2011 · P1 · Q30	D
6.3-M12	May 2011 TZ1 · P1 · Q8	C
6.3-M13	May 2011 TZ2 · P1 · Q31	B
6.3-M14	Nov 2010 · P1 · Q31	B

Part B - Structured questions markscheme

6.3-S1 May 2022 TZ1 · Paper 2 · Q2 parts (a), (b)(i), (b)(iii) · also 7.1, 7.2

2.	**a**		**Alternative 1** put Mg in Zn^{2+} (aq) ✓ Zn/«black» layer forms «on surface of Mg» ✓ **Alternative 2** place both metals in acid ✓ bubbles evolve more rapidly from Mg ***OR*** Mg dissolves faster ✓ **Alternative 3** construct a cell with Mg and Zn electrodes ✓ bulb lights up ***OR*** shows (+) voltage ***OR*** size/mass of Mg(s) decreases <<over time>> ***OR*** size/mass of Zn increases <<over time>> ✓	*Award **[1 max]** for "no reaction when Zn placed in Mg^{2+}(aq)".* *Accept "electrons flow from Mg to Zn".* *Accept Mg is negative electrode/anode* ***OR*** *Zn is positive electrode/cathode* *Accept other correct methods.*	**2**
2.	**b**	**i**	Cell potential: «(–0.45 V – (–2.37 V)» = «+»1.92 «V»✓		**1**
2.	**b**	**iii**	$2\ H_2O + 2\ e^- \rightarrow H_2 + 2\ OH^-$ ✓	*Accept equation with equilibrium arrows.*	**1**

6.3-S2 May 2017 TZ2 · Paper 3 · Q5 parts (b)(i), (b)(ii)

5.	**b**	**i**	Fe/iron ***OR*** Zn/zinc ***OR*** Co/cobalt ***OR*** Cd/cadmium ***OR*** Cr/chromium ✓	*Accept "Mn/manganese".*	**1**
5.	**b**	**ii**	$Ni^{2+}(aq) + Fe(s) \rightarrow Ni(s) + Fe^{2+}(aq)$ ***OR*** $Ni^{2+}(aq) + Zn(s) \rightarrow Ni(s) + Zn^{2+}(aq)$ ***OR*** $Ni^{2+}(aq) + Co(s) \rightarrow Ni(s) + Co^{2+}(aq)$ ***OR*** $Ni^{2+}(aq) + Cd(s) \rightarrow Ni(s) + Cd^{2+}(aq)$ ***OR*** $Ni^{2+}(aq) + Cr(s) \rightarrow Ni(s) + Cr^{2+}(aq)$ ✓	*Accept "$3Ni^{2+}(aq) + 2Cr(s) \rightarrow 3Ni(s) + 2Cr^{3+}(aq)$".* *Do **not** penalize similar equations involving formation of $Fe^{3+}(aq)$, $Mn^{2+}(aq)$ **OR** $Co^{3+}(aq)$.* *Ignore Cl^- ions.* *Accept correctly balanced non-ionic equations eg, "$NiCl_2(aq) + Zn(s) \rightarrow Ni(s) + ZnCl_2(aq)$" etc.* *Do **not** allow ECF from (b) (i).*	**1**

6.3-S3 Nov 2015 · Paper 2 · Q7 parts (d)(i), (d)(ii), (e) · also 7.2

(d) (i) $Pb < Ni < Fe < Zn$;; ***[2]***

*Award **[2]** for the correct order.*
*Award **[1]** for $Zn > Fe > Ni > Pb$ as metals not listed in order of increasing reactivity.*
*Award **[1]** if one error in the order.*

(ii) Pb^{2+}/ lead(II) (ions); ***[1]***

Do not accept Pb/lead.

(e) *Dilute sodium chloride:*

$$H_2O(l) \rightarrow \frac{1}{2}O_2(g) + 2H^+(aq) + 2e^- / 2OH^-(aq) \rightarrow \frac{1}{2}O_2(g) + H_2O(l) + 2e^-;$$

C*oncentrated sodium chloride:*

$$Cl^-(aq) \rightarrow \frac{1}{2}Cl_2(g) + e^-; \quad \textbf{[2]}$$

*Award **[1 max]** if equations are given the wrong way round.*
Accept e instead of e^-.
Ignore state symbols and equilibrium sign.
Accept correct half-equations balanced with different coefficients.

LESSON 7.1

Primary & Secondary Cells

Answers for 57 multiple-choice and 36 structured questions

Part A - Multiple choice answers

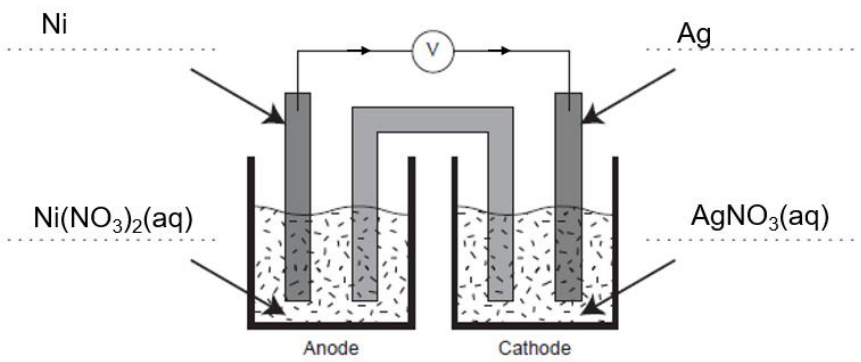
7.1-M1	May 2025 TZ2 · P1A · Q37	D
7.1-M2	May 2025 TZ3 · P1A · Q39	A
7.1-M3	Nov 2024 · P1 · Q29	D
7.1-M4	Nov 2024 · P1 · Q31	C
7.1-M5	May 2024 TZ1 · P1 · Q29	D
7.1-M6	May 2024 TZ2 · P1 · Q31	B
7.1-M7	May 2023 TZ1 · P1 · Q30	B
7.1-M8	May 2023 TZ1 · P1 · Q31	A
7.1-M9	May 2023 TZ2 · P1 · Q29	A
7.1-M10	May 2023 TZ2 · P1 · Q30	A
7.1-M11	Nov 2022 · P1 · Q28	B
7.1-M12	Nov 2022 · P1 · Q30	B
7.1-M13	May 2022 TZ2 · P1 · Q29	C
7.1-M14	May 2022 TZ2 · P1 · Q30	A
7.1-M15	Nov 2021 · P1 · Q29	D
7.1-M16	Nov 2021 · P1 · Q30	D
7.1-M17	May 2021 TZ1 · P1 · Q30	D
7.1-M18	May 2021 TZ2 · P1 · Q30	A
7.1-M19	Nov 2020 · P1 · Q29	D
7.1-M20	May 2019 TZ1 · P1 · Q29	B
7.1-M21	May 2019 TZ1 · P1 · Q31	A
7.1-M22	May 2019 TZ2 · P1 · Q29	C
7.1-M23	May 2019 TZ2 · P1 · Q30	A
7.1-M24	Nov 2018 · P1 · Q29	B
7.1-M25	Nov 2018 · P1 · Q31	D
7.1-M26	May 2017 TZ1 · P1 · Q30	D
7.1-M27	May 2017 TZ2 · P1 · Q30	D
7.1-M28	Nov 2016 · P1 · Q31	D
7.1-M29	May 2016 · P1 · Q31	D
7.1-M30	Nov 2015 · P1 · Q32	C
7.1-M31	May 2015 TZ1 · P1 · Q31	C
7.1-M32	May 2015 TZ1 · P1 · Q33	B
7.1-M33	May 2015 TZ2 · P1 · Q32	C
7.1-M34	Nov 2014 · P1 · Q32	C
7.1-M35	May 2014 TZ1 · P1 · Q33	C
7.1-M36	May 2014 TZ2 · P1 · Q32	A
7.1-M37	May 2014 TZ2 · P1 · Q33	A
7.1-M38	May 2014 TZ2 · P1 · Q34	B
7.1-M39	Nov 2013 · P1 · Q33	B
7.1-M40	May 2013 TZ1 · P1 · Q32	A
7.1-M41	May 2013 TZ1 · P1 · Q33	D
7.1-M42	May 2013 TZ2 · P1 · Q31	D
7.1-M43	Nov 2012 · P1 · Q31	D
7.1-M44	Nov 2012 · P1 · Q32	C
7.1-M45	Nov 2012 · P1 · Q33	C
7.1-M46	May 2012 TZ1 · P1 · Q31	B
7.1-M47	May 2012 TZ1 · P1 · Q32	C
7.1-M48	May 2012 TZ2 · P1 · Q29	C
7.1-M49	May 2012 TZ2 · P1 · Q30	D
7.1-M50	Nov 2011 · P1 · Q32	B
7.1-M51	May 2011 TZ1 · P1 · Q30	A
7.1-M52	May 2011 TZ1 · P1 · Q32	B
7.1-M53	May 2011 TZ2 · P1 · Q32	D
7.1-M54	Nov 2010 · P1 · Q32	A
7.1-M55	May 2010 TZ1 · P1 · Q31	D
7.1-M56	May 2010 TZ2 · P1 · Q31	A
7.1-M57	May 2010 TZ2 · P1 · Q32	D

Part B - Structured questions markscheme

7.1-S1 May 2025 TZ3 · Paper 2 · Q6 parts (a)(i), (a)(ii), (b)

6.	**(a)**	**(i)**	$Li \rightarrow Li^+ + e^-$ ***OR*** $LiC_6 \rightarrow C_6 + Li^+ + e^-$ ✓	*Do **not** accept 6C.* *Do **not** penalize equilibrium arrows.*	**1**
6.	**(a)**	**(ii)**	negative sign ***AND*** large value ✓ oxidation/reverse reaction is «highly» spontaneous/favourable ✓	*Accept Li is a good reductant/reducing agent for M2.*	**2**
6.	**(b)**		arrow from cathode to anode in the external circuit ✓	*Do **not** award mark if any arrows are in the electrolyte.*	**1**
				Accept "stores charge/power for night use".	

7.1-S2 Nov 2024 · Paper 2 · Q8 parts (a), (b)

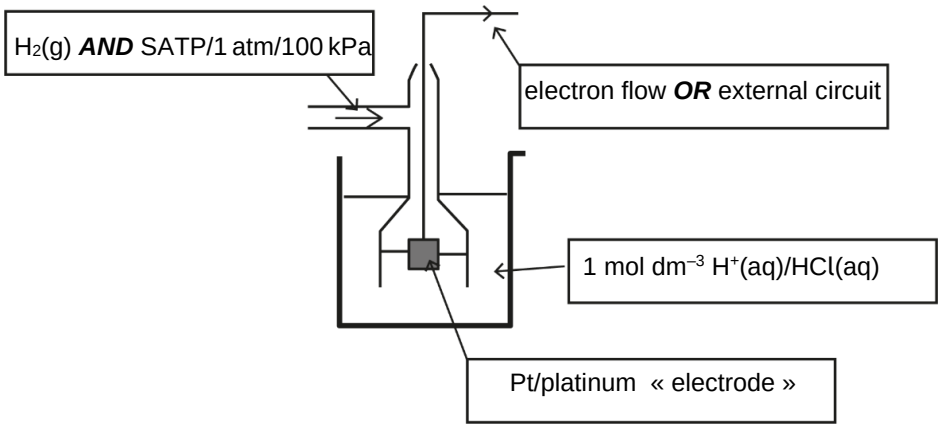
8.	**(a)**		 anode to have Ni ***AND*** $Ni(NO_3)_2/Ni^{2+}$ ***AND*** cathode to have Ag ***AND*** $AgNO_3/Ag^+$ ✓ electrons to flow from anode to cathode/left to right ✓	*Electron flow must be clearly through the external circuit for M2.*	**2**
8.	**(b)**		balances charges/keeps each half-cell **«**electrically**»** neutral ***OR*** completes circuit/allows ion flow **«**between cells**»** ✓		**1**

7.1-S3 Nov 2024 · Paper 3 · Q15 parts (a), (c)

15.	**a**		**«**redox**»** reaction in rechargeable battery is reversible **«**but not in a primary cell**»** ***OR*** rechargeable battery needs to be charged before use ***OR*** rechargeable battery has greater rate of self-discharge ✓	*Accept "rechargeable battery can be recharged **AND** primary cell cannot"* ***OR*** *"rechargeable battery can be used more than once/many times **AND** primary cell can be used once only".*	**1**

15.	**c**		*Chemical natures:* further apart in redox table/activity series «used as electrodes» produces higher voltage/electrical output ✓ *Quantities:* «larger» quantity of materials increase/impact total work/length of time battery can operate ***OR*** «higher» concentrations can influence voltage/electrical output ***OR*** «higher» concentrations influence/increase total energy/voltage/electrical output ✓	*Do **not** accept "higher charge" alone for M1.*	**2**

7.1-S4 May 2024 TZ1 · Paper 2 · Q7 parts (a), (b)(i)

7.	**(a)**		 *Award **[1]** for every two correctly placed answers.* $H_2(g)$ ***AND*** SATP/1 atm/100 kPa electron flow ***OR*** external circuit 1 mol dm^{-3} $H^+(aq)/HCl(aq)$ Pt/platinum « electrode » ✓✓	*Accept current / pd/ potential difference / voltage for external circuit* *Do not accept "electron" or "e^-" only for external circuit*	**2**
7.	**(b)**	**(i)**	$Pt\|H_2\|H^+\|\|Cu^{2+}\|Cu$ ✓	*Do not penalize missing Pt* *Accept fractions / multiples of species eg ½ H_2*	**1**

7.1-S5 May 2024 TZ1 · Paper 3 · Q16 parts (b)(i), (b)(ii)

16.	**b**	**i**	*Positive electrode (cathode):* lead(IV) oxide/PbO_2 ✓ *Negative electrode (anode):* lead/Pb ✓	*Do **not** accept* lead(IV)/Pb^{4+}. *Accept correct reactants shown in half-equations.*	**2**
16.	**b**	**ii**	higher energy density/charge per unit mass ✓	*Do **not** accept lithium ion lighter/smaller without mentioning similar capacity.* *Do **not** accept references to toxicity.* *Accept does not involve a liquid/corrosive electrolyte.* *Accept lower internal resistance.*	**1**

7.1-S6 May 2024 TZ2 · Paper 3 · Q13 parts (a), (b)(i), (c)(i), (c)(ii), (c)(iii) · also 7.3

13.	**a**		energy density ***AND*** requires less storage space ***OR*** gases need **«**extra weight for**»** compression equipment ***OR*** **«**small volume**»** easier to transport/carry ***OR*** high specific energy with low energy density technologies are cost prohibitive ✓		**1**
13.	**b**	**i**	M_r ethylbenzene = 106.18 g mol^{-1} ✓ **«**$\Delta H_c = \frac{-4.135 \times 10^7 \text{ J kg}^{-1}}{1000 \text{ g kg}^{-1}} \times \frac{106.18 \text{ g mol}^{-1}}{1000 \text{ J kJ}^{-1}}$ **=»** –4391 **«**kJ mol^{-1}**»** ✓	*Award **[2]** for correct final answer.* *Award **[1 max]** for missing negative sign in answer.* *Award **[2]** for -4380 **«**kJ mol^{-1}**»** for M2 when 106 g mol^{-1} used for M1.*	**2**

13.	**c**	**i**	1 mol **«**of atoms only**»** 7 g for Li ***AND*** 207 g for Pb ***OR*** lower mass of Li for same number of atoms **«**oxidized**»** ***OR*** lower mass of Li per number of electrons **«**transferred**»** ✓ **«**voltage per cell**»** oxidation potential of Li(s) is +3.04 V ***AND*** Pb(s) is +0.13 V ***OR*** higher voltage for Li ✓	*Accept "higher charge density for Li" for M2.*	**2**
13.	**c**	**ii**	*Graphite:* $Li^+ + e^- \rightarrow Li(s)$ ***OR*** $6C(s, graphite) + Li^+ + e^- \rightarrow LiC_6(s)$ ✓ *Li CoO_2(s):* $LiCoO_2(s) \rightarrow Li^+ + CoO_2(s) + e^-$ ✓	*Award **[1 max]** for reversed half-equations given at the correct electrodes.*	**2**

13.	**c**	**iii**	sensitive to high temperature ***OR*** can overheat/explode/catch fire ***OR*** damaged if overcharged **«**so require circuit protection**»** ***OR*** ruined if completely discharged ***OR*** high cost ***OR*** ageing/loss of battery capacity/limited number of charge/discharge cycles ✓	*Apply List Principle*	**1**

7.1-S7 Nov 2023 TZ1 · Paper 3 · Q15 parts (b)

15.	**b**		*Fuel cells:* refilled ***AND*** with fuel/hydrogen/methanol/reducing agent ✓ *Secondary cells*: recharged ***AND*** by reversing **«**chemical**»** reaction ***OR*** recharged ***AND*** by external power source ✓	*Award **[1 max]** for Fuel cells: refilled **AND** Secondary cells: recharged with no or incorrect details given.*	**2**

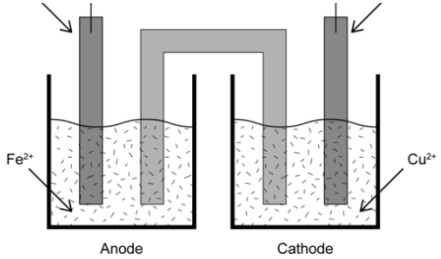
7.1-S8 May 2023 TZ1 · Paper 2 · Q4 parts (a), (b)(i), (b)(ii), (c)(i) · also 6.1

4.	**(a)**		+6/VI ✓	*Do **not** accept 6/6+.*	**1**
4.	**(b)**	**(i)**	Zinc more reactive/ <<better>> reducing agent/ <<more>> easily oxidized/loses electrons <<more>> easily. ✓	*Accept "zinc higher in the activity «series»".* *Accept "*zinc has a negative electrode potential/Cu has a positive electrode potential".	**1**
4.	**(b)**	**(ii)**	Anode (negative electrode): $Zn_{(s)} \rightarrow Zn^{2+}_{(aq)} + 2e^-$ ✓ Cathode (positive electrode): $Cu^{2+}_{(aq)} + 2e^- \rightarrow Cu_{(s)}$ ✓	*Award [**1 max**] for equilibria.* *Award [**1 max**] for equations at the wrong electrodes.* *State symbols not required for mark.*	**2**
4.	**(c)**	**(i)**	«E^{θ}_{cell} = +0.34-(-0.76) = +» 1.10 «V» ✓	*Accept ECF from 4 (b) (ii).*	**1**

7.1-S9 Nov 2022 · Paper 2 · Q3 parts (a), (c), (d), (f)(i), (f)(ii), (f)(iii), (f)(iv), (f)(v) · also 2.1

3.	**(a)**		$1s^2 2s^2 2p^6 3s^2 3p^6 3d^6$ ***OR*** $[Ar]3d^6$ ✓		**1**
3.	**(c)**		Cu^{2+} ***AND*** fewer shielding electrons/less electron-electron repulsion **«**from same nuclear charge**»** ***OR*** Cu^{2+} ***AND*** larger effective nuclear charge ***OR*** Cu^{2+} ***AND*** more energy required to remove electron from positive ion than neutral parent atom ***OR*** Cu^{2+} ***AND*** smaller radius ***OR*** Cu^{2+} ***AND*** electron is being lost from a lower energy/inner/3d orbital ✓		**1**

3.	**(d)**		***Alternative 1*** «$E = 745\ kJ\ mol^{-1} \times \frac{1\ mol}{6.02 \times 10^{23} atoms} =$ » 1.24×10^{-21}«$kJ\ atom^{-1}$» ✓ **«**$E = h\nu$» «$1.24 \times 10^{-21} kJ \times \frac{1000\ J}{1\ kJ} = 6.63 \times 10^{-34} Js \times \nu$**»** **«**$\nu =$ » 1.87×10^{15}«s^{-1}» ✓ ***Alternative 2*** *«E=$h\nu$»* *«745 x 10^3 J mol^{-1} = 6.63 x 10^{-34} J s x υ »* *«υ =» 1.12x 10^{39} «s^{-1} mol^{-1}» ✓* *« =$\frac{1.12\ x\ 10^{39} s^{-1}}{6.02\ x\ 10^{23}}$» = 1.87 x 10^{15} «s^{-1}» ✓*	*Award **[2]** for correct final answer.*	**2**

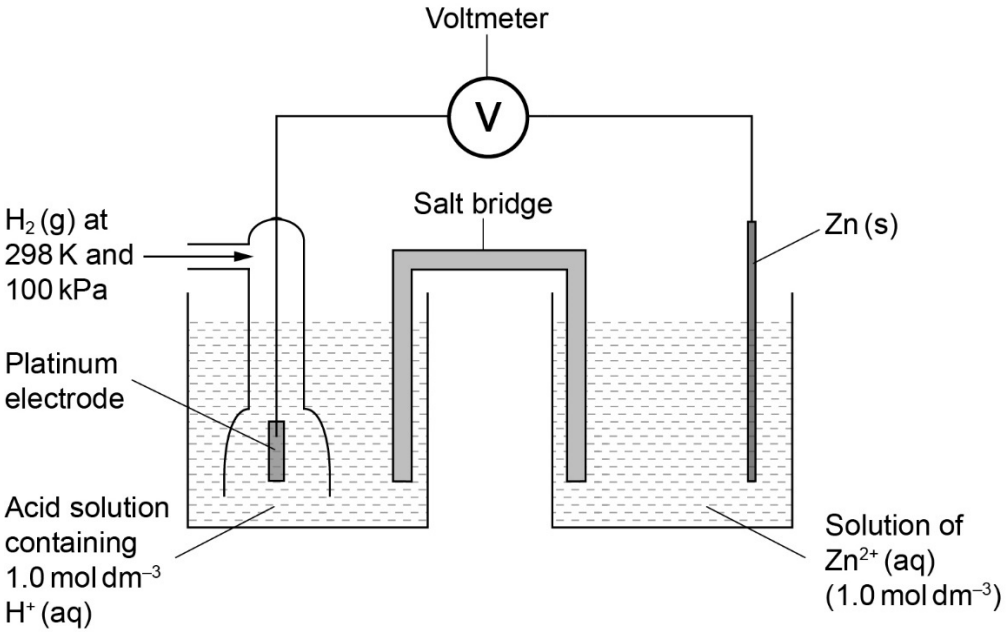
3.	**(f)**	**(i)**	 all 4 species correctly labelled ✓ arrow showing electron flow from anode to cathode in the external circuit ✓	*Accept any soluble salt of copper(II) for Cu^{2+} and any soluble salt of iron(II) for Fe^{2+}.* *Do **not** apply ECF for M2.*	**2**
3.	**(f)**	**(ii)**	$Fe(s) \rightarrow Fe^{2+}(aq) + 2e^-$ ✓	*Accept equilibrium arrow.* *Do **not** award ECF for $Cu(s) \rightarrow Cu^{2+}(aq) + 2e^-$.*	**1**
3.	**(f)**	**(iii)**	**«**keep**»** each half-cell/electrolyte **«**electrically**»** neutral ✓	*Accept balance charges/ions.* *Accept allow ion flow **«**between cells**»**.*	**1**

3.	**(f)**	**(v)**	**«**$E^\ominus$ = + 0.34 V – (– 0.45 V) = **+»** 0.79 **«**V**»** ✓		**1**

7.1-S10 May 2022 TZ2 · Paper 2 · Q3 parts (a)(i), (a)(ii), (b)(i), (b)(ii) · also 7.2

3.	**a**	**i**	«$E^{Ö}_{cell}$ = 1.09 – 0.77 =» 0.32 «V» ✓		**1**
3.	**a**	**ii**	«$2Fe^{2+}(aq) + I_2(s) \rightarrow 2Fe^{3+}(aq) + 2I^-(aq)$ » no/non-spontaneous ***AND*** $E^{Ö}_{cell}$ «= 0.54 – 0.77 »= –0.23 «V»/ $E^{Ö} < 0$ ***OR*** no ***AND*** reduction potential of I_2 lower «than $Fe3^+$ »/ 0.54 <0.77 ✓	*Accept "standard electrode potential of I_2 lower /less positive than iron".*	**1**
3.	**b**	**i**	Cathode (negative electrode): $Zn^{2+} + 2e^- \rightarrow Zn(l)$ ✓ Anode (positive electrode): $2Cl^- \rightarrow Cl_2(g) + 2e^-$ ***OR*** $Cl^- \rightarrow ½\ Cl_2(g) + e^-$ ✓		**2**
3.	**b**	**ii**	$ZnCl_2(l) \rightarrow Zn(l) + Cl_2(g)$ balanced equation ✓ correct state symbols ✓	*Accept ionic equation.*	**2**

7.1-S11 Nov 2021 · Paper 2 · Q8

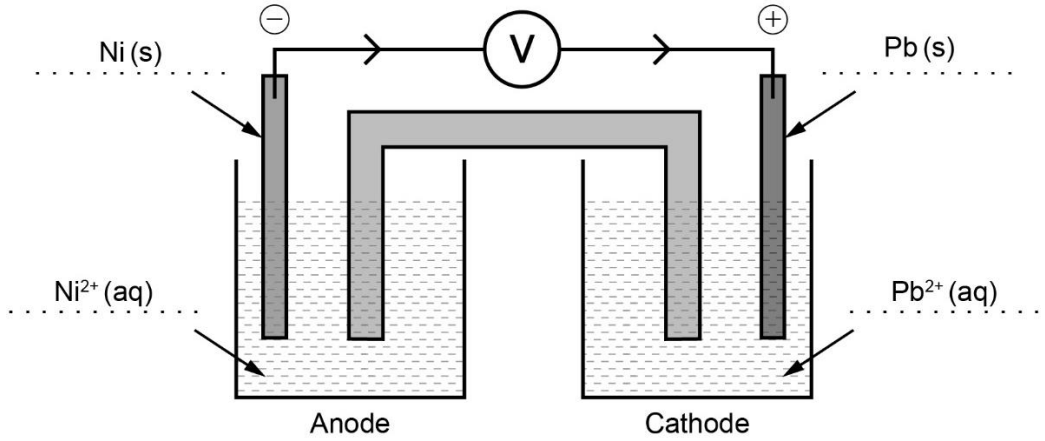
8.			 H_2 (g) entering at «298 K and» 100 kPa ✓ platinum electrode on left ✓ voltmeter connecting electrodes ***AND*** salt bridge connecting electrolytes ✓ 1 mol dm^{-3} H^+ on the left ***AND*** 1 mol dm^{-3} Zn^{2+} on the right ✓	*Voltmeter and salt bridge need to be drawn but not necessarily annotated for **M3.*** *Concentrations, but not state symbols, required for **M4.***	**4**

7.1-S12 May 2021 TZ2 · Paper 2 · Q3 parts (a), (b), (c)

3.	**a**		Al/aluminium **«**electrode**»** ***AND*** aluminium nitrate/$Al(NO_3)_3$/Al^{3+} on left ✓ Sn/tin **«**electrode**»** ***AND*** tin**«**(II)**»** nitrate/$Sn(NO_3)_2$/Sn^{2+} on right ✓ salt bridge ***AND*** voltmeter/V/lightbulb ✓	*Award **[1]** if M1 and M2 are reversed.* *Award **[1]** for two correctly labelled solutions **OR** two correctly labelled electrodes for M1 and M2.* *Accept a specific salt for "salt bridge".* *Accept other circuit components such as ammeter/A, fan, buzzer, resistor/heating element/R/Ω.*	**3**
3.	**b**		$3Sn^{2+}(aq) + 2Al(s) \rightarrow 3Sn(s) + 2Al^{3+}(aq)$ ***OR*** $3Sn(NO_3)_2(aq) + 2Al(s) \rightarrow 3Sn(s) + 2Al(NO_3)_3(aq)$ ✓	*If half-cells are reversed in (a) then the equation must be reversed to award the mark.* *Do **not** penalize equilibrium arrows.*	**1**
3.	**c**		**«**1.66 + (–0.14) = +**»**1.52 **«**V**»** ✓	*Calculation must be consistent with equation given in 3 b.*	**1**

7.1-S13 Nov 2020 · Paper 2 · Q4 parts (c), (d)(i), (d)(ii), (d)(iv) · also 5.2

4.	**c**		hydrogen bonding/bonds **«**and dipole–dipole and London/dispersion forces are present in**»** propan-2-ol ✓ dipole–dipole **«**and London/dispersion are present in**»** propanone ✓ propan-2-ol less volatile ***AND*** hydrogen bonding/bonds stronger **«**than dipole–dipole **»** ***OR*** propan-2-ol less volatile ***AND*** **«**sum of all**»** intermolecular forces stronger ✓		**3**

4.	**d**	**i**			**1**
4.	**d**	**ii**	**«**–0.13 V – (–0.26 V) = +**»** 0.13 **«**V**»** ✓		**1**

4.	**d**	**iv**	Bi/Cu/Ag/Pd/Hg/Pt/Au ✓	*Accept Sb **OR** As.*	**1**

7.1-S14 Nov 2020 · Paper 3 · Q13 parts (a)

13.	**a**		$Mg(s) + Ni^{2+}(aq) \rightarrow Mg^{2+}(aq) + Ni(s)$ ✓	*Accept a balanced molecular equation such as "$Mg + NiSO_4 \rightarrow MgSO_4 + Ni$".*	**1**

7.1-S15 May 2019 TZ1 · Paper 2 · Q6 parts (a), (e), (f)(i) · also 2.1

6.	**a**		$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^6$ ✓		**1**
6.	**e**		lemon juice is the electrolyte ***OR*** lemon juice allows flow of ions ***OR*** each nail/metal forms a half-cell with the lemon juice ✓ *Any one of:* iron is higher than copper in the activity series ***OR*** each half-cell/metal has a different redox/electrode potential ✓ iron is oxidized ***OR*** $Fe \rightarrow Fe^{2+} + 2e^-$ ***OR*** $Fe \rightarrow Fe^{3+} + 3e^-$ ***OR*** iron is anode/negative electrode of cell ✓ copper is cathode/positive electrode of cell ***OR*** reduction occurs at the cathode ***OR*** $2H^+ + 2e^- \rightarrow H_2$ ✓ electrons flow from iron to copper ✓		**2**
6.	**f**	**i**	«$E^\ominus = +0.34\,V - (-0.45\,V) = +$»0.79 «V» ✓		**1**

7.1-S16 May 2019 TZ2 · Paper 3 · Q19 parts (c)

<table>
<tr>
<td>19.</td>
<td>c</td>
<td></td>
<td>Any one of:
<table>
<tr><th>Primary</th><th></th><th>Secondary</th></tr>
<tr><td>«electrode» materials cannot be reused/regenerated</td><td>AND</td><td>«electrode» materials can be regenerated/reused ✓</td></tr>
<tr><td>non-reversible reaction(s)</td><td>AND</td><td>reversible reaction(s) ✓</td></tr>
<tr><td>lower rate of self-discharge</td><td>OR</td><td>higher rate of self-discharge ✓</td></tr>
<tr><td>only lower current demands</td><td>OR</td><td>can deliver stronger current demands ✓</td></tr>
</table></td>
<td>Accept “primary cannot be recharged AND “secondary can be recharged”.</td>
<td>1 max</td>
</tr>
</table>

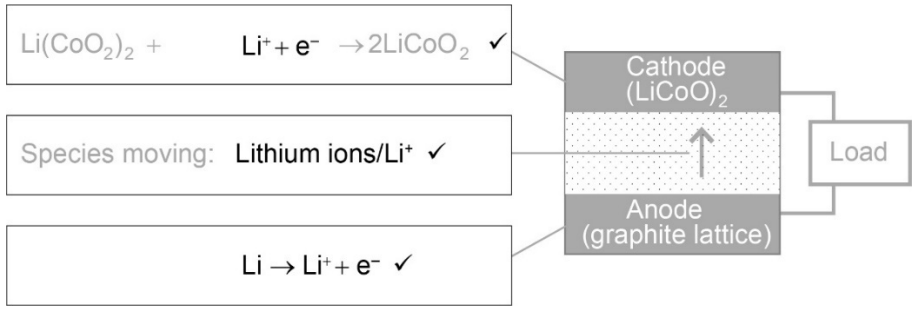
7.1-S17 Nov 2018 · Paper 3 · Q15 parts (a), (b)

15.	**a**		«redox» reaction in rechargeable battery is reversible «but not in a primary cell» ***OR*** secondary cells need to be charged before use ***OR*** secondary cells have greater rate of self-discharge ✓	*Accept "primary cells cannot be recharged/reused", "primary cells can be used only once" **OR** "lithium batteries may explode".*	**1**
15.	**b**		*Anode (negative electrode):* $Li\,(graphite) \rightarrow Li^+\,(electrolyte) + e^-$ ***OR*** $LiC_6\,(s) \rightarrow 6C\,(s) + Li^+\,(electrolyte) + e^-$ ✓ *Cathode (positive electrode):* $Li^+\,(electrolyte) + e^- + MnO_2\,(s) \rightarrow LiMnO_2\,(s)$ ***OR*** $Li^+\,(electrolyte) + e^- + NiO_2\,(s) \rightarrow LiNiO_2\,(s)$ ***OR*** $Li^+\,(electrolyte) + e^- + CoO_2\,(s) \rightarrow LiCoO_2\,(s)$ ***OR*** $2Li^+\,(electrolyte) + 2e^- + 2CoO_2\,(s) \rightarrow Co_2O_3\,(s) + Li_2O\,(s)$ ✓	*Accept "polymer" for "electrolyte".* *Award **[1 max]** if electrodes are reversed.* *Do **not** accept "CO" for "Co".*	**2**

7.1-S18 May 2018 TZ1 · Paper 2 · Q6 parts (a), (b), (c), (d), (e) · also 1.2

6.	**a**		salt bridge ✓ movement of ions ***OR*** balance charge ✓	*Do **not** accept "to complete circuit" unless ion movement is mentioned for M2.*	**2**
6.	**b**		*Positive electrode (cathode):* $Ag^+(aq) + e^- \rightarrow Ag(s)$ ✓ *Negative electrode (anode):* $Mg(s) \rightarrow Mg^{2+}(aq) + 2e^-$ ✓	*Award **[1 max]** if correct equations given at wrong electrodes.*	**2**
6.	**c**		in external wire from left to right ✓		**1**
6.	**d**		«$E = +0.80\ V - (-2.37\ V) = +$» 3.17 «V» ✓		**1**
6.	**e**		«moles of silver $= \frac{0.10\ g}{107.87\ g\ mol^{-1}}$» moles of magnesium $= \frac{0.5 \times 0.10\ «g»}{107.87\ «g\ mol^{-1}»}$ ✓ «loss in mass of magnesium $= \frac{24.31\ g\ mol \times 0.5 \times 0.10\ g}{107.87\ g\ mol^{-1}} =$ » 0.011 «g» ✓	*Award **[2]** for correct final answer.*	**2**

7.1-S19 May 2018 TZ1 · Paper 3 · Q14 parts (a)(i)

14.	**a**	**i**		*Accept any balanced equation which shows Li oxidized to Li^+ for M3, such as* $LiC_6 \rightarrow Li^+ + C_6 + e^-$ *or* $Li_xC_6 \rightarrow xLi^+ + 6C + xe^-$	**3**

7.1-S20 Nov 2017 · Paper 2 · Q7 parts (a), (b), (c), (d), (e) · also 7.2

7.	**a**		$Ni(s) + I_2(aq) \rightarrow 2I^-(aq) + Ni^{2+}(aq)$ ✓		**1**
7.	**b**		electron movement **«**in the wire**»** from Mn(s) to Ni(s) ✓ $E^\ominus$ **«**for reduction**»** of Ni^{2+} is greater/less negative than $E^\ominus$ **«**for reduction**»** of Mn^{2+} ***OR*** Ni^{2+} is stronger oxidizing agent than Mn^{2+} ***OR*** Mn is stronger reducing agent than Ni ✓		**2**
7.	**c**		**«**0.54 V – (–1.18 V) = +**»**1.72 **«**V**»** ✓	*Do **not** accept –1.72 V.*	**1**
7.	**d**		Mn **«**(s)**»** ✓		**1**

7.	**e**		*Positive electrode (anode):* $2Cl^-(aq) \rightarrow Cl_2(g) + 2e^-$ ✓ Cl^- oxidized because higher concentration ***OR*** electrode potential/*E* depends on concentration ***OR*** electrode potential values **«**of H_2O and Cl^-**»** are close ✓ *Negative electrode (cathode):* $2H_2O(l) + 2e^- \rightarrow H_2(g) + 2OH^-(aq)$ ***OR*** $2H^+(aq) + 2e^- \rightarrow H_2(g)$ ✓ H_2O/H^+ reduced because Na^+ is a weaker oxidizing agent ***OR*** Na^+ not reduced to Na in water ***OR*** H^+ easier to reduce than Na^+ ***OR*** H lower in activity series **«**than Na**»** ✓	*Accept* ⇌.	**4**

7.1-S21 May 2017 TZ1 · Paper 2 · Q3 parts (a), (b)(i), (b)(ii), (c)(i) · also 6.1, 6.2

3.	**a**		V_2O_5: +5 ✓ VO^{2+}: +4 ✓	*Do **not** penalize incorrect notation twice.*	**2**
3.	**b**	**i**	H_2SO_3 (aq) ***OR*** Pb (s) ✓		**1**
3.	**b**	**ii**	Zn (s) ✓		**1**
3.	**c**	**i**	VO^{2+} (aq) + V^{2+} (aq) + $2H^+$ (aq) → $2V^{3+}$ (aq) + H_2O (l) ✓	*Accept equilibrium sign.*	**1**

7.1-S22 May 2017 TZ1 · Paper 3 · Q22 parts (a), (b)(i), (b)(ii) · also 7.3

22.	**a**		*Anode:* CH_3OH (aq) + H_2O (l) → CO_2 (aq) + $6H^+$ (aq) + $6e^-$ ✓ *Cathode*: O_2 (aq) + $4H^+$ (aq) + $4e^-$ → $2H_2O$ (l) ✓ *Overall:* $2CH_3OH$ (aq) + $3O_2$ (g) → $2CO_2$ (aq) + $4H_2O$ (l) ✓	*Accept correctly balanced equations with multiples of the coefficients given here.* *Accept reversible or non-reversible arrows for all.*	**3**
22.	**b**	**i**	«portable» sources of electrical energy/electricity ***OR*** convert chemical «potential» energy to electrical energy/electricity ✓		**1**
22.	**b**	**ii**	primary cells involve irreversible reactions ***AND*** rechargeable cells involve reversible reactions ✓	*Accept "primary cells have a limited life before going 'flat' **AND** rechargeable cells can be recharged when 'flat'".*	**1**

7.1-S23 Nov 2016 · Paper 2 · Q4 parts (d)(ii), (i)(i), (i)(ii), (i)(iii), (j), (k)(i), (k)(ii) · also 2.1, 7.2

4.	**d**	**ii**	ionisation energy ✓		**1**

4.	**i**	**i**	*Anode (positive electrode):* $2Cl^- \rightarrow Cl_2(g) + 2e^-$ ✓ *Cathode (negative electrode):* $Mg^{2+} + 2e^- \rightarrow Mg(l)$ ✓	*Penalize missing/incorrect state symbols at Cl_2 and Mg once only.* *Award **[1 max]** if equations are at wrong electrodes.* *Accept Mg (g).*	**2**

4.	**i**	**ii**	reduction ✓		**1**
4.	**i**	**iii**	*Anode (positive electrode)*: oxygen/O_2 ***OR*** hydrogen ion/proton/H^+ ***AND*** oxygen/O_2 ✓ *Cathode (negative electrode)*: hydrogen/H_2 ***OR*** hydroxide **«**ion**»**/OH^- ***AND*** hydrogen/H_2 ✓	*Award **[1 max]** if correct products given at wrong electrodes.*	**2**
4.	**j**		*Any two of:* **«**inert**»** Pt electrode ***OR*** platinum black conductor ✓ $1\,\text{mol}\,\text{dm}^{-3}\ H^+(\text{aq})$ ✓ $H_2(\text{g})$ at 100 kPa ✓	*Accept 1 atm $H_2(g)$.* *Accept 1 bar $H_2(g)$* *Accept a labelled diagram.* *Ignore temperature if it is specified.*	**2 max**
4.	**k**	**i**	$\text{Mg}(\text{s}) + \text{Cu}^{2+}(\text{aq}) \rightarrow \text{Mg}^{2+}(\text{aq}) + \text{Cu}(\text{s})$ ✓		**1**
4.	**k**	**ii**	**«**$+0.34\,\text{V} - (-2.37\,\text{V}) = +$**»**$2.71$ **«**V**»** ✓		**1**

7.1-S24 Nov 2015 · Paper 3 · Q12 parts (a), (b), (c)

12. (a) *Negative electrode (anode):*
cadmium (metal);

Electrolyte:
(aqueous) potassium hydroxide; ***[2]***
Accept "(aqueous) sodium hydroxide" or "(aqueous) lithium hydroxide".

Names required not chemical symbols.

(b) *Negative electrode (cathode):* $Cd(OH)_2(s) + 2e^- \rightarrow Cd(s) + 2OH^-(aq)$
and
Positive electrode (anode): $Ni(OH)_2(s) + OH^-(aq) \rightarrow NiO(OH)(s) + H_2O(l) + e^-$; ***[1]***
Ignore state symbols.
Accept anode half-equation balanced with $2e^-$.
Accept e for e^-.

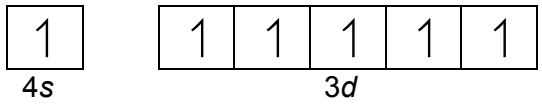
(c) *Any one for **[1 max]** from:*
both convert chemical energy to electrical energy;
Accept "both are voltaic/galvanic cells".

both involve spontaneous reactions;
both have anode acting as negative electrode / both have cathode acting as positive electrode / both have reduction taking place at positive electrode/cathode / both have oxidation taking place at negative electrode/anode;

*Any two for **[2 max]** from:*
fuel cells work non-stop while rechargeable batteries cannot work while recharging;
fuel cells have longer operating life;
fuel cells need a constant supply of reactants/fuel (producing electrical energy as long as fuel is provided to cell) while rechargeable batteries have stored chemical energy providing power until chemicals are used up;
fuel cells convert energy **and** rechargeable batteries store energy;
fuel cell products must be constantly removed (but not for rechargeable batteries);
fuel cells have inert/inactive electrodes/components while rechargeable have active/non-inert electrodes;
fuel cells run at higher temperatures (compared to rechargeable batteries);
fuel cells require pumps/cooling systems (while rechargeable batteries do not);
chemicals in rechargeable batteries are pollutants / chemicals in fuel cells are not pollutants; ***[3 max]***
Accept "fuel cells are more expensive (than rechargeable batteries)".

7.1-S25 May 2015 TZ2 · Paper 2 · Q8 parts (a)(i), (b)(i), (c)(i), (d)(i), (d)(ii), (e)(i), (e)(ii), (e)(iii), (e)(iv), (e)(v), (e)(vi) · also 1.1, 2.1, 6.1, 6.2

(a) (i)



[1]

Accept full-arrows.
Accept boxes in reverse order or at different energy levels.
Do not award the mark if sub-levels are incorrectly labelled or if no boxes are drawn.

(b) (i) chromium(III) oxide; **[1]**
Do not award the mark for chromium oxide.

(c) (i) III / +3; **[1]**
Do not cccept incorrect notation such as 3+/3.

(d) (i) *Half equation:*

$I^-(aq) + 3H_2O(l) \rightarrow IO_3^-(aq) + 6H^+(aq) + 6e^-$;

Accept e instead of e^-.
Accept reversible arrows.

Overall equation:

$Cr_2O_7^{2-}(aq) + I^-(aq) + 8H^+(aq) \rightarrow 2Cr^{3+}(aq) + IO_3^-(aq) + 4H_2O(l)$; **[2]**

Ignore state symbols.

(ii) oxidized **and** increase (in oxidation number) of 6/from –1/–I to +5/+V; **[1]**

(e) (i) potential (of reduction half-reaction) under standard conditions measured relative to standard hydrogen electrode/SHE / *OWTTE*; **[1]**
Accept "solute concentration of 1 mol dm^{-3}" or "1 bar/1 atm (pressure) for gases" instead of "standard conditions".
Accept voltage/emf for potential.

(ii) (+)1.08 (V); **[1]**

(iii) $2Cr(s) + 3Cu^{2+}(aq) \rightarrow 2Cr^{3+}(aq) + 3Cu(s)$; **[1]**
Ignore state symbols.
Do not accept reversible arrows.

(iv) *Negative electrode:* chromium/Cr **and** *Positive electrode:* copper/Cu; **[1]**
*Accept "Cr is the anode **and** Cu the cathode".*

(v) from chromium/Cr to copper/Cu; **[1]**
Accept "from negative electrode/anode to positive electrode/cathode" if electrodes correctly identified in (iv).

(vi) negative ions/anions towards the chromium(III) solution **and** positive ions/cations towards the copper(II) solution / *OWTTE*; **[1]**

7.1-S26 May 2014 TZ2 · Paper 2 · Q2 parts (b), (d)(i), (d)(ii) · also 2.1

(b) $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^1$ / $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1\ 3d^{10}$; ***[1]***
Do not accept upper case letters or numbers as subscripts.

(d) (i) $\left(E^{\ominus}_{\text{cell}} = 0.15 - 0.52 =\right) - 0.37\,(\text{V})$

choosing correct $E^{\ominus}_{\text{cell}}$ values;
combining in correct way; ***[2]***
*Award **[2]** for correct final answer.*
*Award **[1]** for -0.18 (0.34 instead of 0.15) and -0.19 (0.34 instead of 0.52).*

(ii) not spontaneous **because** $E^{\ominus}$ negative / *OWTTE*; ***[1]***

7.1-S27 May 2014 TZ2 · Paper 2 · Q5 parts (a)(ii), (b)(iv), (c)(i), (c)(ii), (d)(i), (d)(ii) · also 5.2, 6.1, 6.2, 6.3

(ii) chlorine more reactive/more powerful oxidizing agent (than bromine);
Accept opposite statements for bromine.
Accept "chloride ion a weaker reducing agent" / "bromide ion a stronger reducing agent".
Accept "chlorine more electronegative than bromine".

$Cl_2(aq) + 2NaBr(aq) \rightarrow Br_2(aq) + 2NaCl(aq)$ /
$Cl_2(aq) + 2Br^-(aq) \rightarrow Br_2(aq) + 2Cl^-(aq)$; ***[2]***
Ignore state symbols.
*Do **not** accept with equilibrium sign.*

(iv) chloric(I) acid has –OH group / hydrogen attached to a very electronegative atom;
Accept polar molecule.

can form hydrogen bonds to water;
hydrogen bonding to water increases its solubility;
(as a weak acid it is) in equilibrium with ions; ***[2 max]***

(c) (i) $ClO^-(aq) + 2H^+(aq) + 2e^- \rightleftharpoons H_2O(l) + Cl^-(aq)$;
$SO_4^{2-}(aq) + 4H^+(aq) + 2e^- \rightleftharpoons SO_2(aq) + 2H_2O(l)$;
Accept $SO_4^{2-}(aq) + 4H^+(aq) + 2e^- \rightleftharpoons H_2SO_3(aq) + H_2O(l)$.

For final equation:
$ClO^-(aq) + SO_2(aq) + H_2O(l) \rightleftharpoons SO_4^{2-}(aq) + 2H^+(aq) + Cl^-(aq)$
Accept $ClO^-(aq) + H_2SO_3(aq) \rightleftharpoons SO_4^{2-}(aq) + 2H^+(aq) + Cl^-(aq)$.
correct reactants and products;
balancing and cancelling e^-, H^+ and H_2O; ***[4]***
Apply ECF if incorrect half-equations written.

Ignore state symbols and absence of equilibrium arrow for all equations and accept inclusion of Na^+ *in any equation.*

(ii) *Award **[2]** for all correct, **[1]** for 2 or 3 correct.*

Element	Initial oxidation number	Final oxidation number
Chlorine	+I / +1;	–I / –1;
Sulfur	+IV / +4;	+VI / +6;

[2]

Remember to apply ECF from final (c) (i) equation.
*Penalise incorrect notation (eg, 4 or 4+ rather than +4) once only, so award **[1]** for a fully correct answer in an incorrect format.*

(d) (i) potential (of reduction half-reaction) under standard conditions measured relative to <u>standard</u> hydrogen electrode/SHE / *OWTTE*; ***[1]***
Allow "solute concentration of 1 mol dm^{-3}*" or "1 bar/1 atm (pressure) for gases" instead of "standard conditions".*

(ii) yes / energetically feasible;
would have a positive E_{cell} / chlorate(V) ion stronger oxidizing agent than dichromate(VI) ion / *OWTTE*; ***[2]***

7.1-S28 May 2014 TZ2 · Paper 3 · Q11 parts (a)(i), (b) · also 7.3

11. (a) (i) *Lead-acid*:

Positive electrode (cathode):

$PbO_2(s) + SO_4^{2-}(aq) + 4H^+(aq) + 2e^- \rightarrow PbSO_4(s) + 2H_2O(l)$ /

$PbO_2(s) + HSO_4^-(aq) + 3H^+(aq) + 2e^- \rightarrow PbSO_4(s) + 2H_2O(l)$;

Negative electrode (anode):

$Pb(s) + SO_4^{2-}(aq) \rightarrow PbSO_4(s) + 2e^-$ /

$Pb(s) + HSO_4^-(aq) \rightarrow PbSO_4(s) + 2e^- + H^+(aq)$; ***[2]***

Allow e for e^- throughout.
Ignore state symbols.
*Award **[1 max]** if correct equations are given but at the wrong electrodes.*

(b) *Similarity*:
(both) convert chemical energy to electrical energy / (both are) voltaic cells;

Differences:
*Award **[2 max]** for any two.*
rechargeable batteries employ reversible reactions while fuel cells have irreversible reactions;
fuel cells work non-stop while rechargeable batteries take time to recharge;
fuel cells need a constant supply of reactants/fuel while rechargeable batteries do not need any other substances;
fuel cells convert energy **and** rechargeable batteries store energy;
fuel cell products must be constantly removed but not for rechargeable batteries;
fuel cells are less polluting/more expensive/weigh less/last longer (than lead-acid rechargeable batteries);
fuel cells have inert/Pt electrodes/components while lead-acid rechargeable batteries have active/non-inert/Pb and PbO_2 electrodes;
fuel cells run at higher temperatures than rechargeable batteries;
fuel cells are less portable than rechargeable batteries / fuel cells require pumps/cooling systems while rechargeable batteries do not; ***[3 max]***
*Award **[2 max]** if three valid points (one similarity and two differences) are given without comparison and **[1 max]** if two valid points are given without comparison.*

7.1-S29 Nov 2013 · Paper 2 · Q6 parts (a), (b)(i), (c), (d)(i), (d)(ii), (d)(iii), (e)(iii), (f)(i), (f)(ii) · also 3.1, 4.2, 6.1

6. (a) potential of the half-cell / reduction half-reaction under standard conditions measured relative to standard hydrogen electrode/SHE; ***[1]***

Allow instead of standard conditions, solute concentration of 1 mol dm^{-3} or 1 bar/1 atm (pressure) for gases.

(b) (i)

Initial oxidation number		Final oxidation number	Oxidized / reduced
IV/+4	**and**	III/+3;	reduced;

[2]

+ sign must be present. Do not award mark for incorrect notation 4, 4+, 3, 3+ etc.

Do not award M2 if inconsistent with M1.

(c) *KI (aq) is added to a solution containing $Ti^{3+}(aq)$ ions:*
no reaction;

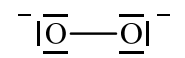
Zn (s) is added to a solution containing $TiO^{2+}(aq)$ and $H^{+}(aq)$ ions:
$Zn(s) + 2TiO^{2+}(aq) + 4H^{+}(aq) \rightarrow Zn^{2+}(aq) + 2Ti^{3+}(aq) + 2H_2O(l)$
correct reactants **and** products;
balanced equation; ***[3]***
Ignore state symbols.

(d) (i) $Fe^{3+}(aq) + e^{-} \rightarrow Fe^{2+}(aq)$;
Ignore state symbols.
Accept equilibrium arrow.

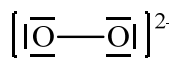
reduction; ***[2]***
Do not apply ECF.

(ii) $(+0.77 - (-0.06)) = (+)0.83(V)$; ***[1]***
Do not accept –0.83 V.

(iii) wire and salt bridge **both** have arrows from B to A; ***[1]***
Accept arrows above or below each provided it is obvious which they refer to.
Apply ECF from part (i).

(iii) 

e-pairs correct;
charges in correct positions; ***[2]***
Accept lines, or pairs of dots or crosses, for electron pairs.

Accept  .

(f) (i) C is sp hybridized **and** Si is sp^3 hybridized;
C–O bond in CO_2 has one σ-bond and one π-bond;
Si–O bond in SiO_2 has one σ-bond only; ***[3]***
*Award **[1 max]** for last two marking points for "C–O double bond and Si–O single bond".*

(ii) silicon-oxygen bonds will have a tetrahedral distribution;
xenon-oxygen bonds will have a square planar distribution;
xenon dioxide has **two** non-bonding/lone pairs of electrons; ***[3]***
Award any of the above marks if clearly indicated in suitable diagrams.

7.1-S30 May 2012 TZ1 · Paper 2 · Q4 parts (a), (b), (c)(i), (c)(ii), (d)(i), (d)(iv) · also 7.2

4. (a) $Mg(s) \rightarrow Mg^{2+}(aq) + 2e^-$;
Fe/iron; ***[2]***
Do not accept Fe/Fe^{2+} half-equation or Fe^{2+}.

(b) $Mg(s) + Fe^{2+}(aq) \rightarrow Mg^{2+}(aq) + Fe(s)$; ***[1]***

(c) (i) (potential of reduction half-reaction) under standard conditions measured relative to standard hydrogen electrode/SHE / difference of standard reduction potential of substance undergoing reduction and standard reduction potential of substance undergoing oxidation / *OWTTE*; ***[1]***

(ii) (+)1.92 (V); ***[1]***

(d) (i) ***S:*** $2H_2O(l) + 2e^- \rightarrow H_2(g) + 2OH^-(aq)$ / $2H^+(aq) + 2e^- \rightarrow H_2(g)$;
T: $2Cl^-(aq) \rightarrow Cl_2(g) + 2e^-$; ***[2]***

(iv) oxygen;

$2H_2O \rightarrow O_2 + 4H^+ + 4e^- \ / \ 4OH^- \rightarrow 2H_2O + O_2 + 4e^-$; ***[2]***

SECTION B

7.1-S31 May 2012 TZ2 · Paper 3 · QC2 parts (b), (c) · also 7.3

(b) *(+) electrode (cathode):* $PbO_2 + 4H^+ + SO_4^{2-} + 2e^- \rightarrow PbSO_4 + 2H_2O$;

(–) electrode (anode): $Pb + SO_4^{2-} \rightarrow PbSO_4 + 2e^-$; ***[2]***

(c) fuel cells produce only water / Cd and Ni are toxic (heavy metals);
fuel cells are more expensive;
fuel cells can operate continuously/do not need recharging;
fuel cells are more unwieldy/less portable/less self-contained/need supply of O_2 and H_2; ***[2 max]***
Accept opposite statements for NiCd cells.

7.1-S32 Nov 2011 · Paper 2 · Q7 parts (a), (b), (c)(i), (c)(ii), (d)(i), (d)(ii), (d)(iii), (e)(iv), (f) · also 1.1, 2.1, 6.1, 6.2, 7.2

7. (a) *Oxidation:* increase in oxidation number **and** *Reduction:* decrease in oxidation number / *OWTTE*; ***[1]***

(b) Cr_2O_3:
chromium(III) oxide;
Do not award mark for chromium oxide.

CrO_3:
chromium(VI) oxide; ***[2]***
Do not award mark for chromium oxide.
Do not award any marks if chromium oxide without Roman numerals is given for both.

(c) (i) substance reduced / causes other substance to be oxidized / increase oxidation number of another species / gains electrons / *OWTTE*; ***[1]***

(ii) *Oxidizing agent:*
$Cr_2O_7^{2-}$ / dichromate (ion);

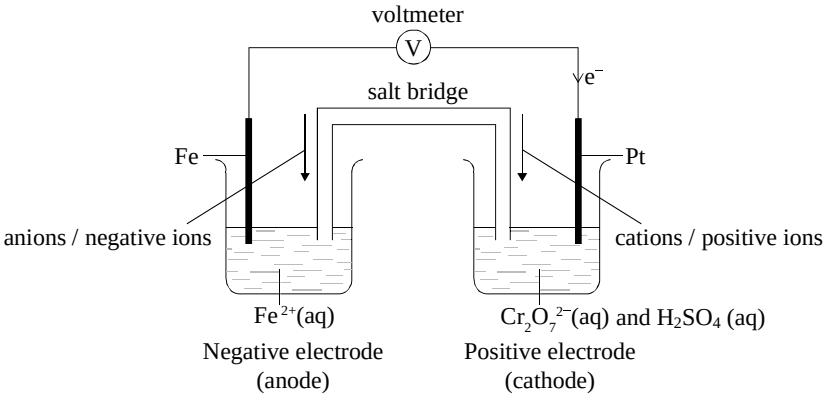
$Cr_2O_7^{2-}(aq) + I^-(aq) + 8H^+(aq) \rightarrow 2Cr^{3+}(aq) + IO_3^-(aq) + 4H_2O(l)$ ***[3]***

*Award **[1]** for coefficients: $Cr_2O_7^{2-}(aq)$, $I^-(aq)$, $2Cr^{3+}(aq)$, $IO_3^-(aq)$.*

*Award **[1]** for coefficients: $8H^+(aq)$, $4H_2O(l)$.*

*Award **[1 max]** if coefficients of reactants only correct i.e. $Cr_2O_7^{2-}$, I^- and $8H^+$.*
*Award **[1 max]** if coefficients of products only correct i.e. $2Cr^{3+}$, IO_3^- and $4H_2O$.*
*Award **[1 max]** for correct reactants and products.*
Ignore state symbols.

(d) (i)



Voltaic cell showing:
labelled positive electrode (cathode) **and** negative electrode (anode);
direction of electrons in external circuit **and** direction of ions in salt bridge;
Award mark if correct direction of electrons is indicated but e^- not labelled in external circuit.
Allow e instead of e^-.
Cations/positive ions and anions/negative ions must be identified in salt bridge.
Allow correct movement of ions in electrolyte instead of movement of ions in salt bridge (e.g. Fe^{2+} from Fe at negative electrode/anode etc.).
If both movement of ions in salt bridge and movement of ions in electrolyte is given but one is incorrect do not award mark.

Positive electrode (cathode):
$Cr_2O_7^{2-}(aq) + 14H^+(aq) + 6e^- \rightarrow 2Cr^{3+}(aq) + 7H_2O(l)$;

Negative electrode (anode):
$Fe(s) \rightarrow Fe^{2+}(aq) + 2e^-$;

Penalize $\rightleftharpoons$ once only.
Penalize once only if electrodes or equations reversed.
For both electrodes allow e instead of e^-.

Overall cell reaction:
$Cr_2O_7^{2-}(aq) + 3Fe(s) + 14H^+(aq) \rightarrow 2Cr^{3+}(aq) + 3Fe^{2+}(aq) + 7H_2O(l)$; ***[5]***
Ignore state symbols throughout (d) (i).

(ii) potential under standard conditions relative to standard hydrogen electrode/SHE; ***[1]***
Reference must be made to standard conditions.
Instead of standard conditions allow either solute concentration of 1 mol dm^{-3}/1 M/1 mol L^{-1} or 100 kPa/10^5 Pa for gases.
Allow 1 bar for 100 kPa/10^5 Pa.
Allow 1 atm/1.01×10^5 Pa.
Allow voltage instead of potential.

(iv)

[1]

Accept half-arrows or full arrows and boxes in reverse order.
Do not penalize if additional sub-levels are shown, if sub-levels are not labelled or if no boxes are drawn (providing system of arrows correct).
Do not award mark if sub-levels are incorrectly labelled.
Orbital diagram may also be represented with sub-levels shown at different relative energy positions.

(f) *Positive electrode (anode):* chromium;
Allow lead/titanium/platinum/graphite.

Negative electrode (cathode): object to be plated;
Allow specific example here e.g. spoon.

Electrolyte: $Cr^{3+}(aq)$; ***[3]***

Allow (mixture of) $Cr^{3+}(aq)$ ***and*** *CrO_4^{2-}(aq)/chromate/chromic acid/H_2CrO_4.*
Ignore state symbols.
Allow any soluble salt of Cr^{3+}.

7.1-S33 May 2011 TZ2 · Paper 2 · Q7 parts (a)(i), (a)(ii), (b)(i), (b)(ii), (b)(iii), (c)(i), (c)(ii), (d)(ii), (d)(iii), (d)(iv), (d)(v), (e)(ii) · also 6.1, 6.2, 7.2

(a) (i) the potential difference/voltage obtained when a half-cell is connected to a standard hydrogen electrode;
under standard conditions / 1.00 mol dm^{-3} solutions, 298 K; ***[2]***

(ii) the electrons flow from the half-cell to the standard hydrogen electrode / the half-cell forms the negative electrode when connected to the standard half-cell / Fe is a better reducing agent than H_2 / Fe is above H_2 in electrochemical series; ***[1]***
Accept "the half reaction is not spontaneous".

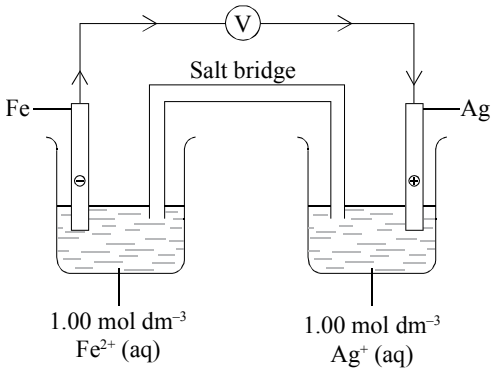
(b) (i) bromine/ Br_2; ***[1]***

(ii) hydrogen/ H_2; ***[1]***

(iii) iron/Fe; ***[1]***

Ignore coefficients for Br_2, H_2 or Fe.

(c) (i)



correct diagram including voltmeter;
No credit if wires to electrodes immersed in the solutions.

labelled salt bridge;
Do not accept name of salt (e.g. potassium nitrate) in place of salt bridge.

correctly labelled (+) and (–) electrodes / cathode and anode;
1 or 1.00 mol dm^{-3} concentrations/298 K;
flow of electrons from Fe to Ag in external circuit; ***[5]***
*Award **[2 max]** if battery shown instead of voltmeter.*

(ii) $Fe + 2Ag^+ \rightarrow Fe^{2+} + 2Ag$;
$E^{\ominus}_{cell}$ (= 0.80 – (– 0.45) =) 1.25 V; ***[2]***
Ignore state symbols.

(ii) +6; ***[1]***

Do not accept 6, 6+ or the use of Roman numerals unless they have already been penalized in (2)(a).

(iii) $Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$; ***[1]***

(iv) $CH_3CH_2OH \rightarrow CH_3CHO + 2H^+ + 2e^-$;

$Cr_2O_7^{2-} + 3CH_3CH_2OH + 8H^+ \rightarrow 2Cr^{3+} + 3CH_3CHO + 7H_2O$ ***[3]***

*For second equation award **[1]** for correct reactants and products and **[1]** for correct balancing.*

(v) H^+ is a reactant / *OWTTE*; ***[1]***

(ii) H^+ ions gain electrons more readily than Na^+ / hydrogen is evolved instead;
hydrogen is below Na in ECS;

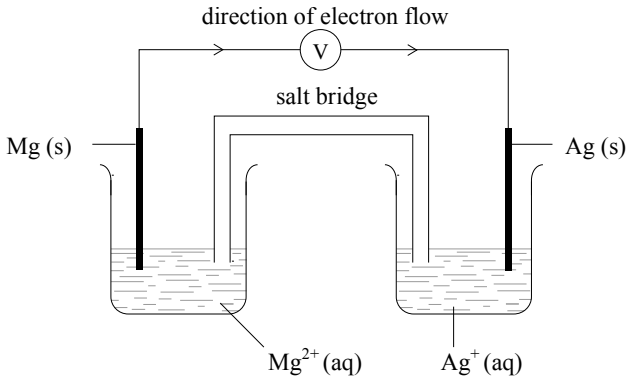
if sodium were to be formed it would react with the water in the solution / *OWTTE*; ***[2 max]***

7.1-S34 Nov 2010 · Paper 3 · QC4 parts (b)

(b) lithium-ion; ***[1]***

7.1-S35 May 2010 TZ1 · Paper 2 · Q6 parts (a)(i), (a)(ii), (a)(iii), (b)(i), (b)(ii), (c), (d)(i), (d)(ii) · also 7.2

6. (a) (i)



correctly labelled electrodes **and** solutions;
labelled salt bridge;
voltmeter;
Allow bulb or ammeter.

direction of electron flow; ***[4]***

(ii) *Oxidation*:
$Mg(s) \rightarrow Mg^{2+}(aq) + 2e^{-}$;

Reduction:
$Ag^{+}(aq) + e^{-} \rightarrow Ag(s)$; ***[2]***

Ignore state symbols.
*Award **[1 max]** if equations not labelled reduction or oxidation or labelled the wrong way round.*
Allow e instead of e^{-}.
Penalize equilibrium sign or reversible arrows once only in parts (a) (ii) and (d) (ii).

(iii) +0.80 – (–2.37) = 3.17 V
correct data;
answer with unit; ***[2]***
*Award **[1]** for –3.17 V or correct working of wrong values.*

(b) (i) Cd/Cd(s); ***[1]***
Do not allow Cd^{2+}.

(ii) $5Cd(s) + 2MnO_4^{-}(aq) + 16H^{+}(aq) \rightarrow 5Cd^{2+}(aq) + 2Mn^{2+}(aq) + 8H_2O(l)$
correct reactants and products;
correct balancing of this equation; ***[2]***
Ignore state symbols.

(c) *Accept suitable diagram with the following indicated:*
Pt electrode;
$[H^+(aq)] = 1$ mol dm^{-3}/0.5 mol dm^{-3} H_2SO_4;

H_2 gas;
at 1 atm / 1.01×10^5 Pa ;
Do not award mark for pressure if no hydrogen gas given.

298 K / 25 °C ; ***[4 max]***

(d) (i) sodium chloride crystals consist of ions in a (rigid) lattice / ions cannot move (to electrodes) / *OWTTE*;
when melted ions free to move / ions move when potential difference/voltage applied; ***[2]***

(ii) positive sodium ions/Na^+ move to the negative electrode/cathode **and** negative chloride ions/Cl^- move to the positive electrode/anode;

electrons are released to positive electrode/anode by negative ions and accepted from negative electrode/cathode by positive ions / reduction occurs at the negative electrode/cathode **and** oxidation occurs at the positive electrode/anode / Na^+ ions are reduced **and** Cl^- ions are oxidized;

(Positive electrode/anode):
$2Cl^- \rightarrow Cl_2 + 2e^-$ / $Cl^- \rightarrow \frac{1}{2}Cl_2 + e^-$;

(Negative electrode/cathode)
$2Na^+ + 2e^- \rightarrow 2Na$ / $Na^+ + e^- \rightarrow Na$; ***[4]***

*Award **[1 max]** if equations not labelled or labelled wrong way round.*
Allow e instead of e^-.
Penalize equilibrium sign or reversible arrows once only in parts (a) (ii) and (d) (ii).

7.1-S36 May 2010 TZ2 · Paper 2 · Q8 parts (a)(i), (a)(ii), (a)(iii), (a)(iv), (b)(ii), (c) · also 6.1, 6.3, 7.2

8. (a) (i) $Pb: 0, PbO_2: +4, PbSO_4: +2$; ***[1]***

Need sign for mark.
Do not accept notations such as 4+, 2+, or IV, II.

(ii) *Negative/–/anode*

$Pb(s) + SO_4^{2-}(aq) \rightarrow PbSO_4(s) + 2e^- / Pb(s) \rightarrow Pb^{2+}(aq) + 2e^-$;

Positive/+/cathode

$PbO_2(s) + 4H^+(aq) + SO_4^{2-}(aq) + 2e^- \rightarrow PbSO_4(s) + 2H_2O(l)$ /

$PbO_2(s) + 4H^+(aq) + 2e^- \rightarrow Pb^{2+}(aq) + 2H_2O(l)$ /

$PbO_2(s) + H_2SO_4(aq) + 2H^+(aq) + 2e^- \rightarrow PbSO_4(s) + 2H_2O(l)$;

Accept $Pb^{4+} + 2e^- \rightarrow Pb^{2+}$.
Ignore state symbols.
Allow e instead of e^-.

oxidizing agent is PbO_2/lead(IV) oxide/lead dioxide **and** reducing agent is Pb/lead;
from negative/–/anode/Pb to positive/+/cathode/PbO_2 (through the external circuit/wire); ***[4]***

(iii) $Pb(s) + Cu^{2+}(aq) \rightarrow Pb^{2+}(aq) + Cu(s)$

$Pb(s) + 2Ag^+(aq) \rightarrow Pb^{2+}(aq) + 2Ag(s)$

$Cu(s) + 2Ag^+(aq) \rightarrow Cu^{2+}(aq) + 2Ag(s)$

Award ***[2]*** *for three correct, award* ***[1]*** *for any two correct, one correct scores no mark.*
Ignore state symbols.
Penalize unbalanced equations once only.

Pb is a stronger reducing agent than Cu and/or Ag / Pb most reactive as it can reduce/displace both Cu^{2+} and Ag^+;
Cu is a stronger reducing agent than Ag but not Pb / Cu in the middle (of the three) as it can reduce/displace Ag^+ but not Pb^{2+};
Accept converse argument.

Decreasing order: Pb, Cu, Ag / Pb > Cu > Ag; ***[5]***
Do not accept Pb^{2+}, Cu^{2+}, Ag^+.

(iv) MnO_4^-;

$2MnO_4^-(aq) + 16H^+(aq) + 10Cl^-(aq) \rightarrow 2Mn^{2+}(aq) + 8H_2O(l) + 5Cl_2(g)$

Accept equation with all coefficients divided by 2 (i.e.
$MnO_4^- + 8H^+ + 5Cl^- \rightarrow Mn^{2+} + 4H_2O + 2\frac{1}{2}Cl_2$*).*
Award ***[1]*** *for correct reactants and products,* ***[1]*** *for correct balancing.*
Ignore state symbols.

$E^{\ominus}_{cell} = (1.51 - 1.36) = (+)\,0.15\,(V)$; ***[4]***

(ii) (choice of Cl^- or H_2O / OH^- to be oxidized), Cl^- oxidized because of concentrated solution/higher concentration / *OWTTE*;
(choice of Na^+ or H_2O / H^+ to be reduced), H_2O / H^+ reduced because Na^+ is a (much) weaker oxidizing agent/ Na^+ not reduced to Na in water / H^+ easier to reduce than Na^+ / *OWTTE*;

positive/+/anode
$2Cl^-(aq) \rightarrow Cl_2(g) + 2e^-$;

negative/–/cathode
$2H_2O(l) + 2e^- \rightarrow H_2(g) + 2OH^-(aq) / 2H^+(aq) + 2e^- \rightarrow H_2(g)$; ***[4]***

Penalize missing or incorrect states once only.
Award only ***[1]*** *out of the last two marks if electrodes not specified or if equations switched.*
Allow e instead of e^-.

(c) *Positive/+/anode*
$2H_2O(l) \rightarrow O_2(g) + 4H^+(aq) + 4e^- / H_2O(l) \rightarrow \frac{1}{2}O_2(g) + 2H^+(aq) + 2e^-$ /
$4OH^-(aq) \rightarrow 2H_2O(l) + O_2(g) + 4e^-$;

Negative/–/cathode
$Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$;
Ignore state symbols.
Award only ***[1]*** *if electrodes not specified or if equations switched.*
Allow e instead of e^-.

Observations: ***[2 max]***
blue colour of $Cu^{2+}(aq)$ fades;
Cu/metal deposited on negative/–/cathode/tin (jewellery);
gas produced/bubbles formed (at positive/+/anode);
pH of solution decreases/acidity increases (observed with indicator/pH paper); ***[4 max]***

LESSON 7.2

Electrolytic Cells

Answers for 29 multiple-choice and 29 structured questions

Part A - Multiple choice answers

7.2-M1	May 2025 TZ2 · P1A · Q39	B
7.2-M2	May 2025 TZ3 · P1A · Q37	A
7.2-M3	May 2024 TZ1 · P1 · Q30	C
7.2-M4	Nov 2023 TZ1 · P1 · Q30	B
7.2-M5	May 2023 TZ1 · P1 · Q29	B
7.2-M6	May 2023 TZ2 · P1 · Q31	A
7.2-M7	May 2022 TZ1 · P1 · Q30	A
7.2-M8	Nov 2021 · P1 · Q31	A
7.2-M9	May 2021 TZ1 · P1 · Q31	C
7.2-M10	May 2021 TZ2 · P1 · Q29	B
7.2-M11	May 2021 TZ2 · P1 · Q31	C
7.2-M12	Nov 2020 · P1 · Q31	C
7.2-M13	Nov 2019 · P1 · Q31	D
7.2-M14	May 2019 TZ1 · P1 · Q28	C
7.2-M15	May 2019 TZ2 · P1 · Q31	D
7.2-M16	May 2018 TZ1 · P1 · Q29	A
7.2-M17	May 2018 TZ1 · P1 · Q30	B
7.2-M18	May 2018 TZ2 · P1 · Q29	A
7.2-M19	May 2018 TZ2 · P1 · Q31	B
7.2-M20	Nov 2017 · P1 · Q31	C
7.2-M21	Nov 2016 · P1 · Q33	A
7.2-M22	May 2016 · P1 · Q32	C
7.2-M23	Nov 2015 · P1 · Q33	B
7.2-M24	Nov 2014 · P1 · Q31	D
7.2-M25	May 2014 TZ1 · P1 · Q32	D
7.2-M26	Nov 2013 · P1 · Q34	D
7.2-M27	May 2013 TZ2 · P1 · Q32	A
7.2-M28	May 2012 TZ1 · P1 · Q33	D
7.2-M29	May 2010 TZ1 · P1 · Q32	C

Part B - Structured questions markscheme

7.2-S1 Nov 2024 · Paper 2 · Q3 parts (b)(i), (b)(ii), (d), (e)(i) · also 2.1, 9.2

3.	**(b)**	**(i)**	*Anode (positive electrode):* $H_2O(l) \rightarrow ½\ O_2 + 2H^+(aq) + 2e^-$ ***OR*** $2OH^-(aq) \rightarrow ½\ O_2 + H_2O(l) + 2e^-$ ✓ *Cathode (negative electrode):* $H_2O(l) + e^- \rightarrow ½\ H_2(g) + OH^-(aq)$ ***OR*** $H^+(aq) + e^- \rightarrow ½\ H_2(g)$ ✓	*Award **[1 max]** for correct equations at wrong electrode.* *Do **not** award a mark for reactions producing chlorine.* *Ignore reversible arrows.*	**2**
3.	**(b)**	**(ii)**	oxygen/O_2 ✓		**1**
3.	**(d)**		successive ionization energies ✓ 3 large increases in **«**ionization**»** energy **«**after first, ninth and seventeenth electron removed**»** ✓	*Award **[2 max]** for a labelled diagram which shows the trend in successive ionization energies.*	**2**

3.	**(e)**	**(i)**	***Alternative 1:*** **«**$\Delta H_f(H_2O(l))$ **=»** −285.8 **«**kJ mol^{-1}**»** ✓ **«**$\Delta H_{reaction}$ = $\Sigma\Delta H_f$(products) – $\Sigma\Delta H_f$(reactants)**»** **«**$\Delta H_{reaction}$ = (2(−481.8) +0) – (0 + 2(−285.8))**»** **«**$\Delta H_{reaction}$ **=»** −392.0 **«**kJ**»** ✓ **«**ΔH **=»** −196.0 **«**kJ mol^{-1}**»** ✓ ***Alternative 2:*** **«**$\Delta H_f(H_2O(l))$ **=»** −285.8 **«**kJ mol^{-1}**»** ✓ $K(s) + H_2O(l) \rightarrow KOH(aq) + \frac{1}{2}H_2(g)$ ***OR*** **«**$\Delta H_{reaction}$ = $\Sigma\Delta H_f$(products) – $\Sigma\Delta H_f$(reactants)**»** **«**$\Delta H_{reaction}$ **=»** (−481.8) – (−285.8) ✓ **«**$\Delta H_{reaction}$ **=»** −196.0 **«**kJ**»** ✓	*Award **[3]** for correct final answer.* *M1 may be awarded from working.*	**3**

7.2-S2 May 2024 TZ1 · Paper 3 · Q3 parts (c)(ii)

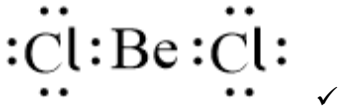
3.	**c**	**ii**	$Cs^+(l) + e^- \rightarrow Cs(l)$ ✓		**1**

7.2-S3 May 2024 TZ2 · Paper 2 · Q7 parts (c), (d), (e), (f), (h) · also 6.3, 7.1, 9.4

7.	**(c)**		*Positive electrode (anode)*: $2Br^- \rightarrow Br_2$ (g) $+ 2e^-$ ✓ *Negative electrode (cathode)*: $Na^+ + e^- \rightarrow Na$ (l) ✓	*Award **[1 max]** if equations are given for wrong electrodes*.	**2**
7.	**(d)**		*Positive electrode (anode):* bromine/Br_2 ✓ *Negative electrode (cathode):* hydrogen gas/H_2 ✓	*Award **[1max] f**or correct products at inverted electrodes.*	**2**
7.	**(e)**		*anode:* Mg ***AND*** *Electrolyte:* Mg^{2+} **«**(aq)**»** ✓ cathode: Zn ***AND*** *Electrolyte:* Zn^{2+} **«**(aq)**»** ✓	*Accept specific salts as electrolytes*.	**2**
7.	**(f)**		**«**½ $\Delta H^{\ominus}_{lBr\text{-}Br}$ = ½ (193 kJ mol^{-1}) =**»** 96.5 **«**kJ mol^{-1}**»** ***OR*** **«**½ $\Delta H^{\ominus}_{atom,Br}$= ½ (30 kJ mol^{-1}) =**»** 15 **«**kJ mol^{-1}**»** ✓ **«**$\Delta H^{\ominus}_{IE,Na}$=**»** 496 **«**kJ mol^{-1}**»** ***AND*** **«=** $\Delta H^{\ominus}_{AE}Br$ =**»** –325 «kJ mol^{-1}**»** ✓ **«**$\Delta H^{\ominus}_{lattice=}$ $\Delta H^{\ominus}_{atom,}$(Na) + $\Delta H^{\ominus}_{atom}$($Br_2$)+ ΔH^{o}(Br-Br) + $\Delta H^{\ominus}_{IE}$ (Na) +$\Delta H^{\ominus}_{EA}$(Br) + $\Delta H^{\ominus}_{formation}$**»** **«**107 + 96.5 + 15 + 496 –325 + 361.5 = **»** 751 **«**kJ mol^{-1}**»** ✓	*Award **[3]** for correct final answer.* *Do **not** accept +754 **«**kJ mol^{-1}**»** the value given in the data booklet.*	**3**

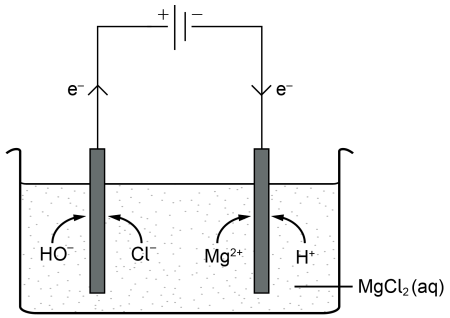
7.	**(h)**		*Iodine:* *n*o reaction ***AND*** I_2/iodine less reactive than Br_2 / bromine ✓ *Chlorine:* produces Br_2 / NaCl ***AND*** Cl_2 / chlorine more reactive ✓		**2**

7.2-S4 Nov 2023 TZ1 · Paper 2 · Q5 parts (b)(i), (b)(ii), (b)(iii), (c)(i), (c)(ii), (d)(i), (f)(i), (f)(ii), (g), (h) · also 2.1, 3.1, 4.2, 7.1

5	**b**	**i**	**«**contains**»** mobile/free moving ions ✓	*Accept has ions that can carry an «electric» current/charge.*	**1**
5	**b**	**ii**	*Electrode:* cathode ***AND*** *Polarity:* negative ✓		**1**
5	**b**	**iii**	$2\ Cl^- \rightarrow Cl_2 + 2\ e^-$ ✓	*Accept* $Cl^- \rightarrow {}^1/_2\ Cl_2 + e^-$. *Accept e for* e^-. *Do **not** apply ECF.*	**1**
5	**c**	**i**	 ✓		**1**
5	**c**	**ii**	**«**Be**»** does not have complete valence shell ✓	*Accept incomplete octet / exception to octet rule / electron deficient.*	**1**
5	**d**	**i**	sp^2 ✓	*Accept the "2" as a subscript or normal character.*	**1**

5	**f**	**i**	E^o **«**= -0.83 – (-1.85)**»** = +1.02 **«**V**»** ✓		**1**
5	**f**	**ii**	spontaneous ***AND*** E^o positive/>0 ✓	*Accept spontaneous **AND** ∆G negative.*	**1**
5	**g**		nuclear charge / number of protons increases **«**for both**»** ✓ Li and Be **«**outer electrons have**»** same subshell/shielding ✓ electron in B lost from p-subshell whereas that in Be lost from s-subshell ✓ **«**outer electron in**»** B/p-subshell experiences greater shielding / has higher energy ✓	*Do **not** accept explanations invoking distance of electrons from nucleus.*	**4**
5	**h**		frequency /wavelength of **«**the radiation at**»** convergence limit **«**is proportional to the ionization energy**»** ✓	*Accept highest frequency/shortest wavelength.*	**1**

7.2-S5 May 2023 TZ2 · Paper 2 · Q3 parts (a)(i), (a)(ii), (a)(iii), (b)(i) · also 1.2, 4.3

3.	**(a)**	**(i)**	 electron flow from anode to battery ***OR*** from battery to cathode ✓ Mg^{2+}/H^+ ions to – electrode ***AND*** Cl^-/OH^- ions to + electrode ✓	*Do not award M1 if electrons are shown in electrolyte.*	**2**
3.	**(a)**	**(ii)**	*Positive electrode:* $H_2O(l) \rightarrow ½\ O_2(g) + 2H^+(aq) + 2e^-$ ***OR*** $2OH^-(aq) \rightarrow ½O_2(g) + H_2O(l) + 2e^-$ ✓ *Negative electrode*: $2H_2O(l) + 2e^- \rightarrow H_2(g) + 2OH^-(aq)$ ***OR*** $2H^+(aq) + 2e^- \rightarrow H_2(g)$ ✓	*Accept $2Cl^-(aq) \rightarrow Cl_2(g) + 2e^-$ for M1.* *Award **[1max]** for correct equations at wrong electrodes.*	**2**

3.	**(a)**	**(iii)**	layers «of carbon atoms in a giant structure» ✓ delocalized electrons «flow along layers» ✓	*Accept suitable diagram for M1.* *Accept two-dimensional network for M1.* *Accept electrons are mobile/flow for M2.*	**2**
3.	**(b)**	**(i)**	«n($S_2O_3^{2-}$) = 0.00500 mol dm^{-3} x 0.0360 dm^3 =» 0.000180 / 1.80 x 10^{-4} «mol» ✓ «n(O_2) =» $\frac{n(S_2O_3^{2-})}{4}$ / 0.0000450 / 4.50 x 10^{-5} «mol» ✓ «[O_2] = $\frac{4.50x10^{-5}\ mol}{0.150\ dm^3}$ =» 0.000300 / 3.00 x 10^{-4} «mol dm^{-3}» ✓	*Award **[3]** for correct final answer.*	**3**

7.2-S6 May 2023 TZ2 · Paper 3 · Q3 parts (b)(i), (b)(iii), (d) · also 4.3, 7.3

3.	**(b)**	**(i)**	Al^{3+} + 3 e^- → Al(l) ✓	*Do **not** penalize if equilibrium arrow used.*	**1**
3.	**(b)**	**(iii)**	high energy consumption «that has environmental implications» ***OR*** large amounts of waste «produced by mining and purification of the ore» ***OR*** mining has negative impact on landscape ***OR*** greenhouse gas/pollution from transport/machinery ✓	*Accept "fluorine/fluorine compounds produced" or their formulas.* ***OR*** *"carbon dioxide/carbon monoxide released".*	**1**
3.	**(d)**		«held together by strong» covalent bonds «and defect free/regular 2D/3D» ✓		**1**

7.2-S7 Nov 2019 · Paper 2 · Q6 parts (a)(i), (a)(ii), (b), (c)(i), (c)(ii), (c)(iii), (c)(iv), (e)(i), (e)(ii) · also 1.2, 2.1, 7.1, 9.2

6.	**a**	**i**	[Ar] $3d^{10}$ ***OR*** $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}$ ✓		**1**
6.	**a**	**ii**	$\Delta H^{\ominus} = \sum \Delta H^{\ominus}_f$ (products) – $\sum \Delta H^{\ominus}_f$ (reactants) ✓ $\Delta H^{\ominus} = 2(-241.8$ **«**kJ mol^{-1}**»**$) - 4(-92.3$ **«**kJ mol^{-1}**»**$) = -114.4$ **«**kJ**»** ✓	*Award **[2]** for correct final answer.*	**2**

6.	**b**		mass of H_2O = «18.360 g – 17.917 g =» 0.443 «g» ***AND*** mass of $CuCl_2$ = «17.917 g – 16.221 g =» 1.696 «g» ✓ moles of $H_2O = \text{«}\frac{0.443\,\text{g}}{18.02\,\text{g mol}^{-1}}\text{=»}$ 0.0246 «mol» ***OR*** moles of $CuCl_2 = \text{«}\frac{1.696\,\text{g}}{134.45\,\text{g mol}^{-1}}\text{= »}$ 0.0126 «mol» ✓ «water : copper(II) chloride = 1.95 : 1» «x =» 2 ✓	*Award **[3]** for correct final answer.* *Accept «x =» 1.95.*	**3**
6.	**c**	**i**	*Wires:* «delocalized» electrons «flow» ✓ *Electrolyte:* «mobile» ions «flow» ✓		**2**
6.	**c**	**ii**	$2Cl^- \rightarrow Cl_2\,(g) + 2e^-$ ***OR*** $Cl^- \rightarrow \frac{1}{2}Cl_2\,(g) + e^-$ ✓	*Accept e for* e^-.	**1**
6.	**c**	**iii**	«electrode» 3 ***AND*** oxygen/O_2 ✓	*Accept chlorine/Cl_2.*	**1**
6.	**c**	**iv**	$2H_2O\,(l) \rightarrow 4H^+\,(aq) + O_2\,(g) + 4e^-$ ✓	*Accept $2Cl^-\,(aq) \rightarrow Cl_2\,(g) + 2e^-$.* *Accept $4OH^- \rightarrow 2H_2O + O_2 + 4e^-$*	**1**

6.	**e**	**i**	$E^\ominus$ = «+0.52 – 0.15 = +» 0.37 «V» ✓		**1**
6.	**e**	**ii**	spontaneous ***AND*** $E^\ominus$ positive ✓		**1**

7.2-S8 Nov 2019 · Paper 3 · Q6 parts (b)(ii)

6.	**b**	**ii**	*Anode (positive electrode):* $2O^{2-} \rightarrow O_2\,(g) + 4e^-$ ***OR*** $2O^{2-} + C \rightarrow CO_2\,(g) + 4e^-$ ✓ *Cathode (negative electrode):* $Al^{3+} + 3e^- \rightarrow Al\,(l)$ ✓ O_2 gas ***AND*** Al liquid ✓	*Award **[1 max]** for M1 and M2 if correct half-equations are given at the wrong electrodes **OR** if incorrect reversed half-equations are given at the correct electrodes.* *Only state symbols of **products** required, which might be written as (g) and (l) in half-equations. Ignore any incorrect or missing state symbols for reactants.*	**3**

7.2-S9 May 2019 TZ1 · Paper 2 · Q7

7.			*Cathode (negative electrode):* $Ag^+\,(aq) + e^- \rightarrow Ag\,(s)$ ✓ *Anode (positive electrode):* $2H_2O\,(l) \rightarrow O_2\,(g) + 4H^+\,(aq) + 4e^-$ ✓	*Accept $4OH^-\,(aq) \rightarrow O_2\,(g) + 2H_2O\,(l) + 4e^-$.* *Accept multiple or fractional coefficients in both half-equations.*	**2**

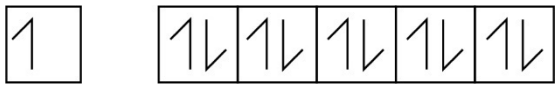
7.2-S10 May 2019 TZ2 · Paper 2 · Q4 parts (b)(ii), (c), (d)(i), (e)(ii), (e)(iii) · also 1.1, 6.2, 6.3, 7.1

4.	**b**	**ii**	electrolyze **«**a solution of /molten**»** rhenium salt/Re^{n+} ✓ graphite as cathode/negative electrode ***OR*** rhenium forms at cathode/negative electrode ✓	*Accept "using rhenium anode" for M1.*	**2**
4.	**c**		place **«**pieces of**»** Re into each solution ✓ if Re reacts/is coated with metal, that metal is less reactive **«**than Re**»** ✓	*Accept other valid observations such as "colour of solution fades" or "solid/metal appears" for "reacts".*	**2**
4.	**d**	**i**	rhenium(III) chloride ***OR*** rhenium trichloride ✓		**1**
4.	**e**	**ii**	$ReO_4^-(aq) + 6H^+(aq) + 3e^- \rightleftharpoons [Re(OH)_2]^{2+}(aq) + 2H_2O(l)$ ✓		**1**
4.	**e**	**iii**	no ***AND*** ReO_4^- is a weaker oxidizing agent than Fe^{3+} ***OR*** no ***AND*** Fe^{3+} is a stronger oxidizing agent than ReO_4^- ***OR*** no ***AND*** Fe^{2+} is a weaker reducing agent than $[Re(OH)_2]^{2+}$ ***OR*** no ***AND*** $[Re(OH)_2]^{2+}$ is a stronger reducing agent than Fe^{2+} ***OR*** no ***AND*** cell emf would be negative/−0.41 V ✓		**1**

7.2-S11 Nov 2018 · Paper 3 · Q4 parts (c)

4.	**c**		«average» oxidation state of C in C_6H_{12}/cyclohexane = –2 ***AND*** in CNTs = 0 ***OR*** oxidation state of C in CNTs is higher than in C_6H_{12}/cyclohexane ***OR*** loss of H's/hydrogens ✓ «oxidation at» positive/+ «electrode»/anode ✓	*Accept "oxidation number" for "oxidation state".*	**2**

7.2-S12 May 2018 TZ2 · Paper 2 · Q3 parts (a)(iii), (a)(iv), (c)(i), (c)(ii), (c)(iii), (c)(iv), (c)(v) · also 2.1

3.	**a**	**iii**	IE «$= \Delta E = h\nu = 6.63 \times 10^{-34}$ J s $\times$ 3.28×10^{15} s^{-1}» $= 2.17 \times 10^{-18}$ «J» ✓		**1**
3.	**a**	**iv**	«$\lambda = \frac{c}{\nu} = \frac{3.00 \times 10^{8}\ ms^{-1}}{3.28 \times 10^{15}\ s^{-1}}$ =» 9.15×10^{-8} «m» ✓		**1**
3.	**c**	**i**			**1**
3.	**c**	**ii**	*Anode (positive electrode):* $Cu(s) \rightarrow Cu^{2+}(aq) + 2e^-$ ✓ *Cathode (negative electrode):* $Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$ ✓	*Accept* $Cu(s) - 2e^- \rightarrow Cu^{2+}(aq)$. *Accept* $\rightleftharpoons$ *for* $\rightarrow$. *Award* ***[1 max]*** *if the equations are at the wrong electrodes.*	**2**

3.	**c**	**iii**	«external» circuit/wire ***AND*** from positive/anode to negative/cathode electrode ✓	*Accept "through power supply/battery" instead of "circuit".*	**1**
3.	**c**	**iv**	no change «in colour» ✓	*Do **not** accept "solution around cathode will become paler and solution around the anode will become darker".*	**1**
3.	**c**	**v**	oxygen/O_2 ✓	*Accept "carbon dioxide/CO_2".*	**1**

7.2-S13 Nov 2017 · Paper 3 · Q1 parts (b)(iii)

1.	**b**	**iii**	ions move «across electrolyte» ✓		**1**

7.2-S14 May 2015 TZ1 · Paper 2 · Q6 parts (d)(i), (d)(ii), (e), (f)(i), (f)(ii), (g) · also 6.1, 7.1, 9.4

(d) (i) *Positive electrode (anode):*

$2Br^- \rightarrow Br_2(g) + 2e^- \; / \; Br^- \rightarrow \frac{1}{2}Br_2(g) + e^-$;

Negative electrode (cathode):

$Na^+ + e^- \rightarrow Na(l)$; **[2]**

*Award **[1 max]** for correct equations at incorrect electrodes.*
Ignore state symbols.
Accept e *instead of* e^-.
Penalize use of equilibrium signs once only.

(ii) *Positive electrode (anode):*
bromine/Br_2;

Negative electrode (cathode):
hydrogen/H_2; **[2]**
Allow sodium hydroxide/NaOH/hydroxide/OH^- formation.

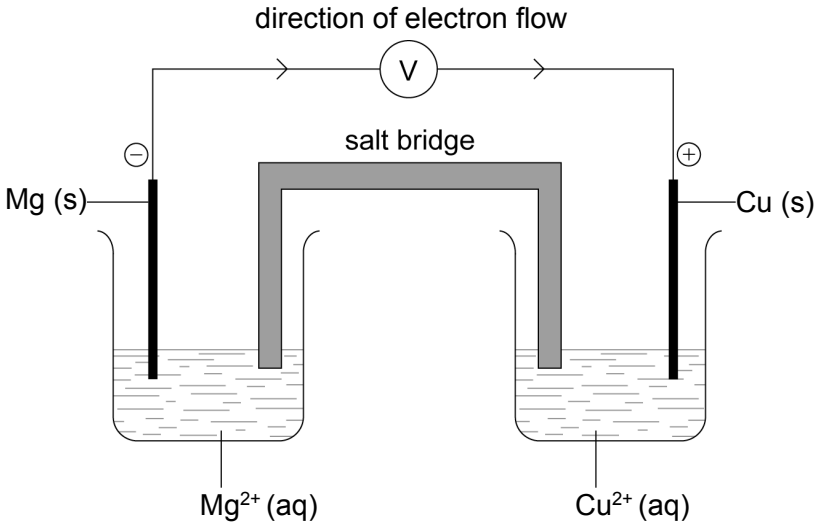
(e) bromine/Br_2; **[1]**

Do not accept bromide/Br^-.

(f) (i) potential of reduction half-reaction under standard conditions measured relative to standard hydrogen electrode/SHE / potential under standard conditions relative to standard hydrogen electrode/SHE; **[1]**

Instead of standard state allow either solute concentration of 1 mol dm^{-3} or 100 kPa/1.00 $\times 10^5$ Pa for gases.
Allow 1 bar for 100 kPa/1.00 $\times 10^5$ Pa.
Allow 1 atm.
Allow voltage instead of potential.

(ii)



correct diagram including (voltmeter), 4 correct species (state symbols not required) and connecting wires;
No credit if wires to electrodes immersed in the solutions.
Accept ammeter/meter/lamp instead of voltmeter.

labelled salt bridge;
Accept an appropriate salt (name or formula) instead of salt bridge (eg, potassium nitrate).

correctly labelled electrodes as +/cathode **and** –/anode;
flow of electrons from Mg to Cu in external circuit; **[4]**

Markscheme for 6(g) not located automatically - see the official May 2015 TZ1 · Paper 2 markscheme.

7.2-S15 May 2015 TZ1 · Paper 3 · Q12 parts (b), (c)

(b) ***K:*** chlorine/Cl_2 **and** ***M:*** sodium hydroxide/NaOH; **[1]**

(c) *Negative electrode (cathode):*
$2H_2O(l) + 2e^- \rightarrow H_2(g) + 2OH^-(aq)$;
Accept $2H^+(aq) + 2e^- \rightarrow H_2(g)$.

Positive electrode (anode):
$2Cl^-(aq) \rightarrow Cl_2(g) + 2e^-$; **[2]**

*Award **[1 max]** for correct equations at incorrect electrodes.*
Ignore state symbols.
Accept e for e^-.
Accept corresponding half-equations involving one e^-.

7.2-S16 May 2015 TZ2 · Paper 2 · Q6 parts (a)

(a) *Positive electrode (anode):*
$2Cl^-(l) \rightarrow Cl_2(g) + 2e^-$ / $Cl^-(l) \rightarrow \frac{1}{2}Cl_2(g) + e^-$;
Negative electrode (cathode):
$Mg^{2+}(l) + 2e^- \rightarrow Mg(l)$;
Accept e *instead of* e^-.
*Award **[1 max]** for correct half-equations given at the wrong electrode.*
Penalize use of reversible arrows once only.

correct state symbols in both equations; **[3]**

7.2-S17 May 2015 TZ2 · Paper 3 · Q13 parts (b)

(b) *Negative electrode (cathode):*
$2H_2O(l) + 2e^- \rightarrow H_2(g) + 2OH^-(aq)$;
Accept $2H^+(aq) + 2e^- \rightarrow H_2(g)$.

Positive electrode (anode):
$2Cl^-(aq) \rightarrow Cl_2(g) + 2e^-$; **[2]**

*Award **[1 max]** if reactions are at the wrong electrode.*
Ignore state symbols and/or reversible arrows.
Accept corresponding half-equations involving one e^-.
Accept e for e^-.

7.2-S18 Nov 2014 · Paper 3 · Q9 parts (a)(ii), (a)(iii)

(ii) *Positive electrode (anode):*
$2O^{2-} \rightarrow O_2(g) + 4e^- / O^{2-} \rightarrow \frac{1}{2}O_2(g) + 2e^- / 2O^{2-} - 4e^- \rightarrow O_2(g) /$
$O^{2-} - 2e^- \rightarrow \frac{1}{2}O_2(g)$;
Allow $C(s) + 2O^{2-} \rightarrow CO_2(g) + 4e^-$.

Negative electrode (cathode):
$Al^{3+} + 3e^- \rightarrow Al(l)$; ***[2]***

Accept e instead of e^-.
Ignore state symbols.
*If correct equations shown at wrong electrodes, award **[1 max]**.*

(iii) the carbon anode oxidises / burns in/reacts with the oxygen (to produce CO_2); ***[1]***

7.2-S19 May 2014 TZ1 · Paper 2 · Q6 parts (a)(i), (a)(ii), (a)(v), (b)(i), (b)(ii), (b)(iii), (b)(iv), (c)(i), (c)(ii), (d)(ii), (d)(iii) · also 6.2, 6.3, 7.1, 7.3

6. (a) (i) $C_4H_9OH(l) \rightarrow C_4H_8O(l) + 2H^+(aq) + 2e^-$; ***[1]***

Ignore state symbols.

(ii) $3C_4H_9OH(l) + Cr_2O_7^{2-}(aq) + 8H^+(aq) \rightarrow 3C_4H_8O(l) + 2Cr^{3+}(aq) + 7H_2O(l)$; ***[1]***

Ignore state symbols.

(v) $C_4H_9OH + 6O_2 \rightarrow 4CO_2 + 5H_2O$ / $(CH_3)_3COH + 6O_2 \rightarrow 4CO_2 + 5H_2O$

correct reactants and products;

correct balancing; ***[2]***

(b) (i) $Z < W < X < Y$;

Accept Y > X > W > Z. ***[1]***

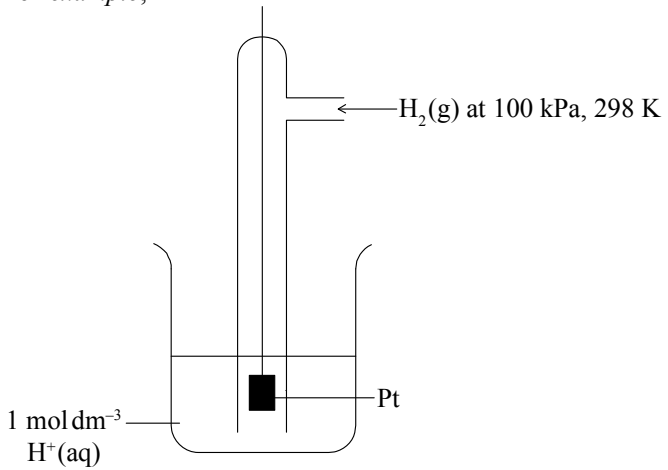
(ii) $X(s) + Z^{2+}(aq) \rightarrow X^{2+}(aq) + Z(s)$; ***[1]***

Ignore state symbols.

Accept X(s) + $ZCl_2(aq)$ → $XCl_2(aq)$ + Z(s).

(iii) $H_2(g)$/hydrogen; ***[1]***

(iv) diagram showing gas, solution and solid electrode;
For example,



*This diagram scores **[3]**.*
1 mol dm^{-3} $H^+(aq)$ **and** Pt;
Allow 1 mol L^{-1} or 1 M.
Allow 1 mol dm^{-3} HCl (aq) or other source of 1 mol dm^{-3} $H^+(aq)$ ions.

100 kPa/10^5 Pa/1 bar ($H_2(g)$ pressure) **and** 298 K / 25 °C; ***[3]***
Ignore state symbols throughout.
Allow 1.01×10^5 Pa/1 atm.

(c) (i) *Positive electrode (anode):*
$I^-(aq) \rightarrow \frac{1}{2} I_2(aq) + e^-$;
Accept correct equation involving 2 mols of I^-.

Negative electrode (cathode):
$H_2O(l) + e^- \rightarrow \frac{1}{2} H_2(g) + OH^-(aq)$ / $H^+(aq) + e^- \rightarrow \frac{1}{2} H_2(g)$ /
$H_3O^+(aq) + e^- \rightarrow H_2O(l) + \frac{1}{2} H_2(g)$; ***[2]***
*Award **[1 max]** if correct equations are given at the wrong electrodes.*
Ignore state symbols.
Allow e instead of e^-.
Penalize equilibrium sign once only.
Accept correct equation involving 2 mols of H^+.

(ii) aluminium will be oxidized (instead of I^-) at positive electrode (anode);
aluminium is a reactive metal / oxidation of aluminium has a positive $E^\ominus$ /
aluminium is higher on the reactivity series than I^- / *OWTTE*; ***[2]***

(ii) blue colour persists in second cell **and** fades in third cell;
pH does not change in second cell **and** decreases in third cell; ***[2]***
*Award **[1 max]** if both colour and pH are correctly stated for one only of either second or third cell.*

(iii) *Colour:*
positive Cu electrode (anode) is oxidized to maintain colour in second cell / $Cu(s) \rightarrow Cu^{2+}(aq) + 2e^{-}$;

pH:
in third cell, H^{+} ions are produced as water is oxidized at positive electrode (anode) / $H_2O(l) \rightarrow \frac{1}{2}O_2(g) + 2H^{+}(aq) + 2e^{-}$ / solution becomes acidic as hydroxide ions are oxidized at positive electrode (anode) / $2OH^{-}(aq) \rightarrow \frac{1}{2}O_2(g) + H_2O(l) + 2e^{-}$; ***[2]***
Ignore state symbols.

7.2-S20 May 2014 TZ2 · Paper 2 · Q8 parts (a), (b)(i), (b)(ii), (c), (d), (g)(i), (g)(ii), (g)(iii), (g)(iv) · also 2.1, 9.4

(a) minimum energy required to remove one electron / energy required to remove most loosely bound/outermost electron;
from gaseous/isolated atom; ***[2]***
Accept "gaseous state".
More extensive definitions involving one mole may be given.

(b) (i) electrons lost in same orbital/valence shell;
(second) electron/electron (being lost from Mg^+ is) closer to the nucleus;
(second) electron/electron (being lost from Mg^+) not subject to e-e repulsion from others in same level; ***[2 max]***
Apply OWTTE for all marking points.
*Do **not** accept "less electrons to share the charge" or answers employing this concept.*

(ii) electron in lower energy level / more stable electron shell;
electron closer to nucleus;
less shielding by complete inner shells / increase in effective nuclear charge; ***[2 max]***
Apply OWTTE for all marking points.

(c) $\Delta H_{at}(\text{Cl}) = \frac{1}{2} \times 243\,(\text{kJ mol}^{-1})$;
Correct calculation of atomization enthalpy of Cl.

$\Delta H_f = +146 + \frac{1}{2}243 + 738 + (-349) + (-790)$;
Correct sign and magnitude of all terms.

$= -134\,(\text{kJ mol}^{-1})$; ***[3]***
*Award **[3]** for correct final answer.*
Final mark involves correct computation of equation the student has produced.
*Award **[2]** for –12 (bond enthalpy of Cl not halved) or +134 (signs wrong).*
*Award **[1]** for +12 (bond enthalpy of Cl not halved and signs wrong).*

(d) $MgF_2 – MgCl_2 – CaCl_2$;
F^- smaller (ionic radius) than Cl^- / Cl^- larger (ionic radius) than F^-;
Mg^{2+} smaller (ionic radius) than Ca^{2+} / Ca^{2+} larger (ionic radius) than Mg^{2+}; ***[3]***
Accept use of atomic radius rather than ionic radius.

(g) (i)

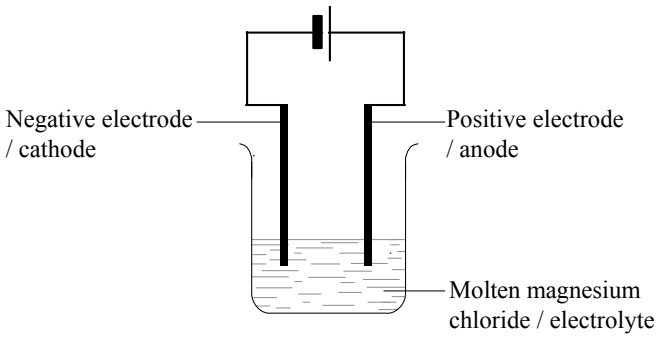


Diagram:
two electrodes connected to a power pack/battery and immersed in an electrolyte;
*Do **not** award mark if salt bridge included in diagram.*

Labelling:
anode/positive electrode, cathode/negative electrode, molten magnesium chloride/$MgCl_2$ (l)/electrolyte correctly labelled; ***[2]***
Check candidates know which end of a battery symbol is which charge.

(ii) *Negative electrode (cathode):* $Mg^{2+}(l) + 2e^- \rightarrow Mg(s)$;
Positive electrode (anode): $2Cl^-(l) \rightarrow Cl_2(g) + 2e^-$; ***[2]***
Accept $Cl^-(l) \rightarrow \frac{1}{2}Cl_2(g) + e^-$.
Ignore state symbols.
Allow e instead of e^-.
*If both correct equations are given for the wrong electrodes award **[1 max]**.*

(iii) *Negative electrode (cathode):*
$2H_2O(l) + 2e^- \rightarrow H_2(g) + 2OH^-(aq)$ / $2H^+(aq) + 2e^- \rightarrow H_2(g)$;
Accept $4H_2O(l) + 4e^- \rightarrow 2H_2(g) + 4OH^-(aq)$ / $4H^+(aq) + 4e^- \rightarrow 2H_2(g)$ / $H_2O(l) + e^- \rightarrow \frac{1}{2}H_2(g) + OH^-(aq)$ / $H^+(aq) + e^- \rightarrow \frac{1}{2}H_2(g)$.

Positive electrode (anode):
$2H_2O(l) \rightarrow O_2(g) + 4H^+(aq) + 4e^-$ / $4OH^-(aq) \rightarrow O_2(g) + 2H_2O(l) + 4e^-$; ***[2]***
Accept $H_2O(l) \rightarrow \frac{1}{2}O_2(g) + 2H^+(aq) + 2e^-$ / $2OH^-(aq) \rightarrow \frac{1}{2}O_2(g) + H_2O(l) + 2e^-$.
State symbols not required.
Allow e instead of e^-.
*If both correct equations are given for the wrong electrodes award **[1 max]**.*

(iv) water/hydrogen ions more easily reduced/better oxidizing agents/have a more positive $E^\ominus$ (than magnesium ions); ***[1]***
Accept converse statements for magnesium ions.
Accept "magnesium is very reactive/high in reactivity series" / OWTTE.

7.2-S21 May 2014 TZ2 · Paper 3 · Q10 parts (a), (b)

10. (a) electrolysis of aqueous/solution of sodium chloride/NaCl; ***[1]***
Accept brine.

(b) *Positive electrode (anode)*:
$2Cl^-(aq) \rightarrow Cl_2(g) + 2e^-$;

Negative electrode (cathode):
$2H^+(aq) + 2e^- \rightarrow H_2(g) / 2H_2O(l) + 2e^- \rightarrow 2OH^-(aq) + H_2(g)$; ***[2]***
Ignore state symbols.
Allow e instead of e^-.

7.2-S22 May 2013 TZ1 · Paper 2 · Q7 parts (a)(ii), (b)(iv), (b)(v), (b)(vi), (c)(i), (c)(ii), (c)(iii), (c)(iv), (c)(v) · also 2.1, 6.1, 6.3

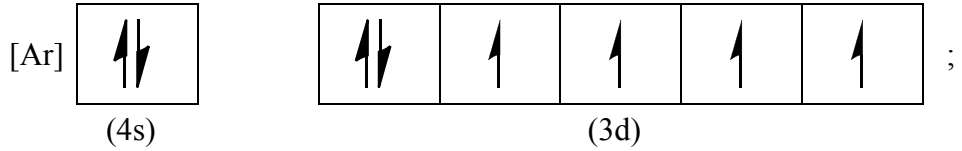
(ii) no reaction with NaCl;
$Br_2(aq) + 2NaI(aq) \rightarrow 2NaBr(aq) + I_2(aq)$; ***[2]***

Accept ionic equation.
Ignore state symbols.

(iv) III / +3; ***[1]***

Penalize incorrect format such as 3+ only if not penalized in 4 (b).

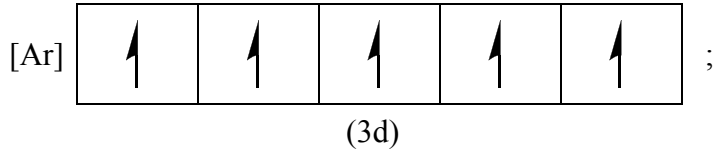
(v)



; ***[1]***

Penalise missing [Ar] only once in (v) and (vi).
Do not accept full orbital diagram; penalise only once in (v) and (vi).
Accept full or half-arrows in (v) and (vi).
Ignore absence of labels 4s and 3d.

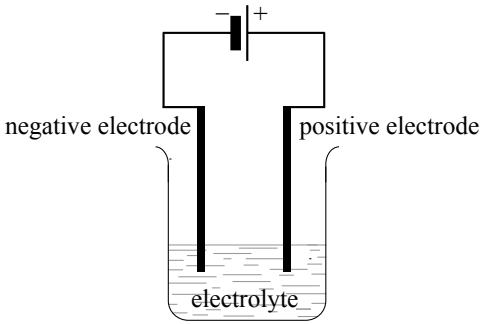
(vi)



; ***[1]***

Accept empty 4s box in (vi).
No ECF from (iv).

(c) (i)



clear diagram containing all elements (power supply, connecting wires, electrodes, container and electrolyte);
Accept power supply if shown as conventional long/short lines (as in diagram above) or clearly labelled DC power supply.

labelled positive electrode/anode **and** negative electrode/cathode;
Accept positive and negative by correct symbols near power supply.

labelled electrolyte/$FeBr_2$ (l)/$FeBr_2$(aq); ***[3]***
State must be included for $FeBr_2$.

(ii) *Electrolyte:* positive ions/cations move to negative electrode/cathode **and** negative ions/anions to positive electrode/anode;
Conductors: electrons flow from negative pole of battery to positive pole of battery / *OWTTE*; ***[2]***
Look at diagram in (i) for possible clarification of electron flow.

*Award **[1 max]** for "electrons in wire/external circuit **and** ions in solution".*

(iii) *Negative electrode/cathode*:
H_2;
$E^{\ominus}(H_2)$ is less negative than $E^{\ominus}$ (Fe) / Fe is more reactive than H_2 / H_2 is lower in reactivity series / H^+ more easily reduced than Fe^{2+} / *OWTTE*;

Positive electrode/anode:
O_2;
$E^{\ominus}(O_2)$ is less positive than $E^{\ominus}(Br_2)$ / in a dilute Br^- solution OH^-/H_2O is preferably discharged / *OWTTE*; ***[4]***
*Award **[3 max]** if electrodes reversed or omitted.*

(iv) Br_2; ***[1]***
Accept Fe.

(v) $2Br^- \rightleftharpoons Br_2 + 2e^-$ shifts to the right; ***[1]***
Accept similar reason for Fe.

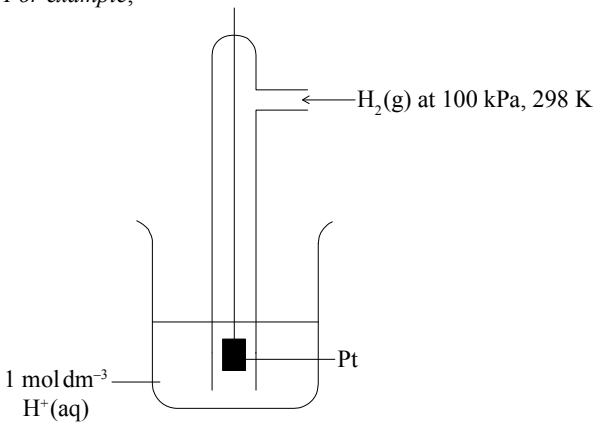
7.2-S23 May 2013 TZ2 · Paper 2 · Q7 parts (a), (b)(i), (b)(ii), (c)(i), (c)(ii), (c)(iii), (d)(i), (d)(ii), (e)(ii), (e)(iii), (e)(iv), (e)(v), (e)(vi), (e)(vii) · also 6.1, 6.2, 7.1

7. (a) increase (in oxidation number); ***[1]***

(b) (i) $Cu(s) + 2NO_3^-(aq) + 4H^+(aq) \rightarrow Cu^{2+}(aq) + 2NO_2(g) + 2H_2O(l)$ /
$Cu(s) + 2HNO_3(aq) + 2H^+(aq) \rightarrow Cu^{2+} + 2NO_2(g) + 2H_2O(l)$;
correct reactants and products;
fully balanced chemical equation; ***[2]***
Ignore state symbols.
M1 can be scored if there are unbalanced electrons in equation.
M2 can only be scored if M1 is correct.
M2 can be scored if there are balanced electrons on both sides of equation.

(ii) *Oxidizing agent:* NO_3^-/nitrate/HNO_3/nitric acid **and** *Reducing agent:* Cu/copper; ***[1]***

(c) (i) Diagram showing gas, solution and solid electrode;
For example,



$1\,mol\,dm^{-3}\ H^+(aq)$ and Pt;
Allow 1 mol L^{-1} or 1 M.
Allow 1 mol dm^{-3} HCl(aq) or other source of 1 mol dm^{-3} $H^+(aq)$ ions.

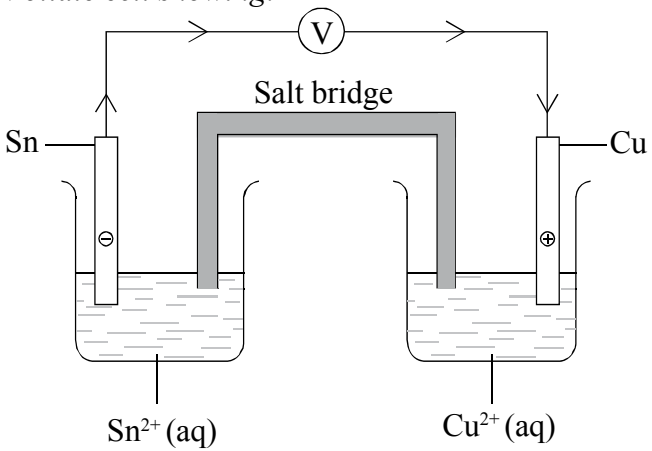
100 kPa/10^5 Pa/1 bar ($H_2(g)$ pressure) **and** 298 K / 25 °C; ***[3]***
Ignore state symbols throughout.
Allow 1.01×10^5 Pa/1 atm.

(ii) potential of reduction half-reaction under standard conditions measured relative to standard hydrogen electrode/SHE / potential under standard conditions relative to standard hydrogen electrode/SHE; ***[1]***
Instead of standard conditions allow either solute concentration of 1 mol dm^{-3} or 100 kPa/10^5 Pa/1 bar (pressure) for gases (allow 1 atm).
Allow voltage/EMF instead of potential.

(iii) $Cu^{2+}(aq) + H_2(g) \rightarrow Cu(s) + 2H^+(aq)$

reactants and products;
fully balanced chemical equation, including state symbols; ***[2]***
M2 cannot be scored if M1 is incorrect.

(d) (i) *Voltaic cell showing:*



Labelled positive electrode (cathode): Cu^{2+}/Cu **and** *negative electrode (anode):* Sn^{2+}/Sn ;
Do not penalize if state symbols are not included (since given in question).

voltmeter and salt bridge;
Voltmeter can be labelled or drawn as a circle with a V.
Allow ammeter/A.
Salt bridge can be labelled, represented with drawing connecting the two half-cells, labelled as potassium nitrate or using its chemical formula (for example, KNO_3) etc.

correct direction of electron movement from Sn to Cu in external circuit; ***[3]***

(ii) (+) 0.48 (V); ***[1]***

(ii) provides ions (to carry current) / water poor conductor (of electricity); ***[1]***
Do not accept electrons instead of ions.

(iii) copper reacts so (nonreactive metal such as) Pt used; ***[1]***
Accept Ag, Au or any named metal less reactive than copper as electrode.
Do not accept Cu reacts with water or graphite as electrode.

(iv) *Positive electrode (anode):*
$2H_2O(l) \rightarrow O_2(g) + 4H^+(aq) + 4e^-$ / $4OH^-(aq) \rightarrow O_2(g) + 2H_2O(l) + 4e^-$;

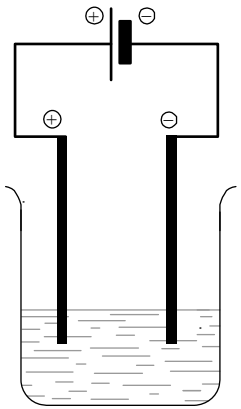
Negative electrode (cathode):
$H^+(aq) + e^- \rightarrow \frac{1}{2}H_2(g)$ / $4H^+(aq) + 4e^- \rightarrow 2H_2(g)$ / $2H^+(aq) + 2e^- \rightarrow H_2$ /
$2H_2O(l) + 2e^- \rightarrow H_2(g) + 2OH^-(aq)$ / $H_2O(l) + e^- \rightarrow \frac{1}{2}H_2(g) + OH^-$; ***[2]***
*Award **[1 max]** if M1 and M2 reversed.*

Ignore state symbols.
Allow e instead of e^-.
Do not penalize use of equilibrium sign instead of $\rightarrow$.
Accept a multiple of the equations.

(v) $2H_2O(l) \rightarrow 2H_2(g) + O_2(g) / H_2O(l) \rightarrow H_2(g) + \frac{1}{2}O_2(g)$; ***[1]***

State symbols required as asked for in question.
Do not penalize use of equilibrium sign instead of $\rightarrow$.
Do not accept any multiple of $2H^+(aq) + 2OH^-(aq) \rightarrow 2H_2(g) + O_2(g)$.

(vi)



electrolytic cell showing solid electrodes immersed in solution;
positive electrode (anode) connected to positive terminal of battery **and** negative electrode (cathode) to negative terminal; ***[2]***
Allow graphite or metal given in e(iii) as electrodes.

(vii) bubbles /gas produced; ***[1]***
Do not accept hydrogen is formed at cathode and oxygen formed at anode.

7.2-S24 May 2013 TZ2 · Paper 3 · QC1 parts (a)(iii)

(iii) *Positive electrode (anode)*:

$2O^{2-} \rightarrow O_2 + 4e^- \ / \ O^{2-} \rightarrow \frac{1}{2}O_2 + 2e^- \ / \ C + 2O^{2-} \rightarrow CO_2 + 4e^-$;

Negative electrode (cathode):

$Al^{3+} + 3e^- \rightarrow Al$; ***[2]***

Allow e instead of e^-.

Accept multiples of the correct equations, such as $2Al^{3+} + 6e^- \rightarrow 2Al$.

*Award **[1 max]** if correct equations but at wrong electrodes.*

Ignore state symbols.

7.2-S25 Nov 2011 · Paper 3 · QC5 parts (a)(ii)

(ii) *Positive electrode (anode)*:

$2Cl^-(aq) \rightarrow Cl_2(g) + 2e^- \ / \ Cl^-(aq) \rightarrow \frac{1}{2}Cl_2(g) + e^-$;

Negative electrode (cathode):

$2H_2O(l) + 2e^- \rightarrow 2OH^-(aq) + H_2(g) \ / \ H_2O(l) + e^- \rightarrow OH^-(aq) + \frac{1}{2}H_2(g)$; ***[2]***

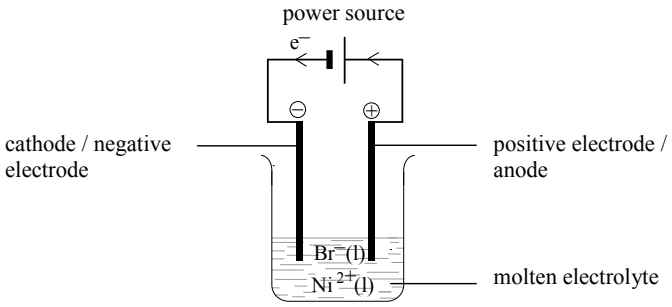
Ignore state symbols.

Allow e instead of e^-.

*Allow **[1]** if correct equations but incorrect electrodes.*

7.2-S26 May 2011 TZ1 · Paper 2 · Q9 parts (a)(i), (a)(ii), (b)(i), (b)(ii), (b)(iii), (c), (e)(i), (e)(ii) · also 6.1, 6.2, 7.1

9. (a) (i)



labelled polarities of positive **and** negative electrodes;
Electrodes can be labelled positive or negative or + and - signs can be used.

direction of electron flow;
e^- does not have to be labelled but arrow essential.

power source **and** molten electrolyte/Ni^{2+} (l) **and** Br^- (l)/$NiBr_2$ (l);
State symbol necessary for M3 unless molten electrolyte stated.
Power source does not need to be labelled if correct symbol used (i.e. short line and long line).

Cathode/negative electrode equation:
$Ni^{2+} + 2e^- \rightarrow Ni$;
Anode/positive electrode equation:
$2Br^- \rightarrow Br_2 + 2e^-$; ***[5]***
Accept balanced half-equation with one e^-.
*Award **[1 max]** for M4 and M5 if electrodes are not identified or if equations are given wrong way round or incorrectly labelled.*
Penalize ⇌ once only in Q.9.
Allow e instead of e^-.
Ignore state symbols for M4 and M5.

(ii) *Dilute sodium chloride:*
$2H_2O \rightarrow O_2 + 4H^+ + 4e^-$ / $4OH^- \rightarrow O_2 + 2H_2O + 4e^-$;

Concentrated sodium chloride:
$2Cl^- \rightarrow Cl_2 + 2e^-$; ***[2]***
Accept alternative balanced half-equations with correct number of electrons.
*Award **[1 max]** if equations are given the wrong way round.*
*Award **[2]** if correct equations are written in order with dilute sodium chloride first and concentrated sodium chloride second but processes not stated explicitly.*
Penalize ⇌ once only in Q.9.
Allow e instead of e^-.
Ignore state symbols

(b) (i) $Sn + Cu^{2+} \rightarrow Sn^{2+} + Cu$; ***[1]***
Ignore state symbols.
Penalize $\rightleftharpoons$ *once only in Q.9.*

(ii) (0.34 – –0.14) = (+) 0.48 V; ***[1]***

(iii) 1.0 mol dm^{-3} solutions **and** 25 °C /298 K; ***[1]***

(c) Cd^{2+} is a stronger oxidizing agent than H_2O **and** will be displaced to produce Cd / *OWTTE*;
Cr^{2+} is a weaker oxidizing agent than H_2O **and** H_2 will displace in preference to Cr / *OWTTE*; ***[2]***
*Award **[1 max]** for stating Cd^{2+} stronger oxidizing agent than H_2O **and** Cr^{2+} weaker oxidizing agent than H_2O / OWTTE.*

(e) (i) *NH_3:* –3;
NO: +2;
N_2: 0; ***[3]***
Penalize incorrect notation such as 3–, III, 2+, 2, II once only.

(ii) *Oxidation:*
$2NH_3 \rightarrow N_2 + 6H^+ + 6e^-$;
Reduction:
$2NO + 4H^+ + 4e^- \rightarrow N_2 + 2H_2O$;
*Award **[1 max]** for M1 and M2 if redox processes are not identified or if equations are given wrong way round.*
Penalize $\rightleftharpoons$ *once only in Q.9.*
Allow e instead of e^-.
Ignore state symbols.

Oxidizing agent: NO; ***[3]***
Allow either formula or name.

7.2-S27 May 2011 TZ1 · Paper 3 · QC4 parts (b)

(b) $2H_2O(l) + 2e^- \rightarrow H_2(g) + 2OH^-(aq)$;

Accept $2H^+(aq) + 2e^- \rightarrow H_2(g)$.

$2Cl^-(aq) \rightarrow Cl_2(g) + 2e^-$; ***[2]***

Ignore any labelling of anode/cathode.
State symbols are not required.

7.2-S28 May 2011 TZ2 · Paper 3 · QC1 parts (a)(ii)

(ii) $2O^{2-} \rightarrow O_2 + 4e^- / O^{2-} \rightarrow \frac{1}{2}O_2 + 2e^-$; ***[1]***

Allow e instead of e^-.

7.2-S29 Nov 2010 · Paper 3 · QC5 parts (a)(iii)

(iii) $Cl^-(aq) \rightarrow \frac{1}{2}Cl_2(g) + e^-$; ***[1]***

Allow correct equations involving multiple coefficients (i.e. $2Cl^-(aq)$ *etc.)*
Ignore state symbols.
Allow e instead of e^-.

LESSON 7.3

Energy From Fuels

Answers for 6 multiple-choice and 52 structured questions

Part A - Multiple choice answers

7.3-M1	May 2025 TZ2 · P1A · Q20	C
7.3-M2	May 2019 TZ2 · P1 · Q2	D
7.3-M3	May 2017 TZ2 · P1 · Q1	C
7.3-M4	May 2015 TZ2 · P1 · Q2	C
7.3-M5	May 2013 TZ2 · P1 · Q33	C
7.3-M6	May 2010 TZ1 · P1 · Q35	B

Part B - Structured questions markscheme

7.3-S1 May 2025 TZ3 · Paper 2 · Q9 parts (b)(i), (b)(ii), (c) · also 8.2, 9.3

9.	**(b)**	**(i)**	*Any two:* «ethanol is» renewable / sustainable resource ✓ «ethanol has» low/zero carbon footprint / produces less CO_2 ✓ less sulfur dioxide «than fossil fuels» ***OR*** less acid rain «than fossil fuels» ✓ less incomplete combustion «than fossil fuels» ***OR*** less carbon monoxide/soot «than fossil fuels» ✓	*Accept "ethanol is biodegradable /less toxic than gasoline".* *Do **not** accept just "less harmful".*	**2 max**
9.	**(b)**	**(ii)**	«bond breaking» 1 C-C + 5 C-H + 1 C-O + 1 O-H + 3 O=O ***OR*** 346 + 5(414) + 358 + 463 + 3(498) ***OR*** 4731 «kJ» ✓ «bond forming» 4 C=O + 6 O-H ***OR*** 4(804) + 6(463) ***OR*** 5994 «kJ» ✓ «ΔH = 4731 – 5994 =» -1263 «kJ mol^{-1}» ✓	*Award **[3]** for correct final answer.*	**3**
9.	**(c)**		«Q = (4.00 / 46.08) x 1367 =» 119 kJ / 119000 J ✓ «ΔT = Q / mc = 119000 / (500.0 x 4.18) =» 56.9 «K» ✓	*Accept 56.8 «K».* *Allow use of ΔH_c calculated in (b)(ii) without penalty. Answer is ΔT = 52.5 / 52.6 «K».*	**2**

7.3-S2 Nov 2024 · Paper 3 · Q14 parts (a)(i), (a)(ii)

14.	**a**	**i**	$C_8H_{18}(l) + 12.5\ O_2(g) \rightarrow 8CO_2(g) + 9H_2O(g)$ ✓ «1 x 8(44.01) ÷ 114.26 =» 3.081«kg CO_2 produced» ✓	*Accept equation with whole number coefficients.* *Award **[2]** for correct final answer.* *Accept "3.09 «kg CO_2 produced»" if whole numbers are used for molar mass.*	**2**
14.	**a**	**ii**	CO_2 «produced by burning ethanol» captured by crops «via photosynthesis» ✓	*Do **not** accept just "renewable" **OR** "carbon neutral" alone"*	**1**

7.3-S3 May 2024 TZ1 · Paper 3 · Q13 parts (a), (b), (c), (f)

13.	**a**		$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(l)$ ✓		**1**
13.	**b**		amount of CH_4 in 1.00 g clathrate «$= 4 \times \frac{1\ g}{478.7\ g\ mol^{-1}}$» $= 0.008356$ «mol» ✓ specific energy «$= 0.008356\ mol\ g^{-1} \times 891\ kJ\ mol^{-1}$» $= 7.45$ «kJ g^{-1}» ✓	*Award **[2]** for correct final answer.*	**2**
13.	**c**		*Lower specific energy:* not all methane clathrate is methane ***OR*** part of methane clathrate does not burn ✓ *Higher energy density:* methane is gas ***AND*** methane clathrate is solid ***OR*** methane has much lower density «than methane clathrate» ***OR*** methane clathrate occupies a much smaller volume than «same mass of» methane ✓	*For M1 accept reference to water molecules in methane clathrate as indicating it is not all methane.*	**2**

13.	**f**		*Support:* overall increase in both the carbon dioxide concentration and the temperature «anomaly» ✓ *Refute:* CO_2 increases between ~1940 and ~1970 without corresponding increase in temperature ***OR*** global temperatures rose between ~1935 and ~1945 despite [CO_2] remaining relatively constant ***OR*** increases in global temperature between ~1900 and ~1940 and between ~1970 and ~2010 are similar, but increase in CO_2 levels much higher in latter period ✓	*For M1, accept there is a good/close correlation «in their shapes».* *For M2, accept the connection may just be coincidental «not causative».* *Do **not** accept comments about fluctuations.* *Accept years that are approximately the same as these listed.*	**2**

7.3-S4 May 2024 TZ1 · Paper 3 · Q14 parts (a)(i), (a)(ii)

14.	**a**	**i**	methane produces less CO_2 per kJ of energy ✓ methane produces fewer pollutants/particulates/CO/VOCs ✓	*For M1 do **not** accept simply methane produces less CO_2.* *For M2 accept gasoline undergoes incomplete combustion.*	**2**
14.	**a**	**ii**	greater energy density ***OR*** longer range «for same volume of fuel»/requires smaller fuel tank ✓	*Accept more concentrated source of energy.* *Accept greater distance between refuelling.* *Accept gasoline more readily available.* *Accept gasoline fuelled cars safer / have lower explosion/auto-ignition risk.* *Accept gasoline easier to handle / store / transport.*	**1**

7.3-S5 Nov 2023 TZ1 · Paper 3 · Q12 parts (c)(ii)

12.	**c**	**ii**	specific energy ***AND*** low density ***OR*** specific energy ***AND*** a small mass occupies a large volume ✓	*Do **not** accept "specific energy **AND** gas/gaseous" alone.*	**1**

7.3-S6 Nov 2023 TZ1 · Paper 3 · Q13 parts (b)

13.	**b**		infrared/IR **«**radiation**»** from earth's surface absorbed **«**by bonds of greenhouse gasses**»** in the **«**lower**»** atmosphere ✓ lower amounts **«**of re-radiated IR**»** reach upper atmosphere ✓	*Do **not** accept "heat for infrared/IR" in M1.* *Do **not** accept "greenhouse gases trap infrared/IR radiation" alone for M1.*	**2**

7.3-S7 Nov 2023 TZ1 · Paper 3 · Q14 parts (b), (c)

14.	**b**		methane/CH_4 «(g)» ✓		**1**
14.	**c**		*Advantage:* *Any one of:* cleaner combustion/less soot/ash ✓ less CO_2 per unit of heat ✓ lower carbon footprint ✓ higher specific energy ✓ *Disadvantage:* *Any one of:* highly combustible/explosive ✓ emits greenhouse gas/methane/CH_4 if leaks occur ✓ lower energy density ✓ transported under pressure ✓ difficult to detect leaks ✓	*Do **not** accept converse of stated advantage for disadvantage.* *Accept "easy to transport/no need to store locally" as an advantage.* *Do **not** accept vague answers such as "less pollution" **OR** "clean fuel" **OR** "less CO_2 produced" for Advantage or "hazardous" for Disadvantage.* *Accept "odourless" as a Disadvantage.*	**2 max**

7.3-S8 May 2023 TZ1 · Paper 3 · Q13 parts (d)(i)

13.	**(d)**	**(i)**	allow H^+ ions to pass through/diffuse/move **«**from anode to cathode» ✓	*Do **not** accept diffusion of electrons or small molecules.*	**1**

7.3-S9 May 2023 TZ2 · Paper 2 · Q6 parts (d)(i), (d)(ii) · also 9.3

6.	**(d)**	**(i)**	$CH_3CH_2OH(l) + 3O_2(g) \rightarrow 2CO_2(g) + 3H_2O(g)$ ✓	*Accept C_2H_6O for ethanol.*	**1**
6.	**(d)**	**(ii)**	«bond breaking» 1 C-C + 5 C-H + 1 C-O + 1 O-H + 3 O=O / 346 + 5(414) + 358 + 463 +3(498) / 4731 «kJ» ✓ «bond forming» 4 C=O + 6 O-H / 4(804) + 6(463) / 5994 «kJ» ✓ ΔH «= 4731– 5994» = −1263 «kJ mol^{-1} » ✓	*Award **[3]** for correct final answer.*	**3**

7.3-S10 May 2023 TZ2 · Paper 3 · Q15 parts (d)(i)

15.	**(d)**	**(i)**	material provides energy in fuel cells ***OR*** fuel continually added in fuel cells ✓	*Accept converse statement.* *Do **not** accept "fuel cells are rechargeable and primary are not easily recharged".*	**1**

7.3-S11 May 2023 TZ2 · Paper 3 · Q16 parts (a)(ii), (c)(i)

16.	**(a)**	**(ii)**	electrical/electricity ✓	*Accept "heat/thermal".*	**1**

16.	**(c)**	**(i)**	***ALTERNATIVE 1*** 5470 «kJ» produced when 8 «mol» produced ✓ «8 x 44.01 =» 352 «g» to produce 5470 «kJ» ✓ 1 «kJ» would release «352/5470 =» 0.0644 «g» ✓ ***ALTERNATIVE 2*** 1 «kJ»/5470«kJ=» 1.828×10^{-4} ✓ 8/5470 «mol =» 0.001463 mol ✓ «8/5470 «mol» x 44.01 «$gmol^{-1}$ =» 0.0644 «g» ✓	*Award **[3]** for correct final answer.* *Accept "0.0643 «g»".*	**3**

7.3-S12 May 2021 TZ1 · Paper 2 · Q5 parts (c)

5.	**c**		carbon monoxide/CO ***AND*** carbon/C/soot ✓		**1**

7.3-S13 May 2021 TZ2 · Paper 2 · Q5 parts (a)(i), (a)(ii) · also 9.2

5.	**a**	**i**	$CH_3CH_2OH\,(l) + 3O_2\,(g) \rightarrow 2CO_2\,(g) + 3H_2O\,(g)$ ✓		**1**
5.	**a**	**ii**	«$n = \frac{56.00\,g}{46.08\,g\,mol^{-1}}$ =» 1.215 «mol» ✓ «$1.215\,mol \times (-1367\,kJ\,mol^{-1})$ =» –1661 «kJ» ✓	*Award **[2]** for correct final answer.* *Award **[1 max]** for "«+»1661 «kJ»".*	**2**

7.3-S14 Nov 2020 · Paper 3 · Q11 parts (a), (c), (d)

11.	**a**		«21 200 kJ dm^{-3} × 5.00 dm^3 =» 106000/1.06 × 10^5 «kJ» ✓		**1**

11.	**c**		*Advantages:* ***[2 max]*** renewable ✓ uses up waste «such as used cooking oil» ✓ lower carbon footprint/carbon neutral ✓ higher flashpoint ✓ produces less SO_x/SO_2 ***OR*** less polluting emissions ✓ has lubricating properties ***OR*** preserves/increases lifespan of engine ✓ increases the life of the catalytic converter ✓ eliminates dependence on foreign suppliers ✓ does not require pipelines/infrastructure «to produce» ✓ relatively less destruction of habitat compared to obtaining petrochemicals ✓ *Disadvantages:* ***[2 max]*** needs conversion/transesterification ✓ takes time to produce/grow plants ✓	*Accept "higher energy density"* ***OR*** *"biodegradable" for advantage.* *Accept "lower specific energy" for disadvantage.*	**4**

			takes up land ***OR*** deforestation ✓ fertilizers/pesticides/phosphates/nitrates **«**used in production of crops**»** have negative environmental effects ✓ biodiversity affected ***OR*** loss of habitats **«**due to energy crop plantations**»** ✓ cannot be used at low temperatures ✓ variable quality **«**in production**»** ✓ high viscosity/can clog/damage engines ✓	*Do **not** accept "lower octane number/rating" for disadvantage.*	

11.	**d**		*Any one of:* uses up fossil fuels more slowly ✓ lower carbon footprint/CO_2 emissions ✓ undergoes more complete combustion ✓ produces fewer particulates ✓ higher octane number/rating ***OR*** less knocking ✓ prevents fuel injection system build up ***OR*** helps keep engine clean ✓	*Accept an example of a suitable advantage even if repeated from 11c.*	**1 max**

7.3-S15 Nov 2019 · Paper 2 · Q3 parts (c)(i)

3.	**c**	**i**	$2C_3H_8O\,(l) + 9O_2\,(g) \rightarrow 6CO_2\,(g) + 8H_2O\,(g)$ ***OR*** $C_3H_8O\,(l) + 4.5O_2\,(g) \rightarrow 3CO_2\,(g) + 4H_2O\,(g)$ ✓		**1**

7.3-S16 Nov 2019 · Paper 3 · Q16 parts (a)

16.	**a**		«similar specific energy and» pentane has «much» larger energy density ✓ *Any two for **[2 max]**:* similar number of bonds/«C and H» atoms in 1 kg «leading to similar specific energy» ***OR*** only one carbon difference in structure «leading to similar specific energy» ✓ pentane is a liquid ***AND*** butane is a gas «at STP» ✓ 1 m^3 of pentane contains greater amount/mass than 1 m^3 of butane ✓	*Accept "both are alkanes" for M2.* *Accept "pentane would be easier to transport".* *Accept "same volume" for "1 m^3" and "more moles" for "greater amount" for M4.*	**3**

7.3-S17 Nov 2019 · Paper 3 · Q19 parts (a)

19.	**a**		$C_2H_5OH\,(l) + 3O_2\,(g) \rightarrow 2CO_2\,(g) + 3H_2O\,(l)$ ✓		**1**

7.3-S18 Nov 2019 · Paper 3 · Q20 parts (a), (b)(i), (c) · also 7.1

20.	**a**		*Anode (negative electrode):* $H_2(g) \rightarrow 2H^+(aq) + 2e^-$ ✓ *Cathode (positive electrode):* $O_2(g) + 4H^+(aq) + 4e^- \rightarrow 2H_2O(l)$ ✓	*Accept any correct integer or fractional coefficients.* *Award **[1 max]** for M1 and M2 if correct half-equations are given at the wrong electrodes **OR** if incorrect reversed half-equations are given at the correct electrodes.*	**2**
20.	**b**	**i**	(+)1.23 **«**V**»** ✓	*Do **not** accept "-1.23 **«**V**»**".*	**1**
20.	**c**		liquid in cell is less/not corrosive ***OR*** does not contain lead/toxic chemicals ***OR*** larger energy density/charge capacity/current per unit mass ***OR*** does not have to be charged prior to use / is always ready for use **«**as long as fuel is available**»** ✓		**1**

7.3-S19 May 2019 TZ1 · Paper 3 · Q17 parts (b)(i), (b)(ii), (b)(iii), (c)

17.	**b**	**i**	Electrode A: $C_2H_6O\,(aq) + 3H_2O\,(l) \rightarrow 12H^+\,(aq) + 12e^- + 2CO_2\,(g)$ ✓ Electrode B: $3O_2\,(g) + 12H^+\,(aq) + 12e^- \rightarrow 6H_2O\,(l)$ ✓	*Accept balanced equations with integer or fractional coefficients.* *Penalize equilibrium arrows once only.*	**2**
17.	**b**	**ii**	*Name:* PEM/proton-exchange membrane/polymer exchange membrane/polymer electrolyte membrane ✓ *Function:* allows the passage of protons/H^+ ions **«**from anode to cathode but not electrons or molecules**»** ✓		**2**
17.	**b**	**iii**	*Any one of:* water is a reactant/allows the cell to operate at a higher concentration of protons/H^+ ions ***OR*** water is a stronger electrolyte and thus produces higher electric current ✓ less dangerous/flammable ✓		**1**
17.	**c**		use of **«**farm**»** land **«**for production**»** ***OR*** deforestation **«**for crop production for fuel**»** ***OR*** can release more NO_x **«**than normal fuel on combustion**»** ✓	*Ignore any reference to cost.*	**1**

7.3-S20 May 2019 TZ2 · Paper 3 · Q15 parts (b)

15.	**b**		specific energy = **«** $\frac{4163\ \text{kJ mol}^{-1}}{86.2\ \text{g mol}^{-1}}$ **=»** 48.3 **«**kJ g^{-1}**»** ✓ energy density = **«**48.3 kJ g^{-1} × 0.660 g cm^{-3} **=»** 31.9 **«**kJ cm^{-3}**»** ✓	*Award **[1 max]** if either or both answers not expressed to three significant figures.*	**2**

7.3-S21 May 2019 TZ2 · Paper 3 · Q17

17.			***Strength*** *Any one of:* less flammable **«**than diesel**»** ✓ recycles carbon **«**lower carbon footprint**»** ***OR*** lower greenhouse gas emissions ✓ easily biodegradable **«**in case of spill**»** ✓ renewable ***OR*** does not deplete fossil fuel reserves ✓ economic security/availability in countries without crude oil ✓ ***Limitation*** *Any one of:* more difficult to ignite inside the engine **«**than diesel**»** ✓ more viscous **«**than diesel**»** ✓ lower energy content/specific energy/energy density ✓ uses food sources ***OR*** uses land that could be used for food ✓ **«**production is**»** more expensive ✓ less suitable in low temperatures ✓ increased NO_x emissions for biodiesel ✓ greenhouse gases still produced ✓	*Accept "**«**close to**»** carbon neutral", "produce less greenhouse gases/CO_2".* *Accept "engines have to be modified if biodiesel used" as limitation.* *Do **not** award marks for strength and limitation that are the same topic/concept.*	**2 max**

7.3-S22 Nov 2018 · Paper 3 · Q13 parts (b), (c)

13.	**b**		«$\frac{141.6}{55.5}$» hydrogen/H_2 produces 2.6 times/more than twice the energy of methane/CH_4 «per mass/g» ***OR*** less mass of hydrogen/H_2 required «to produce same amount of energy» ***OR*** hydrogen/H_2 more energy efficient ✓	*Accept "hydrogen/H_2 produces «nearly» three times more energy than methane/CH_4 «per mass/g»".*	**1**
13.	**c**		m_{octane} «$= 72.0\,\text{dm}^3 \times 703\,\text{g dm}^{-3}$» $= 5.06 \times 10^4$ «g»/50.6 «kg» ✓ $m_{\text{carbon dioxide}}$ «$= \frac{8 \times 44.01}{114.26} \times 50.6$» $= 156$ «kg» ✓	*Award **[2]** for correct final answer.*	**2**

7.3-S23 May 2018 TZ1 · Paper 3 · Q12 parts (a), (b)(i), (b)(ii) · also 5.3

12.	**a**		*Any three of:* different molar masses ***OR*** different strengths of intermolecular forces ✓ different boiling points ✓ temperature in **«**fractionating**»** column decreases upwards ✓ **«**components**»** condense at different temperatures/heights ***OR*** **«**component with**»** lower boiling point leaves column first ✓		**3 max**
12.	**b**	**i**	*specific energy* **«** $= \frac{\text{energy released}}{\text{mass consumed}} = \frac{5470\ \text{kJ mol}^{-1}}{114.26\ \text{g mol}^{-1}}$ **»** $= 47.9$ **«**kJ g^{-1}**»** ✓ *energy density* **«** $= \frac{\text{energy released}}{\text{volume consumed}} = \text{specific energy} \times \text{density} = 47.9\ \text{kJ g}^{-1}$ $\times 0.703\ \text{g cm}^{-3}$**»** $= 33.7$ **«**kJ cm^{-3}**»** ✓	*Do **not** accept "–47.9 **«**kJ g^{-1}**»**".* *Do **not** accept "–33.7 **«**kJ cm^{-3}**»**" unless "–47.9 **«**kJ g^{-1}**»**" already penalized.*	**2**
12.	**b**	**ii**	energy is lost **«**to the surroundings**»** as heat/sound/friction ***OR*** energy is lost to the surroundings **«**as heat/sound/friction**»** ***OR*** incomplete combustion ✓	*Do **not** accept simply "energy is lost".*	**1**

7.3-S24 May 2018 TZ2 · Paper 3 · Q13 parts (a)

13.	**a**		*Any two of:* high energy content/high energy density/high specific energy ***OR*** high enthalpy of combustion/very exothermic enthalpy of combustion ✓ shortage of alternatives ***OR*** alternatives are expensive ***OR*** oil is relatively cheap ***OR*** oil is «still» abundant/common ✓ well-established technology ***OR*** easy for consumers to obtain ***OR*** commonly used ✓ easy to store ***OR*** easy to transport ***OR*** easy to extract ✓ produces energy at a reasonable rate ✓	*Accept "high potential energy" for M1.*	**2 max**

7.3-S25 Nov 2017 · Paper 3 · Q7 parts (e)

7.	**e**		$(–C_2H_3Cl–)_2\,(s) + 5O_2\,(g) \rightarrow 4CO_2\,(g) + 2H_2O\,(l) + 2HCl\,(g)$ correct species in reactants and products ✓ balanced ✓	*Accept "$(–C_2H_3Cl–)_2\,(s) + 5.5O_2\,(g) \rightarrow 4CO_2\,(g) + 3H_2O\,(l) + Cl_2\,(g)$".* *Award M2 only if M1 correct.*	**2**

7.3-S26 Nov 2017 · Paper 3 · Q16 parts (a), (b), (c)

16.	**a**		$M_r\,(C_8H_{18}) = 114.26$ ***AND*** $\Delta H_c^{\ominus} = –5470$ **«**kJ $mol^{–1}$**»** ✓ **«**specific energy $= \frac{5470\text{ kJ}}{0.11426\text{ kg}}$**=»** 4.79×10^4/47873/47900 **«**kJ $kg^{–1}$**»** ✓	*Award **[2]** for correct final answer.* *Accept "48×10^3 **«**kJ $kg^{–1}$**»**" **OR** "47.9×10^3 **«**kJ $kg^{–1}$**»**".*	**2**
16.	**b**		wood is less useful because it requires **«**about three times**»** more mass for same energy ✓	*Accept "octane is more useful because it has higher specific energy".*	**1**
16.	**c**		*Any one of:* wind ✓ tidal/wave ✓ hydro-electric ✓ solar ✓ thermal/geothermal ✓ plant oil ✓	*Accept "biofuel/biodiesel/«bio»ethanol", but **not** just "water" or "fuel cells".*	**1 max**

7.3-S27 Nov 2017 · Paper 3 · Q17 parts (c)

17.	**c**		*Any three of:* incoming solar radiation is short wavelength/high frequency/high energy/UV ✓ radiated/emitted as long wavelength/low frequency/low energy/IR **«**radiation**»** ✓ energy/IR **«**radiation**»** absorbed by **«**bonds in**»** greenhouse gases ✓ energy radiated/emitted as IR **«**radiation**»** some of which returns back to Earth ✓	*Do **not** accept "reflected" **OR** "bounced" **OR** "trapped".*	**3 max**

7.3-S28 Nov 2017 · Paper 3 · Q20 parts (a), (b)

20.	**a**		*Negative electrode (anode):* $CH_3OH\,(aq) + H_2O\,(l) \rightarrow CO_2\,(g) + 6H^+\,(aq) + 6e^-$ ✓ *Positive electrode (cathode):* $O_2\,(g) + 4H^+\,(aq) + 4e^- \rightarrow 2H_2O\,(l)$ ✓ *Overall equation:* $2CH_3OH\,(aq) + 3O_2\,(g) \rightarrow 2CO_2\,(g) + 4H_2O\,(l)$ ✓	*Accept any whole or fractional coefficients in balanced equations.* *Award **[1 max]** for correct half-equations at wrong electrodes for M1 and M2.*	**3**
20.	**b**		*Advantage:* *Any one of:* liquid methanol is easier to transport/store than gaseous hydrogen ***OR*** hydrogen is explosive ✓ longer membrane life «as it operates in aqueous environment» ✓ methanol has greater energy density than hydrogen ✓ *Disadvantage:* *Any one of:* lower voltage ✓ lower power per unit mass «of the cell» ✓ lower efficiency ✓ toxic/can be mistaken for ethanol ✓ lower specific energy ✓	*Ignore any cost references throughout.* *Accept "CO_2/greenhouse gas produced" **OR** "requires a more highly efficient catalyst".* *Do **not** award marks for converse statements for the advantage and disadvantage.*	**2**

7.3-S29 May 2017 TZ1 · Paper 2 · Q4 parts (a), (c)(ii) · also 9.2

4.	**a**		$2NiS\ (s) + 3O_2\ (g) \rightarrow 2NiO\ (s) + 2SO_2\ (g)$ ✓		**1**

4.	**c**	**ii**	$\Delta H^{\ominus}$ **«**$= -633.0 - 4 \times (-110.5)$**»** $= -191$ **«**kJ**»** ✓		**1**

7.3-S30 May 2017 TZ1 · Paper 3 · Q20 parts (c)

20.	**c**		Specific energy **«**$= \frac{12000\,kJ\,mol^{-1}}{299\,g\,mol^{-1}}$**»** $= 40.1$ **«**kJ g^{-1}**»** ✓ Energy density **«**$= 40.1\ kJ\ g^{-1} \times 0.850\ g\ cm^{-3}$**»** $= 34.1$ **«**kJ cm^{-3}**»** ✓	*Award [1] if both are in terms of a unit other than kJ (such as J or MJ).*	**2**

7.3-S31 May 2017 TZ2 · Paper 3 · Q17 parts (a), (b)(i), (b)(ii), (c)(i), (c)(ii), (c)(iii)

17. **a**

Energy source	Advantage	Disadvantage
Biofuels	low carbon footprint ***OR*** sustainable/renewable ***OR*** lower emissions of CO «for biodiesel/ethanol». ***OR*** economic security/availability in countries without crude oil ✓	lower energy content/specific energy ***OR*** high cost (only if a specific example if given *eg*, growing corn for ethanol *etc*.) ***OR*** use agricultural resources/fertilizers/pesticides/water ***OR*** biodiesel has high viscosity/clogs fuel injectors ***OR*** less suitable in low temperatures ***OR*** increased NO_x emissions for biodiesel ***OR*** greenhouse gases/CO_2 «still/also» produced ✓
Fossil fuels	higher energy content/specific energy ***OR*** low cost ***OR*** readily accessible ✓	linked to climate change/global warming/increased release of greenhouse gases ***OR*** not sustainable/renewable ***OR*** greater pollution possibilities ✓

*Do **not** award marks for converse statements for advantage and disadvantage.*

*Points related to greenhouse gases should be counted **only once** for the entire question.*

Biofuels:

Accept "«close to» carbon neutral", "produce less greenhouse gases/CO_2" as an advantage.

Accept "engines have to be modified if biodiesel used" as a disadvantage.

Fossil Fuels:

Accept specific pollution examples (eg, oil spills, toxic substances released when burning crude oil, etc.) as a disadvantage.

4

17.	**b**	**i**	«specific energy =» 142 ✓ kJ g^{-1} ✓	*Accept other correct values with the correct corresponding units.* *M2 can be scored independently.*	**2**
17.	**b**	**ii**	large volumes of hydrogen required ***OR*** hydrogen has lower energy density ✓ not easily transportable «form» as it is a gas ***OR*** heavy containers required to carry ***AND*** compress/regulate «hydrogen» ***OR*** high energy/cost required to compress hydrogen to transportable liquid form ***OR*** atmospheric pollution may be generated during production of hydrogen ***OR*** hydrogen fuel cells do not work at very low temperatures ***OR*** highly flammable when compressed/difficult to extinguish fires ***OR*** leaks not easy to detect ***OR*** high cost of production ***OR*** lack of filling stations/availability to consumer «in many countries» ✓	*Accept "«hydrogen combustion contributes to» knocking in engines"* ***OR*** *"modified engine required" for M2.* *Accept "explosive" but not "more dangerous" for M2.*	**2**

17.	**c**	**i**	*Anode (negative electrode):* $CH_3OH\,(aq) + H_2O\,(l) \rightarrow 6H^+\,(aq) + 6e^- + CO_2\,(g)$ ✓ *Cathode (positive electrode):* $\frac{3}{2}\,O_2\,(g) + 6H^+\,(aq) + 6e^- \rightarrow 3H_2O\,(l)$ ✓	*Award **[1 max]** for correct equations at wrong electrode.* *Accept "e" for "e^-".* *Accept "$O_2\,(g) + 4H^+\,(aq) + 4e^- \rightarrow 2H_2O\,(l)$".*	**2**
17.	**c**	**ii**	allows H^+/ions pass through/diffuse/move «from anode to cathode but not electrons or small molecules» ✓	*Accept "acts as a salt bridge".*	**1**
17.	**c**	**iii**	H^+/ions pass through/diffuse/move from anode/negative electrode «through membrane» to cathode/positive electrode ✓ H^+/ions used to reduce oxygen at cathode/positive electrode ✓	*Oxygen must be mentioned for M2.*	**2**

7.3-S32 May 2017 TZ2 · Paper 3 · Q18 parts (a)(i)

18.	**a**	**i**	<table><tr><th>Type of radiation</th><th>Region</th></tr><tr><td>Incoming radiation from sun</td><td>A «and B»</td></tr><tr><td>Re-radiated from Earth's surface</td><td>B</td></tr><tr><td>Absorbed by CO_2 in the atmosphere</td><td>B ✓</td></tr></table>	*Accept "B" alone for incoming radiation from sun.* *All three correct answers necessary for mark.*	**1**

7.3-S33 Nov 2016 · Paper 3 · Q6 parts (c)(i)

6.	**c**	**i**	due to stability of plastics / strong covalent bonds ***OR*** low volatility preventing good mixing with oxygen **«**gas**»** ***OR*** lack of/insufficient oxygen ***OR*** plastics are often parts of devices with non-combustible components **«**which mechanically prevent the combustion of plastic components**»** ***OR*** PVC already partly oxidized **«**because some C–H bonds are replaced with C–Cl bonds**»**, so it cannot produce enough heat for complete combustion ***OR*** many industrial/household materials contain additives that reduce their flammability/act as flame retardants ✓		**1**

7.3-S34 Nov 2016 · Paper 3 · Q17 parts (a)(i), (a)(ii), (b)(i)

17.	**a**	**i**	**«**$\frac{1.58 \times 10^{7}\,\text{J}}{80.0\,\text{kg}} = \frac{15.8\,\text{MJ}}{80.0\,\text{kg}} =$**»** 1.98×10^{-1} **«**MJ kg^{-1}**»** ✓		**1**
17.	**a**	**ii**	gasoline releases more energy from a given mass of fuel ***OR*** gasoline has higher specific energy ✓	*Do not accept volume in place of mass as question refers to specific energy, not energy density.*	**1**
17.	**b**	**i**	**«**$\frac{15.8\,\text{MJ}}{34.3\,\text{MJ dm}^{-3}}$**»** $= 4.61 \times 10^{-1}$ **«**dm^3**»** ✓		**1**

7.3-S35 May 2016 · Paper 3 · Q1 parts (a), (b)(ii) · also 5.1

1.	**a**		*Ozone:* yes because it absorbs IR/is IR active ✓ *Oxygen:* no because it does not absorb IR/is IR inactive ✓	*Award **[1 max]** for stating "ozone/O_3 is a greenhouse gas but oxygen/O_2 is not".* *Award **[1 max]** for stating "ozone/O_3 absorbs IR/is IR active but oxygen/O_2 does not/is IR inactive".* *Accept "oxygen/O_2 is not a greenhouse gas because it absorbs UV".*	**2**

1.	**b**	**ii**	CCl_4 is symmetrical/dipoles of C–Cl bonds cancel out ***OR*** C–F bond more polar **«**than C–Cl bond**»** ✓ **«**vector**»** sum of bond polarities in CCl_3F non-zero/greater ***OR*** dipoles of **«**three**»** C–Cl bonds do not cancel the dipole of the C–F bond ✓	*Accept suitable diagrams with dipoles represented as vectors illustrating M1 and/or M2.* *Accept "fluorine/F more electronegative **«**than chlorine/Cl**»**" for M1.* *Accept converse statements throughout.* *Accept "dipoles will not cancel out in CCl_3F" for M2.*	**2**

7.3-S36 May 2016 · Paper 3 · Q15 parts (b)(i), (b)(ii)

15.	**b**	**i**	«$\frac{5470}{114.26}$ = » 47.9 «kJ g^{-1}» ✓		**1**

15.	**b**	**ii**	*Advantage:* ethanol does not produce particulates/has less incomplete combustion/CO/HCs/VOCs/is less polluting ***OR*** ethanol has high octane rating ***OR*** ethanol is renewable ***OR*** less environmental risks associated with spills for ethanol ***OR*** less carbon dioxide/CO_2 produced if renewable energy source used ***OR*** economic advantages for countries that cannot produce crude oil ✓ *Disadvantage:* reduces efficiency/lowers specific energy/lowers energy density ***OR*** ethanol is more volatile/evaporates easily «than octane or its isomers» ***OR*** land that could be used for food production used to produce crops for ethanol ***OR*** biodiversity can be affected/loss of habitats «due to energy crop plantations» ***OR*** phosphorus/nitrogen used in production has negative environmental effects ***OR*** modification of current engines «may be required» if ethanol used ✓	*Accept any valid advantage and disadvantage.* *Ignore any mention of cost.* *Ignore any mention of NO_x.* *Accept "if the fuel blend consists of nearly pure ethanol, engine is difficult to start in cold weather".* *Accept for disadvantage any engine-related problem affected by ethanol use (eg. effect on fuel pumps, incorrect fuel quantity indicators, older cars may not be suitable for ethanol use, etc.).*	**2**

7.3-S37 May 2016 · Paper 3 · Q20 parts (a), (b)

20.	**a**		*Negative electrode (anode):* $CH_3OH(aq) + H_2O(l) \rightarrow CO_2(g) + 6H^+(aq) + 6e^-$ ✓ *Positive electrode (cathode):* $O_2(g) + 4H^+(aq) + 4e^- \rightarrow 2H_2O(l)$ ✓	*Award **[1 max]** if correct half-equations are given but at the wrong electrodes.* *Accept e for e–.* *Accept any correct half-equation with fractional coefficients.*	**2**
20.	**b**		*Advantage:* produces continuous supply of electricity «on addition of more raw materials» ***OR*** does not need to be recharged ***OR*** less hazardous if broken/exposed to the environment ***OR*** weighs less for same energy output/has higher energy density/has higher specific energy than lead-acid battery ✓ *Disadvantage:* «more» expensive ***OR*** needs constant supply of fuel ***OR*** methanol/ethanol fuel cells difficult to use in cold weather ***OR*** methanol/ethanol fuel cells produce carbon dioxide ***OR*** storage/transport of gases/hydrogen a problem in hydrogen fuel cell ***OR*** does not produce high current ***OR*** potentially explosive/hydrogen is flammable ✓	*Do **not** accept "water is non-polluting".* *Do **not** accept "weighs less" without reference to energy output/power/capacity etc.* *Do **not** accept "fuel cells are prone to poisoning by impurities in fuel".*	**2**

7.3-S38 Nov 2015 · Paper 2 · Q3 parts (a), (b) · also 9.3

3. (a) $C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(g)$; ***[1]***

Ignore state symbols.

(b) *bonds broken:* 2(C–C)/694 + 8(C–H)/3304 + 5(O=O)/2490 / 6488 (kJ);
bonds made: 6(C=O)/4476 + 8(O–H)/3712 / 8188 (kJ);
(6488 – 8188 =) – 1700 (kJ mol^{-1}); ***[3]***
Ignore signs in M1 and M2.
*Award **[3]** for the correct final answer.*
*Award **[2]** for +1700 (kJ mol^{-1}).*
Accept values from 2016 data booklet to give 6494 (kJ) for M1, 8528 (kJ) for M2, and –2034 (kJ) for M3.

7.3-S39 Nov 2015 · Paper 3 · Q23 parts (b)(i)

(b) (i) (as the concentration/amount of air/oxygen increases) CO/carbon monoxide emissions decrease;
as CO/carbon monoxide is oxidized/reacts with O_2/oxygen (to form carbon dioxide/CO_2) / as more complete combustion occurs (in engine); ***[2]***

7.3-S40 May 2015 TZ2 · Paper 3 · Q14 parts (a)

14. (a) increasing cost of oil (relative to other energy sources);
limited supply (of petroleum);
other sources of energy available / alternative energy sources;
(use as a raw material) reduces/delays greenhouse gas/global warming/climate change problems;
concerns about greenhouse gases/climate change causing changes in behaviour */ OWTTE;*
Do not accept just "greenhouse gases/climate change".
products from raw materials can be recycled / fuels cannot be recycled;
increasing demand as raw material from continued economic growth/demand for wider variety of products;
more profit to be made (by using as raw material);
reduced availability of other sources of hydrocarbons; **[2 max]**
Accept political factors, such as “conflicts disrupting production”.

7.3-S41 May 2015 TZ2 · Paper 3 · Q22 parts (b)(i)

(b) (i) allows (higher frequency) radiation from sun/incoming radiation to pass unhindered;

long wavelength/infrared/IR radiation radiated/emitted by Earth;
*Do **not** accept reflected.*

(some IR radiation is) absorbed by greenhouse gases;
*Do **not** accept "trapped / blocked" or statements that refer to absorption of incoming IR radiation.*

causes (increased) vibration in bonds;
Accept "causes (increased) bond stretching/bending/deformations".

emits/re-radiates IR radiation (some of which returns to Earth); **[3 max]**
*Do **not** accept "heat/IR radiation reflects/bounces".*

7.3-S42 Nov 2014 · Paper 3 · Q21 parts (a)

21. (a) incoming solar radiation is short wavelength/high frequency/higher energy/UV;
(re-)radiated/emitted (by the Earth's surface) as long wavelength/low frequency/low energy/IR radiation;
energy absorbed by (bonds in) greenhouse gases / molecules vibrate when IR radiation absorbed;
energy (re-)radiated/(re-)emitted as IR radiation some of which returns back to Earth; ***[3 max]***
Do not accept reflected, bounced or trapped.

7.3-S43 May 2014 TZ1 · Paper 3 · Q9 parts (a), (b)(iii), (b)(iv), (b)(v)

9. (a) chemical (energy) to electrical (energy); ***[1]***

(iii) catalyst/electrocatalyst / speeds up reaction but not consumed in reaction itself / provides surface for (initial) decomposition of molecules into atoms; ***[1]***

(iv) allows flow of ions/H^+/protons (from anode/negative electrode to cathode/positive electrode) / prevents reactants mixing/moving from one compartment to another / salt bridge / prevents flow of electrons through membrane / *OWTTE*; ***[1]***

(v) storage/transport difficulties of gases / potentially explosive/hydrogen is flammable / needs constant supply of fuel / can contain heavy metal(s) / often operated at high temperature / low power to mass ratio / susceptible to poisoning due to impurities in fuel / *OWTTE*; ***[1]***
Allow a named gas (hydrogen or oxygen) for storage/transport difficulties.
Allow problems related to corrosion.
Accept answers based on ethanol and methanol fuel cells (but needs to be stated) such as difficult to use in cold weather/less clean product (CO_2) formed.

7.3-S44 May 2014 TZ1 · Paper 3 · Q15 parts (c)(ii)

(ii) $2CH_4(g)+3O_2(g)\rightarrow 2CO(g)+4H_2O(g)$ / $CH_4(g)+1.5O_2(g)\rightarrow CO(g)+2H_2O(g)$; ***[1]***
State symbols must be included.
Allow $H_2O(l)$.
Accept any other pollutant, such as methanal: $CH_4(g) + O_2(g) \rightarrow HCHO(g) + H_2O(g)$.

7.3-S45 May 2012 TZ1 · Paper 2 · Q2 parts (b)

(b) *Local pollutant:*
carbon monoxide/CO / volatile organics/VOCs / nitrogen oxide/NO / (unburnt) hydrocarbons;
Do not accept methane/CH_4, ethane/C_2H_6, propane/C_3H_8 or butane/C_4H_{10}.

Global pollutant:
nitrogen oxide/NO / carbon dioxide/CO_2; ***[2]***
Accept nitrogen dioxide/NO_2 / NO_x for both local or global pollutant.
Accept other widely used names for NO such as nitric oxide/nitrogen monoxide/nitrogen(II) oxide or nitrogen(IV) oxide for NO_2.

7.3-S46 May 2012 TZ1 · Paper 3 · QC2 parts (a), (c)

C2. (a) *Similarity:*
both turn chemical energy into electrical energy / use chemical reactions to produce electricity/electrical energy / *OWTTE*;

*Difference **[1 max]**:*
rechargeable batteries have reversible reactions **but** fuel cells do not;
fuel cells consume fuel **but** rechargeable batteries do not require (external) fuel;
rechargeable batteries can be recharged by electricity **but** fuel cells cannot; ***[2 max]***

(c) *Positive electrode:*
$O_2(g) + 4H^+(aq) + 4e^- \rightarrow 2H_2O(l)$;

Negative electrode:
$H_2(g) \rightarrow 2H^+(aq) + 2e^-$; ***[2]***

7.3-S47 Nov 2011 · Paper 3 · QC1 parts (a)(i)

C1. (a) (i) $C(s) + O_2(g) \rightarrow CO_2(g)$; ***[1]***
Ignore state symbols.

7.3-S48 Nov 2011 · Paper 3 · QC2 parts (d)

(d) CO_2 is a greenhouse gas / causes climate change / global warming / formation of soot/particulates / melting of polar ice caps / rising sea levels / *OWTTE*; ***[1]***
Accept CO produced is toxic/poisonous.

7.3-S49 Nov 2011 · Paper 3 · QE2 parts (a)

E2. (a) incoming solar radiation is short(er) wavelength/high(er) frequency/high(er) energy radiation / UV radiation;
(re-)radiated/emitted (by the Earth's surface) as long(er) wavelength/low(er) frequency/ low(er) energy/IR radiation;
the energy is absorbed in bonds in greenhouse gases / the molecules vibrate when IR radiation is absorbed;
the energy is (re-)radiated/(re-)emitted as IR radiation; ***[3 max]***

7.3-S50 May 2011 TZ1 · Paper 3 · QE3 parts (a)

E3. (a) incomplete combustion / air/fuel ratio is low; ***[1]***

7.3-S51 Nov 2010 · Paper 3 · QC1

C1. *Arguments for*: ***[2 max]***
high energy content / high enthalpy of combustion;
shortage of alternatives;
alternatives are expensive / oil relatively cheap;
well-established technology;
easy to store;
easy to transport;
produces energy at a reasonable rate;

Arguments against: ***[2 max]***
chemical feedstock of limited supply/*OWTTE;*
non-renewable;
combustion causes global warming / greenhouse gases;
combustion produces acidic gases; ***[4]***
Apply / OWTTE throughout.

7.3-S52 May 2010 TZ2 · Paper 3 · QE1 parts (a), (b)

E1. (a) methane / CH_4 ;

nitrous oxide / N_2O ;

ozone / O_3 ;

chlorofluorocarbons/CFCs / specific CFC / halocarbon / hydrochlorofluorocarbons/HCFCs;

sulfur hexafluoride / SF_6 ; ***[2 max]***

(b) greenhouse gases/named gas(es) are transparent to/allow to pass through short(er)-wavelength/high(er)-energy radiation/UV light from Sun / *OWTTE*;

greenhouse gases/named gas(es) absorb long(er)-wavelength/IR radiation from Earth / *OWTTE*;

(part of) absorbed radiation is re-radiated to Earth / *OWTTE*; ***[3]***

Do not allow trap or reflected.

LESSON 8.1

Thermochemical Equations

Answers for 26 multiple-choice and 9 structured questions

Part A - Multiple choice answers

8.1-M1	May 2025 TZ2 · P1A · Q18	A
8.1-M2	May 2025 TZ3 · P1A · Q20	D
8.1-M3	May 2024 TZ1 · P1 · Q1	C
8.1-M4	Nov 2023 TZ1 · P1 · Q14	D
8.1-M5	Nov 2023 TZ1 · P1 · Q16	A
8.1-M6	Nov 2022 · P1 · Q2	B
8.1-M7	May 2021 TZ2 · P1 · Q14	D
8.1-M8	May 2021 TZ2 · P1 · Q15	C
8.1-M9	Nov 2020 · P1 · Q15	B
8.1-M10	May 2019 TZ2 · P1 · Q15	C
8.1-M11	Nov 2018 · P1 · Q15	C
8.1-M12	May 2018 TZ1 · P1 · Q15	D
8.1-M13	May 2018 TZ2 · P1 · Q14	A
8.1-M14	May 2017 TZ1 · P1 · Q14	C
8.1-M15	May 2017 TZ2 · P1 · Q14	A
8.1-M16	May 2016 · P1 · Q1	B
8.1-M17	Nov 2015 · P1 · Q14	D
8.1-M18	May 2014 TZ1 · P1 · Q14	B
8.1-M19	May 2014 TZ2 · P1 · Q14	B
8.1-M20	Nov 2013 · P1 · Q15	B
8.1-M21	May 2013 TZ1 · P1 · Q14	A
8.1-M22	May 2013 TZ1 · P1 · Q15	B
8.1-M23	Nov 2011 · P1 · Q15	A
8.1-M24	May 2011 TZ1 · P1 · Q18	A
8.1-M25	May 2010 TZ1 · P1 · Q15	B
8.1-M26	May 2010 TZ2 · P1 · Q14	C

Part B - Structured questions markscheme

8.1-S1 May 2017 TZ2 · Paper 2 · Q5 parts (b)(iv)

5	**b**	**iv**	«$E_{a(rev)} = 226 + 132 =$» 358 «kJ» ✓	*Do not accept –358.*	**1**

8.1-S2 May 2016 · Paper 3 · Q11 parts (b)

11.	**b**		$n(C_6H_{12}O_6)\left(= \frac{15.0}{180.18}\right) = 0.0833$ «mol» ✓ «energy $= 0.0833 \times 2803 =$» 233 «kJ» ✓	*Award **[2]** for correct final answer.* *Accept –233 «kJ».*	**2**

8.1-S3 May 2015 TZ2 · Paper 2 · Q7 parts (a)(iii)

(iii) <u>minimum</u> energy needed (by reactants/colliding particles) to react/start/ initiate a reaction; **[1]**
Accept "energy difference between reactants and transition state".

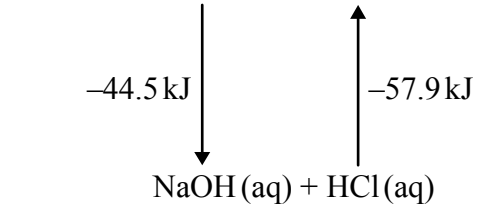
8.1-S4 Nov 2014 · Paper 2 · Q11 parts (e)(i), (e)(ii), (e)(iii), (e)(iv), (f)(ii) · also 2.1, 8.2, 9.1

(e) (i) heat transferred/absorbed/released/enthalpy/potential energy change when 1 mol/molar amounts of reactant(s) react (to form products) / *OWTTE*;
under standard conditions / at a pressure 100 kPa/101.3 kPa/1 atm **and** temperature 298 K/25 °C ; ***[2]***
*Award **[2]** for difference between standard enthalpies of products and standard enthalpies of reactants / $H^{\ominus}$ (products) – $H^{\ominus}$ (reactants).*
*Award **[2]** for difference between standard enthalpies of formation of products and standard enthalpies of formation of reactants / $\Sigma\Delta H_f^{\ominus}$ (products) – $\Sigma\Delta H_f^{\ominus}$ (reactants).*

(ii) $(1.00 \times 0.0500 =) 0.0500\,(\text{mol})$;
$(0.0500 \times 57.9 =) 2.90\,(\text{kJ})$; ***[2]***
Ignore any negative sign.
*Award **[2]** for correct final answer.*
*Award **[1 max]** for 2900 J.*

(iii) $\left(\frac{2.50}{40.00} =\right) 0.0625\,(\text{mol NaOH})$;
$0.0500 \times 4.18 \times 13.3 = 2.78\,(\text{kJ}) / 50.0 \times 4.18 \times 13.3 = 2780\,(\text{J})$;
$\left(\frac{2.78}{0.0625}\right) = -44.5\,(\text{kJ mol}^{-1})$; ***[3]***
*Award **[3]** for correct final answer.*
Negative sign is necessary for M3.
Award M2 and M3 if 52.5 g is used to obtain an enthalpy change of –46.7 (kJ mol-1).

(iv) $-44.5 - 57.9$ / correct Hess's Law cycle (as below) / correct manipulation of equations;



$-102.4\,kJ$; ***[2]***

*Award **[2]** for correct final answer.*

(ii) Fe^{2+}: $1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,3d^6$ /[Ar] $3d^6$ **and** Fe^{3+}: $1s^2\,2s^2\,2p^6\,3s^2\,3p^6\,3d^5$ /[Ar] $3d^5$;
half-full sub-level/$3d^5$ has extra stability;
less repulsion between electrons / electrons singly occupy orbitals / electrons do not have to pair with other electrons; ***[3]***

Accept converse points for Fe^{2+}.

8.1-S5 May 2013 TZ1 · Paper 2 · Q3 parts (a)

3. (a) minimum energy needed to react/start a reaction / energy difference between reactants and transition state; ***[1]***

8.1-S6 May 2013 TZ2 · Paper 2 · Q2 parts (a)

2. (a) minimum energy needed (by reactants/colliding particles) to react/start/initiate a reaction / for a successful collision; ***[1]***

Allow energy difference between reactants and transition state.

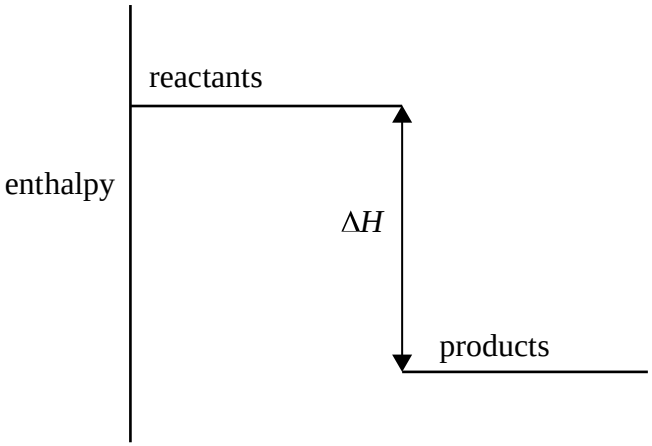
8.1-S7 Nov 2012 · Paper 2 · Q6 parts (b)(i)

(b) (i) minimum/least/smallest energy needed (by reactants/colliding particles) to react/start/initiate a reaction; ***[1]***
Allow energy difference between reactants and transition state.
Minimum/least/smallest required for the mark.

8.1-S8 May 2012 TZ2 · Paper 2 · Q8 parts (a)(i), (a)(ii), (a)(iii), (c) · also 8.2

8. (a) (i) enthalpy on *y*-axis;
Do not accept energy.

reactants higher than products;
ΔH labelled; ***[3]***



Accept appropriate formulas for reactants and products.
Arrow heads not needed.
57.6 is acceptable as an alternative to ΔH .

(ii) products are more stable as they are at a lower enthalpy level / energy has been given off by the reactants / reaction is exothermic / *OWTTE*; ***[1]***

(iii) $n(\text{NaOH}) = 0.125\ \text{mol}$;
change in heat energy = $(-57.6 \times 0.125) = -7.20$ (kJ) / heat released $= (57.6 \times 0.125) = 7.20\ (\text{kJ})$; ***[2]***

(c) $q = (mc\Delta T =) 100.0 \times 4.18 \times 3.50 / 1463\ \text{J} / 1460\ \text{J}$;

$n(\text{NH}_4\text{Cl}) = \frac{5.35}{53.5} / 0.100\ \text{mol}$;

$\Delta H = +14.6 / 14.6\ (\text{kJ mol}^{-1})$; ***[3]***

Accept $q = 105.35 \times 4.18 \times 3.50 / 1541\ J$.

Accept $\Delta H = +15.4 / 15.4\ (kJ\ mol^{-1})$.

8.1-S9 Nov 2011 · Paper 2 · Q8 parts (a), (c)(i), (e)(i) · also 3.1

8. (a) minimum energy needed (by reactants/colliding particles) to react/start/initiate a reaction; ***[1]***

Allow energy difference between reactants and transition state.

(c) (i) (system) absorbs/takes in heat from surroundings / *OWTTE*; ***[1]***

Allow standard enthalpy change/ $\Delta H^{\ominus}$ *positive.*

Allow bond breaking more energetic then bond formation / OWTTE.

Absorbs/takes in heat alone not sufficient for mark.

(e) (i) dative (covalent)/coordinate;

carbon monoxide/CO / hydronium (ion)/ H_3O^+ / ammonium (ion)/ NH_4^+ / aluminium chloride/Al_2Cl_6 / any relevant transition metal complex (*e.g.* $[\text{Ni}(\text{NH}_3)_6]^{2+}$); ***[2]***

Accept $AlCl_3$.

LESSON 8.2

Calorimetry

Answers for 26 multiple-choice and 20 structured questions

Part A - Multiple choice answers

8.2-M1	May 2025 TZ1 · P1A · Q23	C
8.2-M2	May 2025 TZ2 · P1A · Q19	B
8.2-M3	May 2025 TZ3 · P1A · Q21	A
8.2-M4	Nov 2024 · P1 · Q18	D
8.2-M5	May 2024 TZ1 · P1 · Q14	A
8.2-M6	May 2024 TZ2 · P1 · Q16	A
8.2-M7	May 2023 TZ1 · P1 · Q38	C
8.2-M8	May 2023 TZ2 · P1 · Q15	C
8.2-M9	May 2022 TZ1 · P1 · Q14	B
8.2-M10	May 2022 TZ2 · P1 · Q13	A
8.2-M11	May 2021 TZ1 · P1 · Q14	A
8.2-M12	May 2021 TZ1 · P1 · Q38	A
8.2-M13	Nov 2019 · P1 · Q14	C
8.2-M14	Nov 2019 · P1 · Q15	D
8.2-M15	May 2019 TZ1 · P1 · Q14	A
8.2-M16	May 2018 TZ2 · P1 · Q15	B
8.2-M17	May 2017 TZ1 · P1 · Q38	C
8.2-M18	Nov 2016 · P1 · Q17	C
8.2-M19	May 2015 TZ1 · P1 · Q14	A
8.2-M20	May 2015 TZ1 · P1 · Q15	C
8.2-M21	May 2014 TZ2 · P1 · Q19	A
8.2-M22	May 2013 TZ2 · P1 · Q14	A
8.2-M23	Nov 2012 · P1 · Q14	D
8.2-M24	May 2012 TZ1 · P1 · Q14	D
8.2-M25	May 2012 TZ1 · P1 · Q15	D
8.2-M26	Nov 2010 · P1 · Q14	A

Part B - Structured questions markscheme

8.2-S1 May 2024 TZ1 · Paper 2 · Q2 parts (a)(i), (b)(i), (b)(ii), (b)(iii), (c), (d)(ii), (d)(iii), (g)(iii) · also 1.2, 5.2, 7.3

2.	**(a)**	**(i)**	400.00 * 4.18 * 21.0 = 35 112 = 35 100 J ✓	*Accept 35.1 kJ*	**1**

2.	**(b)**	**(i)**	«36.03 + 8.08 + 16.00 = 60.11» 1.80/60.11 = 0.0299 ✓ –35100/0.0299 = –1200 kJ mol^{-1} ✓ *Assumption: (any one from)* - propan-1-ol only ingredient to release heat - gel is pure propan-1-ol - no heat loss - all heat transferred to water - no heat absorbed by copper - complete combustion ✓	*Accept –1000 to –1004 kJ mol^{-1} if 30 000 used.* *M2 can only be awarded for exothermic answer*	**3**
2.	**(b)**	**(ii)**	« $\frac{100(-1200 - -2021)}{-2021}$ = – » 40.6% ✓	*Ignore negative signs* *Accept 50% if 30 kJ used in 2bi*	**1**

(continues...)

2.	**(b)**	**(iii)**	heat loss ***AND*** insulate the calorimeter ✓	*Accept heat loss **AND** decrease distance between calorimeter and flame.* *Accept heat loss **AND** use screens to prevent drafts / OWTTE.* *Accept heat loss and use a closed system / food calorimeter* *Accept heat loss and use a calorimeter with a low heat capacity.* *Do not accept Styrofoam / coffee cup calorimeter* *Do not accept "add a lid"*	**1**
2.	**(c)**		ALTERNATIVE 1 «1g propan-1-ol contains $\frac{1}{60.11}$ = 0.017 mol » « 0.017(–2021) =» –33.62 «kJ» ✓ «1g e1thanol contains $\frac{1}{46.08}$ = 0.022 mol » « 0.022(–1367) =» –29.72 «kJ» ✓ second gel/ethanol less energy ✓	*Accept positive values for M1, M2* *Do not award M3 without M1 and M2.*	**3**
2.	**(d)**	**(iii)**	London/dispersion forces ✓	*Accept instantaneous / induced dipoles.* *Do not accept van der Waals' forces*	**1**
2.	**(g)**	**(iii)**	H_2O ✓	*Accept water*	**1**

8.2-S2 May 2023 TZ1 · Paper 2 · Q6 parts (a)(i), (a)(iii), (b)(i), (b)(ii), (b)(v) · also 5.1, 9.2

6.	**(a)**	**(i)**	«$\Delta H°rxn$ = $\sum\Delta H°$f (Products) – $\sum\Delta H°$f (Reactants) =» -395.8 - (-296.8)» = –99.0«kJ mol^{-1}» ✓		**1**
6.	**(a)**	**(iii)**	significant/large/0.8 difference in electronegativity/oxygen more electronegative ✓ oxygen «dipole partially» negative/sulfur «dipole partially» positive ***OR*** oxygen more negative/higher electron density «around it than sulfur»✓	Accept suitable diagram showing the O–S dipole.	**2**
6.	**(b)**	**(i)**	«q = -mcΔT = 50.00g x 4.18J $K^{-1}g^{-1}$ x (35.0-20.0)°C =» -3140.0 «J» ✓ «(3140/0.1)/1000 =» -31.4 «kJ mol^{-1}» ✓	*Award **[1 max]** for +31.4 kJ mol^{-1}* *Award **[2]** for correct final answer.*	**2**
6.	**(b)**	**(ii)**	Source of systematic error: heat loss «to the surroundings» ✓ Improvement: insulate reaction apparatus/put a lid on the beaker ***OR*** use a bomb/calibrated calorimeter ***OR*** use of windbreak around the dish/apparatus ✓		**2**
6.	**(b)**	**(v)**	« -297 kJ mol^{-1} - -31.4kJ mol^{-1}/-297 kJ mol^{-1}x 100 =» 89.4 «%» ✓ alternate: « -297 kJ mol^{-1} - -50.0 kJ mol^{-1}/-297 kJ mol^{-1}x 100 =» 83.2 «%» ✓		**1**

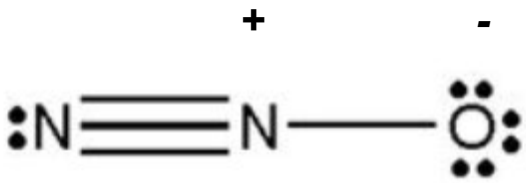
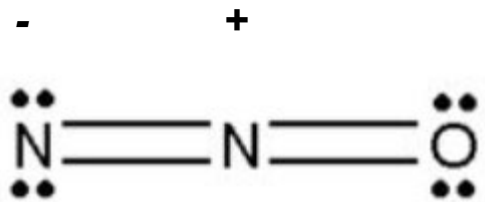
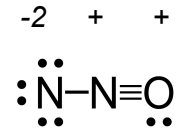
8.2-S3 May 2023 TZ2 · Paper 2 · Q4 parts (c)

4.	**(c)**		«heat energy = 11.98 g x 0.902 J g^{-1} K^{-1} x 22.0 K =» 238 «J» ✓		**1**

8.2-S4 Nov 2022 · Paper 2 · Q1 parts (d)(i), (d)(ii), (d)(iv), (e), (f)(i), (f)(iii), (f)(v) · also 1.2, 3.1, 7.2, 9.2

1.	**(d)**	**(i)**	«$q = mc\Delta T$ = 25.32 g × 4.18 J $g^{-1}K^{-1}$ × (25.2°C – 0.8°C) =» 2580 «J» ✓	*Do **not** accept a negative value.*	**1**
1.	**(d)**	**(ii)**	«$2.58 \times 10^3\ J \times \frac{1\ kJ}{1000\ J} \times \frac{1\ mol}{25.69\ kJ}$ =» 0.100 «mol» ✓ «0.100 mol × 80.06 g mol^{-1} =» 8.01 «g» ✓	*Award **[2]** for the correct final answer.* *Accept range of 8.0 – 8.1 g.* *If 3.11 x 10^3 J used then answer is 9.69 g.*	**2**
1.	**(d)**	**(iv)**	«% error = $\left\lvert\frac{9.50\ g - 8.01\ g}{9.50\ g}\right\rvert \times 100\ \%$ =» 15.7«%» ✓	*Accept range 14.7 – 15.8«%».* *If 6.55 g used then answer is 31.1«%».*	**1**
1.	**(e)**		*Anode:* $H_2O(l) \rightarrow 1/2O_2(g) + 2H^+(aq) + 2e^-$ ✓ *Cathode:* $H^+(aq) + e^- \rightarrow 1/2H_2(g)$ ✓	*Do **not** accept other equations.*	**2**
1.	**(f)**	**(i)**	$NH_4NO_3(s) \rightarrow N_2O(g) + 2H_2O(l)$ ✓		**1**

1.	**(f)**	**(iii)**	2 x – 285.8 **«**kJ mol^{-1}**»** ✓ **«**1mol (82 kJ mol^{-1}) + 2mol (– 285.8 kJ mol^{-1}) – 1mol (– 366 kJ mol^{-1}) =**»** –124 **«**kJ**»** ✓	*Award **[2]** for correct final answer.*	**2**

1.	**(f)**	**(v)**	***Alternative 1*** *Lewis structure:*  ✓ +/1+/+1 on central N atom ***AND*** -/1-/-1 on O atom ✓ ***Alternative 2*** *Lewis structure:*  ✓ +/1+/+1 on central N atom ***AND*** -/1-/-1 on other N atom ✓ *Shape:* linear ✓	*Accept*  . *Formal charges are not needed for M1.* *Allow ECF for both formal charge and shape.* *Only award M3 if the shape corresponds to that expected for the Lewis structure given.*	**3**

8.2-S5 Nov 2021 · Paper 2 · Q7 parts (a)

7.	**a**		«$q = mc\Delta T = 20.0\,\text{g} \times 4.18\,\text{J g}^{-1}\,\text{°C}^{-1} \times 57.3\,\text{°C}$ =» 4790 «J» ✓ «$\Delta H_c = -\dfrac{4790\ \text{J}}{\dfrac{1000}{8.75 \times 10^{-4}\ \text{mol}}}$ =» -5470 «kJ mol^{-1}» ✓	*Award **[2]** for correct final answer.* *Accept answers in the range –5470 to -5480 «kJ mol^{-1}».* *Accept correct answer in any units, e.g. – 5.47 «MJ mol^{-1}» or 5.47 x 10^{6}«J mol^{-1}».*	**2**

8.2-S6 May 2019 TZ1 · Paper 3 · Q2 parts (a)(i), (a)(ii), (b)(i), (b)(ii), (b)(iii), (b)(iv)

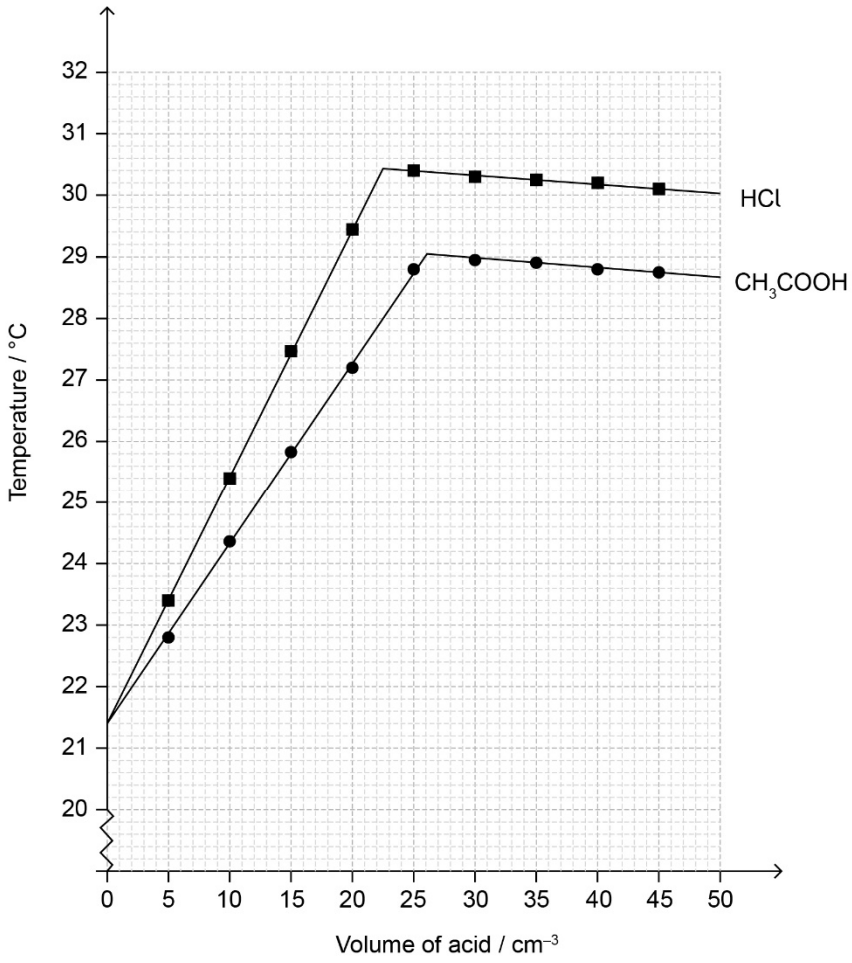
2.	**a**	**i**	100 **«**s**»** ✓	*Accept 90 to 100 s.*	**1**
2.	**a**	**ii**	highest recorded temperature ***OR*** when rate of heat production equals rate of heat loss ✓	*Accept "maximum temperature".* *Accept "completion/end-point of reaction".*	**1**
2.	**b**	**i**	*Maximum temperature:* 73 **«**°C**»** ✓ *Assumption:* «temperature reached if» reaction instantaneous ***OR*** «temperature reached if reaction occurred» without heat loss ✓	*Accept "rate of heat loss is constant"* ***OR*** *"rate of temperature decrease is constant".*	**2**
2.	**b**	**ii**	*Any one of:* copper(II) sulfate ***AND*** mass/amount of zinc is independent variable/being changed ***OR*** copper(II) sulfate ***AND*** with zinc in excess there is no independent variable **«**as amount of copper(II) sulfate is fixed**»** ✓ copper(II) sulfate ***AND*** having excess zinc will not yield different results in each trial ✓ zinc ***AND*** results can be used to see if amount of zinc affects temperature rise «so this can be allowed for» ✓ zinc ***AND*** reduces variables/keeps the amount reacting constant ✓		**1 max**

<table>
<tr>
<td>2.</td>
<td>b</td>
<td>iii</td>
<td>
<table>
<tr><th>Value</th><th>Assumption</th></tr>
<tr><td>$m = 25.00\,\text{g}$</td><td>density of solution is $1.000\,\text{g}\,\text{cm}^{-3}$/same as water
OR
$25.00\,\text{cm}^3$ solution has a mass of 25.00 g
OR
mass of zinc/reactant is negligible
OR
mass of contents was 25.00 g ✓</td></tr>
<tr><td>$c = 4.18\,\text{J}\,\text{g}^{-1}\,\text{K}^{-1}$</td><td>specific heat of solution is $4.18\,\text{J}\,\text{g}^{-1}\,\text{K}^{-1}$ /same as water
OR
zinc/calorimeter/beaker/thermometer absorbs no heat ✓</td></tr>
</table>
</td>
<td>Accept “copper(II) sulfate/zinc sulfate” for “solution”.</td>
<td>2</td>
</tr>
<tr>
<td>2.</td>
<td>b</td>
<td>iv</td>
<td>lower/less exothermic/less negative AND heat loss/some heat not accounted for
OR
lower/less exothermic/less negative AND mass of reaction mixture greater than 25.00 g
OR
greater/more exothermic/more negative AND specific heat of solution less than water ✓</td>
<td>Accept “temperature is lower” instead of “heat loss”.
Accept “similar to theoretical value AND heat losses have been compensated for”.
Accept “greater/more exothermic/more negative AND linear extrapolation overestimates heat loss”.</td>
<td>1</td>
</tr>
</table>

8.2-S7 May 2019 TZ1 · Paper 3 · Q8 parts (c)

<table>
<tr>
<td>8.</td>
<td>c</td>
<td></td>
<td>$q = \text{«}mc\Delta T = 975\,\text{g} \times 4.18\,\text{J}\,\text{g}^{-1}\,\text{K}^{-1} \times 15.0\,\text{K} =\text{»}\ 61\,100\ \text{«J»}$ / 61.1 «kJ» ✓
«heat per gram $= \dfrac{61.1\,\text{kJ}}{3.49\,\text{g}} =\text{»}\ 17.5\ \text{«kJ}\,\text{g}^{-1}\text{»}$ ✓</td>
<td>Award [2] for correct final answer.</td>
<td>2</td>
</tr>
</table>

8.2-S8 Nov 2017 · Paper 2 · Q1 parts (a), (b), (c), (d)(i), (d)(ii) · also 1.2

1.	**a**		 21.4 °C ✓	*Accept values in the range of 21.2 to 21.6* °C. *Accept two different values for the two solutions from within range.*	**1**

1.	**b**		*HCl:* 30.4 **«**°C**»** ✓ *CH_3COOH:* 29.0 **«**°C**»** ✓	*Accept range 30.2 to 30.6* °C. *Accept range 28.8 to 29.2* °C.	**2**
1.	**c**		***ALTERNATIVE 1*** **«**volume CH_3COOH **=»** 26.0 **«**cm^3**»** ✓ **«**$[CH_3COOH] = 0.995 \text{ mol dm}^{-3} \times \frac{50.0 \text{ cm}^3}{26.0 \text{ cm}^3}$ **=»** 1.91 **«**mol dm^{-3}**»** ✓ ***ALTERNATIVE 2*** **«***n*(NaOH) = 0.995 mol dm^{-3} × 0.0500 dm^3 **=»** 0.04975 **«**mol**»** ✓ **«**$[CH_3COOH] = \frac{0.04975}{0.0260} \text{ dm}^3$ **=»** 1.91 **«**mol dm^{-3}**»** ✓	*Accept values of volume in range 25.5 to 26.5 cm^3.* *Award **[2]** for correct final answer.*	**2**
1.	**d**	**i**	**«**total volume = 50.0 + 26.0 **=»** 76.0 cm^3 ***AND*** **«**temperature change 29.0 − 21.4 **=»** 7.6 **«**°C**»** ✓ **«***q* = 0.0760 kg × 4.18 kJ kg^{-1} K^{-1} × 7.6 K **=»** 2.4 **«**kJ**»** ✓	*Award **[2]** for correct final answer.*	**2**
1.	**d**	**ii**	**«***n*(NaOH) = 0.995 mol dm^{-3} × 0.0500 dm^3 **=»** 0.04975 **«**mol**»** ***OR*** **«***n*(CH_3COOH) = 1.91 mol dm^{-3} × 0.0260 dm^3 **=»** 0.04966 **«**mol**»** ✓ **«**$\Delta H = -\frac{2.4 \text{ kJ}}{0.04975 \text{ mol}}$ **=»** −48 / −49 **«**kJ mol^{-1}**»** ✓	*Award **[2]** for correct final answer.* *Negative sign is required for M2.*	**2**

8.2-S9 May 2017 TZ1 · Paper 3 · Q2 parts (a), (b)

2.	**a**		more **«**moles/amount**»** of acid have been added/reacted ***OR*** more of the limiting reagent is present ***OR*** more «of the exothermic» reaction has occurred ✓		**1**
2.	**b**		no more reaction/same energy released ***AND*** cold/colder/cooler liquid added ***OR*** no more reaction/same energy released ***AND*** greater total volume of liquid ✓	*Accept "no more reaction/same energy released **AND** greater heat loss **«**to the surroundings in mixture D**»**".*	**1**

8.2-S10 May 2017 TZ1 · Paper 3 · Q3

3.			volume **«**found by extrapolation of the two best fit lines**»** required to give the highest temperature ***OR*** extrapolate **«**two best fit**»** lines to the point where they meet ✓	*Accept "where lines through the points meet".* *Accept "at maximum temperature".* *Accept "at 35 cm^3 of HCl".*	**1**

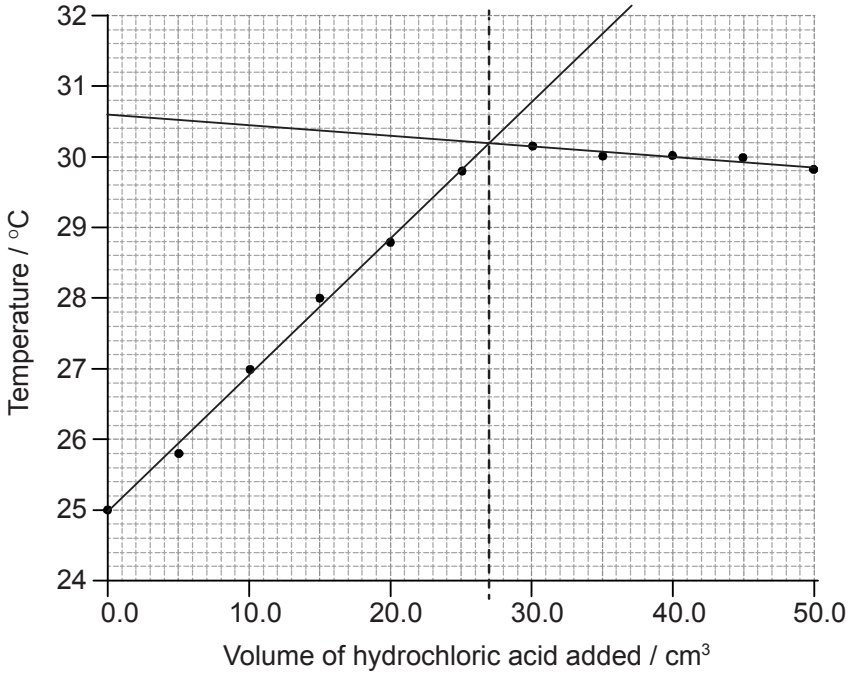
8.2-S11 May 2017 TZ1 · Paper 3 · Q5 parts (a), (b), (c), (d), (f)

5.	**a**		heat change/evolved can be calculated from the **«**maximum**»** temperature increase and the mass of solution ***OR*** $q = mc\Delta T$ ✓ heat **«**evolved**»** gives the number of moles **«**of both acid and alkali present when neutralisation occurs**»** ***OR*** $n = \frac{q}{\Delta H_{neut}}$ ✓ volume **«**of acid and the volume of alkali required to just neutralise each other**»** can be used to calculate the concentration**«**s of both**»** ***OR*** $[NaOH] = \frac{n}{V}$ ✓		**2 max**
5.	**b**		smaller temperature increase/ΔT ***OR*** heat released would **«**appear to**»** be less✓ amount of substance/n **«**calculated is **»** smaller ✓		**2**
5.	**c**		using **«**expanded**»** polystyrene cup ***OR*** insulating beaker ***OR*** putting a lid on beaker ✓	*Accept any other reasonable suggestion.* *Do not accept "calorimeter" by itself.*	**1**
5.	**d**		**«**specific**»** heat capacity of the beaker/container/thermometer is ignored ***OR*** density of the solutions is assumed as 1.00 g cm^{-3}/same as water ***OR*** specific heat capacity of the solutions is assumed as 4.18 J g^{-1} K^{-1}/same as water ✓	*Accept "reaction goes to completion".* *Accept "reaction is conducted under standard conditions".* *Accept "no evaporation occurs".* *Accept any other relevant valid assumption.* *Do **not** accept "heat is not released from other reactions".*	**1**

5.	**f**		temperature rise would be too small **«**to be accurately measured**»** ✓	*Accept "heat released would be too small «to be accurately measured»".*	**1**

8.2-S12 May 2015 TZ2 · Paper 2 · Q1 parts (a)(i), (a)(ii), (b)(i), (b)(ii), (b)(iii), (b)(iv) · also 1.2

(a) (i)



drawing best-fit straight lines to show volume;
There should be approximately the same number of points above and below for both lines.

27.0 (cm^3); **[2]**
Accept any value in the range 26.0 to 28.0 (cm^3) if consistent with student's annotation on the graph.
Accept ECF for volumes in the range 27.0–30.0 cm^3 if it corresponds to maximum temperature of line drawn.
Volume should be given to one decimal place.

(ii) $([HCl]) = \frac{1.00 \times 0.0250}{0.0270}$;
$= 0.926\,\text{mol}\,\text{dm}^{-3}$; **[2]**
Volume of 26.0 gives [HCl] = 0.962 $moldm^{-3}$. Volume of 28.0 gives [HCl] = 0.893 $moldm^{-3}$.
*Award **[2]** for correct final answer with units.*
*Award **[1 max]** for correct concentration without units.*
Accept M, mol L^{-1}, mol/dm^3 as units.

(b) (i) $(30.2 - 25.0 =)(+)5.2\,(^{\circ}\text{C/K})$; **[1]**

Any a*ccepted value must be consistent with student's annotation on the graph but do not accept* $\Delta T < 5.1$.

Accept (+)5.6 $(^{\circ}\text{C/K})$ *(ie, taking into account heat loss and using T when volume = 0.0 cm*^{3}*).*

(ii) $Q = (mc \times \Delta T = (25.0 + 27.0) \times 4.18 \times 5.2 = 1130.272\,\text{J} =) 1.13\,(\text{kJ})$;

$n = (1.00 \times 0.0250 =) 0.0250\,(\text{mol})$;

$\Delta H = (-\frac{Q}{n} = -45210.88\,\text{J mol}^{-1} =) -45\,(\text{kJ mol}^{-1})$; **[3]**

*Award **[3]** for correct final answer.*
*Award **[2]** for +45 (kJ mol* $^{-1}$*).*
Apply ECF for M3 even if both m and ΔT *are incorrect in M1.*
Accept use of c = 4.2 Jg^{-1}*K*$^{-1}$*.*

(iii) $\left(\left| \frac{-45 - (-58)}{(-58)} \right| \times 100 = \right) 22\ (\%)$; **[1]**

Answer must be given to two significant figures.
Ignore sign.

(iv) heat losses;
better (thermal) insulation / using a polystyrene cup / putting a lid on the beaker; **[2]**

Accept other suitable methods for better thermal insulation but do not accept just "use a calorimeter" without reference to insulation.

8.2-S13 May 2014 TZ1 · Paper 2 · Q1 parts (a)(i), (a)(ii), (b)(i), (b)(ii), (c), (d)(ii), (d)(iii), (d)(iv) · also 1.2, 3.1, 4.1, 4.2, 9.1

1. (a) (i) $n(\mathrm{MgSO_4}) = \left(\frac{3.01}{120.37} =\right) 0.0250\,(\mathrm{mol})$; ***[1]***

(ii) energy released $= 50.0 \times 4.18 \times 9.7 = 2027\,(\mathrm{J}) / 2.027\,(\mathrm{kJ})$;

$\Delta H_1 = -81\,(\mathrm{kJ\,mol^{-1}})$; ***[2]***

*Award **[2]** for correct answer.*
*Award **[2]** if 53.01 is used giving an answer of –86 (kJ mol^{-1}).*
*Award **[1 max]** for +81/81/+86/86 (kJ mol^{-1}).*
*Award **[1 max]** for –81000/–86000 if units are stated as J mol^{-1}.*
Allow answers to 3 significant figures.

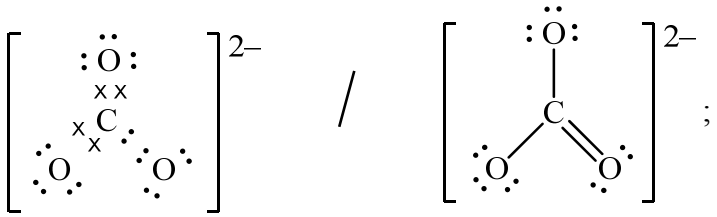
(b) (i) $\Delta H\,(= \Delta H_1 - \Delta H_2) = -99\,(\mathrm{kJ\,mol^{-1}})$; ***[1]***

*Award **[1]** if –86 is used giving an answer of –104 (kJ mol^{-1}).*

(ii) $\frac{(103 - 99)}{103} \times 100 = 3.9\%$; ***[1]***

Accept answer of 2.9% if –100 used but only if a value for (b)(i) is not present.
*Award **[1]** if –104 is used giving an answer of 1.0% .*
Accept correct answers which are not to 1 decimal place.

(c) $\mathrm{MgSO_4}$ not completely anhydrous / *OWTTE*;
$\mathrm{MgSO_4}$ is impure;
heat loss to the atmosphere/surroundings;
specific heat capacity of solution is taken as that of pure water;
experiment was done once only so it is not scientific;
density of solution is taken to be 1 g cm^{-3};
mass of $7\mathrm{H_2O}$ ignored in calculation;
uncertainty of thermometer is high so temperature change is unreliable;
literature values are carried out under standard conditions, but this experiment is not;
all solid not dissolved; ***[2 max]***

(ii)  / ;

Accept crosses, lines or dots as electron pairs.
Accept any correct resonance structure.
*Award **[0]** if structure is drawn without brackets and charge.*
*Award **[0]** if lone pairs not shown on O atoms.*

shape: trigonal/triangular planar;
bond angle: 120° ;
Accept answers trigonal/triangular planar and 120° if M1 incorrect, but no other answers should be given credit. ***[3]***

(iii) (pi/π) electrons are delocalized/spread over more than two nuclei / charge spread (equally) over all three oxygens; ***[1]***

(iv) sp^2 ; ***[1]***

8.2-S14 May 2013 TZ1 · Paper 2 · Q8 parts (a)(i), (a)(ii), (b)(i) · also 9.2

8. (a) (i) $(q = mc\Delta T =)\ 0.0500 \times 4.18 \times 21.3 = 4.45$ (kJ);
Do not accept m = 0.05023 kg.

$(n\ methanol =)\ \dfrac{0.230}{32.05} = 7.18 \times 10^{-3}\,(\text{mol})$;

$\Delta H = \dfrac{4.45}{7.18 \times 10^{-3}}$;

$\Delta H = -6.20 \times 10^{2}\ \text{kJ mol}^{-1}$; ***[4]***

Accept integer values of molar mass.
Final answer must have negative sign and correct units.
*Award **[4]** for correct final answer with correct units.*

(ii) less heat is liberated than theoretically/–726 kJ mol^{-1};
probably due to heat loss/incomplete combustion;
determine heat capacity of calorimeter and take heat absorbed by calorimeter into account / any suitable insulation method / measure temperature with time and extrapolation of graph to compensate heat loss / *OWTTE*; ***[3]***
If the value calculated in (a) (i) is more exothermic than theoretically, allow ECF for M1 and for improvement if consistent.

(b) (i) $CH_3OH + \frac{3}{2}O_2 \rightarrow CO_2 + 2H_2O \qquad \Delta H^{\ominus}_c = -726\ (\text{kJ mol}^{-1})$

$CO + \frac{1}{2}O_2 \rightarrow CO_2 \qquad \Delta H^{\ominus}_c = -283\ (\text{kJ mol}^{-1})$

$H_2 + \frac{1}{2}O_2 \rightarrow H_2O \qquad \Delta H^{\ominus}_c = -286\ (\text{kJ mol}^{-1})$

*Award **[1 max]** for three correct values.*
Mark can be implicit in calculations.

$(\Delta H^{\ominus}_R =)\ 2(-286) + (-283) - (-726)$;

-129 (kJ mol^{-1}); ***[3]***

*Award **[3]** for correct final answer.*
*Award **[2 max]** for +129 (kJ mol^{-1}).*

8.2-S15 Nov 2011 · Paper 3 · QB1

B1. $q = [m \times c \times \Delta T]_{\text{water}} + [C \times \Delta T]_{\text{glass}} = [100.00 \times 4.18 \times 9.4] + [90.2 \times 9.4]\,(\text{J})$;

$= [3929.2 + 847.88] / 4777.08\,(\text{J})$;

$\text{energy value} = \dfrac{[4777.08 \times 100]}{5.00} = 95541.6\,(\text{J}) = 95.5\,(\text{kJ})$; ***[3]***

Apply ECF as follows:

M1 *Likely errors are to omit the $C\Delta T$ expression or use 5.00 or 105.00 instead of 100.00 for the mass. Either or both of these loses M1 but M2 and M3 can be scored by ECF – examples follow:*

M2	*(omitting $C\Delta T$)*	*(using 5.00 g)*	*(omitting $C\Delta T$ and using 5.00 g)*
	3929.2 (J)	*1044.34 (J)*	*196.46 (J)*
M3	$\dfrac{3929.2 \times 100}{5.00} = 78.6\,(kJ)$	$\dfrac{1044.34 \times 100}{5.00} = 20.9\,(kJ)$	$\dfrac{196.46 \times 100}{5.00} = 3.93\,(kJ)$

8.2-S16 May 2011 TZ1 · Paper 2 · Q1 parts (a), (b)(i), (b)(ii), (b)(iii), (c)(i), (c)(ii) · also 9.2

1. (a) $(\Delta H^{\ominus} = \Sigma \Delta H_{\text{f products}}^{\ominus} - \Sigma \Delta H_{\text{f reactants}}^{\ominus})$

$\Delta H_c^{\ominus} = (2 \times -242 + -394) - (-239)$ (kJ mol^{-1});

$\Delta H_c^{\ominus} = -639$ (kJ mol^{-1}); ***[2]***

*Award **[2]** for correct final answer.*
Do not award M2 if M1 incorrect.

(b) (i) m(methanol) = (80.557 – 80.034) = 0.523 (g);

$n(\text{methanol}) = \left(\frac{0.523 \text{ g}}{32.05 \text{ g mol}^{-1}}\right) = 0.0163$ (mol); ***[2]***

*Award **[2]** for correct final answer.*

(ii) ΔT = (26.4 – 21.5) = 4.9 (K);

$q = (mc\Delta T =)\ 20.000 \times 4.18 \times 4.9$ (J) / $20.000 \times 4.18 \times 4.9 \times 10^{-3}$ (kJ);

0.41 (kJ); ***[3]***

*Award **[3]** for correct final answer.*

(iii) $\Delta H_c^{\ominus} = -\frac{0.41 \text{ (kJ)}}{0.0163 \text{ (mol)}} / -25153$ (J mol^{-1});

= –25 (kJ mol^{-1}); ***[2]***

*Award **[2]** for correct final answer.*
*Award **[1]** for (+)25 (kJ mol^{-1}).*

(c) (i) not at standard conditions/1 atm **and** 298 K/25 °C / ΔH_{vap} for water; ***[1]***

(ii) not all heat produced transferred to water / heat lost to surroundings/environment / *OWTTE* / incomplete combustion (of methanol) / water forms as $H_2O(l)$ instead of $H_2O(g)$; ***[1]***

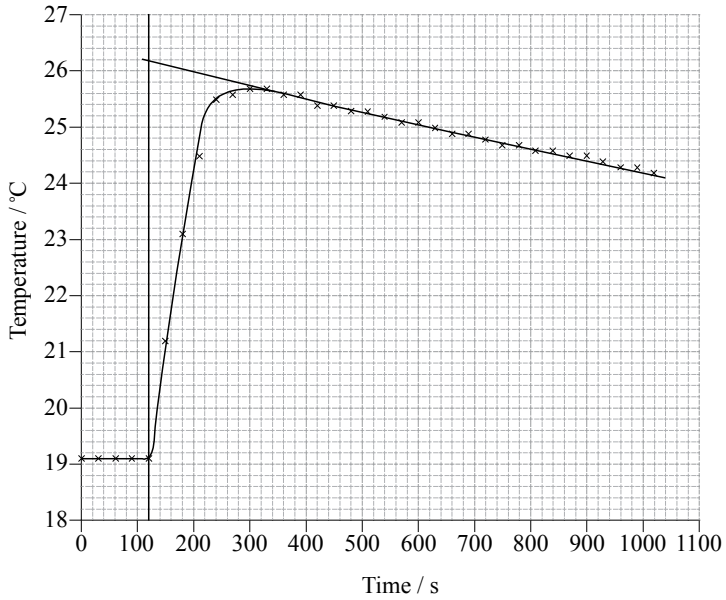
Do not allow just "heat lost".

8.2-S17 May 2011 TZ2 · Paper 2 · Q1 parts (a)(i), (a)(ii), (a)(iii), (a)(iv), (b)(i), (b)(ii), (b)(iii), (c)(i), (c)(ii) · also 1.2, 9.1

(a) (i) $\text{amount} = \frac{3.99}{159.61} = 0.0250\,(\text{mol})$; ***[1]***

(ii) 26.1 (°C);
Accept answers between 26.0 and 26.2 (°C).

temperature rise $= 26.1 - 19.1 = 7.0\,(^\circ\text{C})$; ***[2]***
Accept answers between 6.9 °C and 7.1 (°C).
*Award **[2]** for the correct final answer.*
No ECF if both initial and final temperatures incorrect.



(iii) $\text{heat change} = \frac{50.0}{1000} \times 4.18 \times 7.0\,(\text{kJ}) / 50.0 \times 4.18 \times 7.0\,(\text{J})$;
Accept 53.99 instead of 50.0 for mass.

$= 1.5\,(\text{kJ})$; ***[2]***
Allow 1.6 (kJ) if mass of 53.99 is used.
Ignore sign.

(iv) $\Delta H_1 = -\frac{1.5}{0.0250} = -60\,(\text{kJ mol}^{-1})$; ***[1]***
Value must be negative to award mark.
Accept answers in range –58.0 to –60.0.
Allow –63 (kJ mol^{-1}) if 53.99 g is used in (iii).

(b) (i) (amount of $CuSO_4 \bullet 5H_2O = \frac{6.24}{249.71}$ =) 0.0250 (mol);
(amount of H_2O in 0.0250 mol of $CuSO_4 \bullet 5H_2O = 5 \times 0.0250$ =) 0.125 (mol); ***[2]***

(ii) $(50.0 \times 4.18 \times 1.10 =) 230\ (J)$;
$\left(\frac{229.9}{(1000 \times 0.0250)} = \right) + 9.20\ (kJ)$; ***[2]***
Accept mass of 47.75 or 53.99 instead of 50.00 giving answers of +.8.78 or +9.9.
Do not penalize missing + sign but penalize – sign unless charge already penalized in (a) (iv).

(iii) $(\Delta H_x = \Delta H_1 - \Delta H_2 = -58.4 - (+9.20) =) -67.6\ (kJ\ mol^{-1})$; ***[1]***

(c) (i) $\frac{[-78.0 - (-67.6)]}{-78.0} \times 100 = 13.3\%$; ***[1]***
If 70.0 kJ mol^{-1} is used accept 10.3%.

(ii) the anhydrous copper(II) sulfate had already absorbed some water from the air / *OWTTE*;
the value would be less exothermic/less negative than expected as the temperature increase would be lower / less heat will be evolved when the anhydrous salt is dissolved in water / *OWTTE*; ***[2]***
Do not accept less <u>without</u> a reason.

8.2-S18 Nov 2010 · Paper 2 · Q1 parts (a), (b)(i), (b)(ii), (c), (d) · also 1.2

1. (a) all heat is transferred to water/copper sulfate solution / no heat loss;
specific heat capacity of zinc is zero/negligible / no heat is absorbed by the zinc;
density of water/solution $= 1.0$ / density of solution = density of water;
heat capacity of cup is zero / no heat is absorbed by the cup;
specific heat capacity of solution = specific heat capacity of water;
temperature uniform throughout solution; ***[2 max]***
*Award **[1]** each for any two.*
Accept energy instead of heat.

(b) (i) $T_{\text{final}} = 73.0\,(^{\circ}\text{C})$;
Allow in the range 72 to 74 (°C).

$\Delta T = 48.2\,(^{\circ}\text{C})$; ***[2]***
Allow in the range 47 to 49 (°C).
*Award **[2]** for correct final answer*
Allow ECF if T_{final} or $T_{initial}$ correct.

(ii) $10.1\,(\text{kJ})$; ***[1]***
Allow in the range 9.9 to 10.2 (kJ).

(c) $\left(n_{\text{Zn}} = n_{\text{CuSO}_4} = \dfrac{1.00 \times 50.0}{1000}\right) = 0.0500\,(\text{mol})$; ***[1]***

(d) $-201\,(\text{kJ mol}^{-1})$; ***[1]***
Allow in the range -197 to -206 (kJ mol^{-1}).
Value must be negative to award mark.

8.2-S19 Nov 2010 · Paper 3 · QB1 parts (a)

B1. (a) $\Delta T = 7.35\,(\text{K}/^{\circ}\text{C})$;
$q\,(= mc\Delta T = 200.10\ \text{g} \times 4.18\ \text{J g}^{-1}\ \text{K}^{-1} \times 7.35\ \text{K}) = 6.15 \times 10^{3}\ \text{J}$ (per 0.85 g of glucose heated);
energy value $= 7.2 \times 10^{3}\,(\text{J g}^{-1})$; ***[3]***
*Award **[3]** for correct final answer.*

8.2-S20 May 2010 TZ2 · Paper 3 · QB1

B1. $\Delta T = (29.0 - 20.5) / 8.5\,(^{\circ}\text{C})$;

energy (released by dry bread) $= 600 \times 4.18 \times 8.5 = 21318 / 2.1 \times 10^{4}\,(\text{J})$;

energy (released by 100 g of dried bread) $= 21318 \times 50 = 1065900 / 1.1 \times 10^{6}\,(\text{J})$;

energy released $= \dfrac{1065900}{1000} = 1.1 \times 10^{3}\,(\text{kJ per } 100\text{ g})$; ***[4]***

*Award **[4]** for correct final answer.*

LESSON 9.1

Hess's Law

Answers for 16 multiple-choice and 2 structured questions

Part A - Multiple choice answers

9.1-M1	Nov 2024 · P1 · Q16	D
9.1-M2	Nov 2023 TZ1 · P1 · Q15	C
9.1-M3	May 2023 TZ1 · P1 · Q14	B
9.1-M4	May 2019 TZ1 · P1 · Q15	D
9.1-M5	Nov 2018 · P1 · Q14	A
9.1-M6	May 2018 TZ1 · P1 · Q14	A
9.1-M7	May 2017 TZ1 · P1 · Q15	A
9.1-M8	May 2015 TZ2 · P1 · Q14	C
9.1-M9	Nov 2014 · P1 · Q14	A
9.1-M10	May 2014 TZ1 · P1 · Q15	D
9.1-M11	Nov 2013 · P1 · Q16	A
9.1-M12	May 2013 TZ2 · P1 · Q15	A
9.1-M13	Nov 2012 · P1 · Q15	B
9.1-M14	May 2011 TZ1 · P1 · Q17	A
9.1-M15	Nov 2010 · P1 · Q15	C
9.1-M16	May 2010 TZ2 · P1 · Q15	B

Part B - Structured questions markscheme

9.1-S1 Nov 2018 · Paper 2 · Q7 parts (b), (c), (d) · also 9.2

7.	**b**		«(−286 kJ) + (−1411 kJ) =» −1697 «kJ» ✓		**1**
7.	**c**		«−1697 kJ + 1561 kJ =» −136 «kJ» ***OR*** «$\Delta H^{\ominus} = \Delta H_f^{\ominus}$ (products) − $\Delta H_f^{\ominus}$ (reactants) = −84 kJ − 52 kJ =» −136 «kJ» ✓		**1**

7.	**d**		*Accurate:* no approximations were made in the cycle ***OR*** values are specific to the compounds ***OR*** Hess's law is a statement of conservation of energy ***OR*** method is based on a law ***OR*** data in table has small uncertainties ✓ *Approximate:* values were experimentally determined/had uncertainties ***OR*** each value has been determined to only three/four significant figures ***OR*** different sources have **«**slightly**»** different values for enthalpy of combustion ***OR*** law is valid until disproved ***OR*** law of conservation of energy is now conservation of mass-energy ***OR*** small difference between two quite large terms **«**leads to high percentage uncertainty**»** ✓		**2**

9.1-S2 May 2017 TZ2 · Paper 2 · Q9 parts (a)(i), (a)(ii) · also 8.2

9.	**a**	**i**	$\Delta H = 177.0 - \frac{189.2}{2} - 285.5$ **«**kJ**»** ✓ **«**ΔH **=»** − 203.1 **«**kJ**»** ✓	*Accept other methods for correct manipulation of the three equations.* *Award **[2]** for correct final answer.*	**2**
	a	**ii**	203.1 **«**kJ**»** = 0.850 **«**kg**»** × 4.18 **«**kJ kg^{-1} K^{-1}**»** × ΔT **«**K**»** ***OR*** **«**ΔT **=»** 57.2 **«**K**»** ✓ **«**T_{final} = (57.2 + 21.8)°C **=»** 79.0**«**°C**»** / 352.0 **«**K**»**✓ *If 200.0 kJ was used:* 200.0 **«**kJ**»** = 0.850 **«**kg**»** × 4.18 **«**kJ kg^{-1} K^{-1}**»** × ΔT **«**K**»** ***OR*** **«**ΔT **=»** 56.3**«**K**»** ✓ **«**T_{final} = (56.3 + 21.8)°C **=»** 78.1 **«**°C**»** / 351.1 **«**K**»**✓	*Award **[2]** for correct final answer.* *Accept two, three or four significant figures.* *Unit, if specified, must be consistent with value stated.*	**2**

LESSON 9.2

Enthalpies of Formation & Combustion

Answers for 21 multiple-choice and 13 structured questions

Part A - Multiple choice answers

9.2-M1	May 2025 TZ1 · P1A · Q22	D
9.2-M2	May 2025 TZ1 · P1A · Q24	B
9.2-M3	May 2024 TZ2 · P1 · Q17	D
9.2-M4	May 2023 TZ2 · P1 · Q14	D
9.2-M5	Nov 2022 · P1 · Q14	D
9.2-M6	May 2022 TZ1 · P1 · Q15	C
9.2-M7	Nov 2021 · P1 · Q15	B
9.2-M8	May 2021 TZ1 · P1 · Q15	D
9.2-M9	Nov 2020 · P1 · Q14	C
9.2-M10	Nov 2019 · P1 · Q16	C
9.2-M11	Nov 2017 · P1 · Q16	B
9.2-M12	Nov 2016 · P1 · Q16	C
9.2-M13	May 2016 · P1 · Q16	A
9.2-M14	Nov 2015 · P1 · Q15	D
9.2-M15	May 2015 TZ1 · P1 · Q16	D
9.2-M16	May 2015 TZ2 · P1 · Q16	D
9.2-M17	May 2013 TZ2 · P1 · Q16	B
9.2-M18	Nov 2012 · P1 · Q17	D
9.2-M19	Nov 2011 · P1 · Q16	C
9.2-M20	Nov 2010 · P1 · Q16	A
9.2-M21	May 2010 TZ1 · P1 · Q16	C

Part B - Structured questions markscheme

9.2-S1 May 2025 TZ3 · Paper 2 · Q5 parts (a)(i), (c)(i), (c)(ii) · also 3.1, 4.2

5.	**(a)**	**(i)**	«ΔH^{θ}_{c} values: C_2H_2 = −1301, H_2 = −286, C_2H_6 = −1561 kJ mol^{-1}» «$\sum (\Delta H^{\theta}_{c\ reactants})$ =» (–1301) + 2(–286) / –1873 «kJ mol^{-1}» ***AND*** «$\sum (\Delta H^{\theta}_{c\ products})$ =» −1561 «kJ mol^{-1}» ✓ «$\Delta H^{\theta} = \sum (\Delta H^{\theta}_{c\ reactants}) - \sum (\Delta H^{\theta}_{c\ products})$ = –1873 – (–1561)» ΔH^{θ} = –312 «kJ mol^{-1}» ✓	*Award **[2]** for correct final answer.*	**2**
5.	**(c)**	**(i)**	H – C ≡ C – H ✓	*Accept any combination of dots or crosses to represent electrons, or lines to represent electron pairs.*	**1**
5.	**(c)**	**(ii)**	sp ✓		**1**

9.2-S2 Nov 2023 TZ1 · Paper 2 · Q4 parts (a)(i), (b) · also 3.1

4	**a**	**i**	«$\Sigma\ \Delta H^{\ominus}_{f}$ (reactants) =» +88.7 + 2(-241.8) / -394.9 «kJ mol^{-1}» ***AND*** «$\Sigma\ \Delta H^{\ominus}_{f}$ (products) =» -393.5 + 2(-20.6) / -434.7 «kJ mol^{-1}» ✓ $\Delta H^{\ominus}$ «= -434.7 – (-394.9)» = -39.8 «kJ mol^{-1}» ✓	*Award **[2]** for correct final answer.* *Award **[1]** for +48.2 «kJ mol^{-1}» (obtained using -285.8 kJ mol^{-1} for $H_2O(l)$).*	**2**
4	**b**		*Molecular geometry CS_2:* linear ***AND*** *Molecular geometry H_2S:* bent/V-shaped ✓ *Reason for difference:* «central atom in» H_2S has «two» lone/non-bonding «electron» pairs ***OR*** CS_2 has two ***AND*** H_2S has four electron domains/negative charge centres «around central atom» ✓	*Do **not** accept diagrams for M1 or M2.* *Accept central atom sp hybridized in CS_2 **AND** sp^3 hybridized in H_2S for M2.*	**2**

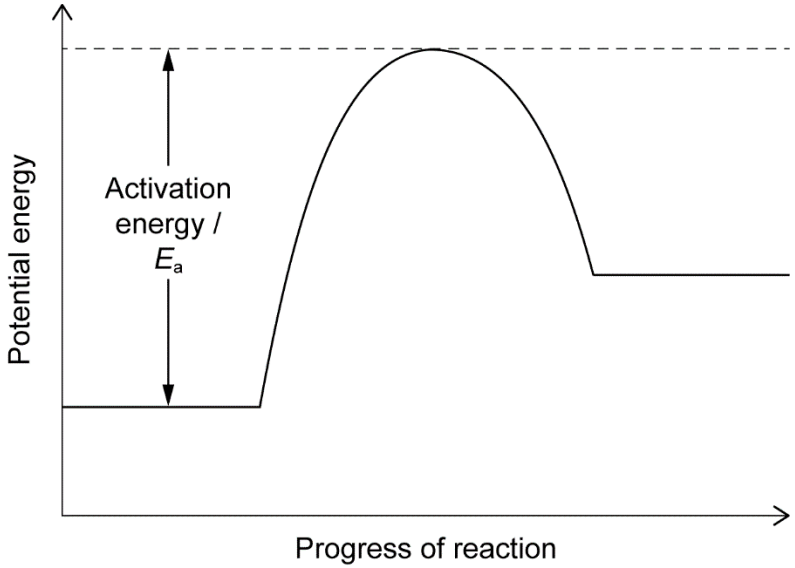
9.2-S3 May 2023 TZ1 · Paper 3 · Q7 parts (a)(ii), (d)(i) · also 7.3

7.	**(a)**	**(ii)**	«n=10.5g/180.18g mol^{-1}=» 0.0583 «mol» ✓ «0.0583 mol x 2810 kJ mol^{-1} =» 164 «kJ» ✓	*Award **[2]** for correct final answer.* *Award **[1 max]** for –164 «kJ».*	**2**

7.	**(d)**	**(i)**	$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ ✓ $C_{57}H_{110}O_6 + 81.5O_2 \rightarrow 57CO_2 + 55H_2O$ ***OR*** $2C_{57}H_{110}O_6 + 163O_2 \rightarrow 114CO_2 + 110H_2O$ ✓		**2**

9.2-S4 May 2021 TZ2 · Paper 2 · Q1 parts (b)(i), (b)(iv), (d)(ii) · also 1.2, 8.1

1.	**b**	**i**	«ΔH =» (–635 «kJ» – 393.5 «kJ») – (–1207 «kJ») ✓ «ΔH = + » 179 «kJ» ✓	*Award **[2]** for correct final answer.* *Award **[1 max]** for –179 kJ.* *Ignore an extra step to determine total enthalpy change in kJ: 179 kJ mol^{-1} x 5.55 mol = 993 kJ.* *Award **[2]** for an answer in the range 990 – 993 « kJ».*	**2**

1.	**b**	**iv**	endothermic sketch ✓ x-axis labelled “extent of reaction/progress of reaction/reaction coordinate/reaction pathway” ***AND*** y-axis labelled “potential energy/ energy/enthalpy✓ activation energy/E_a ✓  	*Do **not** accept “time” for x-axis.*	**3**

1.	**d**	**ii**	«$\frac{2.85}{3.30} \times 100$ =» 86.4 «%» ✓	*Accept answers in the range 86.1-86.4 «%».* *Accept "71.3 %" for using the incorrect given value of 4.00 g.*	**1**

9.2-S5 May 2018 TZ2 · Paper 3 · Q7 parts (e)

7.	**e**		«$\frac{29.9\,\text{g}}{150.15\,\text{g mol}^{-1}}$ =» 0.199 «mol» ✓ «0.199 mol × 205.9 kJ mol^{-1} =» 41.0 «kJ» ✓	*Ignore significant figures in M1.* *Award **[2]** for correct final answer.* *Award **[1 max]** for incorrect significant figures in final answer.*	**2**

9.2-S6 Nov 2017 · Paper 2 · Q5 parts (a)

5.	**a**		$\Delta H^{\ominus}$= [–165.2 + 2(–296.9) + 2(–92.3)] – [–454.7 + 2(–245.7)] ✓ «$\Delta H^{\ominus}$=+»2.5 «kJ» ✓	*Award **[2]** for correct final answer.* *Award **[1]** for –2.5 «kJ».* *Do **not** accept ECF for M2 if more than one error in M1.*	**2**

9.2-S7 May 2017 TZ2 · Paper 2 · Q6 parts (b)

6	**b**		«$\Delta H^{\ominus}$ =» –82.0 «kJ» – 92.3 «kJ» – (–74.0 «kJ») ✓ «$\Delta H^{\ominus}$ =» –100.3 «kJ» ✓	*Award **[2]** for correct final answer.*	**2**

9.2-S8 May 2016 · Paper 2 · Q2 parts (a)(iii)

2.	**a**	**iii**	$\Delta H^{\ominus}$ = –220.1 – (–110.5)✓ $\Delta H^{\ominus}$ = –109.6 «kJ» ✓	*Award **[2]** for correct final answer.* *Award **[1]** for –330.6, or +109.6 «kJ».*	**2**

9.2-S9 May 2015 TZ2 · Paper 2 · Q3 parts (a)

(a) (–239.0 – [–110.5] =) – 128.5 (kJ mol^{-1}) ; **[1]**

9.2-S10 May 2014 TZ1 · Paper 2 · Q4 parts (b)

(b) $\Delta H^{\ominus} = \left(\Sigma \Delta H_f^{\ominus}(\text{products}) - \Sigma \Delta H_f^{\ominus}(\text{reactants}) = -127 - (110 + 0) = \right) - 237\,(\text{kJ mol}^{-1})$; ***[1]***

9.2-S11 May 2014 TZ2 · Paper 2 · Q7 parts (e)(iv), (e)(vi), (e)(vii) · also 5.2, 8.2

(iv) $\Delta H = \sum \Delta H_f(\text{products}) - \sum \Delta H_f(\text{reactants}) / -178 - (-68)$;

$-110\,(\text{kJ mol}^{-1})$; ***[2]***

*Award **[2]** for correct final answer.*

*Award **[1]** for* $+110\,(\text{kJ mol}^{-1})$.

(vi) energy required to heat water $\left(= m \times s \times \Delta T = 1 \times 4.18 \times (100 - 20)\right) = 334.4\,(\text{kJ})$;

Ignore sign of energy change.

amount required $= \dfrac{334.4}{4000} = 0.0836\,(\text{mol})$; ***[2]***

*Award **[2]** for correct final answer.*

(vii) D lower/hexane higher **and** van der Waals'/London/dispersion forces between molecules weaker in D/stronger in hexane;

Accept "intermolecular forces" instead of dispersion forces.

D is more branched / D has a lower surface area / D has a more spherical shape / does not pack as closely; ***[2]***

Accept opposite statements for hexane.

9.2-S12 Nov 2011 · Paper 2 · Q5 parts (b)(i), (b)(ii)

(b) (i) $\Delta H_f^{\ominus}$ of an element (in its most stable state) is zero (since formation of an element from itself is not a reaction) / *OWTTE*; ***[1]***

Do not allow an answer such as because they are elements.

(ii) $\Delta H^{\ominus} (= (1)(-20.6) - (1)(-53.1)) = 32.5\,(\text{kJ mol}^{-1}) / 32500\,(\text{J mol}^{-1})$; ***[1]***

Allow 32.5(kJ) or 3.25×10^4 *(J)*.

9.2-S13 May 2011 TZ2 · Paper 2 · Q6 parts (b)(i), (b)(ii)

(b) (i) by definition $\Delta H^{\ominus}_{f}$ of elements (in their standard states) is zero / no reaction involved / *OWTTE*; ***[1]***

(ii) $\Delta H = -104 - (+20.4)$;

$= -124.4\ (\text{kJ mol}^{-1})$; ***[2]***

*Award **[1 max]** for 124.4 (kJ mol^{-1}).*
*Award **[2]** for correct final answer.*

LESSON 9.3

Bond Enthalpies

Answers for 8 multiple-choice and 15 structured questions

Part A - Multiple choice answers

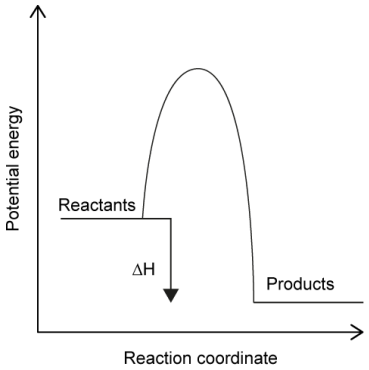
9.3-M1	May 2024 TZ1 · P1 · Q15	C
9.3-M2	May 2022 TZ2 · P1 · Q16	B
9.3-M3	Nov 2021 · P1 · Q14	D
9.3-M4	May 2019 TZ2 · P1 · Q14	C
9.3-M5	May 2016 · P1 · Q17	B
9.3-M6	May 2015 TZ2 · P1 · Q15	B
9.3-M7	Nov 2014 · P1 · Q15	C
9.3-M8	May 2011 TZ2 · P1 · Q17	A

Part B - Structured questions markscheme

9.3-S1 May 2025 TZ1 · Paper 2 · Q8 parts (c)(v), (f)(i), (f)(ii), (f)(iii) · also 5.2, 9.2

8.	**(c)**	**(v)**	stronger forces of attraction with increasing chain length / larger electron cloud /molar mass ✓	*forces for M1.* *Accept "more electrons" for "larger electron cloud" in M2.* *Accept increased surface area.*	**2**
8.	**(f)**	**(i)**	ΔH = «3196 – 3237 = » −41 «kJ » ✓ **ALTERNATIVE 2:** Bonds broken: 1 C=C + 1 O-H / 614 + 463 / 1077 «kJ» ✓ Bonds formed: 1 C-C + 1 C-O + 1 C-H / 346 + 358 + 414 / 1118 «kJ» ✓ ΔH = «1077 – 1118 = » −41 «kJ» ✓	*answer.* *Award [2 max] for +41* «kJ».	**3**
8.	**(f)**	**(ii)**	« $\Delta H^{\ominus} = \Sigma(\Delta H_f^{\ominus}\text{products}) - \Sigma(\Delta H_f^{\ominus}\text{reactants})$ » = −235 – (+52 + (−242)) ✓ = −45 «kJ» ✓	*Award [2] for correct final answer.* *Award **[1 max]** for +45* «kJ»	**2**
8.	**(f)**	**(iii)**	bond enthalpies are average values ***OR*** bond enthalpies vary «slightly» among compounds ✓		**1**

9.3-S2 May 2025 TZ2 · Paper 2 · Q3 parts (a)(iii), (a)(iv) · also 8.1

3	**(a)**	**(iv)**	 exothermic reaction profile ✓ Δ*H* correct ***AND*** labelled reactants and products ✓		**2**

9.3-S3 Nov 2023 TZ1 · Paper 2 · Q2 parts (c)(i), (c)(iii), (g) · also 6.1

2	**c**	**i**	***ALTERNATIVE 1*** «bond breaking» C=O + H-H / 804 + 436 / 1240 «kJ» ✓ «bond forming» C-H + C-O + O-H / 414 + 358 + 463 / 1235 «kJ» ✓ $\Delta H^{\ominus}$ «= 1240 – 1235» = «+»5 «kJ mol^{-1}» ✓ ***ALTERNATIVE 2*** «bond breaking» 2C=O + H-H / 2(804) + 436 / 2044 «kJ» ✓ «bond forming» C=O + C-H + C-O + O-H / 804 + 414 + 358 + 463 / 2039 «kJ» ✓ $\Delta H^{\ominus}$ «= 2044 – 2039» = «+»5 «kJ mol^{-1}» ✓	*Award **[3]** for correct final answer.*	**3**

2	**c**	**iii**	H-H ***AND*** it can only occur in the H_2 molecule ✓	*Accept H-H **AND** does not require averaging.*	**1**

2	**g**		+2 ✓	*Do **not** accept 2 or 2+.*	**1**

9.3-S4 May 2022 TZ1 · Paper 2 · Q3 parts (b)(i), (b)(ii)

3.	**b**	**i**	*bonds broken*: N≡N + 3(H-H) /«1 mol×»945 «kJ mol^{-1}» + 3«mol»×436 «kJ mol^{-1}» / 945 «kJ» + 1308 «kJ» / 2253 «kJ» ✓ *bonds formed*: 6(N-H) / 6«mol»×391 «kJ mol^{-1}» / 2346 «kJ» ✓ ΔH = «2253 kJ - 2346 kJ = » -93 «kJ» ✓	*Award **[2 max]** for (+)93 «kJ».*	**3**
3.	**b**	**ii**	«N-H» bond enthalpy is an average «and may not be the precise value in NH_3» ✓	*Accept ΔH_f data are more accurate / are not average values.*	**1**

9.3-S5 May 2021 TZ1 · Paper 2 · Q4 parts (c)(i), (d)(i), (d)(ii) · also 1.2

4.	**c**	**i**	$CH_4(g) + H_2O(g) \rightleftharpoons CH_3OH(l) + H_2(g)$ ✓	*Accept arrow instead of equilibrium sign.*	**1**

4.	**d**	**i**	$\sum$ bonds broken $= 4 \times 414$ «kJ» $+ 2 \times 463$ «kJ» / 2582 «kJ» ✓ $\sum$ bonds formed $= 1077$ «kJ» $+ 3 \times 436$ «kJ» / 2385 «kJ» ✓ ΔH «$= \sum$ bonds broken $- \sum$ bonds formed $=($ 2582 kJ $-$ 2385 kJ)» $=$ «+»197«kJ» ✓	*Award **[3]** for final correct answer.* Award ***[2 Max]*** for final answer of −197 «kJ»	**3**
4.	**d**	**ii**	bond energies are average values «not specific to the compound» ✓		**1**

9.3-S6 May 2021 TZ2 · Paper 2 · Q4 parts (b), (c), (d) · also 4.1, 4.2

4.	**b**		*Bonds broken:* 2(C–C) + 1(C=C) + 8(C–H) + 6O=O / 2(346) + 1(614) + 8(414) + 6(498) / 7606 **«**kJ**»** ✓ *Bonds formed:* 8(C=O) + 8(O–H) / 8(804) + 8(463) / 10 136 **«**kJ**»** ✓ *Enthalpy change:* **«**Bonds broken – Bonds formed = 7606 kJ – 10 136 kJ =**»** –2530 **«**kJ**»** ✓	*Award **[3]** for correct final answer.* *Award **[2 max]** for **«**+**»** 2530 **«**kJ**»**.*	**3**
4.	**c**		<table><tr><td>*Carbon*</td><td>I</td><td rowspan="2">***AND***</td><td>II</td></tr><tr><td>*Hybridization*</td><td>sp^3</td><td>sp^2 ✓</td></tr></table>		**1**
4.	**d**		*Sigma (σ):* ✓ *Pi (π):* ✓	*Accept any diagram showing end to end/direct overlap of atomic/hybridized orbitals and electron density concentrated between nuclei.* *Accept any diagram showing sideways overlap of unhybridized p/atomic orbitals and electron density above and below plane of bond axis.*	**2**

9.3-S7 Nov 2020 · Paper 2 · Q3 parts (a), (b) · also 9.2

3.	**a**		*Bonds broken*: 8(C–H) + 2(C–C) + 5(O=O) / 8 × 414 **«**kJ mol^{-1}**»** + 2 × 346 **«**kJ mol^{-1}**»** + 5 × 498 **«**kJ mol^{-1}**»** / 6494 **«**kJ**»** ✓ *Bonds formed*: 6(C=O) + 8(O–H) / 6 × 804 **«**kJ mol^{-1}**»** + 8 × 463 **«**kJ mol^{-1}**»** / 8528 **«**kJ**»** ✓ **«**Enthalpy change = bonds broken − bonds formed = 6494 kJ − 8528 kJ =**»** –2034 **«**kJ**»** ✓	*Award **[3]** for correct final answer.*	**3**
3.	**b**		4(–241.8 **«**kJ**»**) ***AND*** 3(–393.5 **«**kJ**»**) ***AND*** **«**1**»**(–105 **«**kJ**»**) ✓ **«**$\Delta H^{\ominus}$ = 4(–241.8 **«**kJ**»**) + 3(–393.5 **«**kJ**»**) **–** **«**1**»**(–105 **«**kJ**»**) =**»** − 2043 **«**kJ**»** ✓	*Award **[2]** for correct final answer.* *Award **[1 max]** for –2219 **«**kJ**»**.*	**2**

9.3-S8 May 2018 TZ1 · Paper 2 · Q3 parts (c)(i), (c)(ii), (c)(iii), (d) · also 4.1, 9.2

3.	**c**	**i**	$\Delta H^{\ominus}$ = bonds broken – bonds formed ✓ **«**$\Delta H^{\ominus}$ = 3(C≡C) – 6(C⎓C)$_{benzene}$ / 3 × 839 – 6 × 507 / 2517 – 3042 =**»** –525 **«**kJ**»** ✓	*Award **[2]** for correct final answer.* *Award **[1 max]** for "+525 **«**kJ**»**".* *Award **[1 max]** for:* ***«**$\Delta H^{\ominus}$ = 3(C≡C) – 3(C–C) – 3(C=C) / 3 × 839 – 3 × 346 – 3 × 614 / 2517 – 2880 =**»** –363 **«**kJ**»**.*	**2**
3.	**c**	**ii**	$\Delta H^{\ominus}$ = $\Sigma\Delta H_f$ (products) – $\Sigma\Delta H_f$ (reactants) ✓ **«**$\Delta H^{\ominus}$ = 49 kJ – 3 × 228 kJ =**»** –635 **«**kJ**»** ✓	*Award **[2]** for correct final answer.* *Award **[1 max]** for "+635 **«**kJ**»".*	**2**
3.	**c**	**iii**	ΔH_f values are specific to the compound ***OR*** bond enthalpy values are averages **«**from many different compounds**»** ✓ condensation from gas to liquid is exothermic ✓	*Accept "benzene is in two different states **«**one liquid the other gas**»**" for M2.*	**2**
3.	**d**		equal C–C bond **«**lengths/strengths**»** ***OR*** regular hexagon ***OR*** **«**all**»** C–C have bond order of 1.5 ***OR*** **«**all**»** C–C intermediate between single and double bonds ✓	*Accept "all C–C–C bond angles are equal".*	**1**

9.3-S9 May 2018 TZ2 · Paper 2 · Q5 parts (a), (b)(i), (b)(ii), (b)(iii) · also 9.2

5.	**a**		*bonds broken:* 4(C–H) + 2(H–O) / 4(414) + 2(463) / 2582 **«**kJ**»** ✓ *bonds made:* 3(H–H) + C≡O / 3(436) + 1077 / 2385 **«**kJ**»** ✓ ΔH **«**= $\sum BE_{(\text{bonds broken})} - \sum BE_{(\text{bonds made})}$ = 2582 − 2385**»** = **«**+**»** 197 **«**kJ**»** ✓	*Award **[3]** for correct final answer.* *Award **[2 max]** for –197 «kJ».*	**3**
5.	**b**	**i**	$\Delta H_f^{\ominus}$ for any element = 0 **«**by definition**»** ***OR*** no energy required to form an element **«**in its stable form**»** from itself ✓		**1**
5.	**b**	**ii**	$\Delta H^{\ominus}$ **«** = $\sum \Delta H_{f\,(\text{products})}^{\ominus} - \sum \Delta H_{f\,(\text{reactants})}^{\ominus}$ = −111 + 0 − [−74.0 + (−242)]**»** = **«**+**»** 205 **«**kJ**»** ✓		**1**
5.	**b**	**iii**	**«**bond enthalpies**»** averaged values **«**over similar compounds**»** ***OR*** **«**bond enthalpies**»** are not specific to these compounds ✓		**1**

9.3-S10 Nov 2016 · Paper 2 · Q1 parts (a)(iii), (b)(i), (b)(ii), (c), (d) · also 5.2, 6.1, 9.2

1.	**a**	**iii**	*Bonds broken*: $2C{\equiv}O + 3(H{-}H)$ / $2(1077\,kJ\,mol^{-1}) + 3(436\,kJ\,mol^{-1})$ / 3462 «kJ» ✓ *Bonds formed*: $2(C{-}O) + 2(O{-}H) + 4(C{-}H) + (C{-}C)$ / $2(358\,kJ\,mol^{-1}) + 2(463\,kJ\,mol^{-1}) + 4(414\,kJ\,mol^{-1}) + 346\,kJ\,mol^{-1}$ / 3644 «kJ» ✓ «Enthalpy change = bonds broken − bonds formed = 3462 kJ − 3644 kJ =» −182 «kJ» ✓	*Award **[3]** for correct final answer.* *Award **[2 max]** for «+»182 «kJ».*	**3**
1.	**b**	**i**	«$\Delta H = \Sigma\,\Delta H_f$ products $- \Sigma\,\Delta H_f$ reactants = $-454.8\,kJ\,mol^{-1} - 2(-110.5\,kJ\,mol^{-1})$ =» −233.8 «kJ» ✓		**1**
1.	**b**	**ii**	in (a)(iii) gas is formed and in (b)(i) liquid is formed ***OR*** products are in different states ***OR*** conversion of gas to liquid is exothermic ***OR*** conversion of liquid to gas is endothermic ***OR*** enthalpy of vapourisation needs to be taken into account ✓	*Accept product is «now» a liquid.* *Accept answers referring to bond enthalpies being means/averages.*	**1**

1.	**c**		*Ethene*: −2 ✓ *Ethane-1,2-diol*: −1 ✓	*Do not accept 2–, 1– respectively.*	**2**

1.	**d**		ethane-1,2-diol can hydrogen bond to other molecules «and ethene cannot» ***OR*** ethane-1,2-diol has «significantly» greater van der Waals forces ✓ hydrogen bonding is «significantly» stronger than other intermolecular forces ✓	*Accept converse arguments.* *Award **[0]** if answer implies covalent bonds are broken.*	**2**

9.3-S11 May 2015 TZ1 · Paper 2 · Q5 parts (b)(i), (b)(ii), (c), (g) · also 4.3, 8.2

(b) (i) energy needed to break (1 mol of) a bond in the gaseous state/phase;
(averaged over) similar compounds; **[2]**
Do not accept "similar bonds" instead of "similar compounds".
Concept of "similar" is important for M2.

(ii) $CH_3CH_2OH + 3O_2 \rightarrow 2CO_2 + 3H_2O$;
Bonds broken:
$347 + (5 \times 413) + 358 + 464 + (3 \times 498) / 4728\,(kJ)$ /
C–C + 5C–H + C–O + O–H + 3O=O;
Bonds made:
$(4 \times 746) + (6 \times 464) = 5768\,(kJ)$ / 4C=O + 6O–H;
$\Delta H = (4728 - 5768 =) -1040\,(kJ\,mol^{-1})$ / bonds broken – bonds formed; **[4]**
*Award **[4]** for correct final answer.*
*Award **[3]** for (+)1040 (kJ mol^{-1}).*

(c) heat loss (to the surroundings); **[1]**

(g) *Allotropes:*
*Any **three** allotropes for **[1]** from:*
diamond
graphite
fullerene
graphene;
Allow (carbon) nanotubes for graphene.
Accept C_{60} /C_{70} /buckminsterfullerene/bucky balls for fullerene.

Structures:
*Any three for **[3]** from:*
Diamond:
tetrahedral arrangement of (carbon) atoms/each carbon bonded to four others / sp^3 **and** 3D/covalent network structure;
Graphite:
each carbon bonded to three others (in a trigonal planar arrangement) / sp^2 **and** 2D / layers of (carbon) atoms;
Fullerene:
each (carbon) atom bonded to three others (in a trigonal arrangement) / sp^2 **and** joined in a ball/cage/sphere/connected hexagons and pentagons;
Accept "trigonal planar" for "each carbon atom bonded to three others" part in M4.
Graphene:
each carbon bonded to three others (in a trigonal arrangement) / sp^2 **and** 2D structure; **[4]**

9.3-S12 Nov 2014 · Paper 2 · Q2 parts (a), (b)(i), (b)(ii), (c) · also 9.2

2. (a) (bonds broken) C=C and O–H / $612 + 464 / 1076$;
(bonds formed) C–C and C–H and C–O / $347 + 413 + 358 / 1118$;

OR

(bonds broken) C=C and two O–H and four C–H / $612 + 4(413) + 2(464) / 3192$;
(bonds formed) C–C and five C–H and C–O and O–H / $347 + 5(413) + 358 + 464 / 3234$;

Ignore signs (+ and –) in M1 and M2. These two marks are awarded for recognizing the correct bonds.

enthalpy change $= -42\,(\text{kJ})$; ***[3]***

Correct sign is necessary for awarding M3.
*Award **[3]** for the correct final answer.*
Do not penalize candidates using the former Data Booklet bond energy values (348, 412 and 463) (final answer will then be -45(kJ)).

(b) (i) heat/enthalpy change when 1 mol of a compound/substance is formed;
from its elements in their standard states/at $100\,\text{kPa}/10^5\,\text{Pa}$; ***[2]***

Allow $1.01\times10^5\,Pa/101\,kPa/1\,atm$ *as an alternative to 100 kPa/10^5 Pa.*
Allow under standard conditions or standard ambient temperature and pressure as an alternative to 100 kPa/10^5 Pa.
Allow "energy needed/absorbed" as an alternative to "heat/enthalpy change".
Temperature is not required in definition, allow if quoted (eg, 298K / 25 °C).

(ii) $(-235)-(52-242)$ / $\Delta H = \Sigma\Delta H_f^{\ominus}(\text{products}) - \Sigma\Delta H_f^{\ominus}(\text{reactants})$;
$-45\,(\text{kJ})$; ***[2]***

*Award **[2]** for the correct final answer.*
*Award **[1]** for +45 or 45.*

(c) value in (b)(ii) (is more accurate) as values used in (a) are average values / value in (b)(ii) (is more accurate) as exact bond enthalpy depends on the surroundings of the bond / *OWTTE*; ***[1]***

9.3-S13 Nov 2012 · Paper 2 · Q5 parts (a), (b)(i), (b)(ii), (b)(iii), (b)(iv), (b)(v), (c)(ii), (c)(iii) · also 3.1, 4.1, 4.2, 9.2

5. (a) energy needed to break (1 mol of) a bond in a gaseous molecule/state/phase;
average calculated from a range of similar compounds / *OWTTE*; ***[2]***
Do not accept similar bonds instead of similar compounds.
M2 can be scored independently.

(b) (i) *Bonds breaking*:
$4 \times (N{=}O) + 4 \times (N{-}N) + 8 \times (C{-}H) + 4 \times (N{-}O) + 8 \times (C{-}N)$ /
$= (4)(607) + (4)(158) + (8)(413) + (4)(201) + (8)(286)$ /
$= 9456\ (\text{kJ mol}^{-1})$;

Bonds forming:
$4 \times (N{\equiv}N) + 4 \times (C{\equiv}O) + 8 \times (O{-}H)$ /
$= (4)(945) + (4)(1072) + (8)(464) / = 11780\ (\text{kJ mol}^{-1})$;

Enthalpy change:
$(9456 - 11780 =) - 2324\ (\text{kJ mol}^{-1})$;
M3 is scored from M1-M2.
*Award **[2 max]** for (+)2324 (kJ mol^{-1}).*
*Award **[3]** for –2324 (kJ mol^{-1}) without working.*

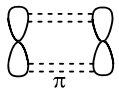
$\left(10.0 \times \frac{-2324}{296.2} =\right) - 78.5\ (\text{kJ})$; ***[4]***
M4 is scored from 10.0/296.2 × M3.
Allow answers of –78.2, –78 and –79 (kJ), but negative sign must be included.
*Award **[4]** for correct final answer.*

(ii) NH_4^+/ammonium / $N_2H_5^+$/hydrazinium / $CH_3NH_3^+$/methylammonium/ methanaminium / $H_2NO_3^+$ /nitrooxonium; ***[1]***

(iii) *Sigma bonds*:

result from head-on/end-on overlap of orbitals / *OWTTE*;
Allow symmetric (orbital) with respect to same plane / OWTTE.

Pi bonds:



result from sideways overlap of orbitals / *OWTTE*;
Allow antisymmetric (orbitals) with respect to (defining) plane (containing at least one atom) / OWTTE.

suitable diagrams showing σ **and** π bonds after formation; ***[3]***

*Award **[1 max]** for correct diagram without description given or **[2 max]** for*

(iv) *σ bonds*: 28;
π bonds: 4; ***[2]***

(v) mixing/combining/merging of atomic orbitals to form new/molecular orbitals (for bonding);
A: sp^2;
B: sp^3; ***[3]***
*Award **[1 max]** for M2 and M3 for sp^2 **and** sp^3 if **A** and **B** are not identified explicitly but do not award M2 and M3 if sp^3 is given for **A** and sp^2 for **B**.*

(ii) heat/enthalpy change when 1 mol of a compound is formed from its elements;
in their standard states / at 100 kPa/10^5 Pa/1bar (pressure); ***[2]***
Allow 1.01×10^5 Pa / 101kPa / 1 atm.
Allow under standard conditions or standard ambient temperature and pressure.
Temperatures not required in definition, allow if quoted (e.g. 298 K/25 °C – most common) but correct pressure value must be stated if given.
Only award M2 for a correct M1 or a near miss at M1.

(iii) $\Delta H^{\ominus} = (-485 + 111 + 239 =) -135$ (kJ mol^{-1}); ***[1]***

9.3-S14 Nov 2011 · Paper 2 · Q3 parts (a), (b), (c), (d) · also 5.2, 7.3

3. (a) energy required to break (1 mol of) a bond in a gaseous molecule/state;
Accept energy released when (1 mol of) a bond is formed in a gaseous molecule/ state / enthalpy change when (1 mol of) bonds are formed or broken in the gaseous molecule/state.

average values obtained from a number of similar bonds/compounds / *OWTTE*; ***[2]***

(b) $CH_3(CH_2)_3OH(l) + 6O_2(g) \rightarrow 4CO_2(g) + 5H_2O(l)$; ***[1]***
Allow C_4H_9OH *or* $C_4H_{10}O$ *for* $CH_3(CH_2)_3OH$.
Ignore state symbols.

(c) *Bonds broken:*
$(6)(O{=}O) + (3)(C{-}C) + (1)(O{-}H) + (1)(C{-}O) + (9)(C{-}H)$ /
$((6)(498) + (3)(347) + (1)(464) + (1)(358) + (9)(413) =) 8568\ (kJ\ mol^{-1})$;

Bonds formed:
$(8)(C{=}O) + (10)(O{-}H)$ / $((8)(746) + (10)(464) =) 10608\ (kJ\ mol^{-1})$;

$\Delta H = (8568 - 10608 =) -2040\ (kJ\ mol^{-1})$; ***[3]***
Award ***[3]*** *for correct final answer.*
Award ***[2]*** *for* $+2040\ (kJ\ mol^{-1})$.

(d) hydrogen bonding in butan-1-ol;
stronger than dipole-dipole attractions in butanal; ***[2]***
Accept converse argument.
Do not penalize dipole-dipole bonding instead of dipole-dipole attractions.

9.3-S15 May 2010 TZ1 · Paper 2 · Q5 parts (a)(i)

5. (a) (i) correct substitution of values and numbers of bonds broken / $(1\times945)+(3\times436)/2253$;

correct substitution of values and numbers of bonds made / $(6\times391)/2346$;

ΔH = (sum of energies of bonds broken) – (sum of energies of bonds formed)

= (2253–2346) −93 (kJ); ***[3]***

Ignore units.

*Award **[3]** for correct final answer.*

*Award **[2 max]** for +93 or 93.*

LESSON 9.4

Born-Haber Cycles

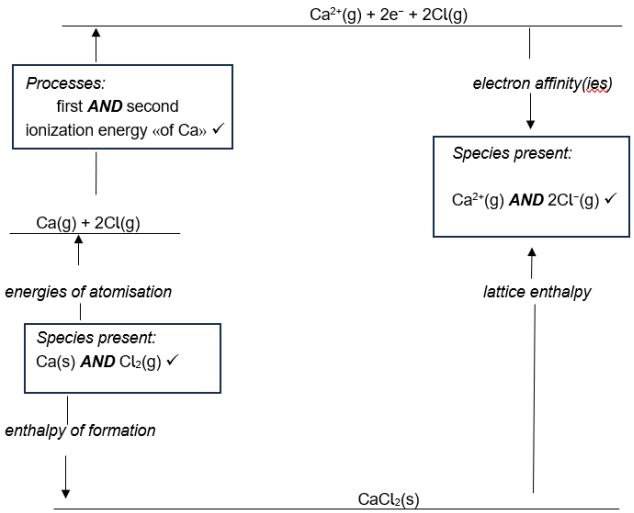
Answers for 35 multiple-choice and 4 structured questions

Part A - Multiple choice answers

9.4-M1	May 2025 TZ3 · P1A · Q22	C
9.4-M2	May 2024 TZ1 · P1 · Q16	D
9.4-M3	May 2024 TZ2 · P1 · Q18	A
9.4-M4	May 2022 TZ1 · P1 · Q16	A
9.4-M5	Nov 2021 · P1 · Q9	C
9.4-M6	Nov 2021 · P1 · Q16	A
9.4-M7	May 2021 TZ1 · P1 · Q17	D
9.4-M8	May 2021 TZ2 · P1 · Q16	B
9.4-M9	May 2019 TZ1 · P1 · Q17	A
9.4-M10	May 2019 TZ2 · P1 · Q16	B
9.4-M11	Nov 2018 · P1 · Q17	D
9.4-M12	May 2018 TZ2 · P1 · Q16	C
9.4-M13	Nov 2017 · P1 · Q18	C
9.4-M14	May 2017 TZ2 · P1 · Q16	B
9.4-M15	Nov 2016 · P1 · Q19	B
9.4-M16	Nov 2015 · P1 · Q16	A
9.4-M17	Nov 2015 · P1 · Q17	A
9.4-M18	May 2015 TZ2 · P1 · Q17	B
9.4-M19	Nov 2014 · P1 · Q17	C
9.4-M20	May 2014 TZ1 · P1 · Q16	A
9.4-M21	May 2014 TZ2 · P1 · Q16	B
9.4-M22	May 2014 TZ2 · P1 · Q18	D
9.4-M23	Nov 2013 · P1 · Q17	D
9.4-M24	May 2013 TZ1 · P1 · Q16	A
9.4-M25	May 2013 TZ2 · P1 · Q18	C
9.4-M26	Nov 2012 · P1 · Q16	D
9.4-M27	May 2012 TZ1 · P1 · Q16	A
9.4-M28	May 2012 TZ2 · P1 · Q16	A
9.4-M29	Nov 2011 · P1 · Q17	B
9.4-M30	May 2011 TZ1 · P1 · Q19	D
9.4-M31	May 2011 TZ2 · P1 · Q16	A
9.4-M32	Nov 2010 · P1 · Q17	D
9.4-M33	May 2010 TZ1 · P1 · Q17	A
9.4-M34	May 2010 TZ1 · P1 · Q18	C
9.4-M35	May 2010 TZ2 · P1 · Q20	A

Part B - Structured questions markscheme

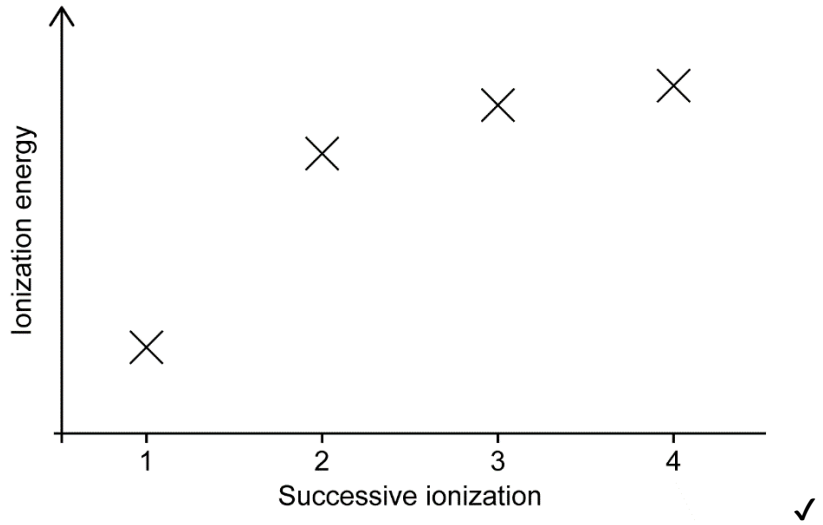
9.4-S1 May 2025 TZ1 · Paper 2 · Q4 parts (c)(i), (c)(ii)

4.	**(c)**	**(i)**		*Do not accept ionization energy for Processes.* *Accept ionization energies for Processes.*	**3**
4.	**(c)**	**(ii)**	Ca^{2+} has larger ionic radius «than Mg^{2+}» ***OR*** greater distance between ion centres «in $CaCl_2$» ✓	*Must be clear reference to ionic for M1*	**1**

9.4-S2 May 2023 TZ2 · Paper 2 · Q8 parts (b), (c)(i) · also 2.1

8.	**(b)**		ionic radius Ba^{2+} is greater than that of Mg^{2+} ✓ weaker attraction between «Ba^{2+} and Cl^-» ions ✓		**2**
8.	**(c)**	**(i)**	$1s^2\ 2s^2 2p^6\ 3s^2 3p^6 3d^7$ ✓		**1**

9.4-S3 May 2019 TZ1 · Paper 2 · Q3 parts (a), (d)(i), (d)(ii), (d)(iii), (f), (g)(i), (g)(ii), (h), (k) · also 1.2, 2.1, 4.1, 6.1, 9.2, 9.3

3.	**a**		 ✓	*Accept curve showing general trend.* *Award mark only if the energy difference between the first two points is larger than that between points 2/3 and 3/4.*	**1**

3.	**d**	**i**	$\frac{1}{2}\ O_2\,(g) \rightarrow O^{2-}(g)$ **«**$\Delta H_{atomisation}$ (O) + 1st EA + 2nd EA = 249 kJ mol^{-1} − 141 kJ mol^{-1} + 753 kJ mol^{-1} **=» «+»**861 **«**kJ mol^{-1}**»** ✓ $Na\,(s) \rightarrow Na^+\,(g)$ **«**$\Delta H_{atomisation}$ (Na) + 1st IE = 107 kJ mol^{-1} + 496 kJ mol^{-1} **=» «+»**603 **«**kJ mol^{-1}**»** ✓		**2**
3.	**d**	**ii**	lattice enthalpy = 861 **«**kJ mol^{-1}**»** + 2 × 603 **«**kJ mol^{-1}**»** −(−414 **«**kJ mol^{-1}**»**) ✓ **«**= +**»** 2481 **«**kJ mol^{-1}**»** ✓	*Award **[2]** for correct final answer.* *If given values are used:* *M1: lattice enthalpy = 850 **«**kJ mol^{-1}**»** + 2 × 600 **«**kJ mol^{-1}**»** −(−414 **«**kJ mol^{-1}**»**)* *M2: **«**= +**»** 2464 **«**kJ mol^{-1}**»***	**2**
3.	**d**	**iii**	K^+ ion is larger than Na^+ ***OR*** smaller attractive force because of greater distance between ion «centres» ✓		**1**

3.	**f**		$n(Na_2O_2)$ theoretical yield **«**= $\frac{5.00\text{ g}}{61.98\text{ g mol}^{-1}}$ **»** = 0.0807/8.07 × 10^{-2} **«**mol**»** ***OR*** mass of Na_2O_2 theoretical yield **«**= $\frac{5.00\text{ g}}{61.98\text{ g mol}^{-1}}$ × 77.98 g mol^{-1}**»** = 6.291 **«**g**»** ✓ % yield **«**= $\frac{5.50\text{ g}}{6.291\text{ g}}$ × 100**»** ***OR*** **«** $\frac{0.0705}{0.0807}$ x 100**»** = 87.4 **«**%**»** ✓	*Award **[2]** for correct final answer.*	**2**

3.	**g**	**i**	$\Sigma\Delta H_f$ products = 2 × (–1130.7) / –2261.4 **«**kJ**»** ✓ $\Sigma\Delta H_f$ reactants = 2 × (–510.9) + 2 × (–393.5) / –1808.8 **«**kJ**»** ✓ ΔH = **«**$\Sigma\Delta H_f$ products – $\Sigma\Delta H_f$ reactants = –2261.4 –(–1808.8) =**»** –452.6 **«**kJ**»** ✓	*Award **[3]** for correct final answer.* *Award **[2 max]** for "+ 452.6 «kJ»".*	**3**
3.	**g**	**ii**	only valid for covalent bonds ***OR*** only valid in gaseous state ✓		**1**
3.	**h**		bond in O_3 has lower enthalpy ***AND*** bond order is 1.5 **«**not 2**»** ✓	*Accept "bond in ozone is longer".*	**1**
3.	**k**		IV ✓		**1**

9.4-S4 May 2012 TZ2 · Paper 2 · Q3 parts (a), (b), (c)(i), (c)(ii)

3. (a) radius of the metal/positive ion/cation increases;
attraction (between ions) is weaker; ***[2]***

(b) higher charge (on cation);
smaller radius (of cation); ***[2]***

(c) (i) (standard enthalpy of) atomization/$\Delta H^{\ominus}_{\text{atomization}}$ / (standard enthalpy of) sublimation/$\Delta H^{\ominus}_{\text{sublimation}}$; ***[1]***

(ii) lattice enthalpy $= +1049\ \text{kJ mol}^{-1}$ (Data Booklet Table 13)
bond enthalpy $\text{F–F} = +158\ \text{kJ mol}^{-1}$ (Data Booklet Table 10)
first ionization energy $(\text{Li}) = +520\ \text{kJ mol}^{-1}$ (Data Booklet Table 7)
$\Delta H^{\ominus}_{\text{electron affinity}}\ (\text{F}) = -328\ \text{kJ mol}^{-1}$ (Data Booklet Table 7)
$\Delta H^{\ominus}_{\text{atomization}}$ of $\text{Li} = +159\ \text{kJ mol}^{-1}$ (given)

$\Delta H_f^{\ominus} = +159 + \frac{1}{2}(158) + 520 + (-328) - (1049)$;
$= -619\ \text{kJ mol}^{-1}$; ***[2]***

Accept use of theoretical value of lattice enthalpy, $+1030\ kJ\ mol^{-1}$:
$\Delta H_f^{\ominus} = +159 + \frac{1}{2}(158) + 520 + (-328) - (1030)$
$= -600\ kJ\ mol^{-1}$